

A VISUAL, SEVEN-DAY PRACTICE GUIDE

OneLab.

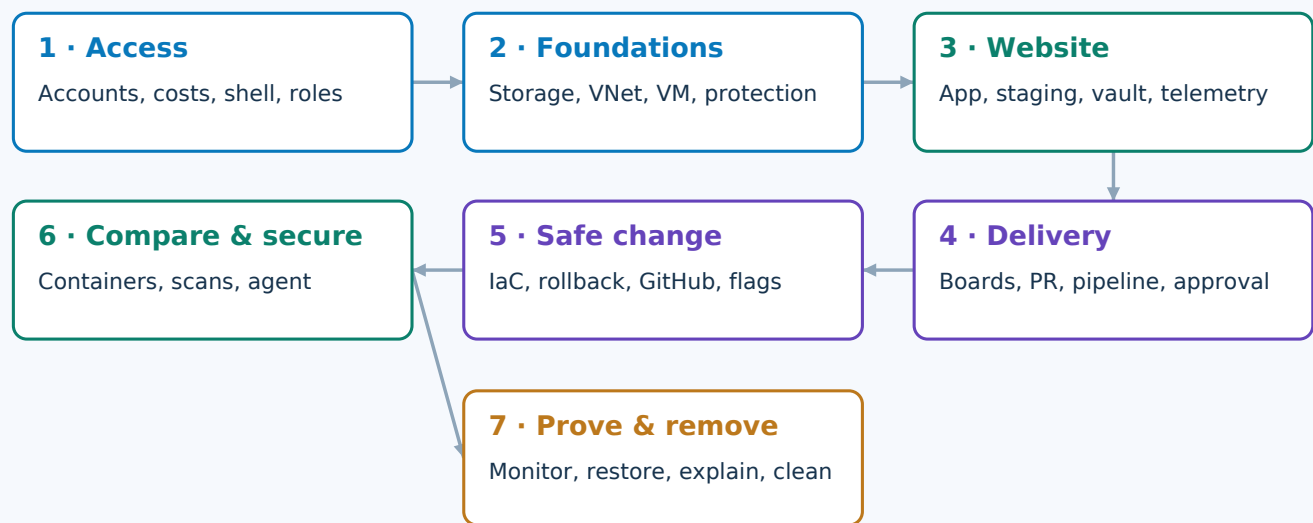
One small step at a time.

AZ-104 + AZ-400

Build one service. Learn how to run it, release it, repair it and remove it. Every action has a visual route, a simple reason and a check.

Your week: one connected service, built in layers

Four blocks each day. Finish one action, check it, then press Next.



Start with the HTML companion

AZ104-AZ400-Visual-OneLab.html shows one action at a time. Use this PDF's Day / Block bookmarks when you prefer a printable guide. Download OneLab-GitHub-Project.zip for the complete repository.

Visual edition · 22 September 2026

108 exercises · 583 illustrated action cards · 7 daily plans

Teaching figures are illustrations, not screenshots of completed Azure deployments. Microsoft reference screenshots are labelled separately. Use your own prefix and account values.

How to use this guide without getting lost

When you see...	Do this
Blue path	Open portal.azure.com, find the named resource, then follow the menu labels from left to right.
Purple path	Open dev.azure.com and your OneLab project. Organization settings and Project settings are different places.
Green terminal box	Use the exact named terminal. Cloud Shell Bash, VM RunShellScript and PowerShell are different execution places.
Numbered fields in a figure	Find the field in your own portal and use the lab value. The drawing is a teaching view; grouping can vary by rollout.
REPLACE_* or PREFIX	Replace the placeholder first. PREFIX is your saved P value from Day 1. Never type a sample subscription ID.
You should see / If stuck	Check the stated result, then use the targeted fix. Do not repeat a destructive command just because a portal is slow.
DESIGN / BLOCKED / WAITING	Record the real status. A diagram, unavailable permission or submitted job is not a completed live operation.

Only open today's block. The extra pages give each picture and command room; they are not additional isolated projects. Use the HTML search or the PDF bookmarks when you need one setting again.

A realistic week

Plan about 7–8 hours each day including breaks and retries. Most daily blocks total about 6–6½ hours of active work. These are estimates, not a timed beginner trial. Permissions, quota, pipeline capacity, backup and replication may extend the calendar time. Begin those checks early.

The lab maps the main skill families from the AZ-104 scope effective 17 April 2026 and AZ-400 scope effective 27 July 2026. Some live features need paid capacity, a licence, a domain or extra rights. Their connected extension or design route is marked. Use the practical challenges to judge independence; a single successful run cannot guarantee a pass. [R01, R02]

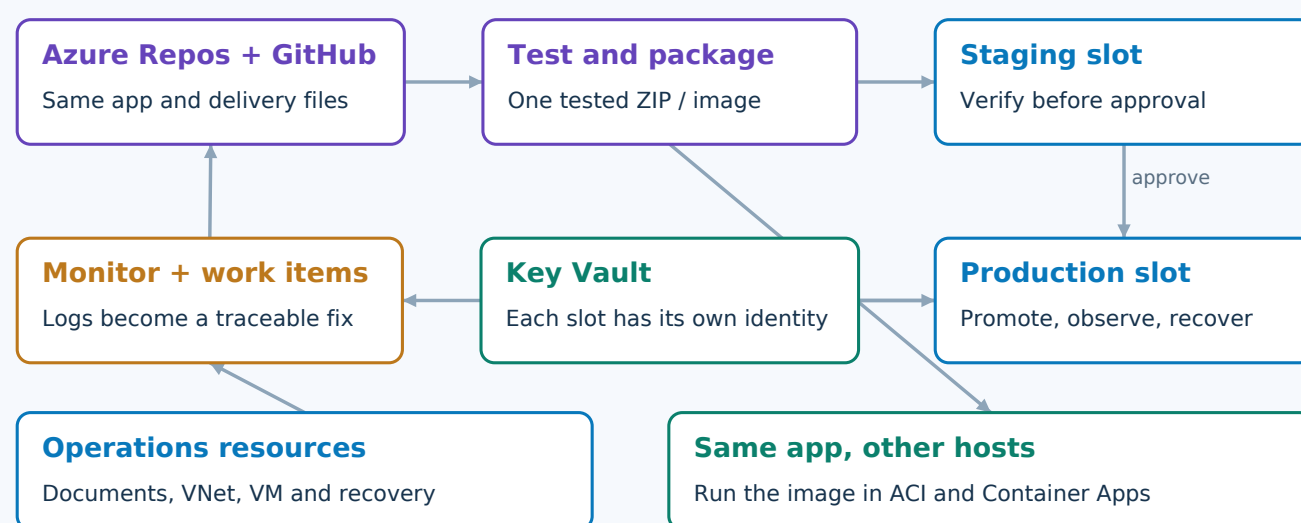
Before Day 1: accounts and your workspace

Open this tab	Purpose
https://portal.azure.com	Create, inspect and remove the lab's Azure resources.
https://dev.azure.com	Boards, Azure Repos and Azure Pipelines.
https://github.com	The same app's second repository and GitHub Actions.
https://entra.microsoft.com	Directory users, groups and application identities.
https://admin.microsoft.com	Licence assignment only if your account already has the required product and rights.

1. Use a computer and a modern browser. Keep private notes and only accounts, resources and email addresses you control.
2. If you do not have Azure, open <https://azure.microsoft.com/free/>. Choose the offered sign-up option. Sign in or create a Microsoft account. Complete the identity and billing checks shown, and read the offer and spending terms before accepting. Return to portal.azure.com when the subscription is active.
3. Day 1 shows how to check the subscription, permissions, costs and build capacity. A subscription Owner role does not automatically grant Entra user or licence administration rights.
4. Open the HTML companion from your downloaded files. Its Download project ZIP button saves the same ZIP supplied separately. The ZIP is also attached inside this PDF; desktop PDF readers often show it in an Attachments/paperclip pane.
5. Create a private note of your current Day, Block and next action. Do not publish notes containing credentials or private identifiers.

One project: OneLab service and its operations

Storage holds project documents; the VM is an operations and recovery target.



Your Day / Block navigation

Use these entries or the PDF outline/bookmarks. Each task keeps its Day / Block label in the action heading. The HTML companion is the easiest way to show only one action.

OneLab.One small step at a time.	1
How to use this guide without getting lost	2
Before Day 1: accounts and your workspace	3
Your Day / Block navigation	4
Get comfortable and set up access	6
1A · Open the accounts and check access	7
1B · Learn the terminal once	25
1C · Add a learner and give limited access	43
1D · Add guardrails and stop for today	65
Store files and build the network	76
2A · Save, recover and share a small file	77
2B · Copy files and compare shared storage	97
2C · Connect two private networks	110
2D · Create the VM and start protection	122
Put your website online	141
3A · Test the network paths safely	142
3B · Run the supplied app and publish it	158
3C · Add staging and protected settings	174
3D · Turn on monitoring	190
Let a pipeline release your changes	198
4A · Plan one change and save the code	199
4B · Protect Git and connect the pipeline	214
4C · Run the full release once	234

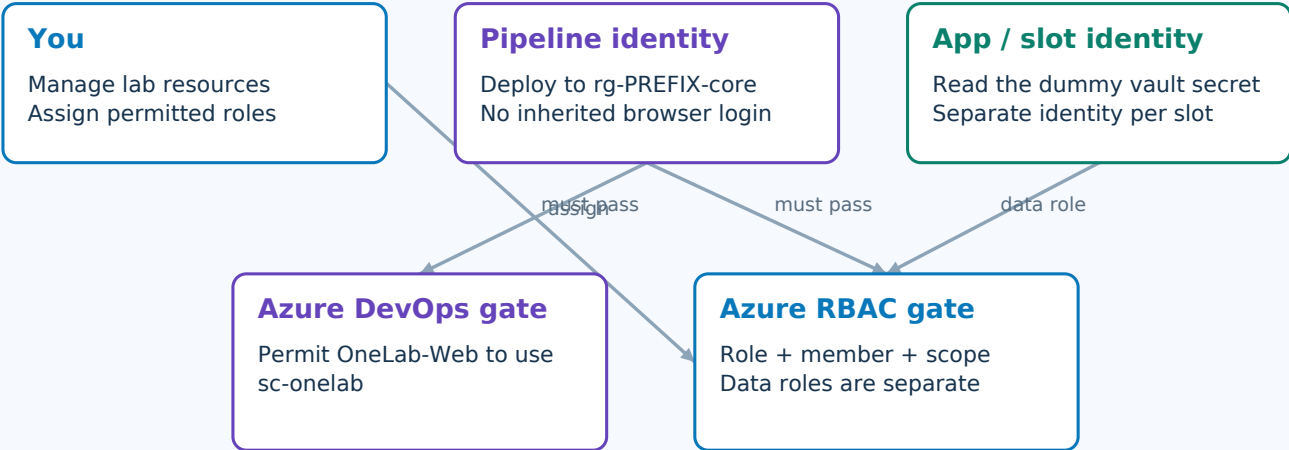
4D · Keep a version you can reuse	250
Practise safe changes in both platforms	258
5A · Preview and apply a small infrastructure file	259
5B · Rollback, flags and database order	272
5C · Use GitHub with the same application	283
5D · Read the pipeline controls	302
Try other hosts and secure the delivery	310
6A · Compare VM choices and repair drift	311
6B · Run OneLab in two container hosts	325
6C · Read and act on security findings	338
6D · See how an agent runs a job	356
Recover, explain and clean up	363
7A · Find the evidence and trigger an alert	364
7B · Restore a copy of the service	382
7C · Rehearse recovery and explain your work	393
7D · Save your work and remove the lab	405
Coverage map: where each skill family is taught	415
Troubleshooting: find the failing boundary	417
Plain-English glossary	419
Source files: everything needed to run the project	421
Publishing the prepared GitHub repository	434
Validation and source notes	435

DAY 1 OF 7

Get comfortable and set up access

You can find your subscription, use the terminal and give a test user the right access.

Three permissions you must keep separate



Start today

Have your own practice Azure account and a browser ready. Keep a private notes file. Use accounts and email addresses you control.

Start early and track the job

Request Azure Pipelines capacity today if it is missing. Quota, tenant permissions and paid-feature access also need checking today.

Block	Today's work	Minutes
A	Open the accounts and check access	90
B	Learn the terminal once	90
C	Add a learner and give limited access	105
D	Add guardrails and stop for today	60

1A · Open the accounts and check access

Planning estimate: 90 minutes. Finish with: Correct Azure account, a budget and the OneLab DevOps project.

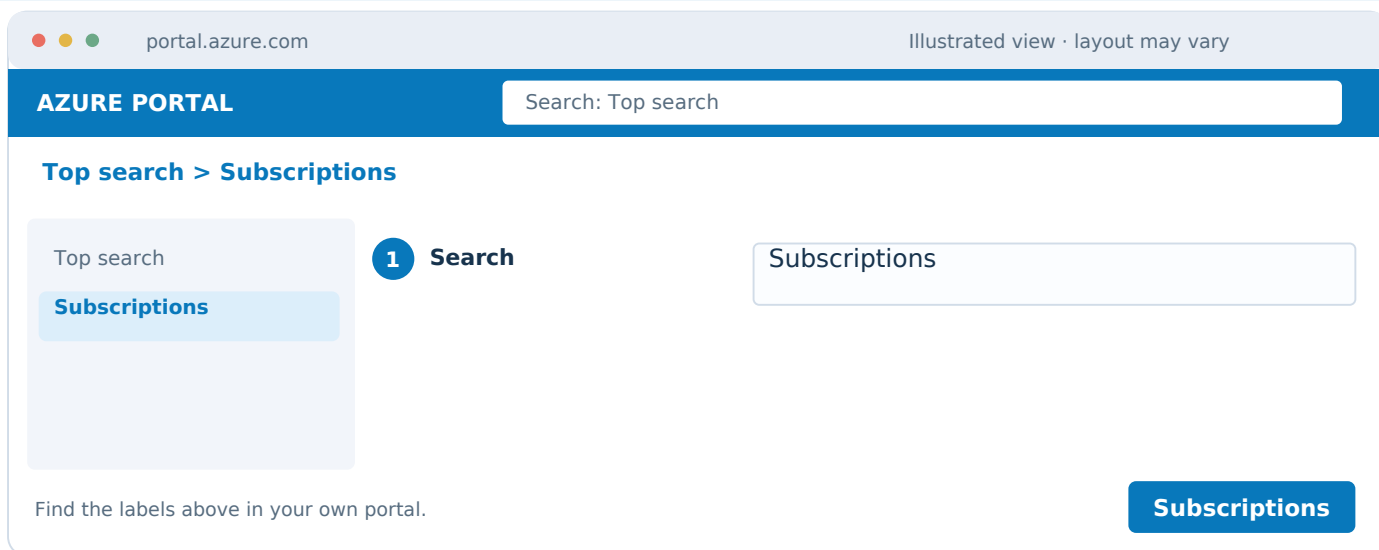
1A.01 Check your Azure subscription

AZ-104 + AZ-400 · CORE

Why this task: A subscription is the billing and resource container. An Entra tenant is the identity directory; they are different things.

1A.01 · ACTION 1 OF 5

AZURE PORTAL · Top search → Subscriptions



Illustrated navigation view · use your own saved names and IDs

Why this action: A subscription is the billing and resource container. An Entra tenant is the identity directory; they are different things.

1. Sign in.
2. In the top search box, type Subscriptions and select the matching service.

1A.01 · ACTION 2 OF 5**AZURE PORTAL · Subscriptions → Your subscription → Overview**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Subscriptions

Subscriptions > Your subscription > Overview

Subscriptions
Your subscription
Overview

1 Subscription name Confirm YOUR practice subscription

2 Subscription ID Copy YOUR value

3 Directory ID Copy YOUR value

Find the labels above in your own portal. **Copy**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Open your practice subscription.
2. On Overview, record its name, Subscription ID and Directory ID in your private notes.

1A.01 · ACTION 3 OF 5**AZURE PORTAL · Your subscription → Access control (IAM) → View my access**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Your subscription

Your subscription > Access control (IAM) > View my access

Your subscription
Access control (IAM)
View my access

1 Current identity Your signed-in user

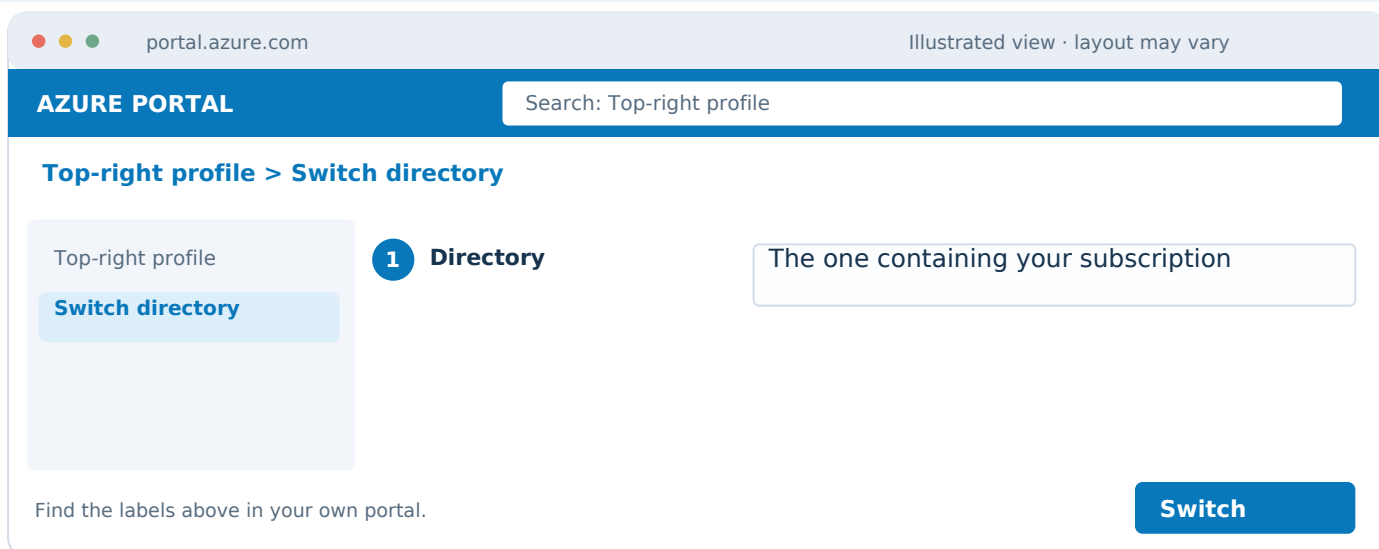
2 Role Review your actual assigned role

Find the labels above in your own portal. **View my access**

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

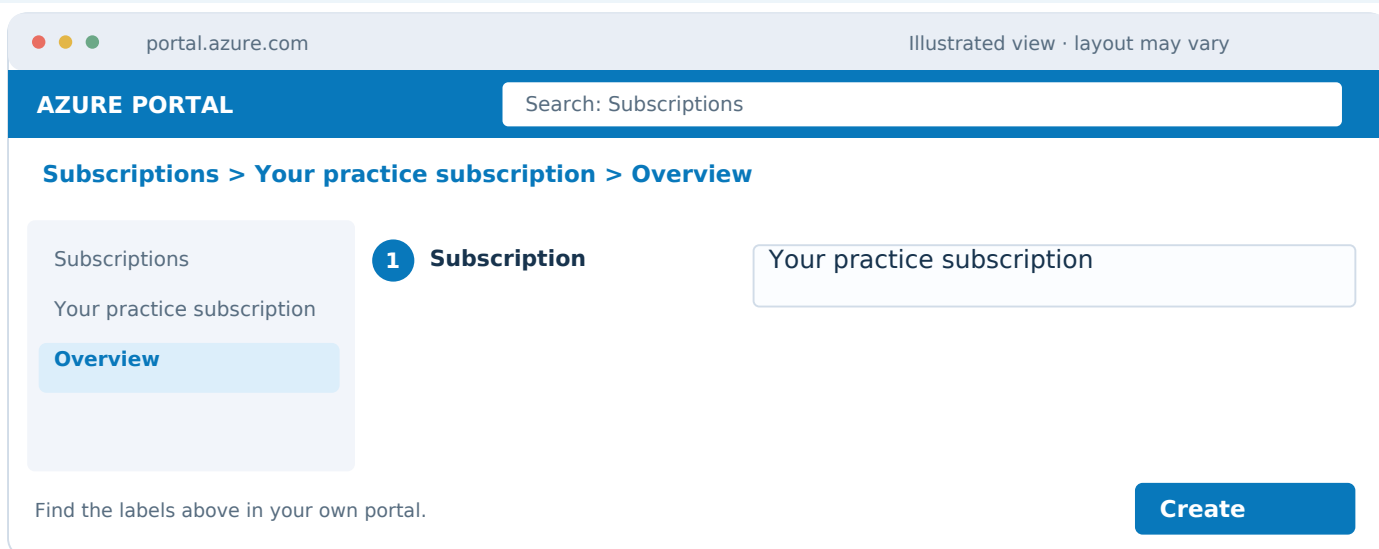
1. Select Access control (IAM) > View my access.
2. For this personal practice subscription, Owner is the simplest setup role.
3. A managed training environment can instead provide Contributor plus appropriate role-assignment rights.

1A.01 · ACTION 4 OF 5**AZURE PORTAL · Top-right profile → Switch directory**

Illustrated navigation view · use your own saved names and IDs

Why this action: A subscription is the billing and resource container. An Entra tenant is the identity directory; they are different things.

1. Open your profile at the top right.
2. Use Switch directory if the expected subscription is missing.
3. Do not create a second subscription just because the wrong directory is selected.

1A.01 · ACTION 5 OF 5**AZURE PORTAL · Subscriptions → Your practice subscription → Overview**

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Use your own practice resources.
2. An Azure subscription role does not automatically grant permission to create Entra users or assign licences.

You should see

You can see the correct subscription and understand your assigned role.

If it does not work

If access is missing, the subscription or tenant administrator must grant the relevant role. Follow the DESIGN route for tenant tasks you cannot perform; record the gap.

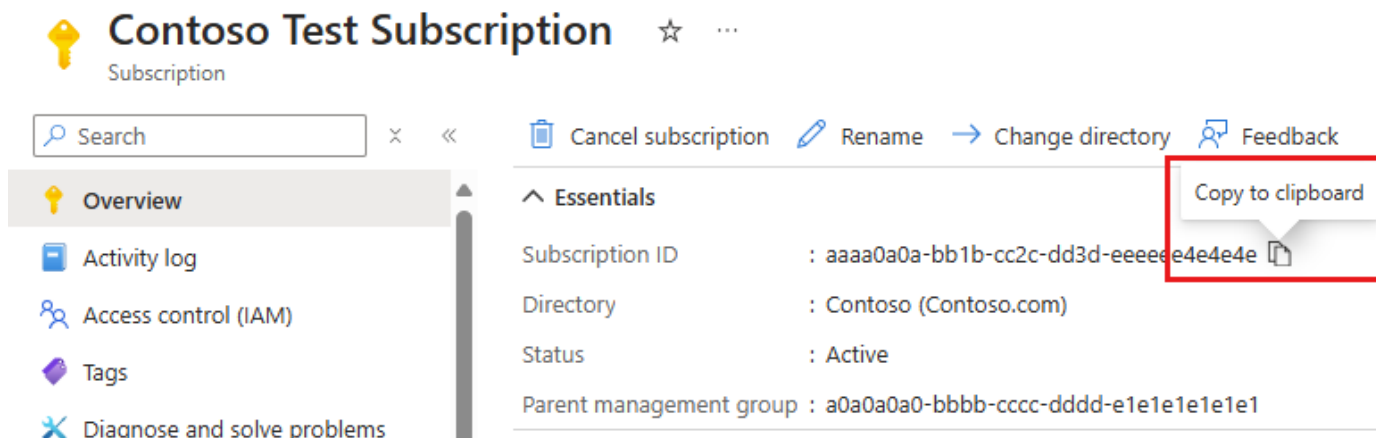
Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

Place	Address and purpose
Azure Portal	https://portal.azure.com - create and manage cloud resources
Microsoft Entra admin center	https://entra.microsoft.com - users, groups and identity settings
Azure DevOps	https://dev.azure.com - Boards, Repos, Pipelines and Artifacts
GitHub	https://github.com - the second automation route
Microsoft 365 admin center	https://admin.microsoft.com - licence assignment, if you have licences
Terminal	Azure Cloud Shell, Bash mode, opened from the Azure Portal toolbar

Microsoft reference screenshot · Screenshot showing the option to copy a subscription ID in the Azure portal.



Microsoft [reference source](#) · Sample values belong to that reference, not OneLab.

Microsoft reference screenshot · Screenshot showing the option to copy a tenant ID in the Azure portal.

Overview

Monitoring

Properties

Recommendations

Setup guides



Basic information

Name

Contoso

Tenant ID

aaaabbbb-0000-cccc-1111-dddd2222eeee

Copy to clipboard



Microsoft reference source · Sample values belong to that reference, not OneLab.

1A.02 Check permission and cost gates

AZ-104 + AZ-400 · CORE

Why this task: Some exam topics cannot be fully practised using only free services.

1A.02 · ACTION 1 OF 6

AZURE PORTAL · Cost Management + Billing → Cost Management → Cost analysis

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL

Search: Cost Management + Billing

Cost Management + Billing > Cost Management > Cost analysis

Cost analysis

Budgets

Cost alerts

1 Scope

Select your practice subscription

2 View

Accumulated costs

Find the labels above in your own portal.

Apply

Illustrated navigation view · use your own saved names and IDs

Why this action: The scope chooses which spending you are looking at. Check it before making a budget.

1. Search Cost Management + Billing.
2. Select your subscription scope and open Cost analysis.
3. Confirm you are looking at the practice subscription.

1A.02 · ACTION 2 OF 6**AZURE PORTAL · Cost Management → Budgets → + Add → Create a budget**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Cost Management

Cost Management > Budgets > + Add > Create a budget

Cost analysis
Budgets
Cost alerts

1 Name OneLab-budget

2 Reset period Monthly

3 Start / expiry Your chosen dates

4 Amount Your affordable amount in shown currency

Find the labels above in your own portal. **Next**

Illustrated navigation view · use your own saved names and IDs

Why this action: The amount is your warning level. Azure does not choose what you can afford.

1. Open Budgets > Add.
2. Name it OneLab-budget; choose a monthly period and an amount you can afford in the displayed billing currency.

1A.02 · ACTION 3 OF 6**AZURE PORTAL · Create a budget → Set alerts**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Create a budget

Create a budget > Set alerts

Cost analysis
Budgets
Cost alerts

1 Alert 1 Actual cost, 50%

2 Alert 2 Actual cost, 80%

3 Alert 3 Actual cost, 100%

4 Email recipients Your own email

Find the labels above in your own portal. **Create**

Illustrated navigation view · use your own saved names and IDs

Why this action: Three warning levels give you time to review and remove resources before your limit.

1. Add actual-cost alerts at 50%, 80% and 100%; add your own email address; review and create.
2. Confirm the budget appears.

1A.02 · ACTION 4 OF 6**YOUR BROWSER · azure.microsoft.com → Pricing calculator**

Your own browser tab Illustrated view · layout may vary

YOUR BROWSER Search: azure.microsoft.com

azure.microsoft.com > Pricing calculator

Cost analysis

Budgets

Cost alerts

- 1 Products**
- 2 Region**
- 3 Usage**
- 4 Currency**

App Service + VM + Storage + Monitor

Your chosen region

Your expected running time

Your billing currency

Find the labels above in your own portal. **Review estimate**

Illustrated navigation view · use your own saved names and IDs

Why this action: The calculator helps you see which services cost money before you create them.

1. Open <https://azure.microsoft.com/pricing/calculator/> in another tab.
2. Estimate an App Service Linux Standard S1 plan, a small Linux VM, storage, public IP, monitoring and any extensions before creating them.

1A.02 · ACTION 5 OF 6**SEE THE IDEA · What keeps billing while you pause?****A SMALL DECISION MAP****What keeps billing while you pause?**

- | | |
|--------------------------|--------------------------------|
| 1 Budget | Warns, does not stop resources |
| 2 VM deallocated | Compute stops |
| 3 Web app stopped | App Service plan still costs |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: A stopped screen or app can still have paid resources behind it.

1. Remember: a budget sends warnings; it does not stop resources.
2. Stopping a web app does not stop its App Service plan charge.
3. Deallocating a VM stops compute billing but leaves chargeable disks and possibly other resources.

1A.02 · ACTION 6 OF 6**SEE THE IDEA · Check the cost before an extension****A SMALL DECISION MAP****Check the cost before an extension**

- | | | |
|----------|------------------------|--|
| 1 | Extra resources | Private endpoints, Bastion, replication |
| 2 | Licences | Identity and code-security features |
| 3 | Decision | Use only the authorized affordable route |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Some parts need money or extra account rights. Check those before starting a long task.

1. Paid extensions include additional VM instances, Bastion tiers, private endpoints, recovery replication, some security features and licences.
2. The relevant task tells you when to create and remove them.

You should see

The budget exists and you know which resources continue billing while idle.

If it does not work

If Budgets is unavailable, check billing permissions. Use Cost analysis regularly and keep resources only for the required sessions. [R03]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Microsoft reference screenshot · Screenshot showing budget creation with monthly cost data.

Budget scoping

The budget you create will be assigned to the selected scope. Use additional filters like resource groups to have your budget monitor with more granularity as needed.

Scope Contoso (Demo)
[Change scope](#)

Filters + Add filter

Budget Details

Give your budget a unique name. Select the time window it analyzes during each evaluation period, its expiration date and the amount.

* Name

* Reset period Monthly

* Creation date 2021 March 1

* Expiration date 2023 February 28

Budget Amount

Give your budget amount threshold

Amount (\$) *

1 Suggested budget: \$148.9K based on forecast.

VIEW OF MONTHLY COST DATA

Sep 2020 - Aug 2021

LAST MONTH **\$97.7K** MAX (PAST 7 MONTH) **\$97.7K** MAX MONTHLY FORECAST **\$148.9K**

Legend: forecast (light green), actual (dark green), Budget (red dashed line)

Previous

Next >

Microsoft reference source · Sample values belong to that reference, not OneLab.

Microsoft reference screenshot · Screenshot showing alert conditions.

✓ Create a budget

✓ Set alerts

Configure alert conditions and send email notifications based on your spend.

*** Alert conditions**

Type	% of budget	Amount
Actual	90	133200
Forecasted	100	148000
Select type	Enter %	-

*** Alert recipients (email)**

Alert recipients (email)

It is recommended to add azure-noreply@microsoft.com to your email white list to ensure alert mails do not go to your spam folder.

Language preference

Select your preferred language for receiving the alert email for all recipients provided above. Default is the language associated to your enrollment.

Languages * Default

1 Your budget evaluation will begin in a few hours. [Learn more](#)

VIEW OF MONTHLY COST DATA

Sep 2020 - Aug 2021

LAST MONTH **\$97.7K** MAX (PAST 7 MONTH) **\$97.7K** MAX MONTHLY FORECAST **\$148.9K**

Legend: forecast (light green), actual (dark green), Budget (red dashed line), 90% thresh... (blue dashed line), 100% thresh... (blue dashed line)

Previous

Create

Microsoft reference source · Sample values belong to that reference, not OneLab.

Read the path → do the action → check the result

15

1A.03 Check region and quota

AZ-104 + AZ-400 · CORE

Why this task: A correct deployment can fail if your subscription lacks capacity or is restricted in that region.

1A.03 · ACTION 1 OF 4

AZURE PORTAL · Quotas → My quotas

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Quotas

Quotas > My quotas

Quotas
My quotas

1 Provider Compute

2 Subscription Your practice subscription

3 Region Your primary region

Find the labels above in your own portal.

Apply

Illustrated navigation view · use your own saved names and IDs

Why this action: A correct deployment can fail if your subscription lacks capacity or is restricted in that region.

1. Search Quotas > My quotas.
2. Select provider Compute, your subscription, and your preferred region.

1A.03 · ACTION 2 OF 4**AZURE PORTAL · My quotas → Compute**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: My quotas

My quotas > Compute

My quotas

Compute

1 Read Total Regional vCPUs

2 Also read Standard DSv5 family or your chosen VM family

Find the labels above in your own portal. [View](#)

Illustrated navigation view · use your own saved names and IDs

Why this action: A correct deployment can fail if your subscription lacks capacity or is restricted in that region.

1. Check Total Regional vCPUs and the VM family you plan to use.
2. The example VM is Standard_D2s_v5; choose an allowed equivalent if unavailable.

1A.03 · ACTION 3 OF 4**SEE THE IDEA · One location for the main lab****A SMALL DECISION MAP****One location for the main lab**

- | | |
|----------------------|--|
| 1 Example | East US |
| 2 Your choice | An allowed region with the needed services |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: A correct deployment can fail if your subscription lacks capacity or is restricted in that region.

1. Use one primary region, for example East US.
2. For disaster recovery, choose a supported different region later.
3. Resource availability and prices vary.

1A.03 · ACTION 4 OF 4**AZURE PORTAL · My quotas → Select quota → Request increase**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: My quotas

My quotas > Select quota > Request increase

My quotas

Select quota

Request increase

1 New limit

Only the capacity needed

2 Wait

Until approved before relying on it

Find the labels above in your own portal.

Submit

Illustrated navigation view · use your own saved names and IDs

Why this action: A correct deployment can fail if your subscription lacks capacity or is restricted in that region.

1. If quota is insufficient, select the relevant quota and Request increase, or choose an allowed region/size.
2. A submitted request is not an approved increase.

You should see

You have a region and VM size that your subscription permits.

If it does not work

Free or student subscriptions can restrict sizes or services. Record the restriction and use the provided design exercise if it cannot be resolved.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

1A.04 Create a private project and check agents

AZ-400 · CORE

Why this task: The organization contains projects; each project groups its Boards, repositories and pipelines.

1A.04 · ACTION 1 OF 6

AZURE DEVOPS · dev.azure.com → New organization

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: dev.azure.com

dev.azure.com > New organization

dev.azure.com

New organization

1 Name Your unique organization name

2 Region Your selected organization region

Find the labels above in your own portal.

Continue

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Open dev.azure.com and sign in with the account linked to your practice directory.
2. If you need an organization, use New organization and record its exact URL.

1A.04 · ACTION 2 OF 6**AZURE DEVOPS · Organization home → New project**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: Organization home

Organization home > New project

Organization home

New project

1 Project name OneLab

2 Visibility Private

3 Version control Git

4 Work item process Agile

Find the labels above in your own portal.

Create

Illustrated navigation view · use your own saved names and IDs

Why this action: The organization contains projects; each project groups its Boards, repositories and pipelines.

1. From the organization home choose New project.
2. Name OneLab; visibility Private; expand Advanced and select Git plus the Agile work-item process.
3. Create.

1A.04 · ACTION 3 OF 6**AZURE DEVOPS · Organization settings → Users**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: Organization settings

Organization settings > Users

Organization settings

Users

1 **User** Your working account

2 **Access level** Basic or another suitable entitlement

Find the labels above in your own portal.

Manage user

Illustrated navigation view · use your own saved names and IDs

Why this action: The organization contains projects; each project groups its Boards, repositories and pipelines.

1. Open Organization settings > Users.
2. Your working user needs Basic or another suitable entitlement for Repos/Pipelines.
3. Stakeholder access is useful for some work tracking but is not full developer access.

1A.04 · ACTION 4 OF 6**AZURE DEVOPS · Organization settings → Billing**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: Organization settings

Organization settings > Billing

Organization settings

Billing

1 **Azure subscription** Your authorized practice subscription

2 **Paid quantities** Review before changing

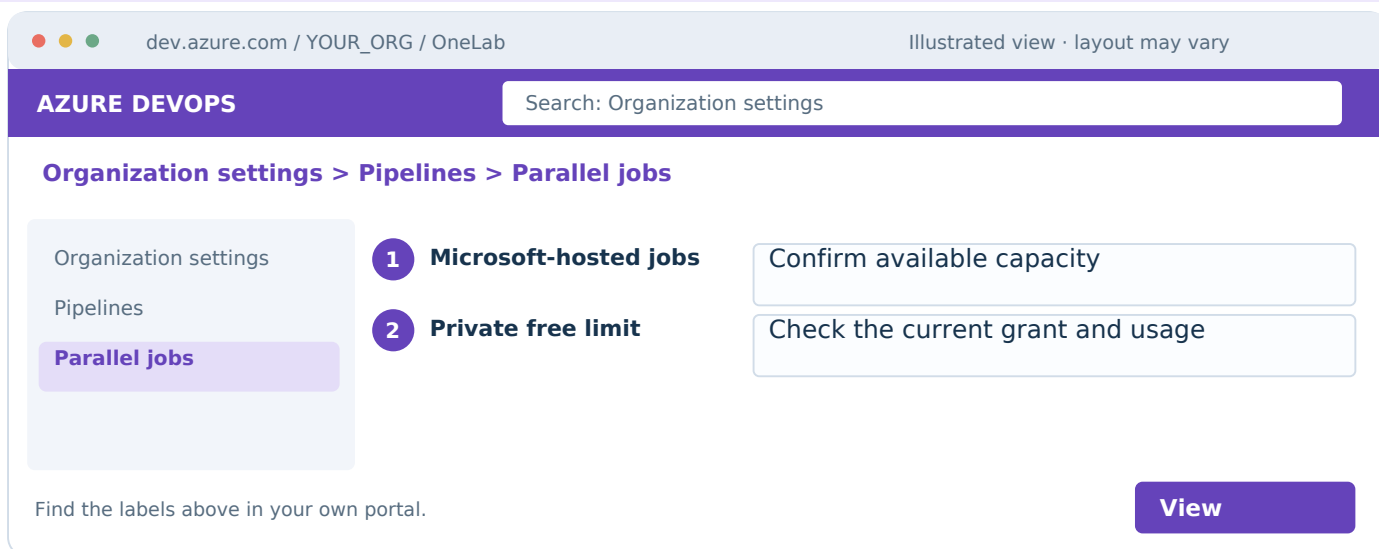
Find the labels above in your own portal.

Set up billing

Illustrated navigation view · use your own saved names and IDs

Why this action: The organization contains projects; each project groups its Boards, repositories and pipelines.

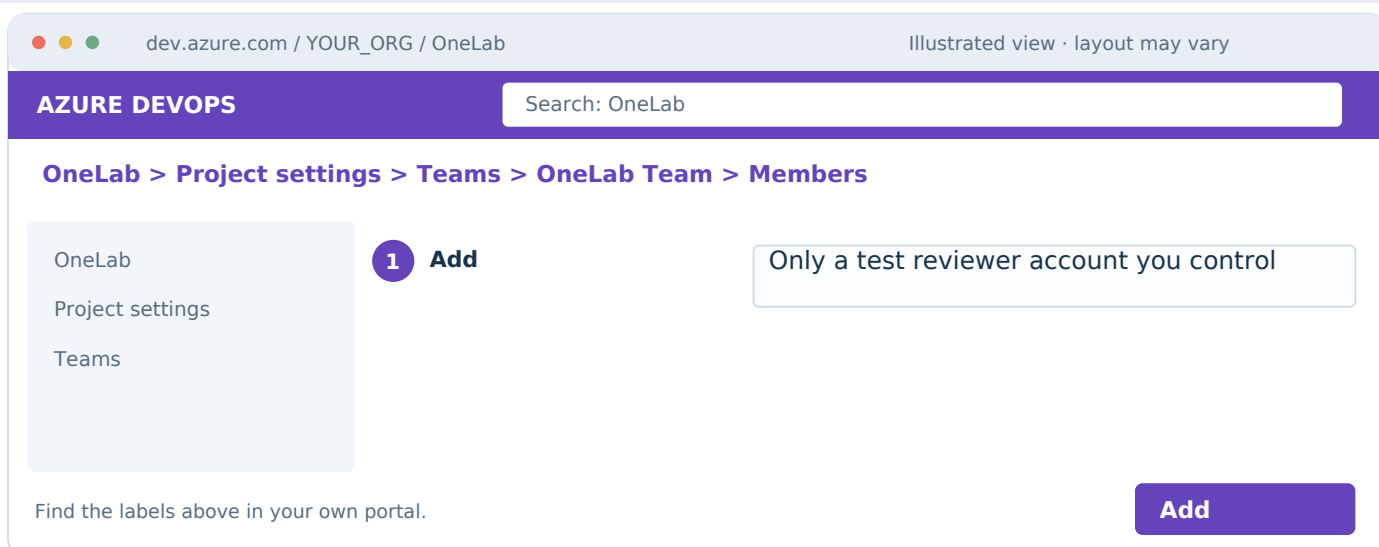
1. Open Organization settings > Billing.
2. If needed, link the authorized Azure subscription to activate the current free private-project parallel-job grant.
3. Review billing before enabling paid quantities.

1A.04 · ACTION 5 OF 6**AZURE DEVOPS · Organization settings → Pipelines → Parallel jobs**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open Organization settings > Pipelines > Parallel jobs.
2. Confirm available Microsoft-hosted parallelism.
3. The current private free allowance is one concurrent job, up to 60 minutes per job and 1,800 minutes per month.

1A.04 · ACTION 6 OF 6**AZURE DEVOPS · OneLab → Project settings → Teams → OneLab Team → Members**

Illustrated navigation view · use your own saved names and IDs

Why this action: The organization contains projects; each project groups its Boards, repositories and pipelines.

1. Open Project settings > Teams.
2. Use the default OneLab Team; open Members and add only a second test/reviewer account you control if available.

You should see

You have one private project and know whether a hosted job can run.

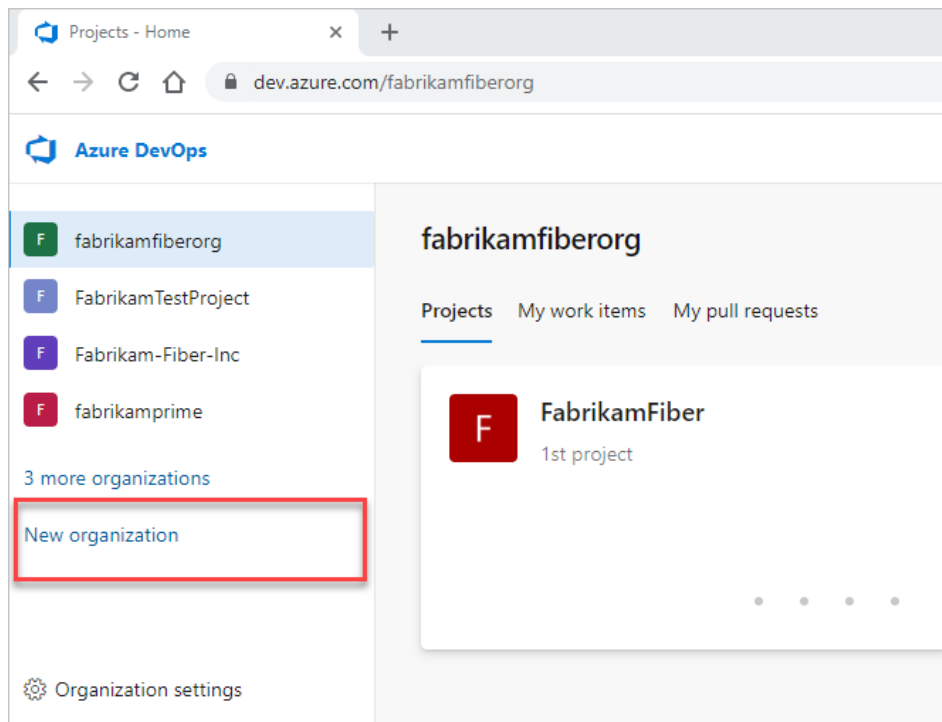
If it does not work

A queued pipeline with no hosted parallelism is an entitlement issue, not a YAML syntax issue. Linking billing can enable the free grant; additional usage may be chargeable. [R31]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Microsoft reference screenshot · Screenshot shows New organization button for selection.



Microsoft reference source · Sample values belong to that reference, not OneLab.

Microsoft reference screenshot · Screenshot of create new project form.

Create new project

×

Project name *

Description

Visibility

 **Private**

Only people you give access to will be able to view this. Want to create a public project? [Try GitHub](#)

▼ **Advanced**

[Microsoft reference source](#) · Sample values belong to that reference, not OneLab.

1B · Learn the terminal once

Planning estimate: 90 minutes. Finish with: A persistent folder, lab.env and a simple troubleshooting record.

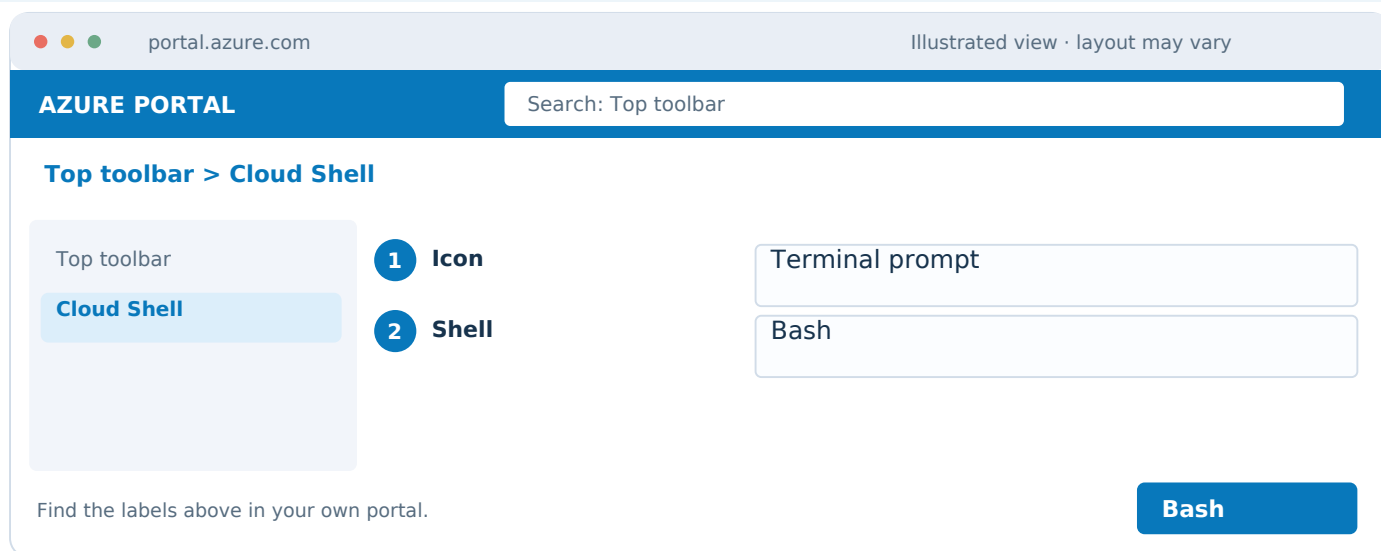
1B.01 Start Bash and keep your files

AZ-104 + AZ-400 · CORE

Why this task: Cloud Shell provides Azure CLI, Git, PowerShell and common tools in a browser.

1B.01 · ACTION 1 OF 5

AZURE PORTAL · Top toolbar → Cloud Shell



Illustrated navigation view · use your own saved names and IDs

Why this action: Cloud Shell provides Azure CLI, Git, PowerShell and common tools in a browser.

1. In the Azure Portal top toolbar, click the Cloud Shell icon, which looks like a terminal prompt.
2. Choose Bash, not PowerShell, for the main lab.

1B.01 · ACTION 2 OF 5**AZURE PORTAL · Cloud Shell → Getting started**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Cloud Shell

Cloud Shell > Getting started

Cloud Shell

Getting started

- Subscription** Practice subscription
- Storage** Mount storage
- Resource group** rg-onelab-shell

Find the labels above in your own portal. **Apply**

Illustrated navigation view · use your own saved names and IDs

Why this action: Cloud Shell provides Azure CLI, Git, PowerShell and common tools in a browser.

1. On the first-run storage screen, choose to mount storage so files can persist.
2. Select the practice subscription.
3. Create a separate resource group named rg-onelab-shell for the shell storage if needed.

1B.01 · ACTION 3 OF 5**AZURE PORTAL · Cloud Shell → Mount storage**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Cloud Shell

Cloud Shell > Mount storage

Cloud Shell

Mount storage

- Storage account** Your shell account
- File share** Your shell share

Find the labels above in your own portal. **Create storage**

Illustrated navigation view · use your own saved names and IDs

Why this action: Cloud Shell provides Azure CLI, Git, PowerShell and common tools in a browser.

1. Choose or create the requested storage account and file share.
2. Complete the storage setup.
3. This shell storage is separate from the lab storage exercises.

1B.01 · ACTION 4 OF 5**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

The dark code panel below is the terminal/editor view for this action.

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

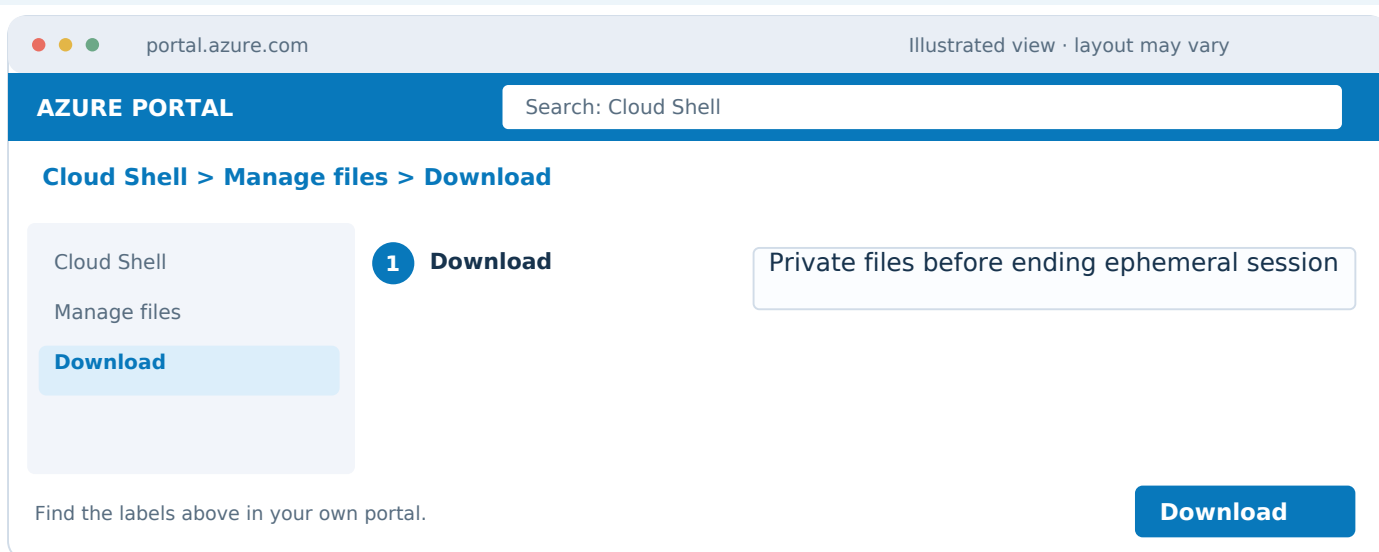
1. Wait until a prompt appears.
2. Run the commands below.
3. The clouddrive directory must exist; it is the persistent file share.

USE IN: BASH · COPY ONLY THE CODE

```
ls ~/clouddrive
mkdir -p ~/clouddrive/one-lab
cd ~/clouddrive/one-lab
pwd
az version
git --version
python --version
```

What it means: ls lists a directory. mkdir -p creates it if needed. cd moves into it. pwd shows where you are. Version commands tell you which tools are installed.

mkdir -p · Create the folder and missing parent folders; keep an existing folder.

1B.01 · ACTION 5 OF 5**AZURE PORTAL · Cloud Shell → Manage files → Download**

Illustrated navigation view · use your own saved names and IDs

Why this action: Cloud Shell provides Azure CLI, Git, PowerShell and common tools in a browser.

1. If your organization only permits an ephemeral session, download your files before ending it and upload them into the next session.
2. Do not assume the next shell remembers your files or variables.

You should see

pwd prints a path ending in clouddrive/one-lab.

If it does not work

If clouddrive is missing, complete the storage mounting setup from Cloud Shell settings before continuing. [R04]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

1B.02 Understand copying and editing

AZ-104 + AZ-400 · CORE

Why this task: A command must run in the correct shell and directory.

1B.02 · ACTION 1 OF 6

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

portal.azure.com

Illustrated view · layout may vary

CLOUD SHELL

Search: Cloud Shell

Cloud Shell > Bash > ~/clouddrive/one-lab

Cloud Shell

Bash

~/clouddrive/one-lab

1 Editor

2 Paste

3 Save

4 Exit

nano

Ctrl+Shift+V or right-click

Ctrl+O, Enter

Ctrl+X

Find the labels above in your own portal.

Open

Illustrated navigation view · use your own saved names and IDs

Why this action: A command must run in the correct shell and directory.

1. Copy only the code inside a command box.
2. Do not copy page numbers, headings, a dollar-sign prompt, or the explanation under it.

1B.02 · ACTION 2 OF 6**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

The screenshot shows the Azure portal interface for the Cloud Shell. The breadcrumb path is 'Cloud Shell > Bash > ~/clouddrive/one-lab'. On the left, a list of items includes 'Cloud Shell', 'Bash', and '~/clouddrive/one-lab'. A red circle with the number '1' and the label 'Paste' points to the '~/clouddrive/one-lab' item. To the right, a text box contains 'Ctrl+Shift+V or right-click'. At the bottom right, there is a green 'View' button. The page title is 'CLOUD SHELL' and the search bar contains 'Search: Cloud Shell'.

Illustrated navigation view · use your own saved names and IDs

Why this action: A command must run in the correct shell and directory.

1. Paste with Ctrl+Shift+V, right-click Paste, or your browser paste shortcut.
2. Press Enter after the full command.

1B.02 · ACTION 3 OF 6**CLOUD SHELL · The characters → and >> write to files**

The screenshot shows the Azure portal interface for the Cloud Shell. The breadcrumb path is 'The characters > and >> write to files'. On the left, a list of items includes 'The characters' and 'and >> write to files'. A red circle with the number '1' and the label 'Locate' points to the 'and >> write to files' item. A second red circle with the number '2' and the label 'Action' points to the same item. To the right, a text box contains 'and >> write to files'. Below it, another text box contains 'A backslash at the end of a Bash line joins it to the next line.' At the bottom right, there is a green 'View' button. The page title is 'CLOUD SHELL' and the search bar contains 'Search: The characters'.

Illustrated navigation view · use your own saved names and IDs

Why this action: A command must run in the correct shell and directory.

1. A backslash at the end of a Bash line joins it to the next line.
2. There must be no spaces after the backslash.
3. The characters > and >> write to files; they are not prompts to type separately.

1B.02 · ACTION 4 OF 6

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: Your note lets you resume here and explain the result later without guessing.

1. To create a file manually, type nano followed by its path, for example nano notes.txt.
2. Paste the file content.
3. Press Ctrl+O, then Enter to save; press Ctrl+X to exit.

USE IN: BASH · COPY ONLY THE CODE

```
nano notes.txt
```

What it means: Type a short note. Ctrl+O then Enter saves; Ctrl+X exits.

Nano keys · Ctrl+O then Enter saves. Ctrl+X exits. Paste only the intended file content.

1B.02 · ACTION 5 OF 6

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: A command must run in the correct shell and directory.

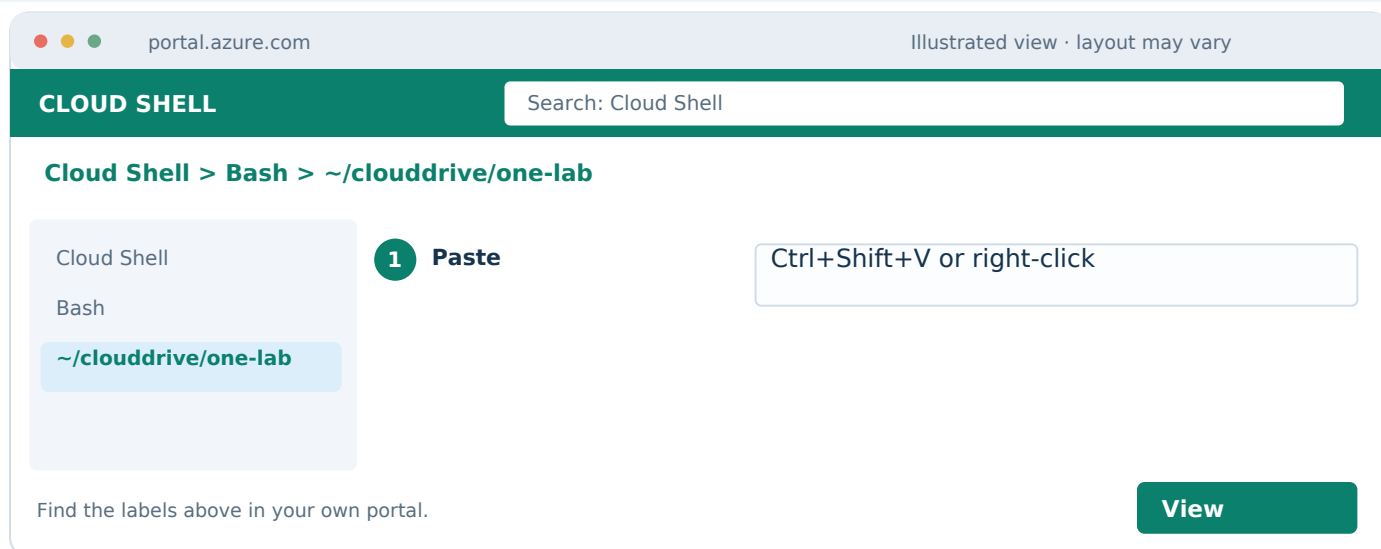
1. For a block beginning cat > file <<'EOF', paste the whole block, including the final EOF on its own line.
2. It creates a file; you do not also need Nano.
3. Run the small file-creation example here once, then read its output.

USE IN: BASH · COPY ONLY THE CODE

```
cat > notes.txt <<'EOF'
OneLab: I can create a file.
EOF
cat notes.txt
```

What it means: cat prints a file. The single quotes around EOF prevent Bash from expanding dollar signs inside the file content.

<<'EOF' ... EOF · Write a literal multi-line block; include the closing EOF alone on its own line.

1B.02 · ACTION 6 OF 6**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

Illustrated navigation view · use your own saved names and IDs

Why this action: A command must run in the correct shell and directory.

1. Press Ctrl+C to stop a running command or leave an unfinished pasted block.
2. Read the last error, fix it, and rerun that command.
3. Do not paste every task at once.

You should see

You can create, save and reopen a text file.

If it does not work

If the prompt is PS, switch Cloud Shell to Bash. Bash variables such as \$P do not work as PowerShell variables automatically.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

1B.03 Create your permanent naming file

AZ-104 + AZ-400 · CORE

Why this task: The same names must be used in every portal and script.

1B.03 · ACTION 1 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: The same names must be used in every portal and script.

1. Run az account list below and copy the ID of the correct subscription.
2. Open nano lab.env.

USE IN: BASH · COPY ONLY THE CODE

```
az account list --query "[].{Name:name,ID:id}" -o table
nano lab.env
```

What it means: -o table makes readable output. --query selects fields without changing Azure.

--query · Select or transform returned fields with a JMESPath expression; this does not itself change the resource.

-o · Azure CLI output format: table, tsv or none.

Nano keys · Ctrl+O then Enter saves. Ctrl+X exits. Paste only the intended file content.

1B.03 · ACTION 2 OF 5

FILE / WIKI EDITOR · Project file → lab.env

The dark code panel below is the terminal/editor view for this action.

Why this action: The same names must be used in every portal and script.

1. Type the file shown below.
2. Replace REPLACE_SUBSCRIPTION_ID with that ID.
3. Replace ali9267 with your own 6-10 lowercase letters/digits, starting with a letter.
4. Keep it the same for the entire lab.
5. Also replace eastus if you selected a different permitted region.
6. Open the portal region field to match the chosen region; use the CLI location name in lab.env.

USE IN: FILE: LAB.ENV · COPY ONLY THE CODE

```
export P=ali9267
export SUB=REPLACE_SUBSCRIPTION_ID
export LOC=eastus
export RG=rg-$P-core
export OPS=rg-$P-ops
export DR=rg-$P-dr
export STORE=${P}data
export COPY=${P}copy
export APP=$P-web
export PLAN=asp-$P
export KV=$P-kv
export ACR=${P}acr
export LAW=law-$P
export CONFIG=$P-config
```


What it means: These are names, not passwords. `export` also makes a value available to child processes. Storage and registry names use letters/digits with no hyphens.

export NAME=value · Set a Bash variable and make it available to child processes. Do not put spaces around the equals sign.

\$NAME / \${NAME} · Read a variable from this Bash session. Source `lab.env` first when these are lab names.

1B.03 · ACTION 3 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: The same names must be used in every portal and script.

1. Save the file, then run `source lab.env`.
2. This loads variables into this shell.
3. Every new shell must source the file again.

USE IN: BASH · COPY ONLY THE CODE

```
source lab.env
```

What it means: Load your saved settings into this Bash session. This does not create Azure resources.

source FILE · Load settings or activate an environment in the current Bash shell.

1B.03 · ACTION 4 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Run `az account set` and `az account show` below.
2. Compare the displayed subscription ID to the one in the portal.
3. The final two commands create the core and operations resource groups.
4. In the Azure Portal search Resource groups and confirm both names appear.

USE IN: BASH · COPY ONLY THE CODE

```
source lab.env
az account set --subscription "$SUB"
az account show --query '{Name:name,ID:id,Tenant:tenantId}' -o table
echo "$RG"
az group create -n "$RG" -l "$LOC" --tags Project=OneLab
az group create -n "$OPS" -l "$LOC" --tags Project=OneLab
```

What it means: Quotes preserve a variable as one argument. `-n` means name, `-l` location, and `-g` will mean resource group. Creating an existing group with the same settings is normally safe.

--subscription · Select the Azure subscription for this command.

--query · Select or transform returned fields with a JMESPath expression; this does not itself change the resource.

--tags · Add labels such as `Project=OneLab` to the resource.

-n · Short form of `--name` in these Azure commands.

-l · Short form of `--location` in these Azure commands.

-o · Azure CLI output format: `table`, `tsv` or `none`.

\$NAME / \${NAME} · Read a variable from this Bash session. Source `lab.env` first when these are lab names.

source FILE · Load settings or activate an environment in the current Bash shell.

1B.03 · ACTION 5 OF 5

SEE THE IDEA · Create your permanent naming file → Explain the decision

A SMALL DECISION MAP

Create your permanent naming file > Explain the decision

- | | | |
|----------|---------------|-----------------------|
| 1 | Locate | Explain the decision |
| 2 | Action | Never commit lab.env. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: The same names must be used in every portal and script.

1. Never commit lab.env.
2. It holds environment identifiers and may later contain private settings.
3. It is ignored by the supplied .gitignore.

You should see

The terminal subscription matches the portal and the printed core group name contains your prefix.

If it does not work

If a globally unique storage or app name is taken, choose a new prefix before creating resources.
After resources exist, do not change the prefix without intentionally starting a separate lab.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

1B.04 Make a session restart routine

AZ-104 + AZ-400 · CORE

Why this task: Your files may persist while exported variables and login context change.

1B.04 · ACTION 1 OF 3

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: Your files may persist while exported variables and login context change.

1. At the start of each new shell session, run the restart block below.
2. The extra Python lines do nothing until you have the source files on Day 3.

USE IN: BASH · COPY ONLY THE CODE

```
cd ~/clouddrive/one-lab
source lab.env
az account set --subscription "$SUB"
# After Day 3, Block B, recreate the temporary Python environment if needed.
if [ -f requirements-dev.txt ]; then
    python -m venv /tmp/onelab-venv
    source /tmp/onelab-venv/bin/activate
    python -m pip install -r requirements-dev.txt
fi
```

What it means: Run this block each time you return. Source stays on clouddrive; the temporary Python environment is safely rebuilt from the dependency file.

--subscription · Select the Azure subscription for this command.

\$NAME / \${NAME} · Read a variable from this Bash session. Source lab.env first when these are lab names.

source FILE · Load settings or activate an environment in the current Bash shell.

python -m MODULE · Run a module using the currently selected Python interpreter.

-r requirements file · Read the dependency list from the named requirements file.

1B.04 · ACTION 2 OF 3**CLOUD SHELL · Cloud Shell → Bash → Session restart**

The screenshot shows the Azure portal interface for the 'Cloud Shell' section. The breadcrumb path is 'Cloud Shell > Bash > Session restart'. On the left, a sidebar lists 'Cloud Shell', 'Bash', and 'Session restart' (highlighted). To the right of the sidebar, two numbered steps are shown: '1 Locate' and '2 Action'. The '2 Action' step is expanded, showing the text 'Session restart' and a description: 'At the end of a work block, record Day/Block, the last finished task and the exact next action in notes.txt.' A 'View' button is at the bottom right. The top of the page shows 'portal.azure.com' and 'Illustrated view · layout may vary'.

portal.azure.com Illustrated view · layout may vary

CLOUD SHELL Search: Cloud Shell

Cloud Shell > Bash > Session restart

Cloud Shell
Bash
Session restart

1 Locate

2 Action

Session restart

At the end of a work block, record Day/Block, the last finished task and the exact next action in notes.txt.

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. At the end of a work block, record Day/Block, the last finished task and the exact next action in notes.txt.
2. Write observations, never passwords.

1B.04 · ACTION 3 OF 3**CLOUD SHELL · Use Cloud Shell Manage files → Download with the path to notes**

The screenshot shows the Azure portal interface for the 'Use Cloud Shell Manage files' section. The breadcrumb path is 'Use Cloud Shell Manage files > Download with the path to notes'. On the left, a sidebar lists 'Use Cloud Shell Manage files' and 'Download with the path to notes' (highlighted). To the right of the sidebar, two numbered steps are shown: '1 Locate' and '2 Action'. The '2 Action' step is expanded, showing the text 'Download with the path to notes' and a description: 'Use Cloud Shell Manage files > Download with the path to notes.txt and lab.env to keep private local copies.' A 'View' button is at the bottom right. The top of the page shows 'portal.azure.com' and 'Illustrated view · layout may vary'.

portal.azure.com Illustrated view · layout may vary

CLOUD SHELL Search: Use Cloud Shell Manage files

Use Cloud Shell Manage files > Download with the path to notes

Use Cloud Shell Manage files
Download with the path to notes

1 Locate

2 Action

Download with the path to notes

Use Cloud Shell Manage files > Download with the path to notes.txt and lab.env to keep private local copies.

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Use Cloud Shell Manage files > Download with the path to notes.txt and lab.env to keep private local copies.
2. Do not publish them.

You should see

You can close and reopen Cloud Shell and resume in the same directory.

If it does not work

If source says file not found, run `pwd` and `ls`. Restore the downloaded file if the session was ephemeral.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED
Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

Naming rule: wherever portal instructions say `PREFIX`, replace it with the `P` value in `lab.env`. For example, if `P=ali9267`, `rg-PREFIX-core` means `rg-ali9267-core` and `PREFIXdata` means `ali9267data`. Do not type a dollar-sign Bash variable into a portal name field.

Name in the guide	Example when P=ali9267
Core resource group	rg-ali9267-core
Operations resource group	rg-ali9267-ops
Storage / registry	ali9267data / ali9267acr
Web app / plan / vault	ali9267-web / asp-ali9267 / ali9267-kv
Workspace / configuration store	law-ali9267 / ali9267-config
DevOps fixed names	Project OneLab; service connection sc-onelab; variable group onelab-settings

1B.05 Use a repeatable diagnosis record

Both exams · Use whenever a step fails

Why this task: A small structured record stops you from changing unrelated settings.

1B.05 · ACTION 1 OF 5

YOUR COMPUTER · notes.txt → Troubleshooting

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER

Search: notes.txt

notes.txt > Troubleshooting

notes.txt

Troubleshooting

1 Record

2 Compare

3 Change

4 Verify

Exact error without secrets

Expected versus actual

One targeted fix

Repeat the same check

Find the labels above in your own portal.

Open

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Create a simple section in notes.txt called Troubleshooting.
2. When an error occurs, copy its exact text without secrets and note the time, resource and operation.

1B.05 · ACTION 2 OF 5

YOUR COMPUTER · notes.txt → Troubleshooting

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER

Search: notes.txt

notes.txt > Troubleshooting

notes.txt

Troubleshooting

1 Compare

Expected versus actual

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: A small structured record stops you from changing unrelated settings.

1. State expected versus actual behavior in one sentence each.

1B.05 · ACTION 3 OF 5

YOUR COMPUTER · notes.txt → Troubleshooting

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER

Search: notes.txt

notes.txt > Troubleshooting

notes.txt

Troubleshooting

1 Locate

2 Action

Troubleshooting

Identify the boundary: account, identity, authorization, network, configuration, application or quota.

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Identify the boundary: account, identity, authorization, network, configuration, application or quota.

1B.05 · ACTION 4 OF 5

YOUR COMPUTER · notes.txt → Troubleshooting

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER

Search: notes.txt

notes.txt > Troubleshooting

notes.txt

Troubleshooting

1 Record

2 Change

Exact error without secrets

One targeted fix

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Make one targeted change, rerun the same check, and record the result.

1B.05 · ACTION 5 OF 5**YOUR COMPUTER · notes.txt → Troubleshooting**

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER

Search: notes.txt

notes.txt > Troubleshooting

notes.txt

Troubleshooting

1 Record

Exact error without secrets

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Write the root cause and how you proved recovery.
2. Link the record to the relevant work item if it came from a release.

You should see

Your notes file has a place to record a future error. You do not need to create an error today.

If it does not work

Avoid retrying a destructive command just because an unrelated portal operation is slow.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

Symptom	Check in order
Subscription missing	Profile directory > subscription filter > account identity > subscription access. Compare portal with az account show.
AuthorizationFailed	Correct subscription/resource scope > actual principal > required management role > deny assignment/policy > propagation delay.
Storage 403	Correct auth method > Blob/Files data role > expiry/permissions of SAS > firewall/public access > private DNS and routing.
Quota/SKU error	Region > VM family/total vCPU quota > subscription offer restrictions > current capacity > allowed alternative.
Policy denied creation	Read the policy assignment and failing property in the deployment error. Correct the property or use an authorized scope; do not delete organization policy.
App 502/503 or startup timeout	Deployment/build log > ZIP root > requirements installation > runtime version > gunicorn app:app > listening on 0.0.0.0:8000.
Secret remains a reference string	Production versus staging identity > correct Key Vault Secrets User role > versionless URI > vault network access > reference refresh.
Git push denied	Exact org/project/repo URL > Git permission > token scope/expiry > branch policy > remote history. Use a PR when main is protected.
Pipeline YAML error	Indentation and spaces > correct nesting (stages/jobs/steps) > template path/case > variable name > task inputs.
Pipeline waits forever	Agent parallel-job entitlement > available pool agent > pending resource authorization > approval/check timeout.
PR does not build	Azure Repos branch Build validation policy, not YAML pr. For GitHub inspect workflow event and target branch.
Federated login denied	Tenant/client ID > exact issuer/subject/audience > GitHub environment versus branch entity > service connection state > Azure role assignment.
Image pull fails	Registry hostname/tag/digest > runtime identity attachment > registry role/mode > network path > architecture compatibility.
VM cannot reach Internet	Explicit outbound method > NSG outbound rules > effective route > DNS > service endpoint/firewall constraints.
Private endpoint seems public	Client is actually in linked network > zone name > VNet link > expected DNS record > endpoint approval/subresource.
Load balancer backend unhealthy	Application listening > correct probe/port/path > NSG probe allowance > backend NIC membership > route and client test location.
KQL table not found	Selected workspace > instrumentation/diagnostic setting/DCR destination > generated traffic > ingestion delay > actual table schema.

Symptom	Check in order
Alert does not send email	Condition fired? > time window/aggregation > action group attached/tested > processing-rule suppression > inbox/junk.
Vault cannot be deleted	Protected items > replica/test cleanup > soft-deleted items > immutability/retention > locks > remaining dependencies.

1C · Add a learner and give limited access

Planning estimate: 105 minutes. Finish with: A test user/group and a Reader assignment, or a precisely recorded tenant restriction.

1C.01 Create a learner and a group

AZ-104 · CORE + permission-gated extension

Why this task: A test user lets you see the effect of permissions without relying on your administrator account.

1C.01 · ACTION 1 OF 6

MICROSOFT ENTRA · Identity → Users → All users → New user → Create new user

entra.microsoft.com

Illustrated view · layout may vary

MICROSOFT ENTRA Search: Identity

Identity > Users > All users > New user > Create new user

Users 1 **Directory** Your practice tenant

Groups

Password reset

Find the labels above in your own portal.

Create new user

Illustrated navigation view · use your own saved names and IDs

Why this action: A test user lets you see the effect of permissions without relying on your administrator account.

1. Open entra.microsoft.com.
2. Check the directory name in the top-right profile.
3. Go to Identity > Users > All users > New user > Create new user.

1C.01 · ACTION 2 OF 6**MICROSOFT ENTRA · Create user → Basics**

entra.microsoft.com Illustrated view · layout may vary

MICROSOFT ENTRA Search: Create user

Create user > Basics

Users
Groups
Password reset

1 User principal name learner1 + tenant domain

2 Display name OneLab Learner

3 Account enabled Yes

Find the labels above in your own portal. **Next**

Illustrated navigation view · use your own saved names and IDs

Why this action: A test user lets you see the effect of permissions without relying on your administrator account.

1. Use user principal name learner1 and the offered verified tenant domain.
2. Display name: OneLab Learner.
3. Leave Account enabled on.

1C.01 · ACTION 3 OF 6**MICROSOFT ENTRA · Create user → Password / Review**

entra.microsoft.com Illustrated view · layout may vary

MICROSOFT ENTRA Search: Create user

Create user > Password / Review

Users
Groups
Password reset

1 Password Keep temporary value private

2 First sign-in Change password

Find the labels above in your own portal. **Review + create**

Illustrated navigation view · use your own saved names and IDs

Why this action: A test user lets you see the effect of permissions without relying on your administrator account.

1. Create a temporary password using the portal.
2. Keep it privately and require the user to change it on first sign-in.
3. Review + create.

1C.01 · ACTION 4 OF 6**MICROSOFT ENTRA · Users → learner1 → Properties → Edit**

The screenshot shows the Microsoft Entra management console interface. At the top, the browser address bar displays 'entra.microsoft.com' and a note says 'Illustrated view · layout may vary'. The main header is 'MICROSOFT ENTRA' with a search bar containing 'Users'. Below the header, the breadcrumb navigation is 'Users > learner1 > Properties > Edit'. On the left, a sidebar lists 'Users' (selected), 'Groups', and 'Password reset'. The main content area has two numbered steps: '1 Department' with a text input field containing 'Training', and '2 Usage location' with a text input field containing 'Your actual country'. At the bottom left, it says 'Find the labels above in your own portal.' and at the bottom right is a blue 'Save' button.

Illustrated navigation view · use your own saved names and IDs

Why this action: A test user lets you see the effect of permissions without relying on your administrator account.

1. Open the new user > Properties.
2. Edit Department to Training and Usage location to the user's actual country.
3. Save.

1C.01 · ACTION 5 OF 6**MICROSOFT ENTRA · Groups → All groups → New group**

entra.microsoft.com Illustrated view · layout may vary

MICROSOFT ENTRA Search: Groups

Groups > All groups > New group

Users
Groups
Password reset

- Group type** Security
- Group name** OneLab-Readers
- Membership type** Assigned
- Members** learner1

Find the labels above in your own portal. **Create**

Illustrated navigation view · use your own saved names and IDs

Why this action: A test user lets you see the effect of permissions without relying on your administrator account.

1. Go to Identity > Groups > All groups > New group.
2. Group type Security; name OneLab-Readers; membership type Assigned.
3. Add learner1 as a member and create.

1C.01 · ACTION 6 OF 6**MICROSOFT ENTRA · Groups → OneLab-Readers → Members**

entra.microsoft.com Illustrated view · layout may vary

MICROSOFT ENTRA Search: Groups

Groups > OneLab-Readers > Members

Users
Groups
Password reset

- Member** learner1
- Then user properties** Copy user Object ID

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open the group and confirm learner1 appears under Members.
2. Copy the user's Object ID into private notes, clearly labelled user object ID.

You should see

The test user is a member of OneLab-Readers.

If it does not work

User creation needs a suitable Entra role such as User Administrator. Azure subscription Owner alone is not sufficient. Dynamic membership requires appropriate licensing; use Assigned in this lab.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED
Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

Entra answers who the person is. Azure RBAC answers what that person can do to a cloud resource. A group lets you grant access to several people together. A guest is an external identity invited into the directory. Use only accounts you control for invitation and email tests.

1C.02 Invite an external test identity

AZ-104 · CORE + permission-gated extension

Why this task: A guest can keep their existing sign-in while receiving limited access to your project.

1C.02 · ACTION 1 OF 4

MICROSOFT ENTRA · Identity → Users → New user → Invite external user

entra.microsoft.com

Illustrated view · layout may vary

MICROSOFT ENTRA

Search: Identity

Identity > Users > New user > Invite external user

Identity

Users

New user

1 Email

2 Display name

3 User type

Second email YOU control

OneLab Guest

Guest

Find the labels above in your own portal.

Invite

Illustrated navigation view · use your own saved names and IDs

Why this action: A guest can keep their existing sign-in while receiving limited access to your project.

- 1. Go to Users > New user > Invite external user.
- 2. Enter a second email account that you control.

1C.02 · ACTION 2 OF 4

MICROSOFT ENTRA · Identity → Users → New user → Invite external user

entra.microsoft.com

Illustrated view · layout may vary

MICROSOFT ENTRA

Search: Identity

Identity > Users > New user > Invite external user

Identity

Users

New user

1 Display name

2 User type

OneLab Guest

Guest

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: A guest can keep their existing sign-in while receiving limited access to your project.

- 1. Use display name OneLab Guest.
- 2. Review the invitation message and send the invitation.

1C.02 · ACTION 3 OF 4

YOUR BROWSER · Your second mailbox → Invitation → Accept invitation

Your own browser tab

Illustrated view · layout may vary

YOUR BROWSER

Search: Your second mailbox

Your second mailbox > Invitation > Accept invitation

Your second mailbox

Invitation

Accept invitation

1 Sign in as

The invited test identity

Find the labels above in your own portal.

Accept

Illustrated navigation view · use your own saved names and IDs

Why this action: A guest can keep their existing sign-in while receiving limited access to your project.

- 1. Open that mailbox yourself, redeem the invitation, and sign in with the invited identity.

1C.02 · ACTION 4 OF 4**MICROSOFT ENTRA · Users → OneLab Guest → Overview**

entra.microsoft.com Illustrated view · layout may vary

MICROSOFT ENTRA Search: Users

Users > OneLab Guest > Overview

Users
OneLab Guest
Overview

1 User type Guest

2 Invitation state Accepted after redemption

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Back in the main account, open the guest and check User type Guest and invitation redemption status.
2. Add the guest to OneLab-Readers only if you will test its access.

You should see

The guest has redeemed the invitation; it is not automatically a subscription administrator.

If it does not work

Invitation permissions and external collaboration settings can block this. Record the restriction rather than sending invitations to unrelated people.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

1C.03 Assign a licence if one is available

AZ-104 · CORE + permission-gated extension

Why this task: A licence enables a product feature; a role grants permission. They are not interchangeable.

1C.03 · ACTION 1 OF 4

MICROSOFT 365 ADMIN · Billing → Licenses → Your available product

The image shows a browser window with the URL `admin.microsoft.com`. The page title is "MICROSOFT 365 ADMIN" and the search bar contains "Billing". The breadcrumb navigation is "Billing > Licenses > Your available product". On the left sidebar, "Your available product" is selected. The main content area has three numbered steps: 1. Assigned to (learner1 or OneLab-Readers), 2. Usage location (Your actual country), and 3. Available seats (At least one unused seat). An "Assign licenses" button is at the bottom right.

admin.microsoft.com

Illustrated view · layout may vary

MICROSOFT 365 ADMIN Search: Billing

Billing > Licenses > Your available product

Billing

Licenses

Your available product

1 Assigned to learner1 or OneLab-Readers

2 Usage location Your actual country

3 Available seats At least one unused seat

Find the labels above in your own portal.

Assign licenses

Illustrated navigation view · use your own saved names and IDs

Why this action: A licence enables a product feature; a role grants permission. They are not interchangeable.

1. Go to `admin.microsoft.com` as an authorized licence administrator.
2. Open Billing > Licenses.
3. Select an already available product.

1C.03 · ACTION 2 OF 4**MICROSOFT 365 ADMIN · Billing → Licenses → Your available product**

admin.microsoft.com

Illustrated view · layout may vary

MICROSOFT 365 ADMIN Search: Billing

Billing > Licenses > Your available product

Billing
Licenses
Your available product

1 Locate Your available product

2 Action Select Assign licenses and choose the OneLab-Readers group or learner1.

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: A licence enables a product feature; a role grants permission. They are not interchangeable.

1. Select Assign licenses and choose the OneLab-Readers group or learner1.
2. Review enabled services and save.

1C.03 · ACTION 3 OF 4**MICROSOFT 365 ADMIN · Billing → Licenses → Your product → Errors & issues**

admin.microsoft.com

Illustrated view · layout may vary

MICROSOFT 365 ADMIN Search: Billing

Billing > Licenses > Your product > Errors & issues

Billing
Licenses
Your product

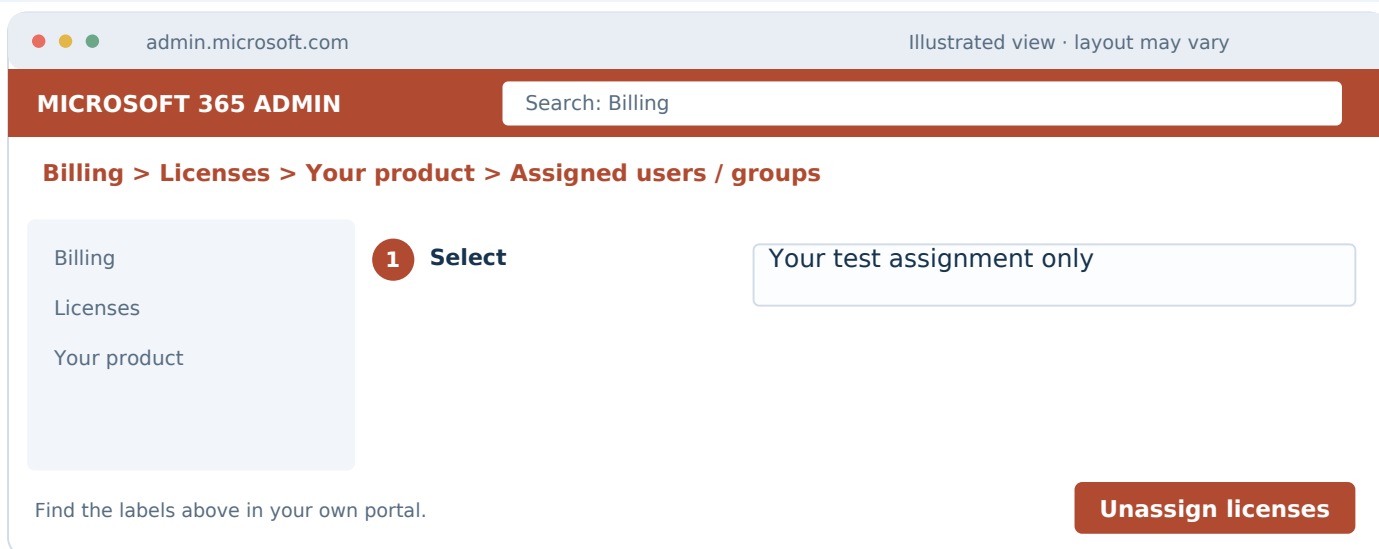
1 Check Assignment status, unused seat and usage location

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Check the assignment result and any Errors & issues.
2. Confirm usage location is set and an unused seat exists.

1C.03 · ACTION 4 OF 4**MICROSOFT 365 ADMIN · Billing → Licenses → Your product → Assigned users / groups**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. After testing, unassign only the test assignment you created.
2. Do not purchase a subscription simply to clear this checkpoint.

You should see

You can explain why licence assignment can fail even when the user exists.

If it does not work

Without a paid seat, use DESIGN: record seat availability, usage location, service-plan conflicts and group membership as the checks. Licence assignment now uses Microsoft 365 admin center, not the old Entra licence-assignment screen. [R05]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

1C.04 Configure self-service password reset

AZ-104 · CORE + permission-gated extension

Why this task: SSPR lets an eligible user reset a forgotten password after proving their identity.

1C.04 · ACTION 1 OF 6

MICROSOFT ENTRA · Password reset → Properties

entra.microsoft.com

Illustrated view · layout may vary

MICROSOFT ENTRA

Search: Password reset

Password reset > Properties

Properties

Authentication methods

Registration

1 Scope

2 Group

3 Methods

4 Registration

Selected

OneLab-Readers

Allowed tenant methods

Required for test users

Find the labels above in your own portal.

Save

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Search Password reset.
2. Under Properties select Selected, then choose OneLab-Readers and Save.
3. Use a test tenant where you may change this setting.

1C.04 · ACTION 2 OF 6**MICROSOFT ENTRA · Password reset → Authentication methods**

entra.microsoft.com Illustrated view · layout may vary

MICROSOFT ENTRA Search: Password reset

Password reset > Authentication methods

Properties
Authentication methods
Registration

1 Allowed methods Your tenant-supported choices

2 Required number Choose the test policy value

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: SSPR lets an eligible user reset a forgotten password after proving their identity.

1. Under Authentication methods choose permitted methods and the required number of methods.
2. The available settings depend on the tenant's authentication-method policy.

1C.04 · ACTION 3 OF 6**MICROSOFT ENTRA · Password reset → Registration**

entra.microsoft.com Illustrated view · layout may vary

MICROSOFT ENTRA Search: Password reset

Password reset > Registration

Properties
Authentication methods
Registration

1 Require users to register Your test-group policy

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Under Registration enable registration at sign-in for the test scope if appropriate.
2. Save.

1C.04 · ACTION 4 OF 6**YOUR BROWSER · Private window → aka.ms/ssprsetup**

Your own browser tab

Illustrated view · layout may vary

YOUR BROWSER Search: Private window

Private window > aka.ms/ssprsetup

Properties

Authentication methods

Registration

1 Signed-in user

learner1

2 Recovery methods

Register the required methods

Find the labels above in your own portal.

Complete registration

Illustrated navigation view · use your own saved names and IDs

Why this action: SSPR lets an eligible user reset a forgotten password after proving their identity.

1. Sign in as learner1 in a private browser window.
2. Register the required recovery methods at <https://aka.ms/ssprsetup>.

1C.04 · ACTION 5 OF 6**YOUR BROWSER · passwordreset.microsoftonline.com**

Your own browser tab

Illustrated view · layout may vary

YOUR BROWSER Search: passwordreset.microsoftonline.com

passwordreset.microsoftonline.com

Properties

Authentication methods

Registration

1 Account

learner1 at your tenant domain

2 Verification

Your registered recovery method

Find the labels above in your own portal.

Reset

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Go to <https://passwordreset.microsoftonline.com> and test recovery for learner1.
2. Sign in with the new password.

1C.04 · ACTION 6 OF 6

MICROSOFT ENTRA · Password reset → Properties

entra.microsoft.com Illustrated view · layout may vary

MICROSOFT ENTRA Search: Password reset

Password reset > Properties

Properties

Authentication methods

Registration

1 Locate

2 Action

Properties

Record the difference: changing a known password is not the same operation as resetting a forgotten one.

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Record the difference: changing a known password is not the same operation as resetting a forgotten one.

You should see

The eligible test user can complete reset using registered methods.

If it does not work

SSPR availability depends on licences and whether identities are cloud-only or synchronized.

Password writeback to an on-premises directory has additional requirements. If unavailable, record the policy and registration flow as DESIGN. [R06]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

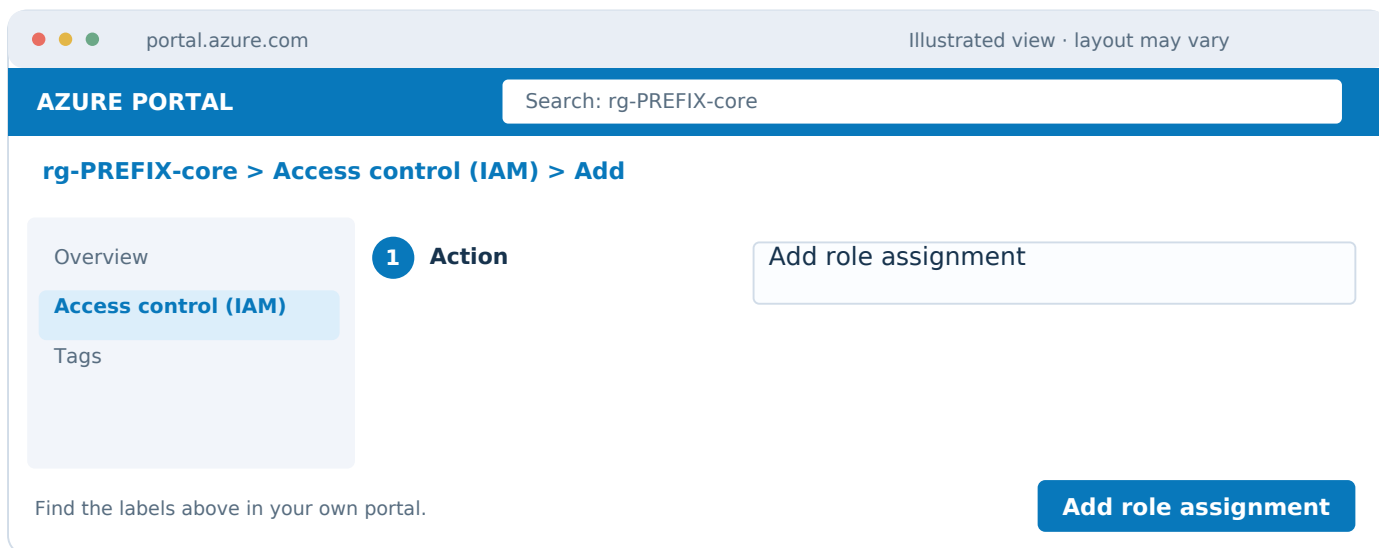
1C.05 Give the group only viewing access

AZ-104 + AZ-400 · CORE

Why this task: Access should match the job and the smallest useful scope.

1C.05 · ACTION 1 OF 6

AZURE PORTAL · rg-PREFIX-core → Access control (IAM) → Add



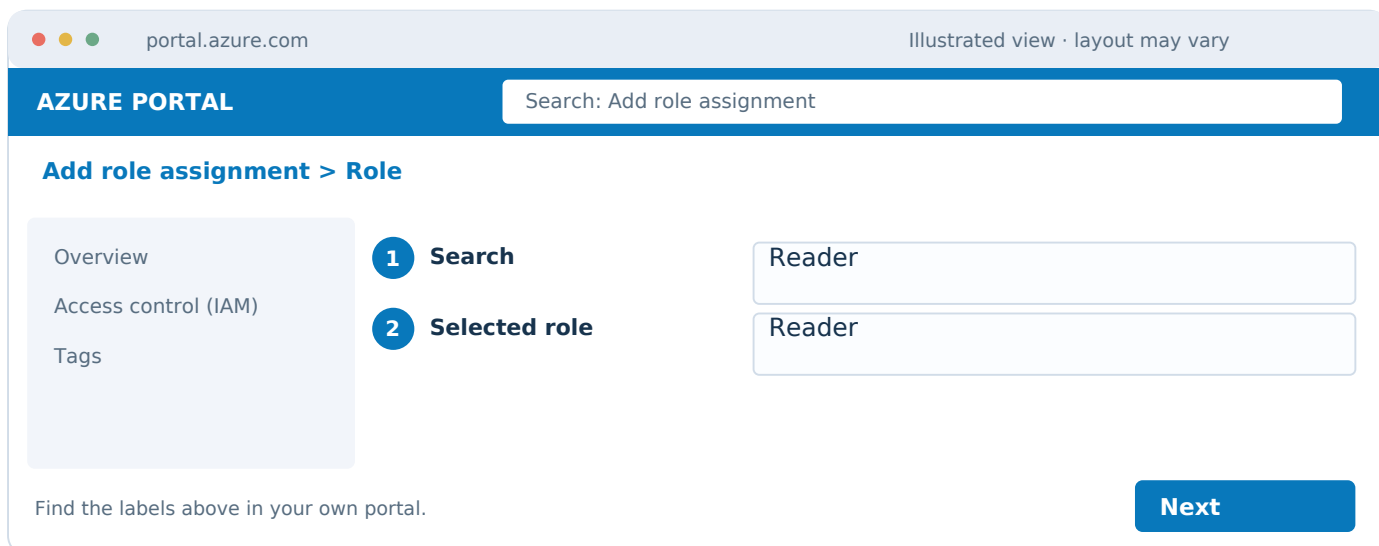
Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Search Resource groups and open your rg-PREFIX-core group.
2. Select Access control (IAM) > Add > Add role assignment.

1C.05 · ACTION 2 OF 6

AZURE PORTAL · Add role assignment → Role



Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. On Role, search Reader and select the built-in role.
2. Click Next.

1C.05 · ACTION 3 OF 6**AZURE PORTAL · Add role assignment → Members**

The screenshot shows the Azure Portal interface for adding a role assignment. The browser address bar shows 'portal.azure.com'. The page title is 'AZURE PORTAL'. A search bar contains 'Add role assignment'. The breadcrumb is 'Add role assignment > Members'. On the left, a navigation pane lists 'Overview', 'Access control (IAM)', and 'Tags'. The main content area has two numbered steps: '1 Assign access to' with a text input 'User, group, or service principal', and '2 Selected member' with a text input 'OneLab-Readers'. A 'Select members' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Access should match the job and the smallest useful scope.

1. On Members, select User, group, or service principal > Select members.
2. Search OneLab-Readers, select it, and click Select.

1C.05 · ACTION 4 OF 6**AZURE PORTAL · Add role assignment → Review + assign**

The screenshot shows the Azure Portal interface for reviewing and assigning a role. The browser address bar shows 'portal.azure.com'. The page title is 'AZURE PORTAL'. A search bar contains 'Add role assignment'. The breadcrumb is 'Add role assignment > Review + assign'. On the left, a navigation pane lists 'Overview', 'Access control (IAM)', and 'Tags'. The main content area has three numbered steps: '1 Role' with a text input 'Reader', '2 Member' with a text input 'OneLab-Readers', and '3 Scope' with a text input 'rg-PREFIX-core'. A 'Review + assign' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Click Review + assign.
2. Open Role assignments and search the group to verify the assignment and scope.

1C.05 · ACTION 5 OF 6**YOUR BROWSER · Private browser → Azure Portal → rg-PREFIX-core**

Your own browser tab

Illustrated view · layout may vary

YOUR BROWSER Search: Private browser

Private browser > Azure Portal > rg-PREFIX-core

Overview
Access control (IAM)
Tags

1 Signed-in user learner1

2 Expected Can view this group

3 Expected limit Cannot create a resource here

Find the labels above in your own portal. **Inspect**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Sign in as learner1 in a private browser window.
2. Open the same group.
3. Confirm it can view but cannot create or delete resources.

1C.05 · ACTION 6 OF 6**AZURE PORTAL · Resource groups → rg-PREFIX-core → Access control (IAM)**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Resource groups

Resource groups > rg-PREFIX-core > Access control (IAM)

Overview
Access control (IAM)
Tags

1 Role Reader

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

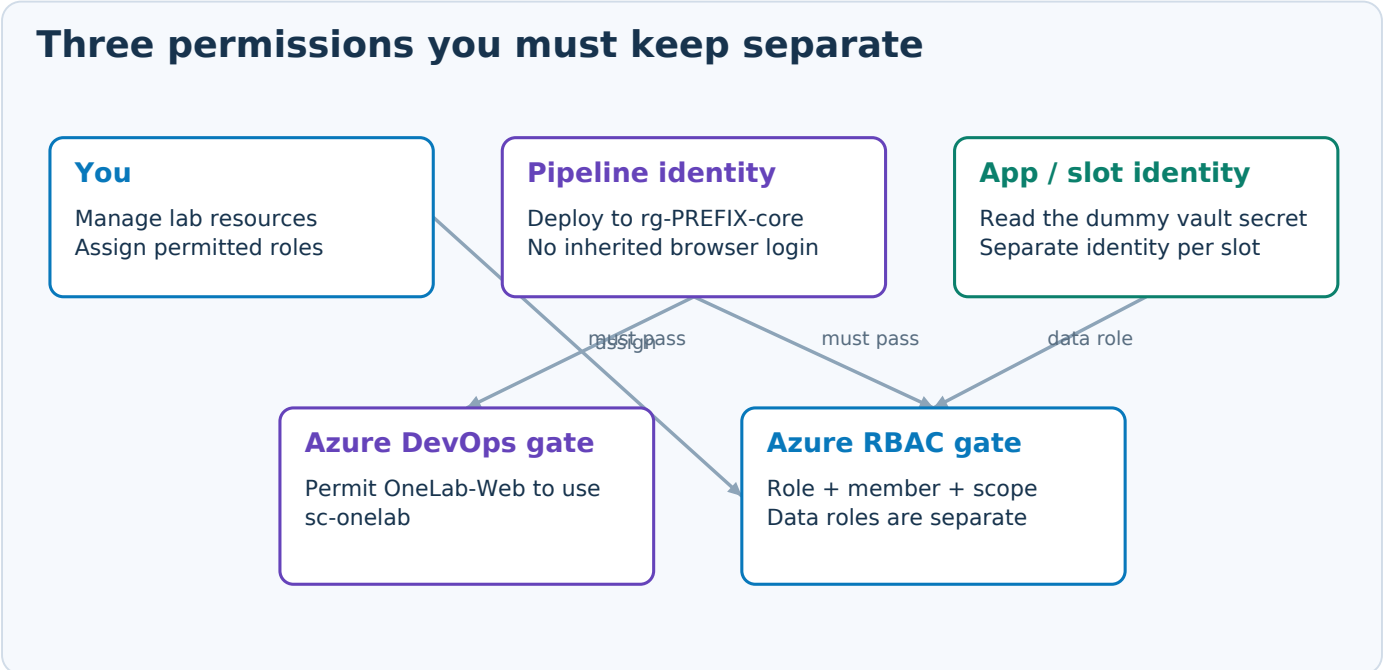
1. If the test user already has a broader inherited role, remove only your own test assignment or use a fresh test user.
2. An inherited Contributor role would invalidate this test.

You should see
The reader can view the group but cannot change it.

If it does not work
Use Check access in IAM to see direct and inherited assignments. A role assignment can take several minutes to become effective.

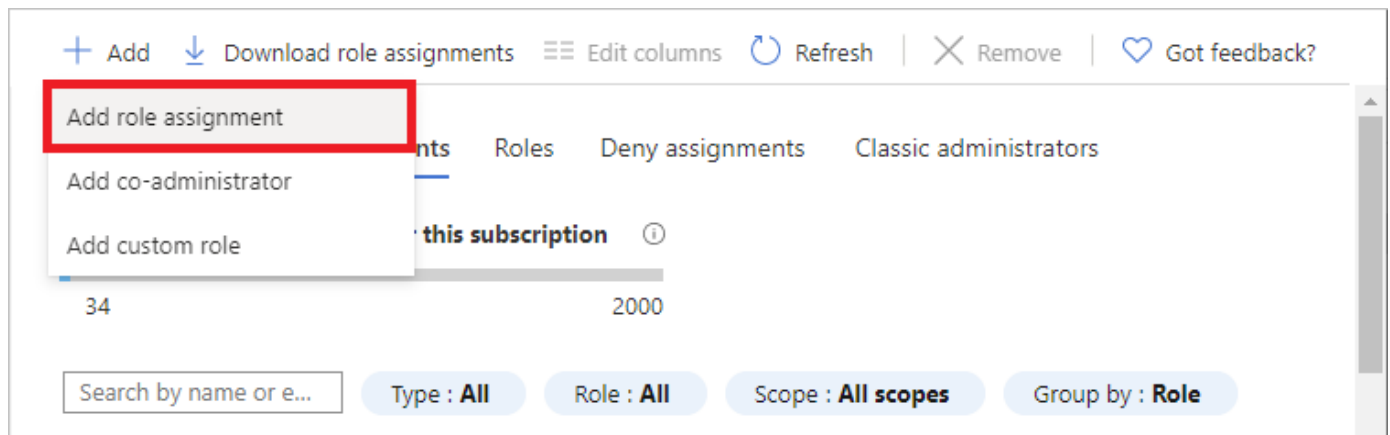
Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED
Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result



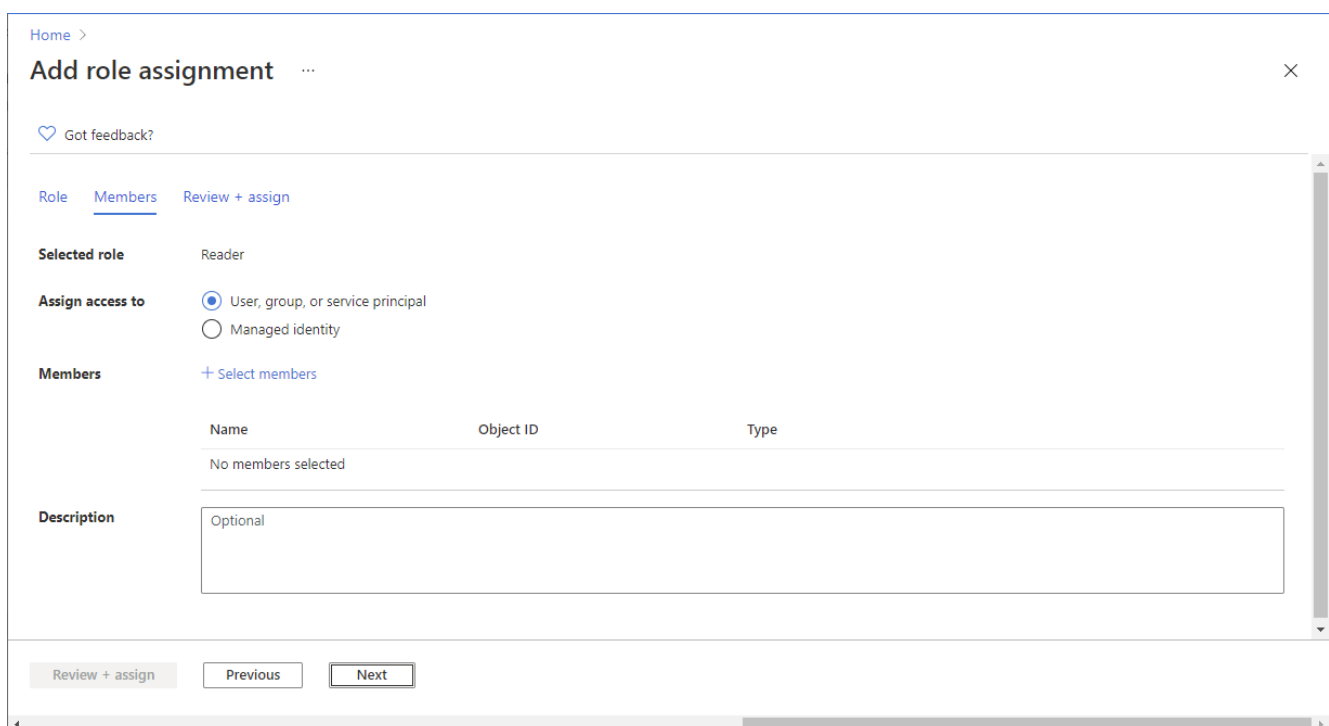
Role or concept	Plain meaning
Reader	View resource settings; does not grant Blob data access.
Contributor	Manage most resources; cannot normally assign Azure roles.
Owner	Manage resources and access assignments.
Storage Blob Data Contributor	Read/write/delete Blob data at the assigned scope.
Entra directory role	Administer identity objects; separate from subscription RBAC.
Inherited role	An assignment at a parent scope also affects child resources.

Microsoft reference screenshot · Screenshot of Add > Add role assignment menu.



[Microsoft reference source](#) · Sample values belong to that reference, not OneLab.

Microsoft reference screenshot · Screenshot of Add role assignment page with Members tab.



[Microsoft reference source](#) · Sample values belong to that reference, not OneLab.

Microsoft reference screenshot · Screenshot of Assign a role page with Review + assign tab.

Home >

Add role assignment

Got feedback?

Role Members Conditions (optional) Review + assign

Role Storage Blob Data Reader

Scope /subscriptions/.../resourceGroups/example-group

Members	Name	Object ID	Type
	Alain	...	User

Description Ticket #7243

Condition None

[Review + assign](#) [Previous](#)

Microsoft reference source · Sample values belong to that reference, not OneLab.

1C.06 Read permissions from the terminal

AZ-104 + AZ-400 · CORE

Why this task: The CLI lets you inspect the same resource scope and role information.

1C.06 · ACTION 1 OF 3

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: The CLI lets you inspect the same resource scope and role information.

1. Run the block below after sourcing lab.env.

USE IN: BASH · COPY ONLY THE CODE

```
RGID=$(az group show -n "$RG" --query id -o tsv)
az role assignment list --scope "$RGID" --include-inherited \
  --query '[].{Principal:principalName,Role:roleDefinitionName,Scope:scope}' \
  -o table
```

What it means: The first line reads the resource group ID. --include-inherited includes assignments from parent scopes.

--query · Select or transform returned fields with a JMESPath expression; this does not itself change the resource.

--scope · The Azure resource ID where the role assignment applies.

--include-inherited · Also show role assignments inherited from parent scopes.

-n · Short form of --name in these Azure commands.

-o · Azure CLI output format: table, tsv or none.

\$NAME / \${NAME} · Read a variable from this Bash session. Source lab.env first when these are lab names.

\$(command) · Run the inner command and use its output as the value.

Backslash at line end · Continue the same Bash command on the next line. No spaces may follow the backslash.

1C.06 · ACTION 2 OF 3**CLOUD SHELL · Cloud Shell → Bash → Read role assignments**

portal.azure.com

Illustrated view · layout may vary

CLOUD SHELL Search: Cloud Shell

Cloud Shell > Bash > Read role assignments

Cloud Shell

Bash

Read role assignments

1 Scope Resource ID of rg-PREFIX-core

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Read principalName, roleDefinitionName and scope.
2. An assignment ending at the subscription scope has broader reach than one ending at a resource.

1C.06 · ACTION 3 OF 3**CLOUD SHELL · Cloud Shell → Bash → Read role assignments**

portal.azure.com

Illustrated view · layout may vary

CLOUD SHELL Search: Cloud Shell

Cloud Shell > Bash > Read role assignments

Cloud Shell

Bash

Read role assignments

1 Locate Read role assignments

2 Action Keep object IDs distinct: a tenant ID identifies a directory, subscription ID identifies a subscription, and a principal object ID identifies a user, group or service principal.

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: The CLI lets you inspect the same resource scope and role information.

1. Keep object IDs distinct: a tenant ID identifies a directory, subscription ID identifies a subscription, and a principal object ID identifies a user, group or service principal.

You should see

You can identify a role and the scope where it was assigned.

If it does not work

A blank principal name can occur when directory lookup is unavailable. Use the object ID to identify it in Entra; do not guess from similar display names.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

Explain back: if learner1 can open a storage account but cannot download a private blob, why? Viewing the account is a management-plane permission. Downloading blob contents also requires the appropriate data-plane role and a permitted network path.

1D · Add guardrails and stop for today

Planning estimate: 60 minutes. Finish with: Tags, an audit policy, a lock exercise and a cost review.

1D.01 Use tags, a temporary lock and an audit policy

AZ-104 · CORE + management-group extension

Why this task: Tags describe ownership. Locks prevent accidental operations. Policy checks whether resource settings meet a rule.

1D.01 · ACTION 1 OF 6

AZURE PORTAL · rg-PREFIX-core → Tags

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: rg-PREFIX-core

rg-PREFIX-core > Tags

Tags

Locks

Policy

1 Project OneLab

2 Environment Lab

Find the labels above in your own portal.

Apply

Illustrated navigation view · use your own saved names and IDs

Why this action: Tags describe ownership. Locks prevent accidental operations. Policy checks whether resource settings meet a rule.

1. Open rg-PREFIX-core > Tags.
2. Add Project=OneLab and Environment=Lab.
3. Apply.
4. Tags on a group do not automatically appear on every child resource.

1D.01 · ACTION 2 OF 6**AZURE PORTAL · rg-PREFIX-core → Locks → Add**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: rg-PREFIX-core

rg-PREFIX-core > Locks > Add

Tags

Locks

Policy

1 Name keep-lab

2 Lock type Delete

Find the labels above in your own portal.

OK

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Open Locks > Add.
2. Name keep-lab; type Delete; add a note and save.
3. Try deleting the empty group only far enough to observe the lock error; keep your resources.

1D.01 · ACTION 3 OF 6**AZURE PORTAL · rg-PREFIX-core → Locks → keep-lab**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: rg-PREFIX-core

rg-PREFIX-core > Locks > keep-lab

Tags

Locks

Policy

1 Remove Only this practice lock

Find the labels above in your own portal.

Delete

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Remove keep-lab before later deletion or recovery exercises.
2. ReadOnly locks can also block operations that look like reads but use a write API, so use Delete for this lesson.

1D.01 · ACTION 4 OF 6**AZURE PORTAL · Policy → Definitions → Search**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Policy

Policy > Definitions > Search

Tags
Locks
Policy

1 Definition Audit VMs that do not use managed disks

2 Effect Audit

3 Scope rg-PREFIX-ops

Find the labels above in your own portal. **Assign**

Illustrated navigation view · use your own saved names and IDs

Why this action: Tags describe ownership. Locks prevent accidental operations. Policy checks whether resource settings meet a rule.

1. Search Policy > Definitions.
2. Search Audit VMs that do not use managed disks.
3. Select the built-in definition, inspect its Audit effect, then select Assign.
4. Use scope rg-PREFIX-ops only and name onelab-audit-managed-disks.

1D.01 · ACTION 5 OF 6**AZURE PORTAL · Policy → Assignment → Parameters**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Policy

Policy > Assignment > Parameters

Tags
Locks
Policy

1 Effect
2 Scope
3 Enforcement

Audit
Your lab group
Do not change company assignments

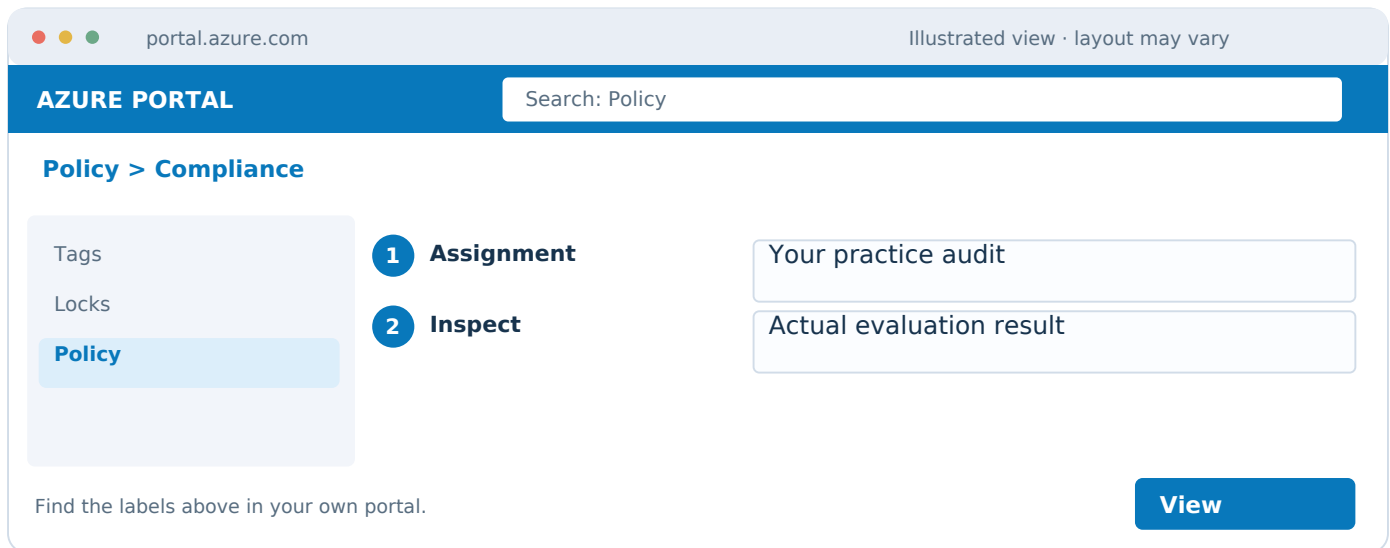
Find the labels above in your own portal.

Review + create

Illustrated navigation view · use your own saved names and IDs

Why this action: Tags describe ownership. Locks prevent accidental operations. Policy checks whether resource settings meet a rule.

1. On the assignment wizard keep the definition defaults.
2. Open Review + create, then Create.
3. There may be no matching VM yet, so no applicable resources is expected today.
4. Revisit Compliance after creating vm-ops on Day 2.
5. Audit reports a mismatch; it does not block the request.
6. A Deny policy, such as Allowed locations, can block creation.

1D.01 · ACTION 6 OF 6**AZURE PORTAL · Policy → Compliance**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Open Policy > Compliance after evaluation.
2. Record compliant/noncompliant resources.
3. Delete your practice assignment after the exercise if it would block later lab extensions.

You should see

You can explain the separate purposes of tags, locks, RBAC and policy.

If it does not work

A policy result is not always immediate. Deny blocks a new request; Audit reports it. Modify or DeployIfNotExists remediation needs an assignment identity and permissions. [R07]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

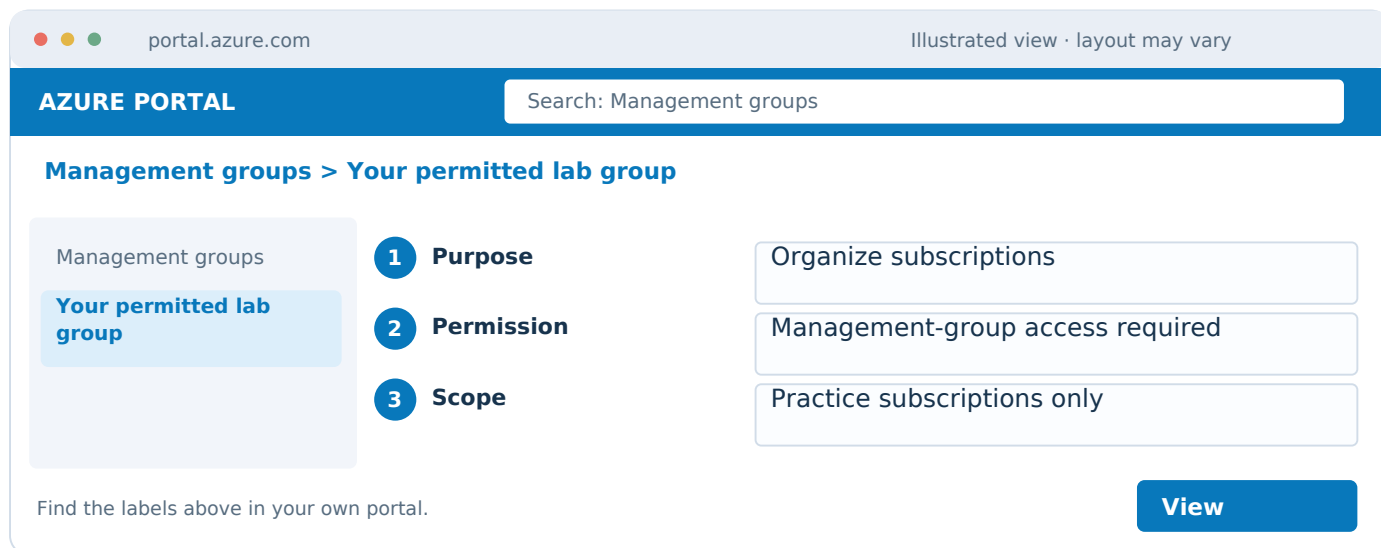
1D.02 Inspect subscriptions and management groups

AZ-104 · CORE + management-group extension

Why this task: A management group applies common rules above multiple subscriptions.

1D.02 · ACTION 1 OF 5

AZURE PORTAL · Management groups → Your permitted lab group



Illustrated navigation view · use your own saved names and IDs

Why this action: A management group applies common rules above multiple subscriptions.

1. Search Management groups.
2. Inspect the tenant root and existing hierarchy; do not move an organization subscription without authorization.

1D.02 · ACTION 2 OF 5**AZURE PORTAL · Management groups → Add management group**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Management groups

Management groups > Add management group

Management groups

Add management group

1 ID onelab-training

2 Display name OneLab Training

Find the labels above in your own portal. **Submit**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. In a personal test tenant where creation is permitted, select Add management group.
2. Use ID onelab-training and display name OneLab Training.

1D.02 · ACTION 3 OF 5**AZURE PORTAL · Management groups → OneLab Training → Add subscription**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Management groups

Management groups > OneLab Training > Add subscription

Management groups

OneLab Training

Add subscription

1 Subscription Dedicated practice subscription only

2 First Review inherited rules

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: A management group applies common rules above multiple subscriptions.

1. If you own a dedicated practice subscription and may reorganize it, use Add subscription to place it under this group.
2. Review inherited policy and access first.

1D.02 · ACTION 4 OF 5**AZURE PORTAL · OneLab Training → Access control (IAM) / Policy**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL

OneLab Training > Access control (IAM) / Policy

OneLab Training

Access control (IAM) / Policy

1 Role for inspection

Reader

2 Scope

This lab management group

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open the group's IAM and Policy to see where company-wide rules could be assigned.
2. Use Reader for a harmless test, then remove the test assignment.

1D.02 · ACTION 5 OF 5**SEE THE IDEA · Inspect subscriptions and management groups → Explain the decision****A SMALL DECISION MAP****Inspect subscriptions and management groups > Explain the decision**

- | | |
|-----------------|---|
| 1 Locate | Explain the decision |
| 2 Action | DESIGN if you cannot create or move: draw Tenant root > Training management group > Practice subscription > OneLab resource groups. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: A management group applies common rules above multiple subscriptions.

1. DESIGN if you cannot create or move: draw Tenant root > Training management group > Practice subscription > OneLab resource groups.
2. Explain which children inherit an assignment.

You should see

You know the difference between a billing subscription and a group used to organize subscriptions.

If it does not work

Subscription directory changes can affect identities and role assignments. Do not perform a directory transfer as a beginner lab shortcut.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

1D.03 Make cost review a daily habit

AZ-104 · CORE + management-group extension

Why this task: A small forgotten service can keep charging after the exercise ends.

1D.03 · ACTION 1 OF 4

AZURE PORTAL · Cost Management → Cost analysis

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Cost Management

Cost Management > Cost analysis

Cost analysis

- Budgets
- Advisor

1 Scope Your practice subscription

2 Group by Resource group

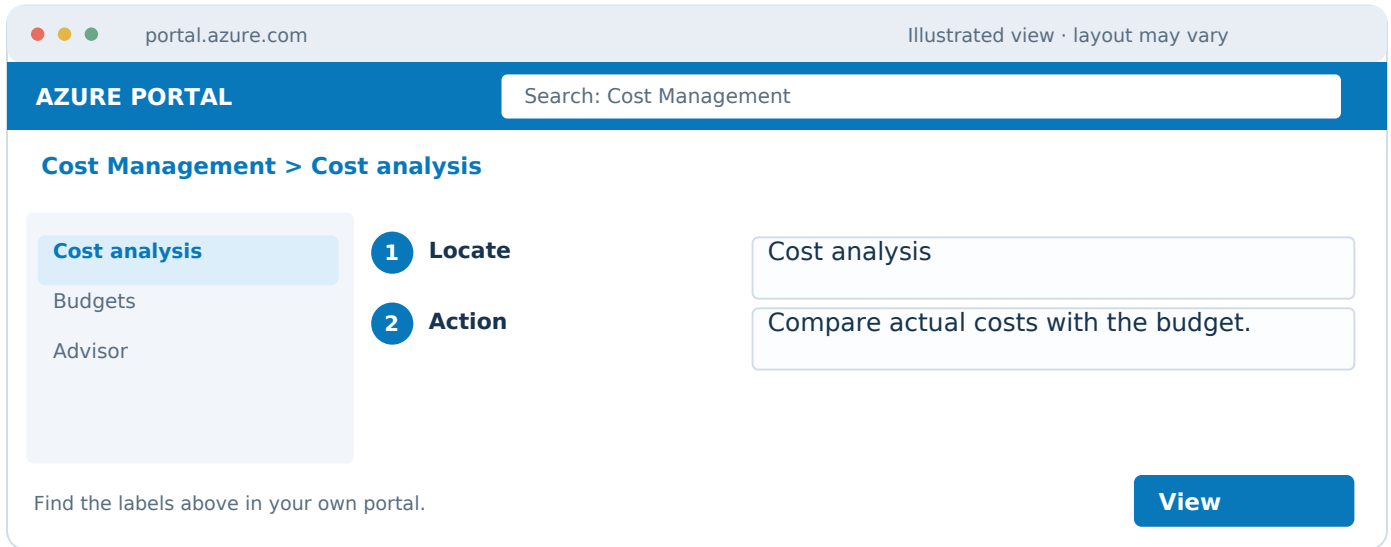
3 Filter Your OneLab groups

Find the labels above in your own portal. **Apply**

Illustrated navigation view · use your own saved names and IDs

Why this action: A small forgotten service can keep charging after the exercise ends.

1. Open Cost Management > Cost analysis at subscription scope.
2. Group by Resource or Resource group and filter your OneLab groups.

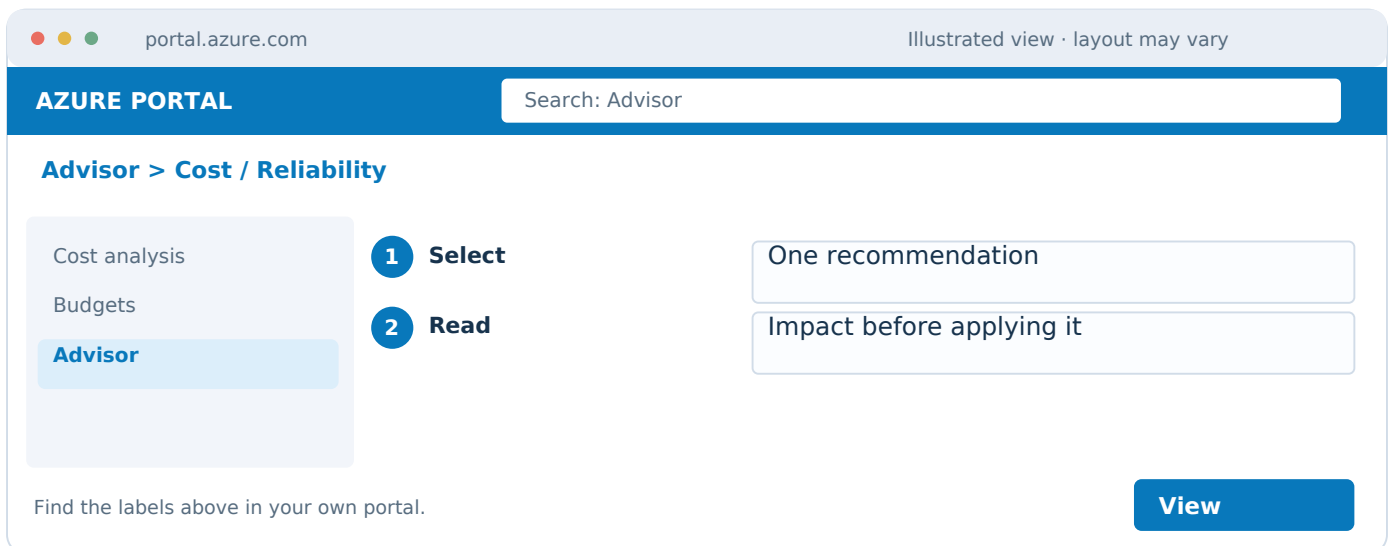
1D.03 · ACTION 2 OF 4**AZURE PORTAL · Cost Management → Cost analysis**

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page header includes the 'AZURE PORTAL' logo and a search bar containing 'Cost Management'. The main navigation pane on the left shows 'Cost Management > Cost analysis'. Under 'Cost analysis', there are three sub-items: 'Cost analysis' (highlighted), 'Budgets', and 'Advisor'. To the right of the navigation pane, there are two numbered steps: '1 Locate' and '2 Action'. The '1 Locate' step shows a text box with 'Cost analysis'. The '2 Action' step shows a text box with 'Compare actual costs with the budget.'. At the bottom right, there is a blue 'View' button. Below the screenshot, a note says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: A small forgotten service can keep charging after the exercise ends.

1. Compare actual costs with the budget.
2. Data can arrive with a delay; no visible cost yet does not mean a service is free.

1D.03 · ACTION 3 OF 4**AZURE PORTAL · Advisor → Cost / Reliability**

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page header includes the 'AZURE PORTAL' logo and a search bar containing 'Advisor'. The main navigation pane on the left shows 'Advisor > Cost / Reliability'. Under 'Cost / Reliability', there are three sub-items: 'Cost analysis', 'Budgets', and 'Advisor' (highlighted). To the right of the navigation pane, there are two numbered steps: '1 Select' and '2 Read'. The '1 Select' step shows a text box with 'One recommendation'. The '2 Read' step shows a text box with 'Impact before applying it'. At the bottom right, there is a blue 'View' button. Below the screenshot, a note says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: A small forgotten service can keep charging after the exercise ends.

1. Search Advisor > Cost and Reliability.
2. Open a recommendation and explain what would change before applying anything.

1D.03 · ACTION 4 OF 4**AZURE PORTAL · Cost Management → Cost analysis**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Keep a note of resources that must be deleted rather than merely stopped: App Service plans, Bastion, private endpoints and recovery replicas are common examples.

You should see

You can identify the main ongoing costs of your lab.

If it does not work

Reservations and savings plans create commitments; compare them in DESIGN only for this short practice project.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Finish Day 1

Save your place and check cost

Save your prefix, subscription, directory and next task privately. Check Cost analysis. Keep the shell storage and two lab groups.

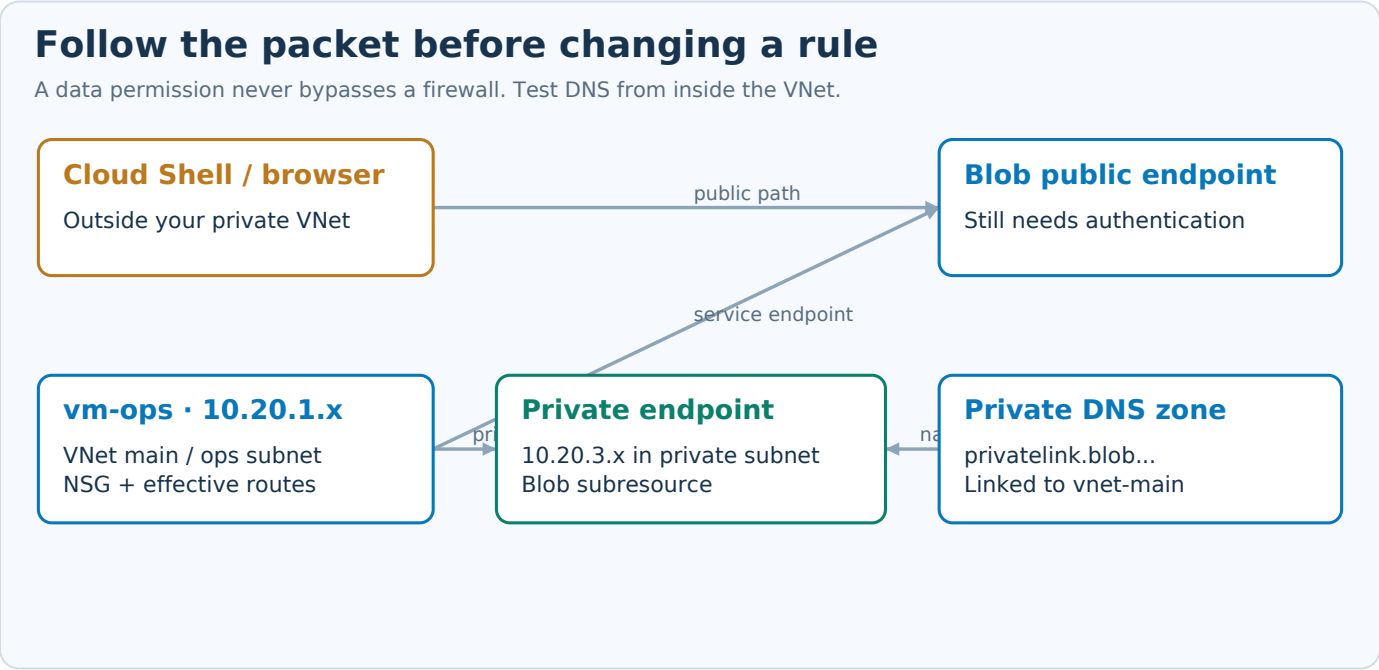
Say it in your own words: Why can someone view a storage account but still be unable to read a private file?

Check your explanation: Viewing resource settings and reading the stored data use different permissions. The network path must also allow the request.

DAY 2 OF 7

Store files and build the network

Your project has recoverable sample files, two connected networks and an operations VM.



Start today

Run the restart routine from Day 1, Block B. Keep lab.env unchanged. Check whether the agent-capacity request has been approved.

Start early and track the job

Start VM backup and regional replication in Block D. Record job IDs/status; their completion is checked on later days.

Block	Today's work	Minutes
A	Save, recover and share a small file	100
B	Copy files and compare shared storage	90
C	Connect two private networks	75
D	Create the VM and start protection	120

2A · Save, recover and share a small file

Planning estimate: 100 minutes. Finish with: A restored Blob version and a read-only temporary SAS.

2A.01 Create the main storage account

AZ-104 · CORE

Why this task: Blob Storage will hold the website's sample documents and recovery exercises.

2A.01 · ACTION 1 OF 6

AZURE PORTAL · Storage accounts → Create

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Storage accounts

Storage accounts > Create

Basics
Advanced
Networking
Data protection
Encryption

- 1 Resource group** rg-PREFIX-core
- 2 Name** PREFIXdata
- 3 Performance** Standard
- 4 Redundancy** LRS
- 5 Account kind** StorageV2
- 6 Hierarchical namespace** Disabled

Find the labels above in your own portal.

Review + create

Illustrated navigation view · use your own saved names and IDs

Why this action: Blob Storage will hold the website's sample documents and recovery exercises.

1. Search Storage accounts > Create.
2. Subscription: your practice subscription; resource group: rg-PREFIX-core; account name: PREFIXdata; region: your chosen primary region.

2A.01 · ACTION 2 OF 6**AZURE PORTAL · Create storage account → Basics**

The screenshot shows the Azure Portal interface for creating a storage account. The browser address bar shows 'portal.azure.com'. The page title is 'AZURE PORTAL'. A search bar contains 'Create storage account'. The breadcrumb is 'Create storage account > Basics'. On the left, a sidebar lists 'Basics', 'Advanced', and 'Networking'. The main content area has three numbered steps: 1. Performance (Standard), 2. Redundancy (Locally-redundant storage (LRS)), and 3. Account kind (StorageV2 when offered). A 'Next: Advanced' button is at the bottom right.

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Create storage account

Create storage account > Basics

Basics
Advanced
Networking

1 Performance Standard

2 Redundancy Locally-redundant storage (LRS)

3 Account kind StorageV2 when offered

Find the labels above in your own portal. **Next: Advanced**

Illustrated navigation view · use your own saved names and IDs

Why this action: Blob Storage will hold the website's sample documents and recovery exercises.

1. Choose Standard performance, StorageV2 where shown, and locally redundant storage (LRS) for this small exercise.

2A.01 · ACTION 3 OF 6**AZURE PORTAL · Create storage account → Advanced**

The screenshot shows the Azure Portal interface for creating a storage account, advanced settings. The browser address bar shows 'portal.azure.com'. The page title is 'AZURE PORTAL'. A search bar contains 'Create storage account'. The breadcrumb is 'Create storage account > Advanced'. On the left, a sidebar lists 'Basics', 'Advanced', 'Networking', 'Data protection', and 'Encryption'. The main content area has four numbered steps: 1. Secure transfer (Required), 2. Minimum TLS (1.2), 3. Anonymous Blob access (Disabled), and 4. Hierarchical namespace (Disabled). A 'Next: Networking' button is at the bottom right.

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Create storage account

Create storage account > Advanced

Basics
Advanced
Networking
Data protection
Encryption

1 Secure transfer Required

2 Minimum TLS 1.2

3 Anonymous Blob access Disabled

4 Hierarchical namespace Disabled

Find the labels above in your own portal. **Next: Networking**

Illustrated navigation view · use your own saved names and IDs

Why this action: HTTPS protects traffic. Turning anonymous access off prevents public reading of files.

1. Keep secure transfer required and minimum TLS 1.2.
2. Disable anonymous Blob access.
3. Keep hierarchical namespace off for the versioning exercise.

2A.01 · ACTION 4 OF 6**AZURE PORTAL · Create storage account → Networking**

The screenshot shows the Azure Portal interface for creating a storage account. The browser address bar shows 'portal.azure.com'. The page title is 'AZURE PORTAL' with a search bar containing 'Create storage account'. The breadcrumb is 'Create storage account > Networking'. On the left, a sidebar lists 'Basics', 'Advanced', and 'Networking' (which is highlighted). The main content area shows three numbered steps: 1. Public network access (Enabled), 2. Network scope (All networks for this lab), and 3. Data access (Authenticated only). At the bottom, there is a text prompt 'Find the labels above in your own portal.' and a blue button labeled 'Next: Data protection'.

Illustrated navigation view · use your own saved names and IDs

Why this action: Reachable on the Internet and readable without a login are different settings.

1. For now allow public network access from all networks, but keep containers private and require authentication.
2. A public endpoint does not mean the data is anonymous.

2A.01 · ACTION 5 OF 6**AZURE PORTAL · Create storage account → Data protection**

The screenshot shows the Azure Portal interface for creating a storage account, specifically the 'Data protection' tab. The browser address bar shows 'portal.azure.com'. The page title is 'AZURE PORTAL' with a search bar containing 'Create storage account'. The breadcrumb is 'Create storage account > Data protection'. On the left, a sidebar lists 'Basics', 'Advanced', 'Networking', 'Data protection' (which is highlighted), and 'Encryption'. The main content area shows four numbered steps: 1. Blob soft delete (Enabled), 2. Container soft delete (Enabled), 3. Blob versioning (Enabled), and 4. Retention (7 days if allowed). At the bottom, there is a text prompt 'Find the labels above in your own portal.' and a blue button labeled 'Review + create'.

Illustrated navigation view · use your own saved names and IDs

Why this action: These recovery settings let you undo your next harmless file mistake.

1. Under Data protection enable Blob soft delete, container soft delete, and Blob versioning.
2. Use a short allowed retention such as 7 days.
3. Review + create.

2A.01 · ACTION 6 OF 6**AZURE PORTAL · PREFIXdata → Access control (IAM) → Add role assignment**

The screenshot shows the Azure Portal interface for adding a role assignment. The browser address bar shows 'portal.azure.com'. The page title is 'AZURE PORTAL'. A search bar contains 'PREFIXdata'. The breadcrumb navigation is 'PREFIXdata > Access control (IAM) > Add role assignment'. On the left, there is a sidebar with 'Basics', 'Advanced', and 'Networking' sections. The main content area has three numbered steps: 1. Role (Storage Blob Data Contributor), 2. Member (Your own user), and 3. Scope (This account). At the bottom, there is a 'Review + assign' button and a note: 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Managing an account does not automatically let you read or write the data inside it.

1. Open the account > IAM and give your own signed-in user Storage Blob Data Contributor at this account.
2. Wait for the assignment to propagate.

You should see

The account exists with private data access and recovery features enabled.

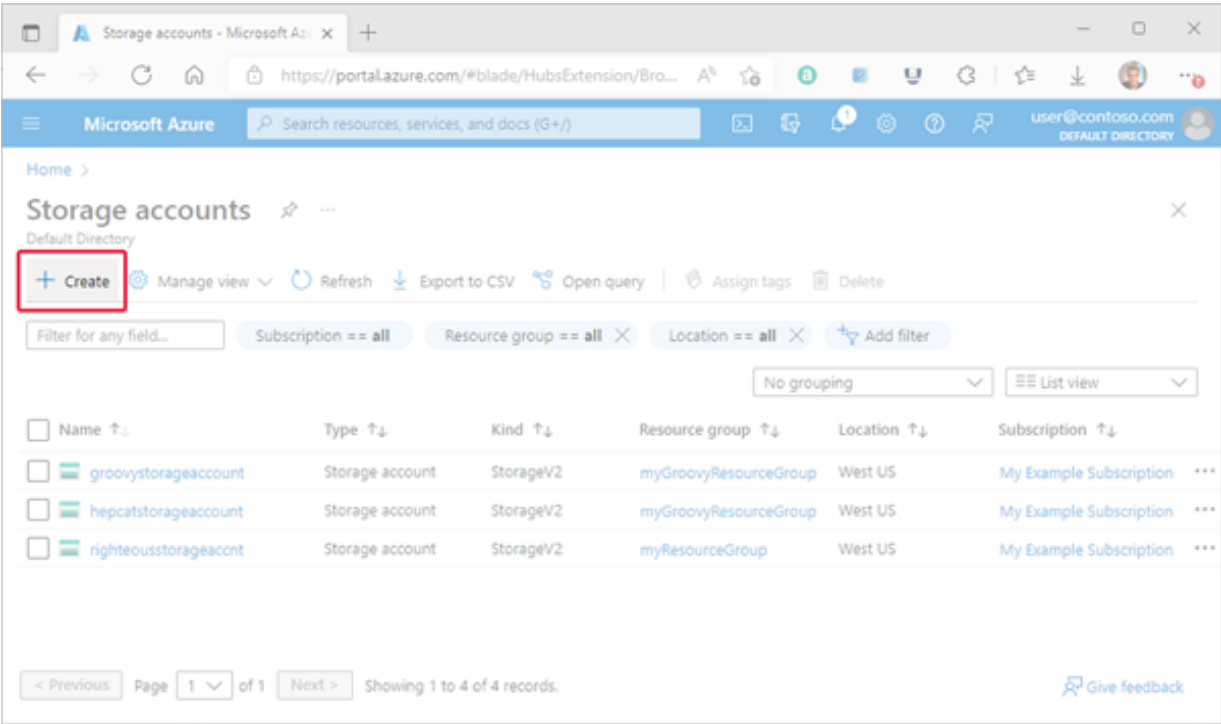
If it does not work

If you can manage the account but cannot browse blobs, check the data role. A Reader role alone is insufficient. [R08]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

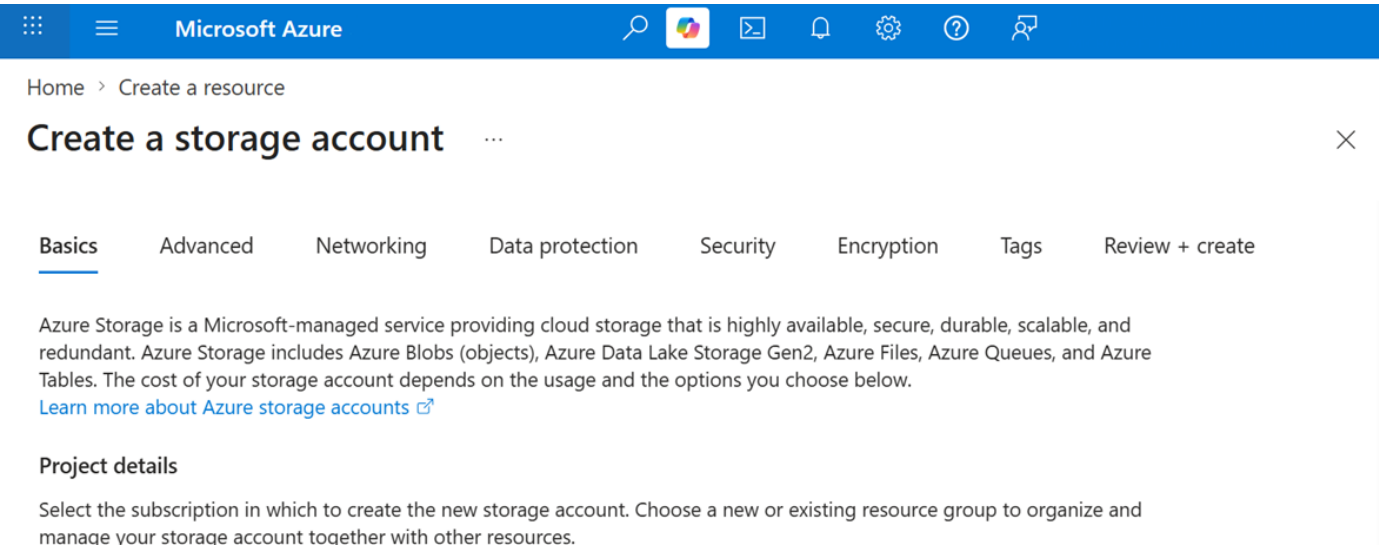
Worked: ☐ Independently ☐ With the guide

Microsoft reference screenshot · Screenshot of the Create button on the Storage accounts page in the Azure portal.



Microsoft reference source · Sample values belong to that reference, not OneLab.

Microsoft reference screenshot · Screenshot of the tabs that contain each storage account setting.



Microsoft reference source · Sample values belong to that reference, not OneLab.

2A.02 Upload a document and create a second version

AZ-104 · CORE

Why this task: Changing an object gives you something real to restore.

2A.02 · ACTION 1 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. Run the commands below to create a private container and upload two versions of the same file.

USE IN: BASH · COPY ONLY THE CODE

```
az storage container create --name documents \  
  --account-name "$STORE" --auth-mode login \  
printf 'OneLab version one\n' > welcome.txt \  
az storage blob upload --account-name "$STORE" --auth-mode login \  
  -c documents -n welcome.txt -f welcome.txt --overwrite \  
printf 'OneLab version two\n' > welcome.txt \  
az storage blob upload --account-name "$STORE" --auth-mode login \  
  -c documents -n welcome.txt -f welcome.txt --overwrite
```

What it means: `--auth-mode login` uses your Entra identity. `-c` is container, `-n` is blob name, and `-f` is the local file. `--overwrite` replaces the current object while versioning preserves the old version.

--name · The name of this resource or object.

--account-name · The storage account to use.

--auth-mode · `login` uses your signed-in Entra identity for this data operation.

--overwrite · Replace the existing blob content; with versioning enabled, older content is retained as a version.

-n · Short form of `--name` in these Azure commands.

blob upload -c / -n / -f · `-c` is the container, `-n` is the blob name and `-f` is the local file to upload.

\$NAME / \${NAME} · Read a variable from this Bash session. Source `lab.env` first when these are lab names.

Backslash at line end · Continue the same Bash command on the next line. No spaces may follow the backslash.

2A.02 · ACTION 2 OF 5**AZURE PORTAL · PREFIXdata → Containers → documents → welcome.txt → Versions**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > Containers > documents > welcome.txt > Versions

Overview
Containers
Data protection

1 Current content OneLab version two

2 Older content OneLab version one

Find the labels above in your own portal.

Versions

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. In the portal open the account > Data storage > Containers > documents.
2. Open welcome.txt and inspect Versions.

2A.02 · ACTION 3 OF 5**AZURE PORTAL · welcome.txt → Versions → Earlier version**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: welcome.txt

welcome.txt > Versions > Earlier version

Overview
Containers
Data protection

1 Action Make current version or supported copy/promote operation

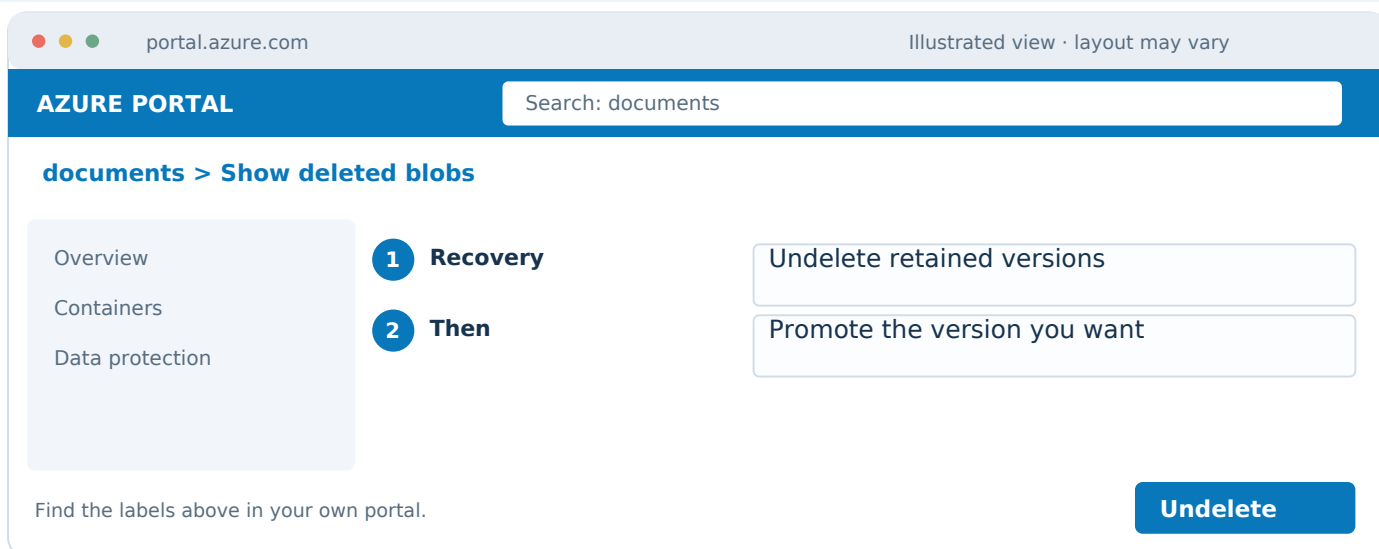
Find the labels above in your own portal.

Make current version

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Download the current version.
2. Confirm it says version two.
3. Select the older version and use Make current version or promote/copy it to the current blob using the available action.

2A.02 · ACTION 4 OF 5**AZURE PORTAL · documents → Show deleted blobs**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Delete the current blob.
2. Enable Show deleted blobs and inspect the retained versions.
3. Undelete the blob if required, then promote the desired previous version.
4. With versioning enabled, undelete alone may not restore a current version.

2A.02 · ACTION 5 OF 5**SEE THE IDEA · Two kinds of recovery****A SMALL DECISION MAP****Two kinds of recovery**

- | | |
|----------------------|-------------------------------|
| 1 Versioning | Keeps old content |
| 2 Soft delete | Keeps deleted data for a time |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Record the result: versioning preserves previous content; soft delete retains recoverable deleted data for a retention period.

You should see

You recover the intended text and can tell a previous version from the current one.

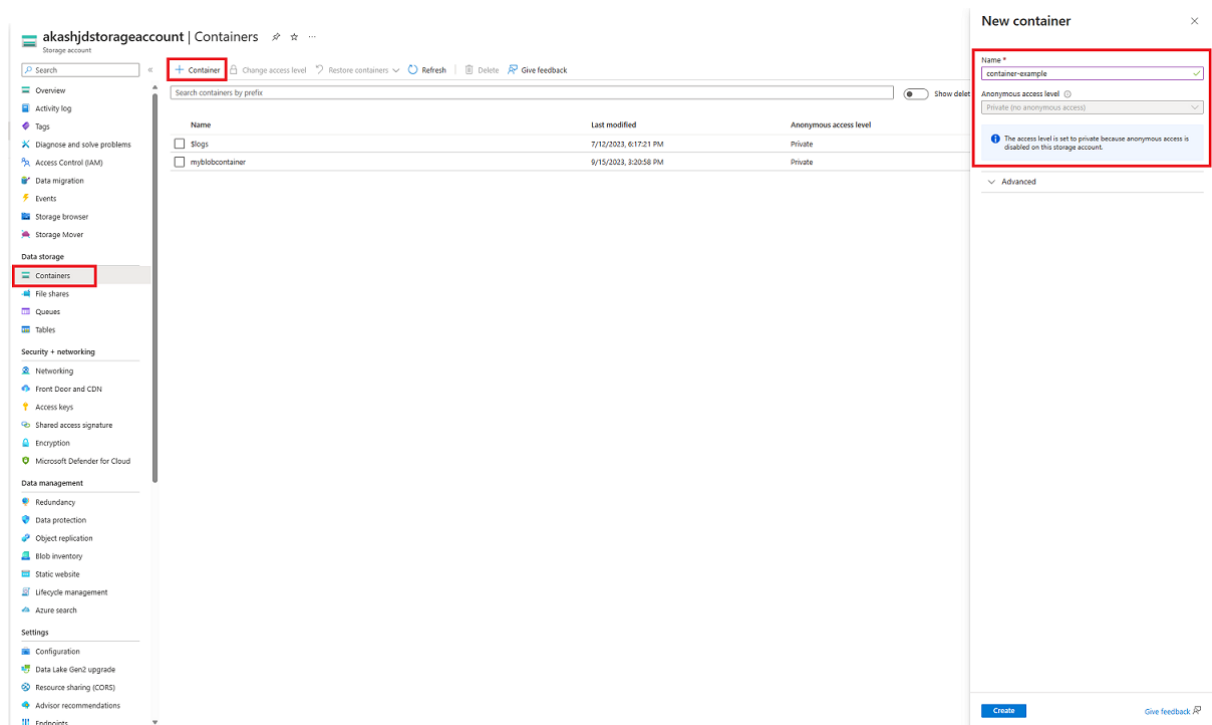
If it does not work

Do not delete the storage account for this exercise. Blob recovery features do not by themselves recover a deleted account.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Microsoft reference screenshot · Screenshot showing how to create a container within the Azure portal.



Microsoft reference source · Sample values belong to that reference, not OneLab.

2A.03 Set tiers and a lifecycle rule

AZ-104 · CORE

Why this task: Less frequently used data may cost less to store but more to retrieve.

2A.03 · ACTION 1 OF 5

AZURE PORTAL · documents → welcome.txt → Change tier

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: documents

documents > welcome.txt > Change tier

Containers
Lifecycle management
Data protection

1 Current tier Hot

2 Choice for this lab Keep Hot

Find the labels above in your own portal. **Cancel**

Illustrated navigation view · use your own saved names and IDs

Why this action: Less frequently used data may cost less to store but more to retrieve.

1. Open welcome.txt > Change tier.
2. Compare Hot, Cool and Cold where available.
3. Keep this frequently read file Hot.

2A.03 · ACTION 2 OF 5

AZURE PORTAL · PREFIXdata → Lifecycle management → Add a rule

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > Lifecycle management > Add a rule

Containers
Lifecycle management
Data protection

1 Name old-documents

2 Blob type Block blobs

3 Prefix documents/archive/

Find the labels above in your own portal. **Next**

Illustrated navigation view · use your own saved names and IDs

Why this action: Less frequently used data may cost less to store but more to retrieve.

1. Go to Data management > Lifecycle management > Add rule.
2. Name it old-documents; apply to block blobs with prefix documents/archive/.

2A.03 · ACTION 3 OF 5**AZURE PORTAL · Add a rule → Base blobs**

The screenshot shows the Azure Portal interface for adding a rule. The browser address bar displays 'portal.azure.com'. The page title is 'AZURE PORTAL' with a search bar containing 'Add a rule'. The main heading is 'Add a rule > Base blobs'. On the left, a sidebar lists 'Containers', 'Lifecycle management', and 'Data protection'. The main content area shows two steps: '1 Condition' with the value 'Days after last modification > 30' and '2 Action' with the value 'Move to cool storage'. At the bottom, there is a 'Next' button and a note: 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Add a rule to move matching base blobs to Cool after 30 days since last modification.
2. Add a separate previous-version cleanup rule only if you understand what it deletes.

2A.03 · ACTION 4 OF 5**AZURE PORTAL · Lifecycle management → Rule review**

The screenshot shows the Azure Portal interface for reviewing a rule. The browser address bar displays 'portal.azure.com'. The page title is 'AZURE PORTAL' with a search bar containing 'Lifecycle management'. The main heading is 'Lifecycle management > Rule review'. On the left, a sidebar lists 'Containers', 'Lifecycle management' (highlighted), and 'Data protection'. The main content area shows two steps: '1 Scope' with the value 'documents/archive/' and '2 Timing' with the value 'Asynchronous, after age condition'. At the bottom, there is an 'Add' button and a note: 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Less frequently used data may cost less to store but more to retrieve.

1. Review the rule and save.
2. It will not move a new blob immediately; this is a configuration exercise.

2A.03 · ACTION 5 OF 5**SEE THE IDEA · Hot, Cool, Cold and Archive****A SMALL DECISION MAP****Hot, Cool, Cold and Archive**

- | | |
|----------------------|-------------------------------|
| 1 Hot | Read frequently |
| 2 Cool / Cold | Read less often |
| 3 Archive | Rehydrate before normal reads |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Less frequently used data may cost less to store but more to retrieve.

1. Explain Archive: it is offline for normal reads and needs rehydration.
2. It has latency and early deletion considerations.
3. Do not archive the file you still need for the lab.

You should see

Only the intended prefix is targeted and you understand that lifecycle processing is asynchronous.

If it does not work

Some redundancy/account combinations do not support every tier. Do not infer a tier is supported just because another account offers it.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

Choice	Use it when
LRS	You need copies in one data-center location at lower cost.
ZRS	You need synchronous copies across zones in one region.
GRS / GZRS	You also need asynchronous replication to another region; GZRS adds zone resilience in the primary region.
RA variants	You need read access to the secondary endpoint before account failover.
Object replication	You need selected block blobs copied to another account according to a rule; this is not the same as account redundancy.

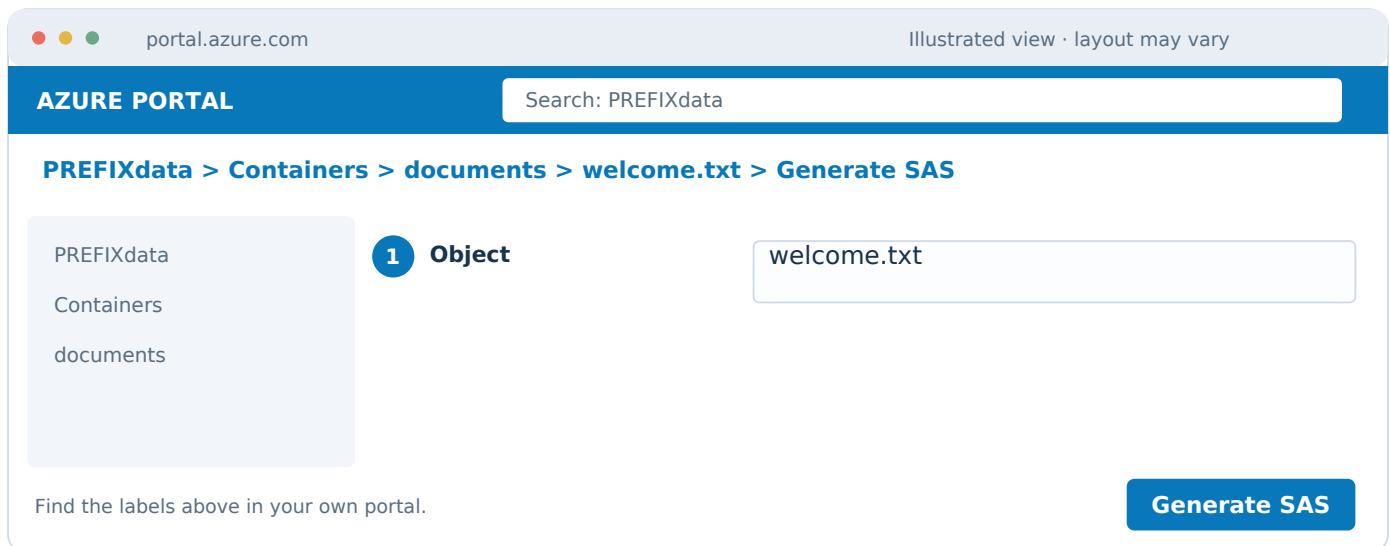
2A.04 Share one object briefly with a SAS

AZ-104 · CORE + identity extension

Why this task: A shared access signature is a limited, time-bound access ticket. Anyone holding it can use its permissions.

2A.04 · ACTION 1 OF 5

AZURE PORTAL · PREFIXdata → Containers → documents → welcome.txt → Generate SAS



Illustrated navigation view · use your own saved names and IDs

Why this action: A shared access signature is a limited, time-bound access ticket. Anyone holding it can use its permissions.

1. Open PREFIXdata > Containers > documents > welcome.txt.
2. Choose Generate SAS.

2A.04 · ACTION 2 OF 5**AZURE PORTAL · Generate SAS → Options**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Generate SAS

Generate SAS > Options

Generate SAS

Options

- 1 Signing**
- 2 Permission**
- 3 Protocol**
- 4 Expiry**

User delegation key

Read only

HTTPS

About 15 minutes

Find the labels above in your own portal.

Generate SAS token and URL

Illustrated navigation view · use your own saved names and IDs

Why this action: A shared access signature is a limited, time-bound access ticket. Anyone holding it can use its permissions.

1. Choose User delegation key if available for your identity.
2. Select Read only, HTTPS only, and an expiry about 15 minutes from now.
3. Give the start time a small allowance for clock differences.

2A.04 · ACTION 3 OF 5**YOUR BROWSER · Private browser window → Paste the full SAS URL**

YOUR BROWSER

Search: Private browser window

Private browser window > Paste the full SAS URL

Private browser window

Paste the full SAS URL

1 Expected

2 Private

File downloads

URL contains a temporary credential

Find the labels above in your own portal.

Open

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Generate the Blob SAS URL.
2. Open it in a private browser window and confirm the file downloads.
3. Keep the URL private; it contains the credential.

2A.04 · ACTION 4 OF 5**YOUR BROWSER · Private browser window → Blob URL without ?query**

YOUR BROWSER

Search: Private browser window

Private browser window > Blob URL without ?query

Private browser window

Blob URL without ?query

1 Expected

Anonymous request denied

Find the labels above in your own portal.

Open

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Copy only the blob URL without its query string into another private window.
2. Confirm anonymous access fails.

2A.04 · ACTION 5 OF 5**YOUR BROWSER · Private browser window → Retry expired SAS**

Your own browser tab

Illustrated view · layout may vary

YOUR BROWSER

Search: Private browser window

Private browser window > Retry expired SAS

Private browser window

1 Expected

Access denied after expiry

Find the labels above in your own portal.

Refresh

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. After expiry, retry the SAS URL.
2. Confirm the ticket no longer works.
3. A SAS grants permissions but does not bypass storage firewall rules.

You should see

The object is readable through the valid ticket and unreadable anonymously.

If it does not work

A user delegation SAS needs permission to request a delegation key as well as data access. A stored access policy applies to a service SAS, not to a user delegation SAS or account SAS. [R09]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

2A.05 Inspect stored access policies, keys and encryption

AZ-104 · CORE + identity extension

Why this task: Different authentication methods have different ways to revoke access.

2A.05 · ACTION 1 OF 5

AZURE PORTAL · PREFIXdata → Containers → documents → Access policy

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page title is 'AZURE PORTAL'. A search bar contains 'PREFIXdata'. The breadcrumb navigation path is 'PREFIXdata > Containers > documents > Access policy'. On the left, a sidebar lists 'Access policy' (selected), 'Access keys', and 'Encryption'. The main content area has three numbered steps: 1. Identifier (read-window), 2. Permissions (Read), and 3. Start / expiry (Your short test window). At the bottom right, there is an 'Add policy' button. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > Containers > documents > Access policy

Access policy

Access keys

Encryption

1 Identifier read-window

2 Permissions Read

3 Start / expiry Your short test window

Find the labels above in your own portal.

Add policy

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Open the documents container > Access policy > Add policy.
2. ID: read-window; permissions: Read; choose a short valid time window and save.
3. This policy only affects a service SAS that references its ID.

2A.05 · ACTION 2 OF 5**AZURE PORTAL · PREFIXdata → Security + networking → Access keys**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > Security + networking > Access keys

Access policy
Access keys
Encryption

1 key1 Do not reveal or copy into Git

2 key2 Do not reveal or copy into Git

Find the labels above in your own portal. **Inspect**

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Go to the storage account > Security + networking > Access keys.
2. Notice there are two keys.
3. Do not copy them into Git or screenshots.

2A.05 · ACTION 3 OF 5**SEE THE IDEA · Rotate the key without losing clients****A SMALL DECISION MAP****Rotate the key without losing clients**

- | | |
|----------------|--------------------------------|
| 1 First | Move clients to key2 |
| 2 Then | Rotate key1 and update clients |
| 3 Last | Rotate key2 after verification |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. DESIGN key rotation: move clients to key2, rotate key1, update clients to the replacement key1, then rotate key2 after verifying no client still uses it.
2. Regenerating a key breaks tickets and applications that rely on that old key.

2A.05 · ACTION 4 OF 5

AZURE PORTAL · PREFIXdata → Encryption

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > Encryption

- Access policy
- Access keys
- Encryption**

- 1 Key type**
 - Microsoft-managed for this lab
- 2 Customer-managed option**
 - Needs a key identity and availability plan

Find the labels above in your own portal. **Inspect**

Illustrated navigation view · use your own saved names and IDs

Why this action: Different authentication methods have different ways to revoke access.

1. Open Encryption.
2. Keep Microsoft-managed keys for the core lab.
3. Explain that customer-managed keys require a Key Vault/key identity arrangement and careful key availability.

2A.05 · ACTION 5 OF 5

SEE THE IDEA · Inspect stored access policies, keys and encryption → Explain the decision

A SMALL DECISION MAP

Inspect stored access policies, keys and encryption > Explain the decision

- 1 Locate** Explain the decision
- 2 Action** Infrastructure encryption is an additional encryption layer selected where supported; it is separate from HTTPS in transit.

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Different authentication methods have different ways to revoke access.

1. Infrastructure encryption is an additional encryption layer selected where supported; it is separate from HTTPS in transit.
2. Changing a SAS cannot enable encryption.

You should see

You know which credential a revocation action affects.

If it does not work

An account key is broad access. Prefer identity or a narrow user delegation SAS where supported. Do not rotate keys used by Cloud Shell storage; this is your separate PREFIXdata account.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

2B · Copy files and compare shared storage

Planning estimate: 90 minutes. Finish with: A verified transfer, share snapshot and replication result or pending status.

2B.01 Copy documents using AzCopy and Storage Explorer

AZ-104 · CORE + identity extension

Why this task: The portal is useful for a few files; a transfer tool is more practical for many files.

2B.01 · ACTION 1 OF 6

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: The correct identity needs permission at the place where it will do the work.

1. Run `azcopy --version`.
2. Sign in with `azcopy login`; follow the device-code sign-in shown by the tool using the same user with Blob data permissions.

USE IN: BASH · COPY ONLY THE CODE

```
azcopy --version
azcopy login
```

What it means: Read the installed version, then complete the device-code sign-in in your browser.

2B.01 · ACTION 2 OF 6

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Run the copy command below.
2. Inspect its completed-transfer summary, then confirm the new blob in the portal.

USE IN: BASH · COPY ONLY THE CODE

```
azcopy copy ./welcome.txt \
"https://$STORE.blob.core.windows.net/documents/azcopy-welcome.txt"
```

What it means: The source is a local file and the destination is one blob. No SAS is included because AzCopy signed in using your identity.

\$NAME / \${NAME} · Read a variable from this Bash session. Source `lab.env` first when these are lab names.

Backslash at line end · Continue the same Bash command on the next line. No spaces may follow the backslash.

2B.01 · ACTION 3 OF 6**YOUR BROWSER · azure.microsoft.com → Storage Explorer → Download**

Your own browser tab Illustrated view · layout may vary

YOUR BROWSER

azure.microsoft.com > Storage Explorer > Download

azure.microsoft.com
Storage Explorer
Download

1 Platform

Download

Find the labels above in your own portal.

Illustrated navigation view · use your own saved names and IDs

Why this action: The portal is useful for a few files; a transfer tool is more practical for many files.

1. On your computer open <https://azure.microsoft.com/products/storage/storage-explorer/> and install Microsoft Azure Storage Explorer for your operating system.

2B.01 · ACTION 4 OF 6**YOUR COMPUTER · Storage Explorer → Account icon → Add an account**

Local file or desktop application Illustrated view · layout may vary

YOUR COMPUTER

Storage Explorer > Account icon > Add an account

Storage Explorer
Account icon
Add an account

1 Cloud

2 Account

3 Subscription

Sign in

Find the labels above in your own portal.

Illustrated navigation view · use your own saved names and IDs

Why this action: The portal is useful for a few files; a transfer tool is more practical for many files.

1. Open Storage Explorer > account icon > Add an account.
2. Select Azure and sign in.
3. Select the practice subscription.

2B.01 · ACTION 5 OF 6

YOUR COMPUTER · Storage Explorer → Storage Accounts → PREFIXdata → Blob Containers → documents

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER

Search: Storage Explorer

Storage Explorer > Storage Accounts > PREFIXdata > Blob Containers > documents

Storage Explorer

Storage Accounts

PREFIXdata

1 Upload

2 Download

3 Check

Harmless text file

The same file

Compare contents

Find the labels above in your own portal.

Upload

Illustrated navigation view · use your own saved names and IDs

Why this action: The portal is useful for a few files; a transfer tool is more practical for many files.

1. Expand Storage Accounts > PREFIXdata > Blob Containers > documents.
2. Upload a harmless text file and download it again.
3. Compare its contents.

2B.01 · ACTION 6 OF 6

CLOUD SHELL · Cloud Shell → Bash → AzCopy

portal.azure.com

Illustrated view · layout may vary

CLOUD SHELL

Search: Cloud Shell

Cloud Shell > Bash > AzCopy

Cloud Shell

Bash

AzCopy

1 Sign-in

Your Blob-authorized account

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Use your Entra sign-in rather than pasting the storage account key.
2. If your organization blocks installation, complete the AzCopy transfer and record the Storage Explorer operation as DESIGN.

Read the path → do the action → check the result

99

You should see

The transferred file appears under the intended name and its contents match.

If it does not work

A successful sign-in does not prove network permission or data-role access. Check both if a transfer returns 403. [R10]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

2B.02 Create a share, snapshot and recover a file

AZ-104 · CORE + identity extension

Why this task: Azure Files provides shared folders using file protocols, unlike Blob object storage.

2B.02 · ACTION 1 OF 5

AZURE PORTAL · PREFIXdata → File shares → teamdocs

The screenshot shows the Azure Portal interface for creating a file share. The browser address bar shows 'portal.azure.com'. The page title is 'AZURE PORTAL'. A search bar contains 'PREFIXdata'. The breadcrumb navigation is 'PREFIXdata > File shares > teamdocs'. On the left, a sidebar menu has 'File shares' selected, with 'Snapshots', 'Browse', and 'Connect' below it. The main content area has four numbered steps: 1. Name (teamdocs), 2. Billing model (Inspect the offered model), 3. Sample data (One harmless text file), and 4. Recovery (Share snapshot). A 'Create' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > File shares > teamdocs

File shares

- Snapshots
- Browse
- Connect

- 1 Name** teamdocs
- 2 Billing model** Inspect the offered model
- 3 Sample data** One harmless text file
- 4 Recovery** Share snapshot

Find the labels above in your own portal. **Create**

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure Files provides shared folders using file protocols, unlike Blob object storage.

1. Open PREFIXdata > File shares > + File share.
2. Name it teamdocs.
3. Choose a small standard pay-as-you-go share if offered; inspect the selected billing model before creating it.

2B.02 · ACTION 2 OF 5**AZURE PORTAL · PREFIXdata → File shares → teamdocs → Browse**

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page header includes 'AZURE PORTAL' and a search bar containing 'PREFIXdata'. The breadcrumb navigation path is 'PREFIXdata > File shares > teamdocs > Browse'. On the left sidebar, 'File shares' is selected. The main content area features a numbered list: '1 File' and '2 Authorization'. To the right of these are two input fields: 'Harmless text file' and 'Supported lab authorization method'. At the bottom right, there is an 'Upload' button. A note at the bottom left states: 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open teamdocs and upload a harmless text file using Browse.
2. If asked for a portal authorization method, use the supported account-key route for this test account or a suitable data role.

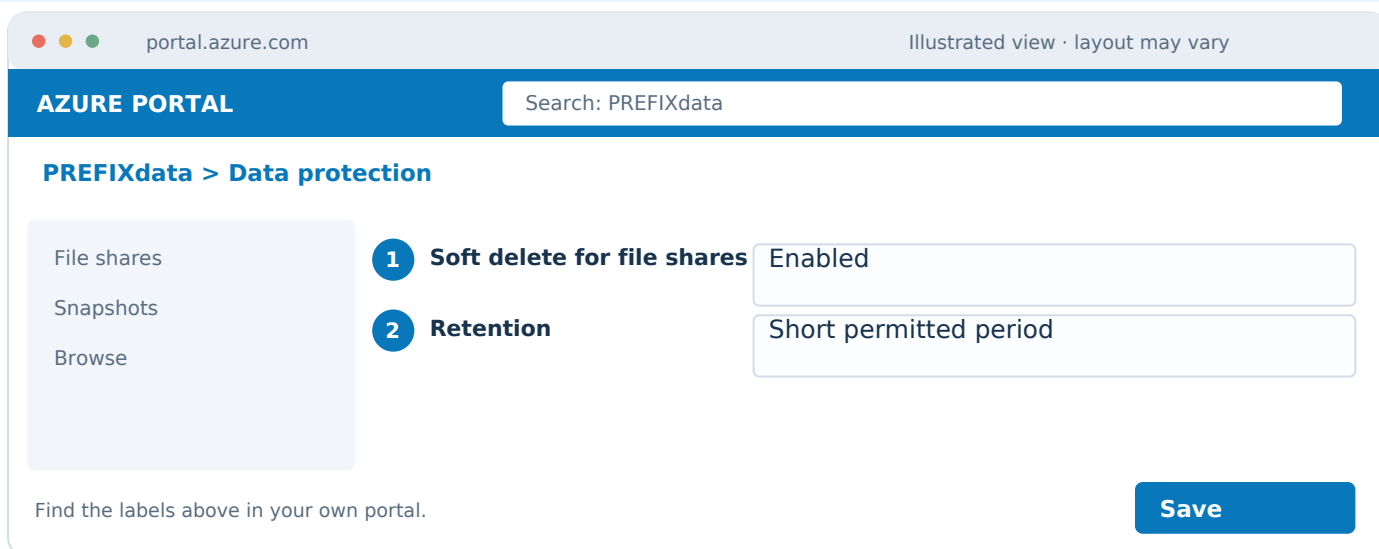
2B.02 · ACTION 3 OF 5**AZURE PORTAL · teamdocs → Snapshots → Add snapshot**

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page header includes 'AZURE PORTAL' and a search bar containing 'teamdocs'. The breadcrumb navigation path is 'teamdocs > Snapshots > Add snapshot'. On the left sidebar, 'Snapshots' is selected. The main content area features a numbered list: '1 Before change', '2 Then', and '3 Recover'. To the right of these are three input fields: 'Create snapshot', 'Change or delete test file', and 'Open snapshot and restore file'. At the bottom right, there is an 'Add snapshot' button. A note at the bottom left states: 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Use Snapshots > Add snapshot.
2. Change or delete the test file, open the snapshot, and restore or copy back that file.

2B.02 · ACTION 4 OF 5**AZURE PORTAL · PREFIXdata → Data protection**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > Data protection

File shares
Snapshots
Browse

1 Soft delete for file shares Enabled

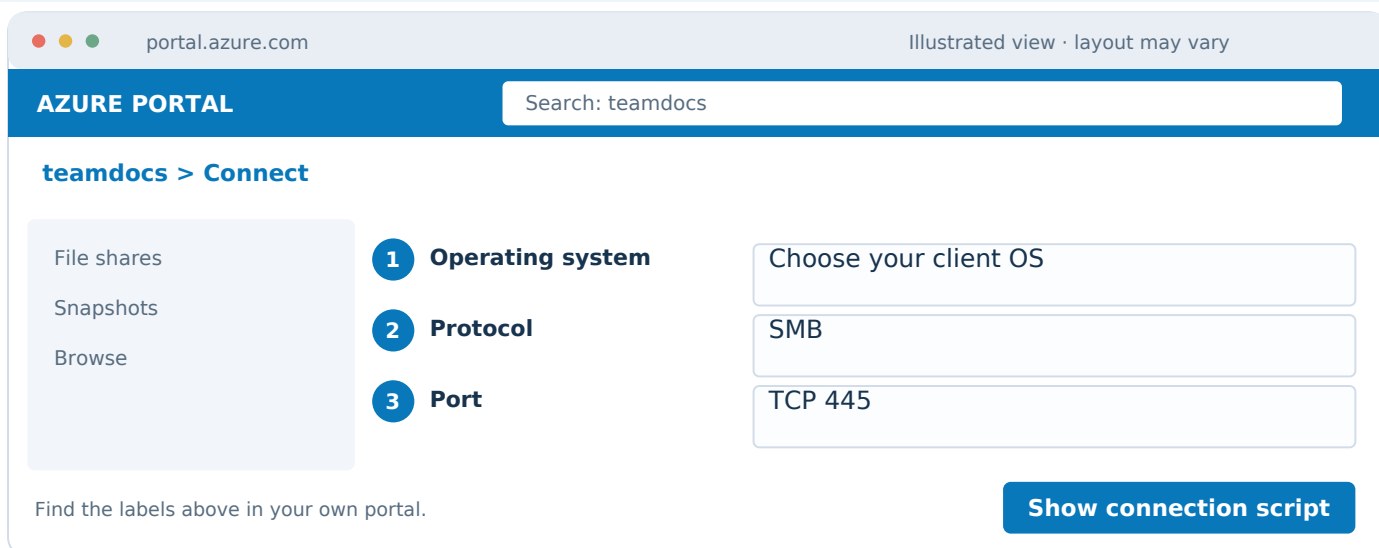
2 Retention Short permitted period

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. At the account > Data protection, confirm soft delete for file shares is enabled with a short allowed retention.
2. This protects a deleted share; it is not the same as a snapshot of a file.

2B.02 · ACTION 5 OF 5**AZURE PORTAL · teamdocs → Connect**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: teamdocs

teamdocs > Connect

File shares
Snapshots
Browse

1 Operating system Choose your client OS

2 Protocol SMB

3 Port TCP 445

Find the labels above in your own portal. **Show connection script**

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure Files provides shared folders using file protocols, unlike Blob object storage.

1. Open Connect, select Windows or Linux, and inspect the generated mount script.
2. Explain that SMB normally uses TCP 445 and a blocked port can prevent a mount.
3. Do not publish generated scripts containing keys.

You should see

You recover the file from a share snapshot and can distinguish share soft delete from snapshots.

If it does not work

A Blob data role does not grant SMB file access. The share, identity source, share-level role, file ACL and network path must all be correct.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

2B.03 Design identity-based SMB access

AZ-104 · CORE + identity extension

Why this task: Real team shares should authorize named people rather than distribute a storage key.

2B.03 · ACTION 1 OF 5

AZURE PORTAL · PREFIXdata → File shares → Identity-based access

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page header includes 'AZURE PORTAL' and a search bar containing 'PREFIXdata'. The breadcrumb navigation path is 'PREFIXdata > File shares > Identity-based access'. On the left, a navigation pane lists 'PREFIXdata', 'File shares', and 'Identity-based access' (which is highlighted). The main content area features a list of requirements: '1 Need' and '2 Also need'. To the right of these are two input fields: 'Supported identity source' and 'Eligible Windows client'. At the bottom left, a note says 'Find the labels above in your own portal.' and at the bottom right is a blue 'Configure' button.

Illustrated navigation view · use your own saved names and IDs

Why this action: Real team shares should authorize named people rather than distribute a storage key.

1. Open the account > File shares > Identity-based access.
2. Inspect the available identity sources: AD DS, Microsoft Entra Domain Services, or Microsoft Entra Kerberos.

2B.03 · ACTION 2 OF 5**SEE THE IDEA · Identity-based SMB prerequisites****A SMALL DECISION MAP****Identity-based SMB prerequisites**

- | | |
|--------------------------|---|
| 1 Identity source | AD DS, Entra Domain Services, or supported Entra Kerberos |
| 2 Client | Meets that source's requirements |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Real team shares should authorize named people rather than distribute a storage key.

1. Choose the method that matches the company's existing identities.
2. The account uses one identity source at a time.
3. Cloud-only Entra Kerberos support has additional documented client, region and configuration requirements.

2B.03 · ACTION 3 OF 5**AZURE PORTAL · PREFIXdata → File shares → Identity-based access**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL

PREFIXdata > File shares > Identity-based access

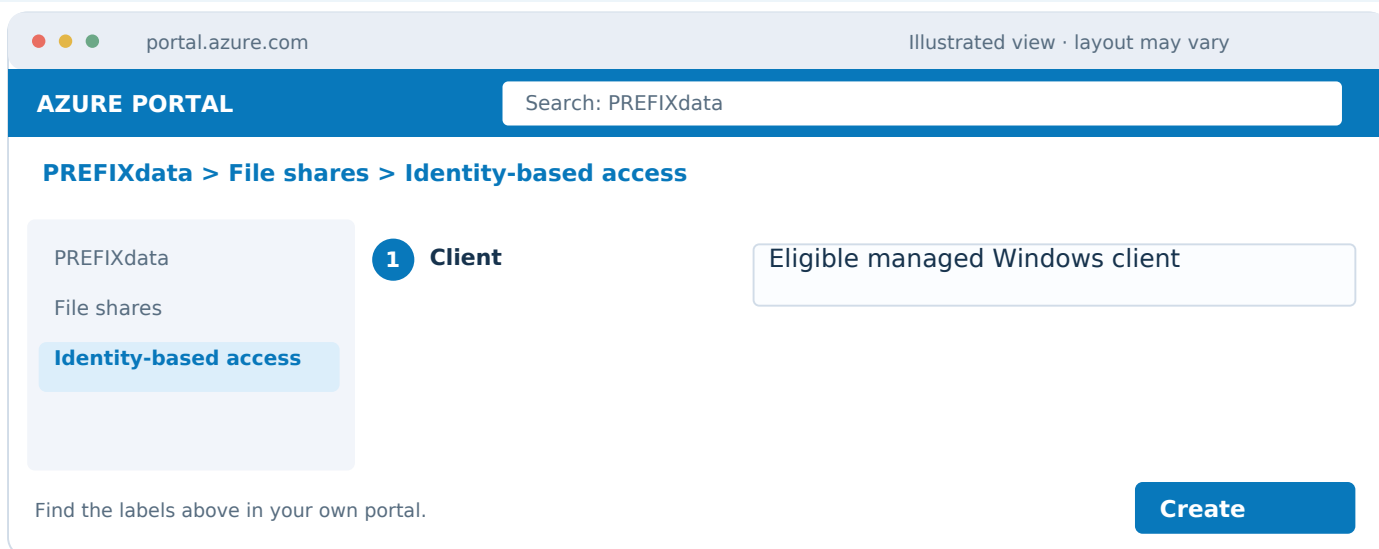
PREFIXdata File shares Identity-based access	1 Identity source	One supported company source
	2 Share role	Storage File Data SMB Share Contributor
	3 File permission	Folder/file ACL
	4 Client	Eligible managed Windows client

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

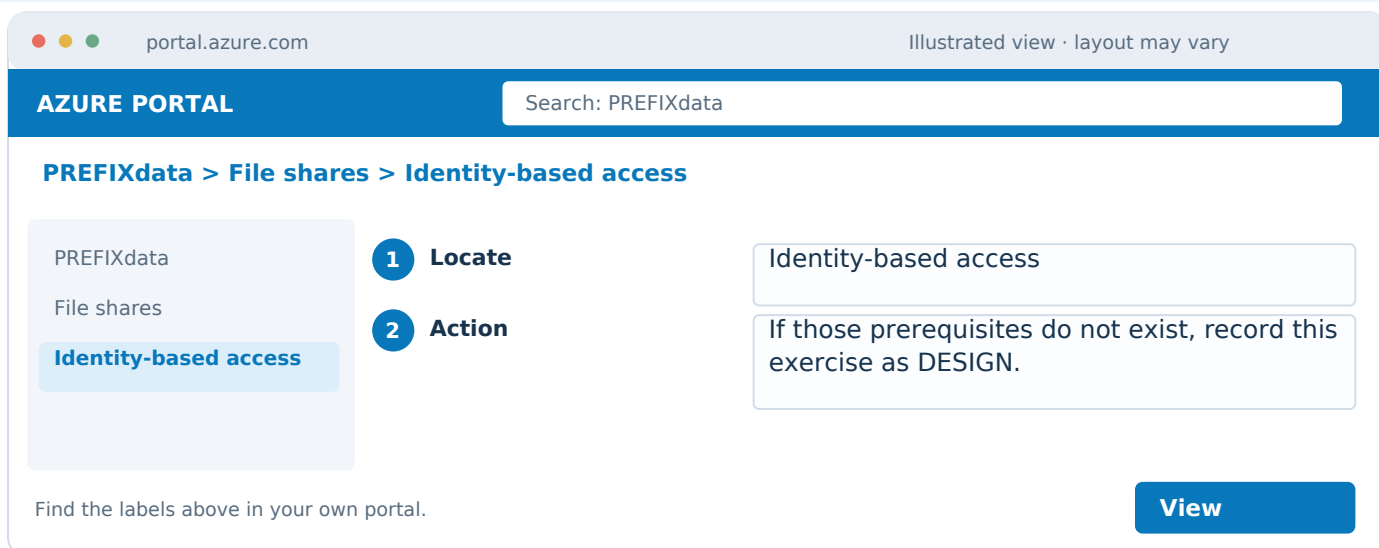
1. If you already have an eligible, managed Windows client and identity setup, enable that identity source, grant the test group Storage File Data SMB Share Contributor at share scope, and set folder/file ACLs for that group.

2B.03 · ACTION 4 OF 5**AZURE PORTAL · PREFIXdata → File shares → Identity-based access**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. From that eligible client, mount the share using the signed-in identity.
2. Create one file as the allowed user and verify a user without permission is denied.

2B.03 · ACTION 5 OF 5**AZURE PORTAL · PREFIXdata → File shares → Identity-based access**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. If those prerequisites do not exist, record this exercise as DESIGN.
2. Do not mistake the Connect script's account-key mount for a completed identity-based access test.

You should see

Your record names the identity source, share role, ACL and client/network requirements.

If it does not work

Cloud Shell is not a domain-joined Windows SMB client. Creating a user in Entra alone is insufficient to complete this scenario. [R11]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

2B.04 Replicate the project documents to a second account

AZ-104 · CORE + identity extension

Why this task: A replication rule teaches selected-object copying independently of the account redundancy choice.

2B.04 · ACTION 1 OF 6

AZURE PORTAL · Storage accounts → Create

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Storage accounts

Storage accounts > Create

Storage accounts

Create

1 Name PREFIXcopy

2 Group rg-PREFIX-core

3 Redundancy Supported replication arrangement

Find the labels above in your own portal.

Review + create

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. Create a second Standard StorageV2 account named PREFIXcopy in rg-PREFIX-core.
2. Keep hierarchical namespace off and containers private.
3. Enable Blob versioning.

2B.04 · ACTION 2 OF 6**AZURE PORTAL · PREFIXdata → Data protection, then PREFIXcopy → Containers**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > Data protection, then PREFIXcopy > Containers

PREFIXdata
Data protection, then PREFIXcopy
Containers

- 1 Source change feed** Enabled
- 2 Source versioning** Enabled
- 3 Destination container** documents-copy

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. Enable change feed and versioning on the source PREFIXdata account.
2. Create a documents-copy container on the destination.

2B.04 · ACTION 3 OF 6**AZURE PORTAL · PREFIXdata → Object replication → Create replication rules**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > Object replication > Create replication rules

PREFIXdata
Object replication
Create replication rules

- 1 Destination account** PREFIXcopy
- 2 Source container** documents
- 3 Destination container** documents-copy

Find the labels above in your own portal. **Next**

Illustrated navigation view · use your own saved names and IDs

Why this action: A replication rule teaches selected-object copying independently of the account redundancy choice.

1. On the source account open Data management > Object replication > Create replication rules.
2. Select the destination account and map documents to documents-copy.

2B.04 · ACTION 4 OF 6**AZURE PORTAL · Create replication rules → Copy scope / Review**

The screenshot shows the Azure Portal interface for the 'Create replication rules' wizard. The breadcrumb trail is 'Create replication rules > Copy scope / Review'. On the left, a sidebar lists 'Create replication rules' and 'Copy scope / Review' (which is highlighted). The main content area shows two numbered steps: '1 Include' and '2 Policy'. Under '1 Include', there is a text input field containing 'Chosen new or existing object time range'. Under '2 Policy', there is a text input field containing 'Linked on both accounts'. At the bottom right, there is a blue 'Save' button. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Choose the copy scope/time offered by the wizard and record whether existing objects are included.
2. Save.
3. Both accounts must receive the linked policy; the same-tenant portal wizard can configure this for you.

2B.04 · ACTION 5 OF 6**AZURE PORTAL · PREFIXcopy → Containers → documents-copy**

The screenshot shows the Azure Portal interface for the 'PREFIXcopy' wizard. The breadcrumb trail is 'PREFIXcopy > Containers > documents-copy'. On the left, a sidebar lists 'PREFIXcopy', 'Containers', and 'documents-copy' (which is highlighted). The main content area shows two numbered steps: '1 First' and '2 Then'. Under '1 First', there is a text input field containing 'Upload a new source blob'. Under '2 Then', there is a text input field containing 'Wait and verify destination content'. At the bottom right, there is a blue 'Refresh' button. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Upload a new small blob to the source.
2. Wait for asynchronous replication and inspect its destination copy and replication status.

2B.04 · ACTION 6 OF 6**AZURE PORTAL · PREFIXdata → Object replication → Your lab policy**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > Object replication > Your lab policy

PREFIXdata
Object replication
Your lab policy

1 Remove

The practice replication rule after verification

Find the labels above in your own portal. **Delete**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Delete the replication rule after testing if you do not need it.
2. Do not directly edit the destination container while a policy makes it read-only.

You should see

A new source object appears in the mapped destination container.

If it does not work

Source change feed and versioning on both accounts are prerequisites. Replication delay is normal; this is not a synchronous backup guarantee. [R12]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

2C · Connect two private networks

Planning estimate: 75 minutes. Finish with: Connected peerings, private DNS links and an NSG.

2C.01 Create two networks and their subnets

AZ-104 · CORE

Why this task: The operations VM and the application integration path need separate address ranges.

2C.01 · ACTION 1 OF 6

AZURE PORTAL · Virtual networks → Create → Basics

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Virtual networks

Virtual networks > Create > Basics

Basics
IP addresses
Security

1 Group rg-PREFIX-ops

2 Name vnet-main

3 Region Primary region

Find the labels above in your own portal. **Next**

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Search Virtual networks > Create.
2. Choose rg-PREFIX-ops; name vnet-main; region your primary region.
3. On IP addresses set address space 10.20.0.0/16.

2C.01 · ACTION 2 OF 6**AZURE PORTAL · Create virtual network → IP addresses**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Create virtual network

Create virtual network > IP addresses

Basics
IP addresses
Security

1 Address space 10.20.0.0/16

2 Subnet name ops

3 Subnet range 10.20.1.0/24

Find the labels above in your own portal. **Add**

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Create subnet ops with range 10.20.1.0/24.
2. Remove an unwanted overlapping default subnet if the wizard created one.
3. Review + create.

2C.01 · ACTION 3 OF 6**AZURE PORTAL · vnet-main → Subnets → + Subnet**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: vnet-main

vnet-main > Subnets > + Subnet

Basics
IP addresses
Security

1 Name integration

2 Address range 10.20.2.0/24

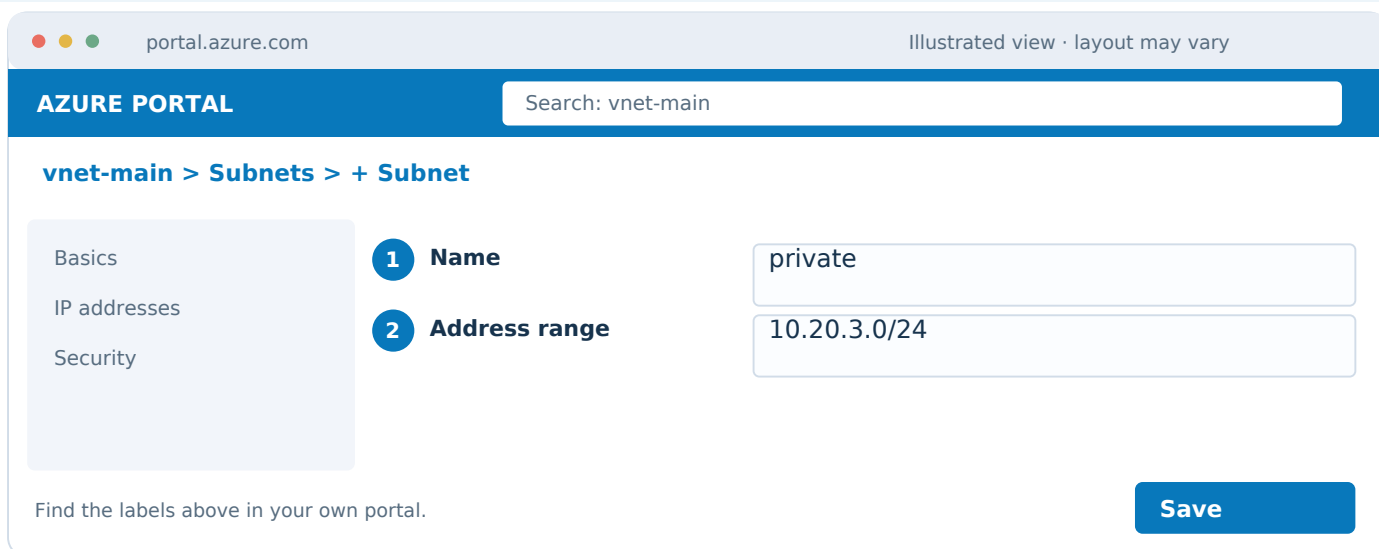
3 Delegation Microsoft.Web/serverFarms

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Open vnet-main > Subnets > + Subnet.
2. Add integration with range 10.20.2.0/24 and delegation Microsoft.Web/serverFarms for App Service outbound integration.

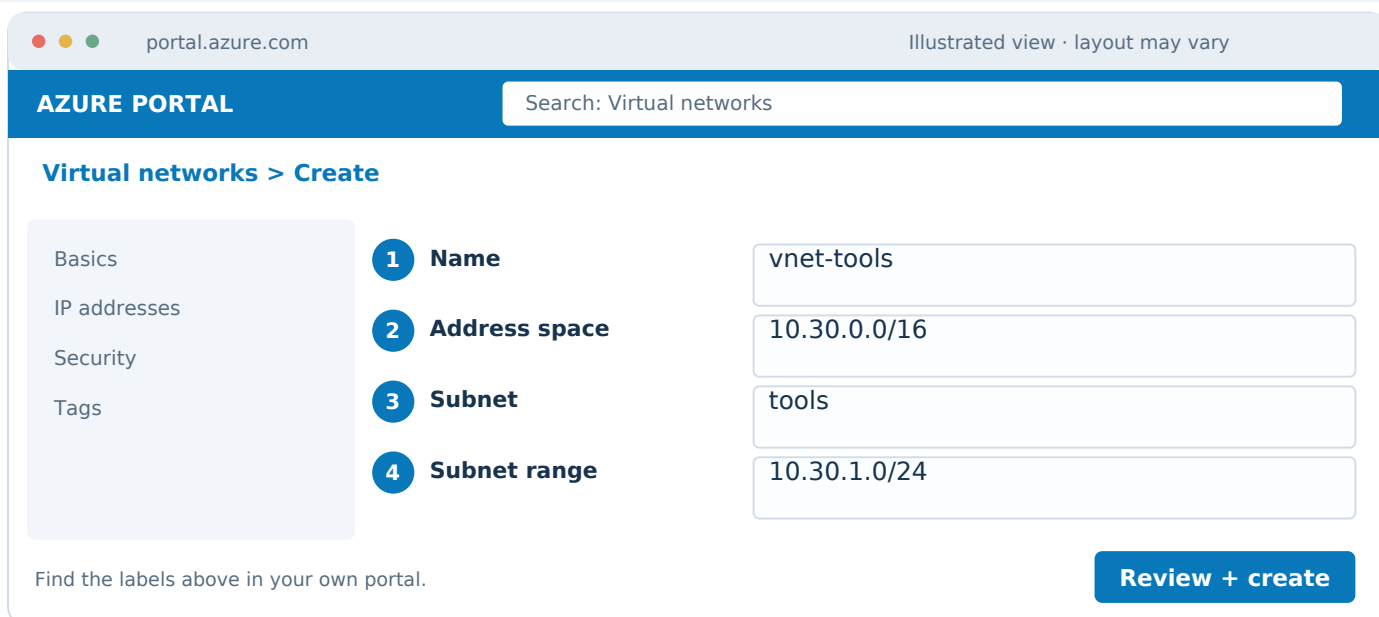
2C.01 · ACTION 4 OF 6**AZURE PORTAL · vnet-main → Subnets → + Subnet**

The screenshot shows the Azure Portal interface for creating a new subnet. The browser address bar displays 'portal.azure.com'. The page title is 'AZURE PORTAL'. A search bar contains 'vnet-main'. The breadcrumb navigation is 'vnet-main > Subnets > + Subnet'. On the left, a sidebar lists 'Basics', 'IP addresses', and 'Security'. The main form has two numbered steps: 1. Name (value: 'private') and 2. Address range (value: '10.20.3.0/24'). A 'Save' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Create another subnet private with range 10.20.3.0/24 for the private endpoint exercise.

2C.01 · ACTION 5 OF 6**AZURE PORTAL · Virtual networks → Create**

The screenshot shows the Azure Portal interface for creating a new virtual network. The browser address bar displays 'portal.azure.com'. The page title is 'AZURE PORTAL'. A search bar contains 'Virtual networks'. The breadcrumb navigation is 'Virtual networks > Create'. On the left, a sidebar lists 'Basics', 'IP addresses', 'Security', and 'Tags'. The main form has four numbered steps: 1. Name (value: 'vnet-tools'), 2. Address space (value: '10.30.0.0/16'), 3. Subnet (value: 'tools'), and 4. Subnet range (value: '10.30.1.0/24'). A 'Review + create' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Create vnet-tools in the same group and region using 10.30.0.0/16, with subnet tools at 10.30.1.0/24.

2C.01 · ACTION 6 OF 6**SEE THE IDEA · Create two networks and their subnets → Explain the decision****A SMALL DECISION MAP****Create two networks and their subnets > Explain the decision**

- | | | |
|----------|---------------|--|
| 1 | Locate | Explain the decision |
| 2 | Action | Never give two peered networks overlapping address ranges. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Never give two peered networks overlapping address ranges.
2. The browser portal and Cloud Shell are not automatically inside these networks.

You should see

You have two nonoverlapping VNets and the named subnets.

If it does not work

If the IP range is rejected, check overlap and that the subnet sits within the VNet address space.
[R13]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

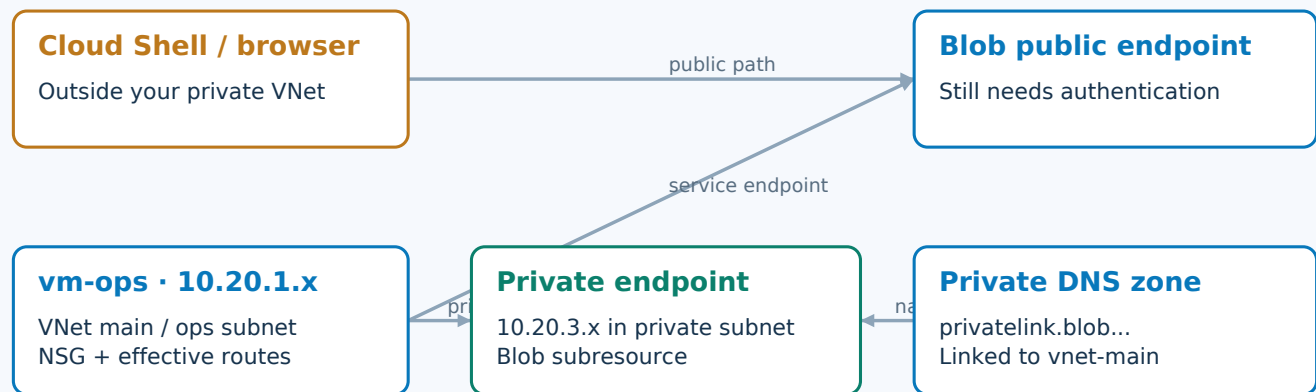
Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

An IP address identifies a network interface. A subnet is a smaller range inside a virtual network. /24 means 256 addresses in the range, but Azure reserves five in each subnet. A route chooses where a packet goes; an NSG decides whether traffic is allowed. DNS translates a name into an address.

Follow the packet before changing a rule

A data permission never bypasses a firewall. Test DNS from inside the VNet.



2C.02 Peer the networks and add private name resolution

AZ-104 · CORE

Why this task: Peering allows private-IP traffic between the two VNets; DNS still needs its own configuration.

2C.02 · ACTION 1 OF 6

AZURE PORTAL · vnet-main → Peerings → Add

The screenshot shows the Azure Portal interface for adding a peering to 'vnet-main'. The breadcrumb navigation is 'vnet-main > Peerings > Add'. The left sidebar shows 'Peerings' as the active section, with 'Subnets' and 'Private DNS links' below it. The main area contains three numbered steps:

- 1 Local link**: Input field contains 'main-to-tools'.
- 2 Remote VNet**: Input field contains 'vnet-tools'.
- 3 Remote link**: Input field contains 'tools-to-main'.

Below the input fields, there is a note: 'Find the labels above in your own portal.' and an 'Add' button.

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Open vnet-main > Peerings > Add.
2. Local link name main-to-tools; remote virtual network vnet-tools; remote link name tools-to-main.

2C.02 · ACTION 2 OF 6**AZURE PORTAL · Peerings → Link settings**

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page header includes 'AZURE PORTAL' and a search bar containing 'Search: Peerings'. The main content area is titled 'Peerings > Link settings'. On the left, a sidebar lists 'Peerings', 'Subnets', and 'Private DNS links', with 'Peerings' selected. The main area contains two numbered steps: '1 Allow VNet access' with a value of 'Enabled both ways', and '2 Gateway transit' with a value of 'Not needed here'. At the bottom left, a note says 'Find the labels above in your own portal.' and at the bottom right is a blue 'Add' button.

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Enable normal virtual-network access in both directions.
2. Leave gateway transit, remote gateway and forwarded traffic options off for this direct lab path.
3. Save.

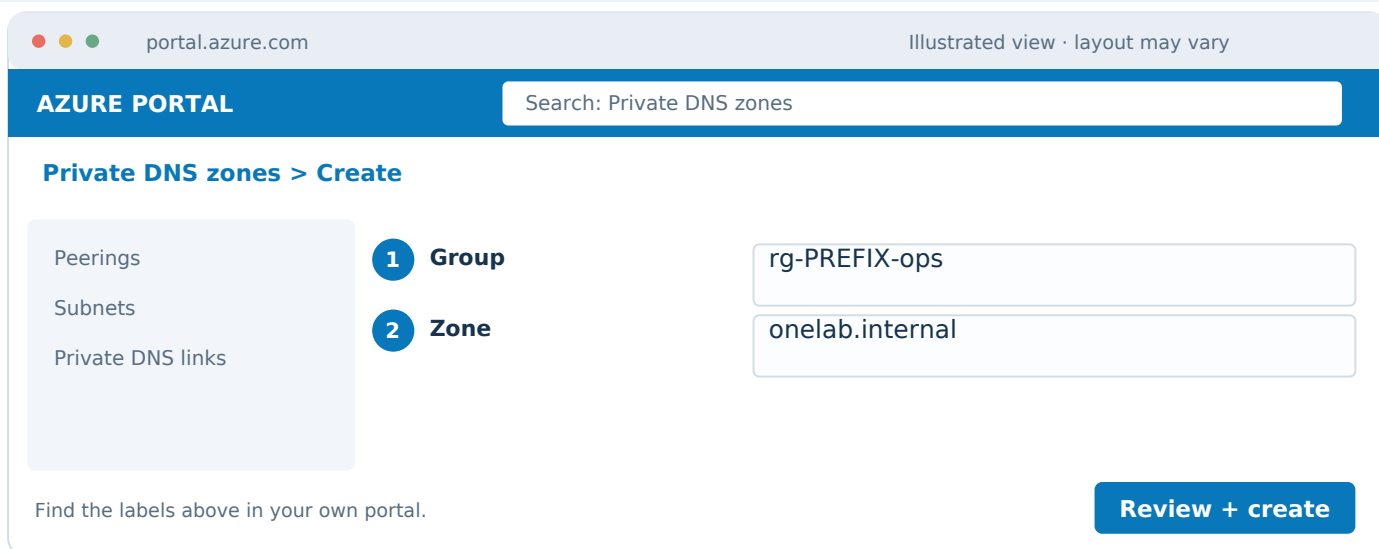
2C.02 · ACTION 3 OF 6**AZURE PORTAL · Each VNet → Peerings**

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page header includes 'AZURE PORTAL' and a search bar containing 'Search: Each VNet'. The main content area is titled 'Each VNet > Peerings'. On the left, a sidebar lists 'Peerings', 'Subnets', and 'Private DNS links', with 'Peerings' selected. The main area contains two numbered steps: '1 Status' with a value of 'Connected', and '2 Address overlap' with a value of 'None'. At the bottom left, a note says 'Find the labels above in your own portal.' and at the bottom right is a blue 'Refresh' button.

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open Peerings on both networks.
2. Confirm Connected.

2C.02 · ACTION 4 OF 6**AZURE PORTAL · Private DNS zones → Create**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Private DNS zones

Private DNS zones > Create

Peerings
Subnets
Private DNS links

1 Group rg-PREFIX-ops

2 Zone onelab.internal

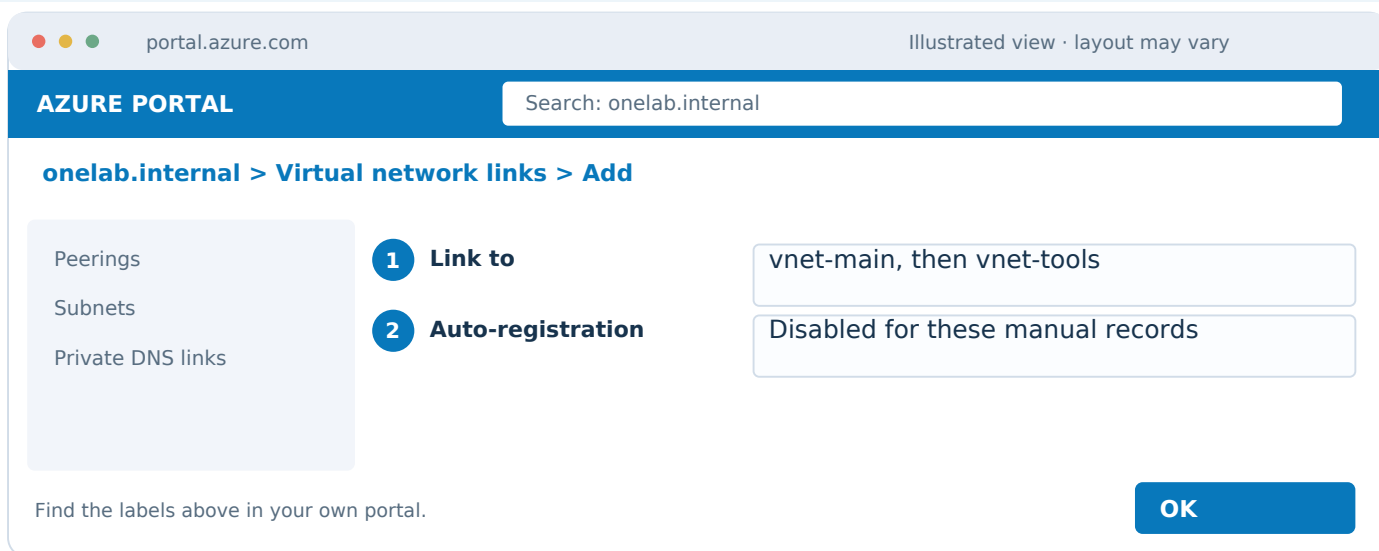
Find the labels above in your own portal.

Review + create

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Search Private DNS zones > Create.
2. Use rg-PREFIX-ops and zone onelab.internal.
3. Create it.

2C.02 · ACTION 5 OF 6**AZURE PORTAL · onelab.internal → Virtual network links → Add**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: onelab.internal

onelab.internal > Virtual network links > Add

Peerings
Subnets
Private DNS links

1 Link to vnet-main, then vnet-tools

2 Auto-registration Disabled for these manual records

Find the labels above in your own portal.

OK

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Open the zone > Virtual network links > Add.
2. Link vnet-main with auto-registration off; repeat for vnet-tools.

2C.02 · ACTION 6 OF 6**SEE THE IDEA · Peer the networks and add private name resolution → Explain the decision****A SMALL DECISION MAP****Peer the networks and add private name resolution > Explain the decision**

- | | | |
|----------|---------------|--|
| 1 | Locate | Explain the decision |
| 2 | Action | You will add an A record for the VM after it exists. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. You will add an A record for the VM after it exists.
2. Peering is not transitive: main-to-tools and tools-to-third do not automatically make main-to-third work.

You should see

Both peering links are connected and both VNets are linked to the private DNS zone.

If it does not work

A public DNS zone is different from a private zone. Public delegation requires control of the domain registrar settings. [R14]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

2C.03 Create a network security group and application group

AZ-104 · CORE

Why this task: You can control traffic by port and group NICs by application role.

2C.03 · ACTION 1 OF 5

AZURE PORTAL · Network security groups → nsg-ops → Subnets

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL

Search: Network security groups

Network security groups > nsg-ops > Subnets

Network security groups

nsg-ops

Subnets

1 NSG

2 VNet

3 Subnet

4 ASG

5 Public SSH / RDP

nsg-ops

vnet-main

ops

asg-ops

No new allow rule

Find the labels above in your own portal.

Associate

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Search Network security groups > Create.
2. Name nsg-ops; use rg-PREFIX-ops and the primary region.

2C.03 · ACTION 2 OF 5**AZURE PORTAL · nsg-ops → Subnets → Associate**

The screenshot shows the Azure Portal interface for associating a subnet with a network security group. The breadcrumb navigation is 'nsg-ops > Subnets > Associate'. On the left, a sidebar shows 'nsg-ops', 'Subnets', and 'Associate' (highlighted). The main area has two numbered steps: 1. VNet (with a dropdown menu showing 'vnet-main') and 2. Subnet (with a dropdown menu showing 'ops'). At the bottom right is an 'OK' button.

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: nsg-ops

nsg-ops > Subnets > Associate

nsg-ops
Subnets
Associate

1 VNet vnet-main
2 Subnet ops

Find the labels above in your own portal. **OK**

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Open nsg-ops > Subnets > Associate.
2. Choose vnet-main and ops.

2C.03 · ACTION 3 OF 5**AZURE PORTAL · Application security groups → Create**

The screenshot shows the Azure Portal interface for creating an application security group. The breadcrumb navigation is 'Application security groups > Create'. On the left, a sidebar shows 'Application security groups' and 'Create' (highlighted). The main area has three numbered steps: 1. Name (with a text input field containing 'asg-ops'), 2. Group (with a text input field containing 'rg-PREFIX-ops'), and 3. Region (with a dropdown menu showing 'Primary region'). At the bottom right is a 'Review + create' button.

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Application security groups

Application security groups > Create

Application security groups
Create

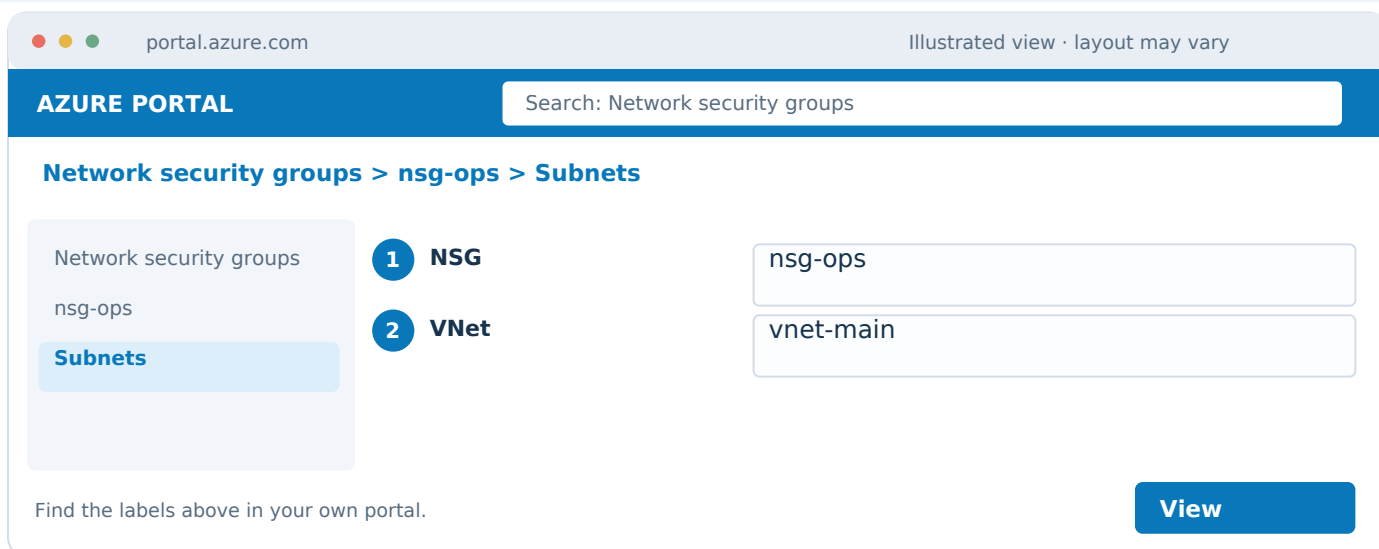
1 Name asg-ops
2 Group rg-PREFIX-ops
3 Region Primary region

Find the labels above in your own portal. **Review + create**

Illustrated navigation view · use your own saved names and IDs

Why this action: You can control traffic by port and group NICs by application role.

1. Search Application security groups > Create.
2. Name asg-ops, same group and region.
3. You will attach the VM NIC to it in Day 2, Block D, Day 6, Block A.

2C.03 · ACTION 4 OF 5**AZURE PORTAL · Network security groups → nsg-ops → Subnets**

Illustrated navigation view · use your own saved names and IDs

Why this action: You can control traffic by port and group NICs by application role.

1. Inspect default NSG rules.
2. AllowVNetInBound and AllowAzureLoadBalancerInBound precede the final deny.
3. Custom lower-numbered priorities are evaluated first.

2C.03 · ACTION 5 OF 5**SEE THE IDEA · Create a network security group and application group → Explain the decision****A SMALL DECISION MAP****Create a network security group and application group > Explain the decision**

- 1 **Locate** Explain the decision
- 2 **Action** Do not add Internet-to-SSH or RDP rules for this lab.

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: You can control traffic by port and group NICs by application role.

1. Do not add Internet-to-SSH or RDP rules for this lab.
2. You will use Run command, and optionally Bastion, to administer the VM.

You should see

nsg-ops is associated with ops and no unnecessary public administration rule exists.

If it does not work

If both subnet and NIC have NSGs, traffic must be permitted by both. A broad allow in one does not override a deny in the other.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

2D · Create the VM and start protection

Planning estimate: 120 minutes. Finish with: A working VM health page; backup/replication submitted and tracked.

2D.01 Create the operations VM

AZ-104 · CORE + availability extension

Why this task: This VM hosts an internal health page and is used for disk, network, monitoring and backup exercises.

2D.01 · ACTION 1 OF 7

AZURE PORTAL · Virtual machines → Create → Azure virtual machine

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Virtual machines

Virtual machines > Create > Azure virtual machine

Basics
Disks
Networking
Management

- 1 Resource group** rg-PREFIX-ops
- 2 Name** vm-ops
- 3 Image** Ubuntu Server 24.04 LTS x64
- 4 Size** Quota-approved size
- 5 Username** azureuser
- 6 Inbound ports** None

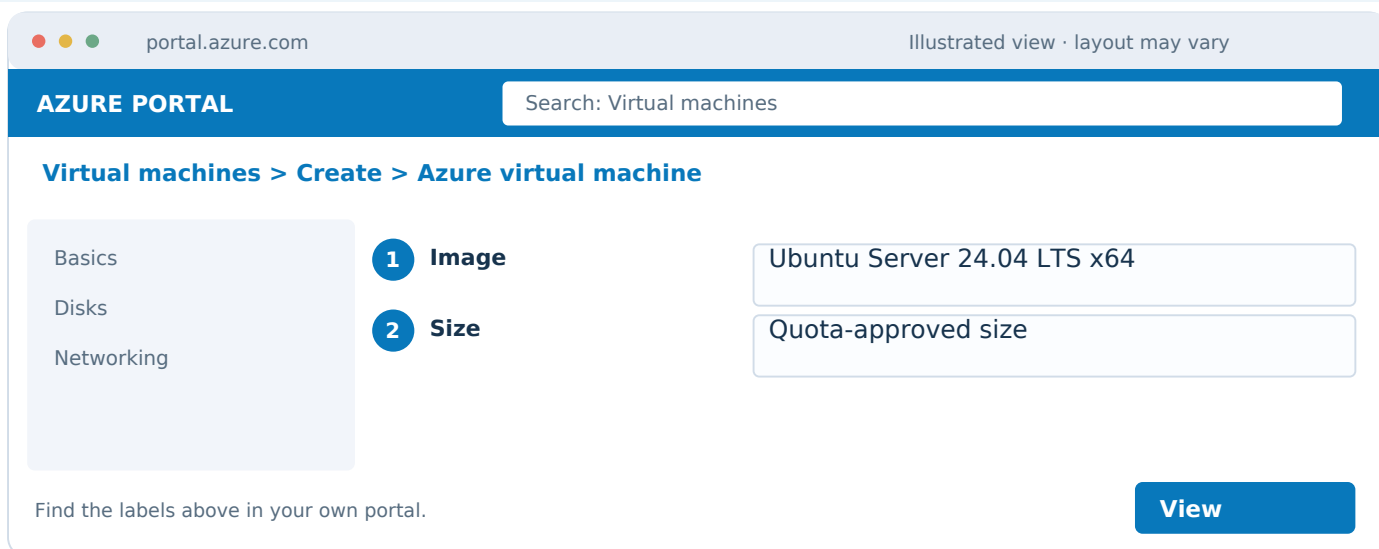
Find the labels above in your own portal.

Review + create

Illustrated navigation view · use your own saved names and IDs

Why this action: This VM hosts an internal health page and is used for disk, network, monitoring and backup exercises.

1. Search Virtual machines > Create > Azure virtual machine.
2. Choose rg-PREFIX-ops; name vm-ops; primary region.

2D.01 · ACTION 2 OF 7**AZURE PORTAL · Virtual machines → Create → Azure virtual machine**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Virtual machines

Virtual machines > Create > Azure virtual machine

Basics
Disks
Networking

1 Image Ubuntu Server 24.04 LTS x64

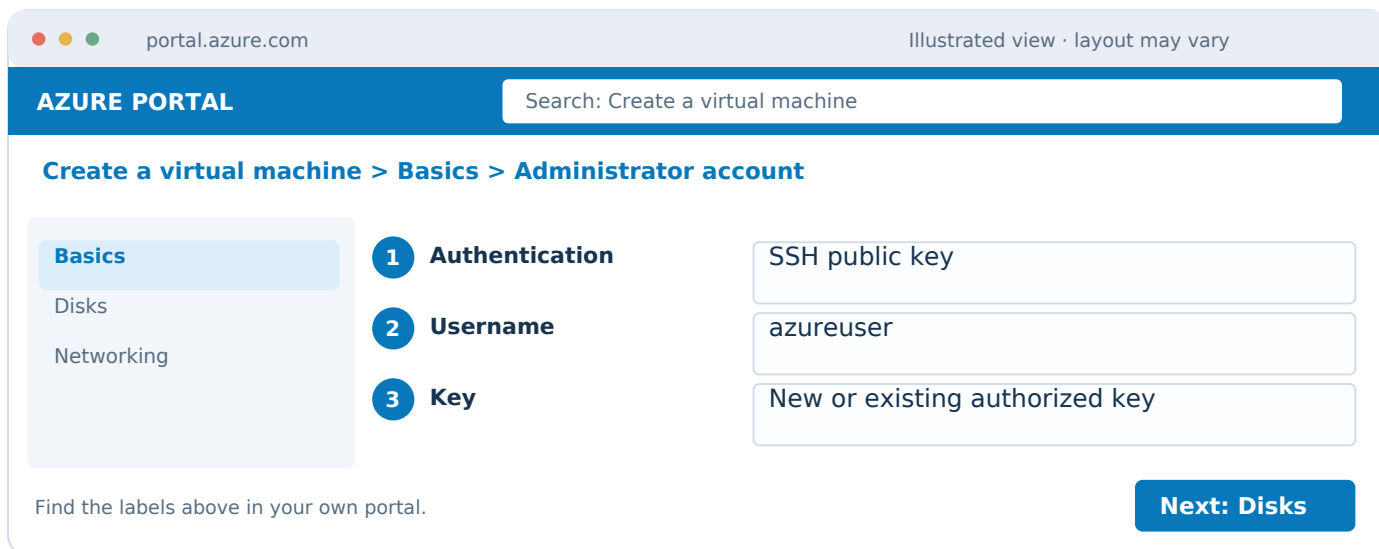
2 Size Quota-approved size

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Choose Ubuntu Server 24.04 LTS x64, a supported Standard_D2s_v5 or quota-approved equivalent, and an available zone if the region/size supports one.
2. Record the chosen zone.

2D.01 · ACTION 3 OF 7**AZURE PORTAL · Create a virtual machine → Basics → Administrator account**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Create a virtual machine

Create a virtual machine > Basics > Administrator account

Basics
Disks
Networking

1 Authentication SSH public key

2 Username azureuser

3 Key New or existing authorized key

Find the labels above in your own portal. **Next: Disks**

Illustrated navigation view · use your own saved names and IDs

Why this action: This VM hosts an internal health page and is used for disk, network, monitoring and backup exercises.

1. Use SSH public key authentication with username azureuser.
2. Generate a new key pair if needed and save the private key locally when prompted; never upload it to Git.

2D.01 · ACTION 4 OF 7**AZURE PORTAL · Create a virtual machine → Disks**

The screenshot shows the Azure Portal interface for creating a virtual machine. The browser address bar displays 'portal.azure.com'. The page title is 'AZURE PORTAL' with a search bar containing 'Create a virtual machine'. The breadcrumb navigation is 'Create a virtual machine > Disks'. On the left, a sidebar lists 'Basics', 'Disks' (selected), and 'Networking'. The main content area shows three numbered steps: 1. OS disk (Standard SSD if allowed), 2. Encryption at host (Only if supported), and 3. Inbound ports (None on Basics). A 'Next: Networking' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Create a virtual machine

Create a virtual machine > Disks

Basics
Disks
Networking

1 **OS disk** Standard SSD if allowed

2 **Encryption at host** Only if supported

3 **Inbound ports** None on Basics

Find the labels above in your own portal. **Next: Networking**

Illustrated navigation view · use your own saved names and IDs

Why this action: This VM hosts an internal health page and is used for disk, network, monitoring and backup exercises.

1. For inbound ports choose None.
2. Under Disks, choose Standard SSD if allowed for the learning workload.
3. Review Encryption at host; enable it only if the subscription and size support it.

2D.01 · ACTION 5 OF 7**AZURE PORTAL · Create a virtual machine → Networking**

The screenshot shows the Azure Portal interface for creating a virtual machine, now on the Networking tab. The browser address bar displays 'portal.azure.com'. The page title is 'AZURE PORTAL' with a search bar containing 'Create a virtual machine'. The breadcrumb navigation is 'Create a virtual machine > Networking'. On the left, a sidebar lists 'Basics', 'Disks', 'Networking' (selected), and 'Management'. The main content area shows four numbered steps: 1. VNet (vnet-main), 2. Subnet (ops), 3. Public IP (Standard (explicit outbound)), and 4. Public inbound rule (None). A 'Next: Management' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Create a virtual machine

Create a virtual machine > Networking

Basics
Disks
Networking
Management

1 **VNet** vnet-main

2 **Subnet** ops

3 **Public IP** Standard (explicit outbound)

4 **Public inbound rule** None

Find the labels above in your own portal. **Next: Management**

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Under Networking choose vnet-main, subnet ops, and an existing or new Standard public IP.
2. Its purpose here is explicit outbound Internet access.
3. Keep inbound access restricted by the NSG.

2D.01 · ACTION 6 OF 7**AZURE PORTAL · Create a virtual machine → Management**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Create a virtual machine

Create a virtual machine > Management

Basics
Disks
Networking

1 Auto-shutdown After your daily lab session

2 Estimate Review before creation

Find the labels above in your own portal.

Review + create

Illustrated navigation view · use your own saved names and IDs

Why this action: This VM hosts an internal health page and is used for disk, network, monitoring and backup exercises.

1. On Management configure auto-shutdown at a time after your practice session.
2. Review the hourly estimate and create.

2D.01 · ACTION 7 OF 7**AZURE PORTAL · vm-ops → Networking → Network interface → Application security groups**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: vm-ops

vm-ops > Networking > Network interface > Application security groups

Basics
Disks
Networking

1 Application security group asg-ops

Find the labels above in your own portal.

Save

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Open Overview.
2. Record private IP and public IP.
3. Open Networking > the NIC > Application security groups and add asg-ops.

You should see

The VM is Running and has a private IP in 10.20.1.0/24.

If it does not work

Newer VNets require an explicit outbound method. An attached Standard public IP supplies one; an NSG still controls incoming access. Do not rely on old implicit outbound behavior. [R15]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

2D.02 Install a private health page with Run command

AZ-104 · CORE + availability extension

Why this task: Run command executes inside the VM through the Azure VM agent without opening SSH.

2D.02 · ACTION 1 OF 5

VM TERMINAL IN AZURE PORTAL · Azure Portal → vm-ops → Run command → RunShellScript

The dark code panel below is the terminal/editor view for this action.

Why this action: Run command executes inside the VM through the Azure VM agent without opening SSH.

1. Open vm-ops > Operations > Run command > RunShellScript.
2. Paste the Linux script below and select Run.

USE IN: TERMINAL: INSIDE VM THROUGH RUNSHELLSCRIPT · COPY ONLY THE CODE

```
sudo apt-get update
sudo apt-get install -y nginx
printf 'OneLab operations\n' | sudo tee /var/www/html/index.html
sudo systemctl enable --now nginx
curl -f http://127.0.0.1
```

What it means: apt installs software. systemctl enables and starts the service. 127.0.0.1 means this VM itself, so the curl check does not yet prove remote network access.

curl · Send an HTTP request. -f fails on HTTP error status; -s quiets the progress meter; -v shows connection details when used.

systemctl enable --now · Set the service to start at boot and start it now; stop/start act on the running service.

sudo · Run that Linux operation with administrator privileges inside the named VM.

2D.02 · ACTION 2 OF 5**AZURE PORTAL · vm-ops → Operations → Run command → RunShellScript**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: vm-ops

vm-ops > Operations > Run command > RunShellScript

vm-ops
Operations
Run command

1 Locate RunShellScript

2 Action Wait for success.

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Wait for success.
2. Read output showing a running nginx service and an HTTP response.

2D.02 · ACTION 3 OF 5**AZURE PORTAL · Private DNS zones → onelab.internal → Recordsets → Add**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Private DNS zones

Private DNS zones > onelab.internal > Recordsets > Add

Private DNS zones
onelab.internal
Recordsets
Add

1 Name ops

2 Type A

3 TTL 300

4 IP address Your VM private IP

Find the labels above in your own portal.

Add

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Open the private zone onelab.internal > Recordsets > Add.
2. Name ops; type A
3. TTL 300; address the VM's private IP; save.

2D.02 · ACTION 4 OF 5**VM TERMINAL IN AZURE PORTAL · Azure Portal → vm-ops → Run command → RunShellScript**

The dark code panel below is the terminal/editor view for this action.

Why this action: Run command executes inside the VM through the Azure VM agent without opening SSH.

1. Return to RunShellScript and run getent hosts ops.onelab.internal.
2. It should resolve to that private IP.

USE IN: VM RUNSHELLSCRIPT · COPY ONLY THE CODE

```
getent hosts ops.onelab.internal
```

What it means: Run on vm-ops after saving the DNS record. Compare the returned IP with its private IP.

2D.02 · ACTION 5 OF 5**SEE THE IDEA · Install a private health page with Run command → Explain the decision****A SMALL DECISION MAP****Install a private health page with Run command > Explain the decision**

- | | |
|-----------------|--|
| 1 Locate | Explain the decision |
| 2 Action | Do not run apt commands in Cloud Shell for this step; they belong inside vm-ops. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Run command executes inside the VM through the Azure VM agent without opening SSH.

1. Do not run apt commands in Cloud Shell for this step; they belong inside vm-ops.

You should see

The VM returns OneLab operations from its local web server and resolves the private name.

If it does not work

If apt cannot connect, check the public IP or other explicit outbound route, NSG outbound rules, DNS and VM agent status.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

2D.03 Add a data disk safely

AZ-104 · CORE + availability extension

Why this task: The OS disk, temporary disk and persistent data disk have different purposes.

2D.03 · ACTION 1 OF 5

AZURE PORTAL · vm-ops → Disks → Create and attach a new disk

The screenshot shows the Azure Portal interface for creating a new disk. The breadcrumb navigation is 'vm-ops > Disks > Create and attach a new disk'. On the left, a sidebar shows 'vm-ops' and 'Disks' with 'Create and attach a new disk' highlighted. The main area contains four numbered steps: 1. Disk name (input: disk-ops-data), 2. Type (input: Small supported Standard SSD), 3. Linux device (input: Identify using lsblk), and 4. OS disk (input: Do not format). A 'Save' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: vm-ops

vm-ops > Disks > Create and attach a new disk

vm-ops
Disks
Create and attach a new disk

1 Disk name disk-ops-data

2 Type Small supported Standard SSD

3 Linux device Identify using lsblk

4 OS disk Do not format

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: The OS disk, temporary disk and persistent data disk have different purposes.

1. Open vm-ops > Disks > Create and attach a new disk.
2. Name disk-ops-data; choose a small supported Standard SSD size
3. Save.

2D.03 · ACTION 2 OF 5

VM TERMINAL IN AZURE PORTAL · Azure Portal → vm-ops → Run command → RunShellScript

The dark code panel below is the terminal/editor view for this action.

Why this action: The OS disk, temporary disk and persistent data disk have different purposes.

1. Run lsblk -o NAME,SIZE,FSTYPE,MOUNTPOINTS inside the VM.
2. Identify the new empty disk by its size and lack of filesystem.
3. Do not assume its Linux name.

USE IN: VM RUNSHELLSCRIPT · COPY ONLY THE CODE

```
lsblk -o NAME,SIZE,FSTYPE,MOUNTPOINTS
```

What it means: Read disk names, sizes and existing filesystems. This command does not format anything.

2D.03 · ACTION 3 OF 5**AZURE PORTAL · vm-ops → Disks → Create and attach a new disk**

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page header includes 'AZURE PORTAL' and a search bar containing 'vm-ops'. The breadcrumb navigation path is 'vm-ops > Disks > Create and attach a new disk'. On the left sidebar, 'vm-ops' and 'Disks' are listed, with 'Create and attach a new disk' highlighted. The main content area features a numbered step '1 OS disk' and a text input field containing 'Do not format'. At the bottom right, there is a 'View' button. A note at the bottom left states 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: The OS disk, temporary disk and persistent data disk have different purposes.

1. For this beginner exercise, stop after identification unless you can positively identify the empty data disk.
2. In DESIGN, explain that partitioning/formatting the OS disk destroys the VM.

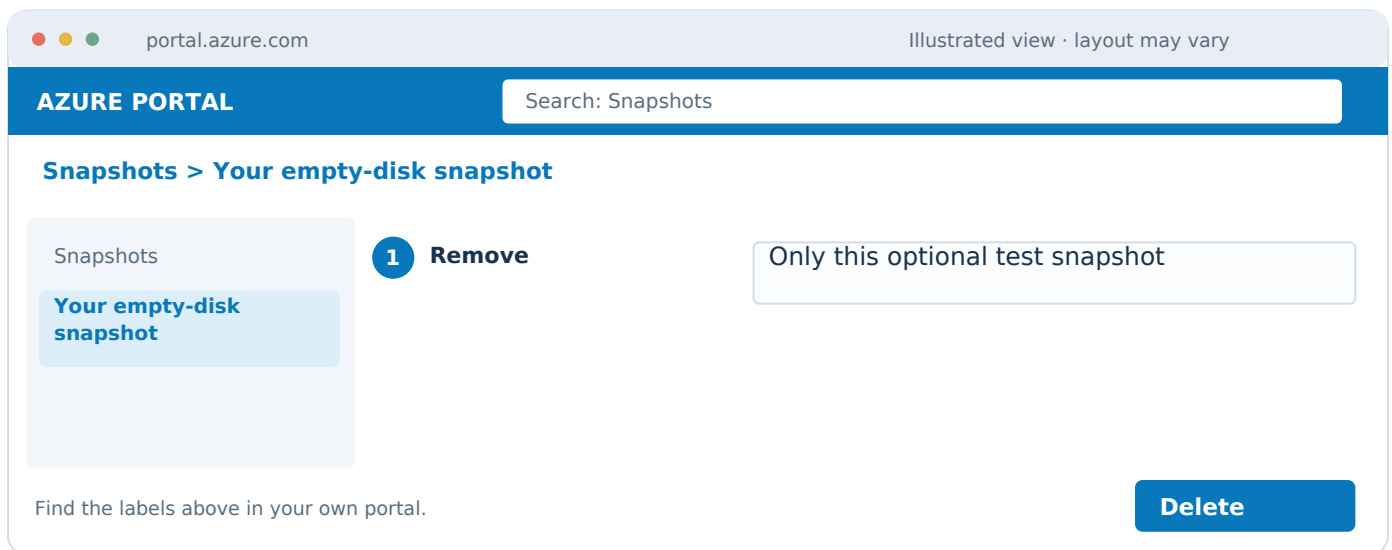
2D.03 · ACTION 4 OF 5**AZURE PORTAL · Disks → disk-ops-data → Overview**

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page header includes 'AZURE PORTAL' and a search bar containing 'Disks'. The breadcrumb navigation path is 'Disks > disk-ops-data > Overview'. On the left sidebar, 'Disks' and 'disk-ops-data' are listed, with 'Overview' highlighted. The main content area features two numbered steps: '1 Inspect' and '2 Snapshot'. To the right of '1 Inspect' is a text input field containing 'Size + performance'. To the right of '2 Snapshot' is a text input field containing 'Empty lab data disk only'. At the bottom right, there is a 'Create snapshot' button. A note at the bottom left states 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open the attached managed disk resource.
2. Inspect Size + performance and the Create snapshot action.
3. Create a snapshot of this empty test disk if you want to practise disk-level recovery.

2D.03 · ACTION 5 OF 5**AZURE PORTAL · Snapshots → Your empty-disk snapshot**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. A disk snapshot is not the same as a full application-consistent backup policy.
2. Delete the empty snapshot after inspection.

You should see

You identify the new disk without modifying the OS disk.

If it does not work

Never copy a destructive mkfs command using an unverified device name. The disk exercise is complete when you can identify attachment, persistence and snapshot scope.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

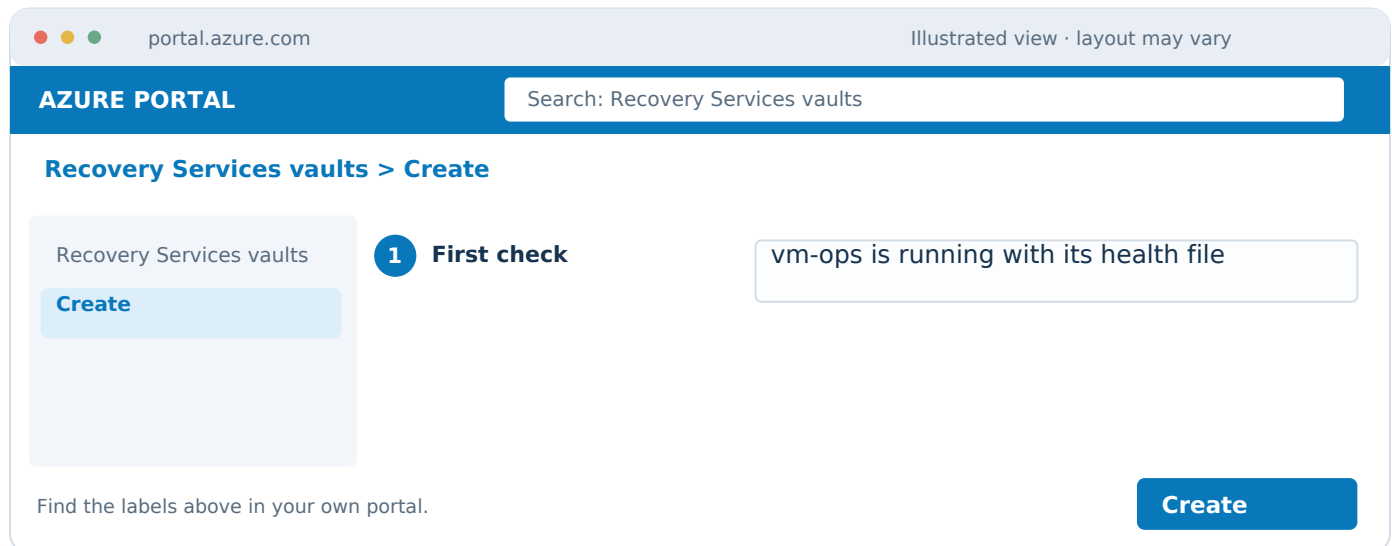
2D.04 Protect vm-ops in a Recovery Services vault

AZ-104 · EXTENSION - backup storage and restore costs

Why this task: A policy schedules backups and determines recovery-point retention.

2D.04 · ACTION 1 OF 6

AZURE PORTAL · Recovery Services vaults → Create



Illustrated navigation view · use your own saved names and IDs

Why this action: A policy schedules backups and determines recovery-point retention.

1. Ensure vm-ops is running and the internal health file still exists.
2. Search Recovery Services vaults > Create.

2D.04 · ACTION 2 OF 6**AZURE PORTAL · Create Recovery Services vault → Basics**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Create Recovery Services vault

Create Recovery Services vault > Basics

Create Recovery Services vault

Basics

- 1 Name** rsv-PREFIX
- 2 Group** rg-PREFIX-ops
- 3 Region** Same as vm-ops
- 4 Before protection** Review redundancy

Find the labels above in your own portal.

Review + create

Illustrated navigation view · use your own saved names and IDs

Why this action: A policy schedules backups and determines recovery-point retention.

1. Use rg-PREFIX-ops, name rsv-PREFIX, and the same region as vm-ops.
2. Review backup storage redundancy before enabling protection.

2D.04 · ACTION 3 OF 6**AZURE PORTAL · the vault → Backup**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: the vault

the vault > Backup

the vault

Backup

- 1 Workload location** Azure
- 2 Workload** Virtual machine
- 3 Policy** daily-lab

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: A policy schedules backups and determines recovery-point retention.

1. Open the vault > Backup.
2. Workload location: Azure; what to back up: Virtual machine.
3. Choose a supported policy type for this VM.

2D.04 · ACTION 4 OF 6**AZURE PORTAL · rsv-PREFIX → Backup policies → Add**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: rsv-PREFIX

rsv-PREFIX > Backup policies > Add

rsv-PREFIX
Backup policies
Add

1 Name daily-lab

2 Schedule Convenient daily UTC time

3 Retention Smallest suitable supported value

Find the labels above in your own portal. **Create**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Create policy daily-lab with a convenient daily UTC schedule and the smallest suitable retention permitted by that policy.
2. Record the actual retention.
3. Do not guess that every policy allows one day.

2D.04 · ACTION 5 OF 6**AZURE PORTAL · rsv-PREFIX → Backup items → Azure Virtual Machine → vm-ops**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: rsv-PREFIX

rsv-PREFIX > Backup items > Azure Virtual Machine > vm-ops

rsv-PREFIX
Backup items
Azure Virtual Machine

1 First Enable protection

2 Then Backup now

Find the labels above in your own portal. **Backup now**

Illustrated navigation view · use your own saved names and IDs

Why this action: A policy schedules backups and determines recovery-point retention.

1. Select vm-ops and enable backup.
2. Then open Backup items > Azure Virtual Machine > vm-ops > Backup now.

2D.04 · ACTION 6 OF 6**AZURE PORTAL · rsv-PREFIX → Backup jobs → Your job**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: rsv-PREFIX

rsv-PREFIX > Backup jobs > Your job

rsv-PREFIX

Backup jobs

Your job

1 Record

Job ID and current status

2 DONE requires

Completed + visible recovery point

Find the labels above in your own portal. **Refresh**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Open Backup jobs, record the job ID and status, then continue to replication setup while the job runs.
2. Recheck on Day 3: mark backup DONE only when the job is Completed and a recovery point is visible.
3. Save its timestamp and consistency type.

You should see

Backup is submitted and tracked as WAITING; later change it to DONE only after a completed job and visible recovery point.

If it does not work

The vault and VM must meet location/support requirements. Do not enable locked immutability casually in a short-lived lab; protected retention can prevent immediate cleanup. [R57]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

Backup answers: can I retrieve an earlier copy? Replication answers: can I start a service elsewhere after a site/region failure? RPO is how much recent data loss you can tolerate. RTO is how long you can tolerate being unavailable. A successful backup job is only part of the evidence; restoration must also be tested.

2D.05 Prepare replication for vm-ops

AZ-104 · PAID EXTENSION; DESIGN if unavailable

Why this task: The operations service may need to start in another region after a regional incident.

2D.05 · ACTION 1 OF 6

AZURE PORTAL · vm-ops → Operations → Disaster recovery

The screenshot shows the Azure Portal interface for configuring disaster recovery for a resource named 'vm-ops'. The breadcrumb navigation is 'vm-ops > Operations > Disaster recovery'. On the left, a sidebar lists 'vm-ops', 'Operations', and 'Disaster recovery' (which is selected). The main area displays five numbered steps: 1. Target region (Supported different region), 2. Target group (rg-PREFIX-dr), 3. Test VNet (vnet-dr-test), 4. Address space (10.40.0.0/16), and 5. Test subnet (10.40.1.0/24). At the bottom right, there is a blue button labeled 'Enable replication'. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: vm-ops

vm-ops > Operations > Disaster recovery

vm-ops

Operations

Disaster recovery

- 1 Target region Supported different region
- 2 Target group rg-PREFIX-dr
- 3 Test VNet vnet-dr-test
- 4 Address space 10.40.0.0/16
- 5 Test subnet 10.40.1.0/24

Find the labels above in your own portal.

Enable replication

Illustrated navigation view · use your own saved names and IDs

Why this action: The operations service may need to start in another region after a regional incident.

1. Review the supported OS/kernel, region and disk requirements shown in Site Recovery before enabling it.
2. Use only the disposable lab VM.

2D.05 · ACTION 2 OF 6**AZURE PORTAL · Resource groups → Create, then Virtual networks → Create**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Resource groups

Resource groups > Create, then Virtual networks > Create

Resource groups
Create, then Virtual networks
Create

1 Group rg-PREFIX-dr

2 Region Supported secondary region

3 Test VNet vnet-dr-test, 10.40.0.0/16

4 Subnet test, 10.40.1.0/24

Find the labels above in your own portal. **Review + create**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Create rg-PREFIX-dr in a supported secondary region and an isolated test VNet there, for example vnet-dr-test with 10.40.0.0/16 and subnet test 10.40.1.0/24.

2D.05 · ACTION 3 OF 6**AZURE PORTAL · vm-ops → Operations → Disaster recovery**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: vm-ops

vm-ops > Operations > Disaster recovery

vm-ops
Operations
Disaster recovery

1 Target region Supported different region

Find the labels above in your own portal. **Next**

Illustrated navigation view · use your own saved names and IDs

Why this action: The operations service may need to start in another region after a regional incident.

1. Open vm-ops > Operations > Disaster recovery.
2. Select a different target region.

2D.05 · ACTION 4 OF 6

AZURE PORTAL · Enable replication → Target settings

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL

Search: Enable replication

Enable replication > Target settings

Enable replication

Target settings

1 Group

2 Target network

3 Test network

4 Vault

rg-PREFIX-dr

Recovery network offered by wizard

vnet-dr-test stays isolated

Supported source/target arrangement

Find the labels above in your own portal.

Review

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Inspect target resource group, target network/subnet, target size, cache storage, replication policy and Recovery Services vault.
2. Use a vault in a supported region for the source/target arrangement; do not force the VM-backup vault from Day 2, Block D, Day 7, Block B if unsuitable.

2D.05 · ACTION 5 OF 6

AZURE PORTAL · Enable replication → Review + Start replication

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL

Search: Enable replication

Enable replication > Review + Start replication

Enable replication

Review + Start replication

1 Record

2 Status

Initial job ID

WAITING until protected

Find the labels above in your own portal.

Enable replication

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Review estimated replica/storage costs and enable replication.
2. Record the job/status as WAITING.
3. Continue with the next independent task and recheck on Day 3; do not repeatedly press Enable replication.

2D.05 · ACTION 6 OF 6**AZURE PORTAL · Site Recovery vault → Replicated items → vm-ops**

The image shows a simulated Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com' and a status bar indicates 'Illustrated view · layout may vary'. The main header is 'AZURE PORTAL' with a search bar containing 'Site Recovery vault'. Below the header, the breadcrumb path is 'Site Recovery vault > Replicated items > vm-ops'. On the left, a sidebar lists 'Site Recovery vault', 'Replicated items', and 'vm-ops' (which is highlighted). The main content area shows three numbered steps: 1. Health (Healthy), 2. Recovery point (Record real timestamp), and 3. Lag (Record observed value). Each step has a corresponding input field. At the bottom right, there is a 'Refresh' button. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Once initial replication completes, confirm healthy protection, record the latest recovery point and observed lag, and mark DONE.
2. A low RPO target is not a promise of zero data loss.

You should see

Replication is submitted and tracked. DONE requires healthy protection and a recovery point; an unsupported prerequisite stays BLOCKED or DESIGN.

If it does not work

Initial replication can take considerable time. Leaving protection enabled retains billable resources even when you are not running a failover. [R60]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Finish Day 2

Save your place and check cost

Save the VM private IP and recovery job details. Recheck the sample SAS after expiry. Check cost. Follow the VM auto-shutdown time you chose.

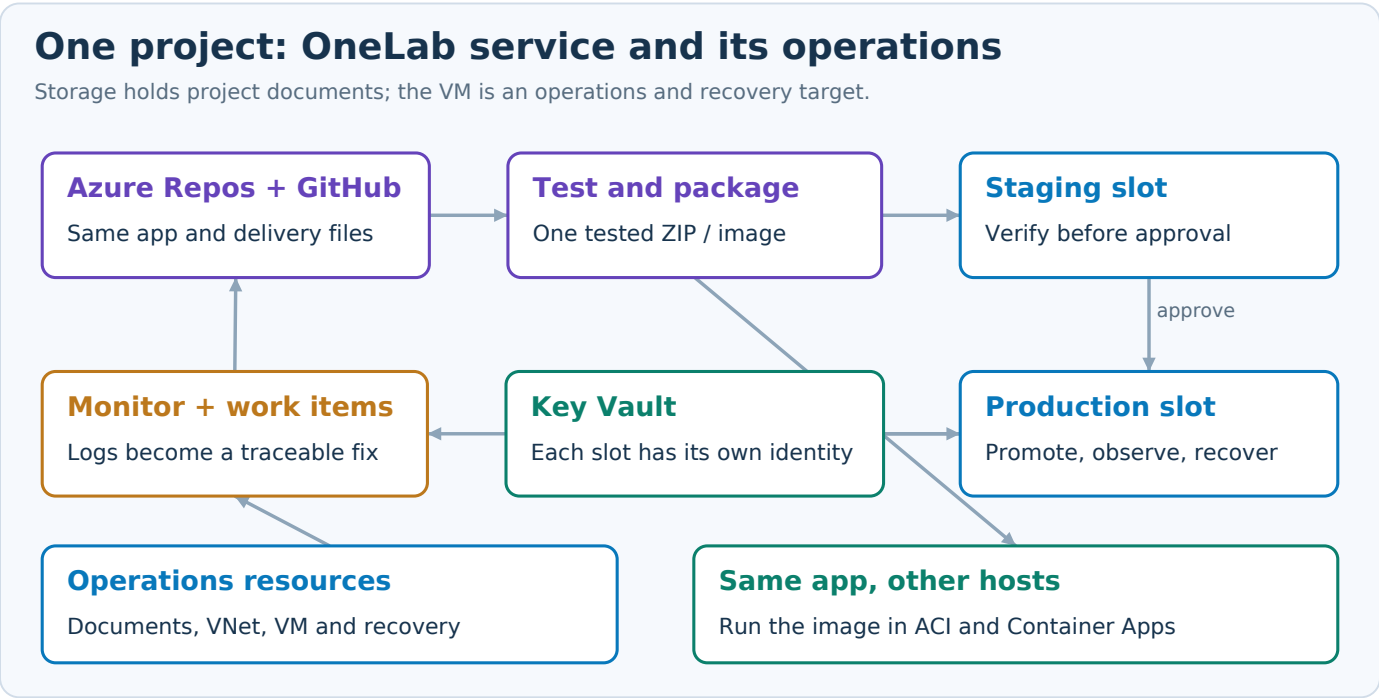
Say it in your own words: Does a valid SAS let a file download through a storage firewall that blocks the client?

Check your explanation: No. The SAS gives data permission; the firewall still decides whether that client can reach the data endpoint.

DAY 3 OF 7

Put your website online

OneLab works locally and on Azure, with a staging slot, secret access and monitoring.



Start today

Run the restart routine. Start vm-ops in Azure Portal if stopped. Review yesterday's backup, object-copy and replication jobs; record real completion, not just submission.

Start early and track the job

Monitoring starts today so you can inspect real data later. Keep the application and both slots for the pipeline exercises.

Block	Today's work	Minutes
A	Test the network paths safely	90
B	Run the supplied app and publish it	105
C	Add staging and protected settings	100
D	Turn on monitoring	60

3A · Test the network paths safely

Planning estimate: 90 minutes. Finish with: A private endpoint check and a repaired route fault; optional load-balancer evidence.

3A.01 Use a service endpoint to restrict storage

AZ-104 · CORE + paid network extensions

Why this task: A service endpoint permits an approved subnet to reach the public service endpoint without making the service private.

3A.01 · ACTION 1 OF 5

AZURE PORTAL · vnet-main → Subnets → ops → Service endpoints

portal.azure.comIllustrated view · layout may vary

AZURE PORTAL

Search: vnet-main

vnet-main > Subnets > ops > Service endpoints

vnet-main

Subnets

ops

1 Service

2 Allowed storage subnet

3 After the exercise

Microsoft.Storage

vnet-main / ops

Restore the stated access setting

Find the labels above in your own portal.

Save

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Open vnet-main > Subnets > ops.
2. Add the Microsoft.Storage service endpoint and save.

3A.01 · ACTION 2 OF 5**AZURE PORTAL · PREFIXdata → Networking → Public access**

The screenshot shows the Azure Portal interface for configuring 'Public access' under the 'Networking' section of a resource named 'PREFIXdata'. The breadcrumb navigation is 'PREFIXdata > Networking > Public access'. On the left, a sidebar shows the navigation tree with 'Public access' selected. The main content area has two numbered steps: '1 Public network access' and '2 Allowed subnet'. Step 1 has a text input field labeled 'Selected networks'. Step 2 has a text input field containing 'vnet-main / ops'. At the bottom right is a blue 'Save' button. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > Networking > Public access

PREFIXdata
Networking
Public access

1 Public network access Selected networks

2 Allowed subnet vnet-main / ops

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Open PREFIXdata > Networking.
2. Change public network access to selected virtual networks/IP addresses.
3. Add vnet-main/ops.

3A.01 · ACTION 3 OF 5**AZURE PORTAL · PREFIXdata → Networking → Firewall**

The screenshot shows the Azure Portal interface for configuring 'Firewall' under the 'Networking' section of a resource named 'PREFIXdata'. The breadcrumb navigation is 'PREFIXdata > Networking > Firewall'. On the left, a sidebar shows the navigation tree with 'Firewall' selected. The main content area has two numbered steps: '1 Client IP' and '2 Cloud Shell'. Step 1 has a text input field labeled 'Add your browser public IP'. Step 2 has a text input field labeled 'May use a different IP'. At the bottom right is a blue 'Save' button. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > Networking > Firewall

PREFIXdata
Networking
Firewall

1 Client IP Add your browser public IP

2 Cloud Shell May use a different IP

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: A service endpoint permits an approved subnet to reach the public service endpoint without making the service private.

1. Add your browser's current public IP using the portal button if you need browser data access.
2. Cloud Shell may use a different egress IP and can now fail; that is expected.

3A.01 · ACTION 4 OF 5

VM TERMINAL IN AZURE PORTAL · Azure Portal → vm-ops → Run command → RunShellScript

The dark code panel below is the terminal/editor view for this action.

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Inside vm-ops use Run command to check DNS for PREFIXdata.blob.core.windows.net.
2. It still resolves using the public service endpoint path.

USE IN: VM RUNSHELLSCRIPT · COPY ONLY THE CODE

```
getent hosts REPLACE_PREFIXdata.blob.core.windows.net
```

What it means: Replace REPLACE_PREFIX with your saved prefix. Run inside vm-ops.

3A.01 · ACTION 5 OF 5

AZURE PORTAL · PREFIXdata → Networking → Public access

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > Networking > Public access

PREFIXdata
Networking
Public access

1 Lab reset All networks for authenticated lab access

2 Blob anonymous access Still disabled

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Restore all-network public endpoint access after recording the restriction, so subsequent Cloud Shell operations continue.
2. The blobs remain private and require authentication.

You should see

You can explain why network access and data permissions are two separate checks.

If it does not work

The trusted Microsoft services exception is not an allow-all for every Azure service. Do not assume it includes your Cloud Shell session. [R17]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

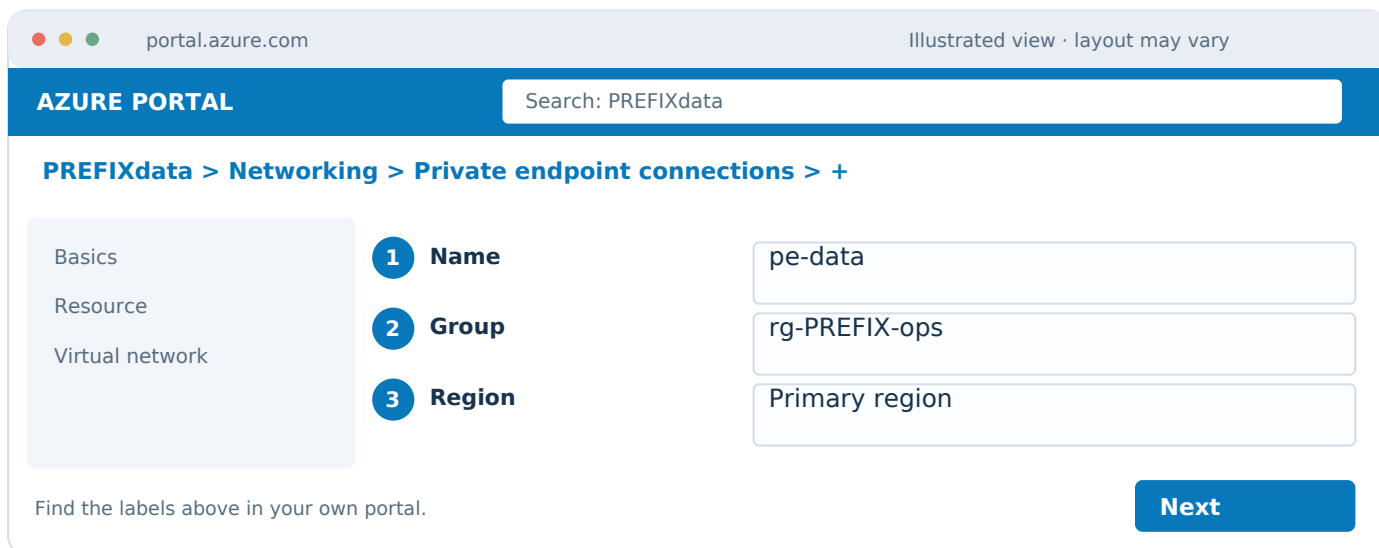
3A.02 Create and test a private endpoint

AZ-104 · CORE + paid network extensions

Why this task: A private endpoint gives a service a private IP in your VNet.

3A.02 · ACTION 1 OF 6

AZURE PORTAL · PREFIXdata → Networking → Private endpoint connections → +



portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > Networking > Private endpoint connections > +

Basics
Resource
Virtual network

1 Name pe-data

2 Group rg-PREFIX-ops

3 Region Primary region

Find the labels above in your own portal. **Next**

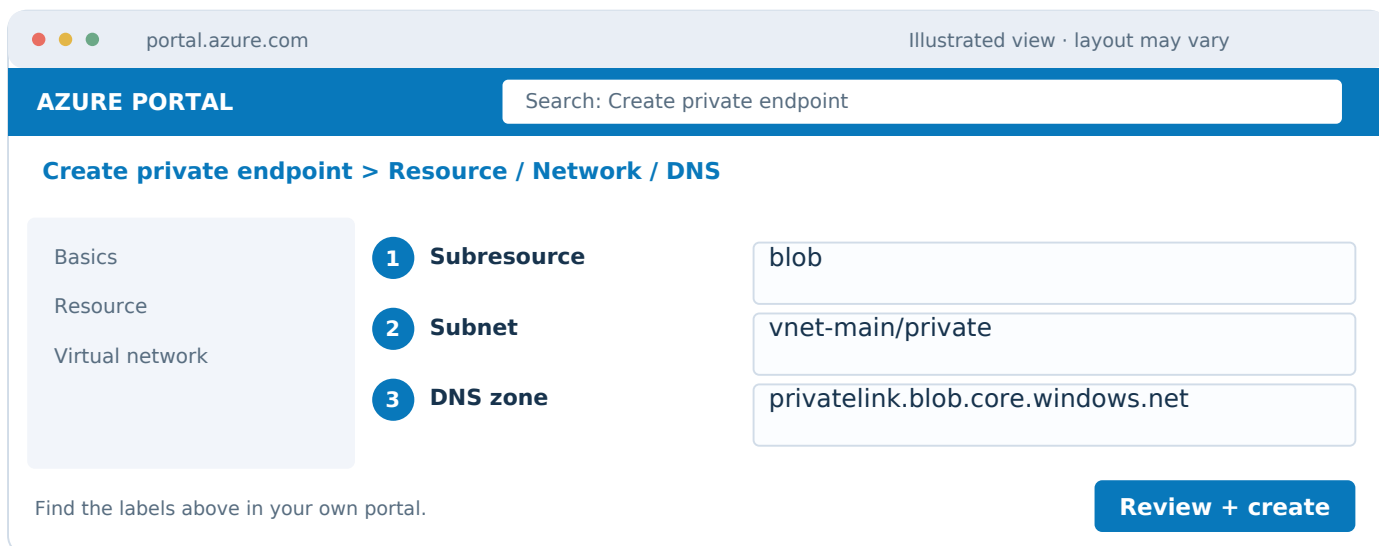
Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Open PREFIXdata > Networking > Private endpoint connections > + Private endpoint.
2. Use rg-PREFIX-ops and the primary region; name pe-data.

3A.02 · ACTION 2 OF 6

AZURE PORTAL · Create private endpoint → Resource / Network / DNS



portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Create private endpoint

Create private endpoint > Resource / Network / DNS

Basics
Resource
Virtual network

1 Subresource blob

2 Subnet vnet-main/private

3 DNS zone privatelink.blob.core.windows.net

Find the labels above in your own portal. **Review + create**

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Select the storage account and target subresource blob.
2. Choose vnet-main/private.
3. Enable private DNS integration using privatelink.blob.core.windows.net and create.

3A.02 · ACTION 3 OF 6**AZURE PORTAL · pe-data → Overview / Network interface**

The screenshot shows the Azure Portal interface for the 'pe-data' resource. The breadcrumb navigation is 'pe-data > Overview / Network interface'. On the left, a sidebar lists 'Basics', 'Resource', and 'Virtual network'. The main content area shows two steps: '1 Connection' and '2 IP'. Under '1 Connection', there is a text box containing 'Approved'. Under '2 IP', there is a text box containing 'Record the private address'. At the bottom right, there is a blue 'View' button. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Open the private endpoint and inspect the network interface IP and connection status Approved.

3A.02 · ACTION 4 OF 6**VM TERMINAL IN AZURE PORTAL · Azure Portal → vm-ops → Run command → RunShellScript**

The dark code panel below is the terminal/editor view for this action.

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Inside vm-ops run getent hosts PREFIXdata.blob.core.windows.net, replacing PREFIX.
2. Confirm resolution reaches the private endpoint IP through the private DNS zone.

USE IN: VM RUNSHELLSCRIPT · COPY ONLY THE CODE

```
getent hosts REPLACE_PREFIXdata.blob.core.windows.net
```

What it means: Use your storage name. The address should match pe-data, not an Internet IP.

3A.02 · ACTION 5 OF 6**AZURE PORTAL · PREFIXdata → Networking → Public access**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > Networking > Public access

Basics
Resource
Virtual network

- 1 Public network access** Disabled briefly
- 2 Inside client** vm-ops
- 3 Outside client** Browser / Cloud Shell

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Temporarily disable public network access on the storage account.
2. Compare access inside the network with Cloud Shell/browser access outside it.

3A.02 · ACTION 6 OF 6**AZURE PORTAL · PREFIXdata → Networking → Public access**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXdata

PREFIXdata > Networking > Public access

Basics
Resource
Virtual network

- 1 Restore** All networks with authentication
- 2 Then delete** pe-data and unused DNS zone/link

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Restore public network access before continuing the core lab.
2. Delete pe-data and its unused private DNS zone/link after the exercise if you will not keep the feature.

You should see

The VM resolves the normal service hostname to a private IP; an outside shell is not automatically connected.

If it does not work

A blob private endpoint does not cover the file subresource. Private DNS links and endpoint approval are common missing steps. [R18]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

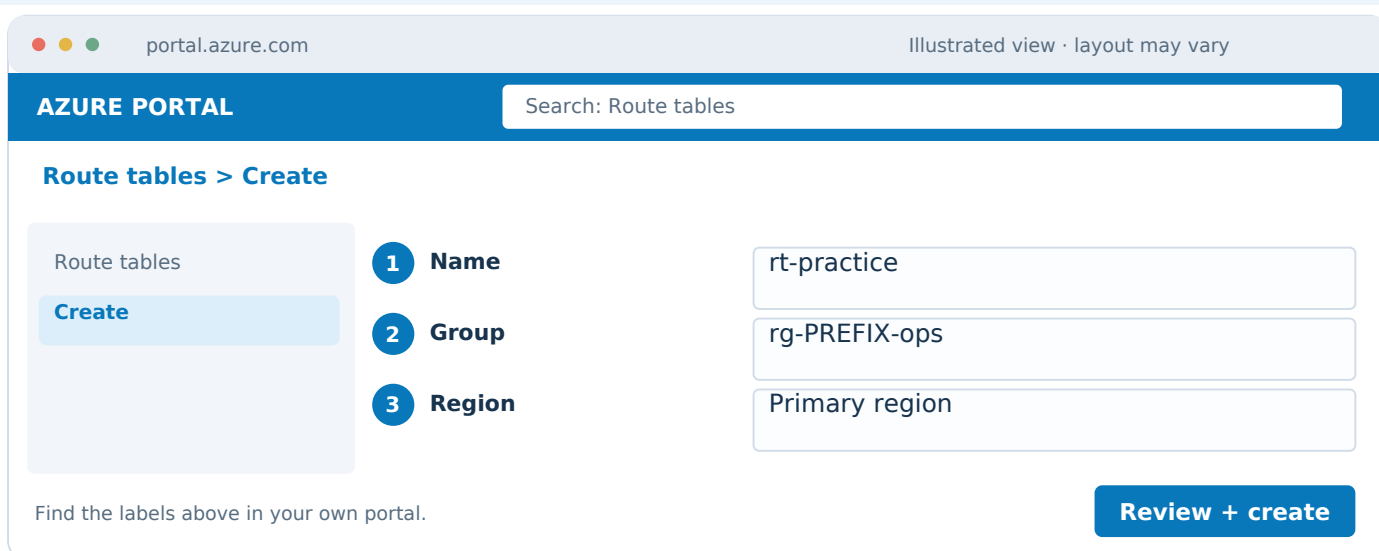
3A.03 Introduce and repair a route fault

AZ-104 · CORE + paid network extensions

Why this task: A route can silently discard traffic even when the NSG allows it.

3A.03 · ACTION 1 OF 6

AZURE PORTAL · Route tables → Create



portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Route tables

Route tables > Create

Route tables

Create

1 Name rt-practice

2 Group rg-PREFIX-ops

3 Region Primary region

Find the labels above in your own portal.

Review + create

Illustrated navigation view · use your own saved names and IDs

Why this action: A route can silently discard traffic even when the NSG allows it.

1. Search Route tables > Create.
2. Name rt-practice; use rg-PREFIX-ops and primary region.

3A.03 · ACTION 2 OF 6**AZURE PORTAL · rt-practice → Routes → Add**

The screenshot shows the Azure Portal interface for adding a new route. The breadcrumb navigation is 'rt-practice > Routes > Add'. On the left, a sidebar shows 'rt-practice', 'Routes', and 'Add' (highlighted). The main area contains four numbered steps: 1. Name (block-tools), 2. Address prefix (10.30.0.0/16), 3. Next hop (None), and 4. Then associate (vnet-main/ops). A blue 'Add' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: rt-practice

rt-practice > Routes > Add

rt-practice
Routes
Add

1 Name block-tools

2 Address prefix 10.30.0.0/16

3 Next hop None

4 Then associate vnet-main/ops

Find the labels above in your own portal. **Add**

Illustrated navigation view · use your own saved names and IDs

Why this action: A route can silently discard traffic even when the NSG allows it.

1. Open Routes > Add.
2. Route name block-tools; destination 10.30.0.0/16; next hop type None.
3. Associate the route table only with vnet-main/ops.

3A.03 · ACTION 3 OF 6**AZURE PORTAL · vm-ops → Networking → NIC → Effective routes**

The screenshot shows the Azure Portal interface for viewing effective routes. The breadcrumb navigation is 'vm-ops > Networking > NIC > Effective routes'. On the left, a sidebar shows 'vm-ops', 'Networking', and 'NIC'. The main area contains three numbered steps: 1. Find (10.30.0.0/16), 2. Source (User), and 3. Next hop (None). A blue 'View' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: vm-ops

vm-ops > Networking > NIC > Effective routes

vm-ops
Networking
NIC

1 Find 10.30.0.0/16

2 Source User

3 Next hop None

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Open the VM NIC > Effective routes.
2. Observe the user route for the tools network.
3. This demonstrates a drop path; do not associate a 0.0.0.0/0 blackhole route.

3A.03 · ACTION 4 OF 6**AZURE PORTAL · rt-practice → Routes → block-tools**

The image shows a simulated Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com' and a note 'Illustrated view · layout may vary'. Below this is a blue header bar with 'AZURE PORTAL' on the left and a search bar containing 'rt-practice'. The main content area features a breadcrumb trail: 'rt-practice > Routes > block-tools'. On the left, a navigation pane lists 'rt-practice', 'Routes', and 'block-tools' (which is highlighted in blue). To the right of the navigation pane, there are two numbered steps: '1 Delete' and '2 Then check'. Further right, there are two text boxes: 'This one practice route' and 'NIC effective routes'. At the bottom right, there is a blue 'Delete' button. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: rt-practice

rt-practice > Routes > block-tools

rt-practice
Routes
block-tools

1 Delete
2 Then check

This one practice route
NIC effective routes

Find the labels above in your own portal. **Delete**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Remove the block-tools route and verify the effective route returns to the peering path.

3A.03 · ACTION 5 OF 6**AZURE PORTAL · Network Watcher → IP flow verify**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL

Network Watcher > IP flow verify

Network Watcher

IP flow verify

1 VM

2 Direction

3 Protocol

4 Local port

5 Remote IP

6 Remote port

vm-ops

Inbound

TCP

22

Your own public IP

50000

Find the labels above in your own portal. **Check**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open Network Watcher > IP flow verify.
2. Choose vm-ops/NIC, inbound TCP and test a hypothetical source/destination port.
3. Read the named NSG rule that permits or denies the packet.

3A.03 · ACTION 6 OF 6**AZURE PORTAL · Network Watcher → Connection troubleshoot**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL

Network Watcher > Connection troubleshoot

Network Watcher

Connection troubleshoot

1 Source

vm-ops

2 Destination

Your app hostname

3 Port

443

Find the labels above in your own portal. **Check**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. For Connection troubleshoot, choose the VM as source and an endpoint you control as destination.
2. Record reachability and whether the issue is DNS, routing or filtering.

You should see

You can distinguish a routing drop from an NSG deny and remove the test fault.

If it does not work

Effective routes and effective security rules are read from the NIC. A successful DNS lookup alone does not prove a port is reachable. [R19]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

3A.04 Inspect Bastion and build a load balancer

AZ-104 · CORE + paid network extensions

Why this task: Bastion is an administration path. A load balancer distributes application traffic.

3A.04 · ACTION 1 OF 7

AZURE PORTAL · vm-ops → Connect → Bastion

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: vm-ops

vm-ops > Connect > Bastion

Frontend IP

Backend pools

Health probes

1 Tier

Inspect available Developer or dedicated option

2 Dedicated subnet

AzureBastionSubnet, /26 or larger

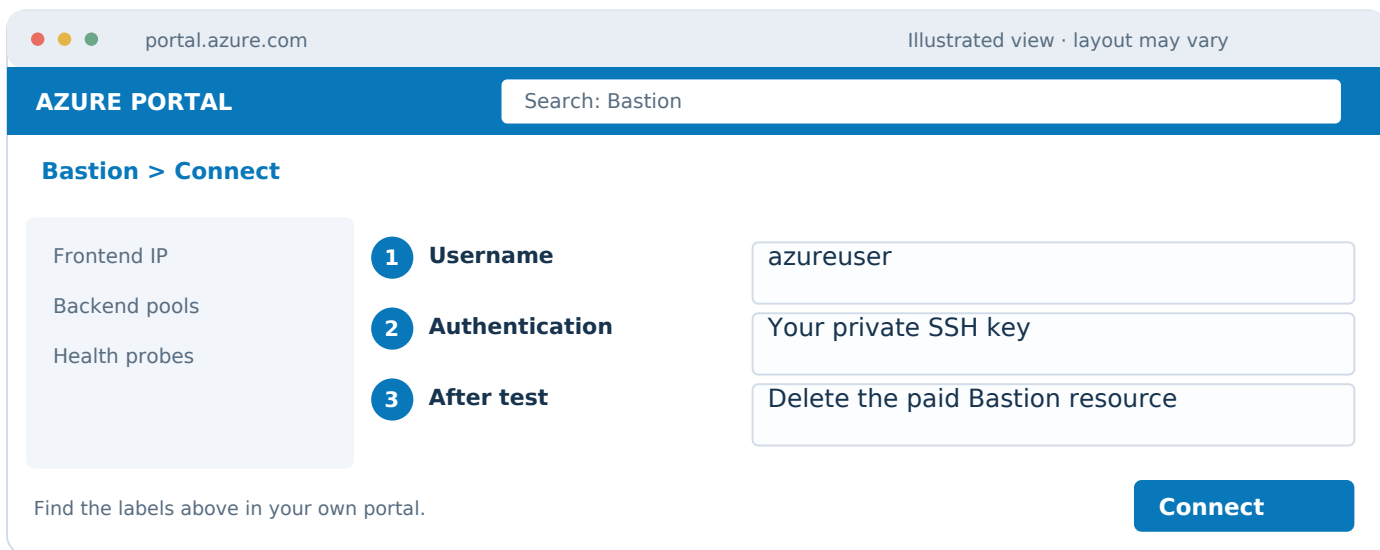
Find the labels above in your own portal.

Review

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. For Bastion, open vm-ops > Connect > Bastion.
2. If the region offers a suitable Developer option, inspect its limitations; otherwise a dedicated paid deployment uses AzureBastionSubnet with the required address size and a public IP for applicable tiers.

3A.04 · ACTION 2 OF 7**AZURE PORTAL · Bastion → Connect**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Bastion

Bastion > Connect

Frontend IP
Backend pools
Health probes

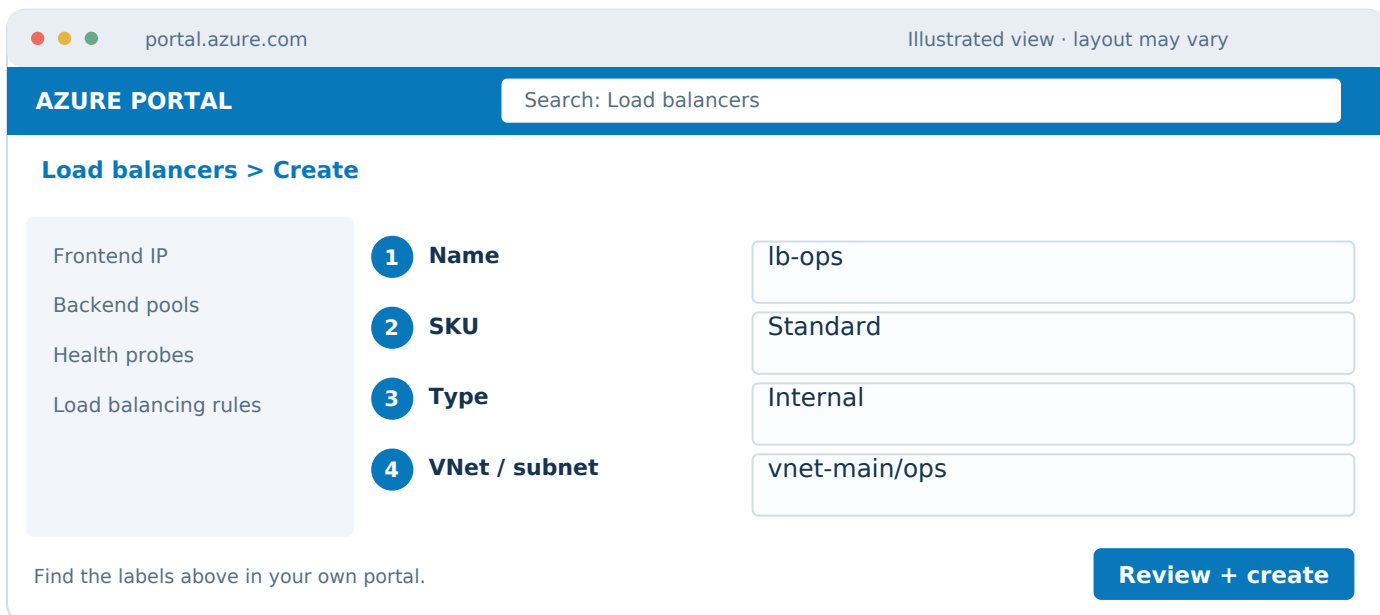
- 1 Username** azureuser
- 2 Authentication** Your private SSH key
- 3 After test** Delete the paid Bastion resource

Find the labels above in your own portal. **Connect**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Create only after reviewing the displayed price.
2. Use the VM username and SSH key to connect through the browser.
3. Then delete the practice Bastion resource when finished; closing its browser does not stop billing.

3A.04 · ACTION 3 OF 7**AZURE PORTAL · Load balancers → Create**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Load balancers

Load balancers > Create

Frontend IP
Backend pools
Health probes
Load balancing rules

- 1 Name** lb-ops
- 2 SKU** Standard
- 3 Type** Internal
- 4 VNet / subnet** vnet-main/ops

Find the labels above in your own portal. **Review + create**

Illustrated navigation view · use your own saved names and IDs

Why this action: Bastion is an administration path. A load balancer distributes application traffic.

1. For a load-balancing exercise, create a Standard internal Load Balancer lb-ops in rg-PREFIX-ops, vnet-main/ops, with a private frontend IP.

3A.04 · ACTION 4 OF 7

AZURE PORTAL · lb-ops → Backend pools / Health probes / Rules

portal.azure.comIllustrated view · layout may vary

AZURE PORTAL

Search: lb-ops

lb-ops > Backend pools / Health probes / Rules

Frontend IP

Backend pools

Health probes

1 Backend

2 Probe

3 Rule

vm-ops NIC

TCP 80

TCP frontend 80 to backend 80

Find the labels above in your own portal.

Add

Illustrated navigation view · use your own saved names and IDs

Why this action: Bastion is an administration path. A load balancer distributes application traffic.

- 1. Add a backend pool containing vm-ops's NIC.
- 2. Add a TCP probe on port 80 and a TCP rule frontend 80 to backend 80 using that probe.

3A.04 · ACTION 5 OF 7**AZURE PORTAL · Virtual machines → vm-client → Run command → RunShellScript**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL

Virtual machines > vm-client > Run command > RunShellScript

Frontend IP
Backend pools
Health probes

1 Client VNet
2 Command

vnet-tools/tools
curl -v --max-time 10
http://REPLACE_LB_PRIVATE_IP

Find the labels above in your own portal. **Run**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. To test the internal load balancer, create a temporary client VM named vm-client in rg-PREFIX-ops, vnet-tools/tools, using the same VM wizard choices as Day 2, Block D: Ubuntu, supported small size, username azureuser, no inbound ports, and a Standard public IP for outbound.
2. Open vm-client > Run command > RunShellScript and run curl -v --max-time 10 http://REPLACE_LB_PRIVATE_IP.
3. Use the frontend private IP shown by lb-ops.
4. Delete vm-client and its disks/NIC/IP after the test.
5. Without this client, mark the load-balancer traffic check DESIGN.

3A.04 · ACTION 6 OF 7**VM TERMINAL IN AZURE PORTAL · Azure Portal → vm-ops → Run command → RunShellScript**

The dark code panel below is the terminal/editor view for this action.

Why this action: The network needs a valid path and a name that points to the intended destination.

1. On nsg-ops allow the required client subnet to asg-ops on port 80.
2. Keep the AzureLoadBalancer probe source permitted.
3. Check probe and backend health, then stop nginx to observe an unhealthy backend; start nginx to recover.
4. In nsg-ops > Inbound security rules > Add, use source IP Addresses 10.30.1.0/24, source ports *, destination Application security group asg-ops, service Custom, destination port 80, protocol TCP, action Allow, priority 200 and name allow-tools-http.
5. Save.

USE IN: VM RUNSHELLSCRIPT · COPY ONLY THE CODE

```
sudo systemctl stop nginx
curl --max-time 5 http://127.0.0.1
# After observing the failed check, restore the service:
sudo systemctl start nginx
curl -f http://127.0.0.1
```

What it means: Run on vm-ops, then retest the load balancer from vm-client. A probe needs time to detect each change.

--max-time · Stop a curl request after this many seconds.

curl · Send an HTTP request. -f fails on HTTP error status; -s quiets the progress meter; -v shows connection details when used.

systemctl enable --now · Set the service to start at boot and start it now; stop/start act on the running service.

sudo · Run that Linux operation with administrator privileges inside the named VM.

3A.04 · ACTION 7 OF 7**SEE THE IDEA · Inspect Bastion and build a load balancer → Explain the decision****A SMALL DECISION MAP****Inspect Bastion and build a load balancer > Explain the decision**

- | | |
|-----------------|---|
| 1 Locate | Explain the decision |
| 2 Action | DESIGN public alternative: use a Standard public frontend IP and carefully limited inbound rules. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Bastion is an administration path. A load balancer distributes application traffic.

1. DESIGN public alternative: use a Standard public frontend IP and carefully limited inbound rules.
2. The load balancer is Layer 4
3. HTTP host/path routing and WAF belong to Layer 7 services.

You should see

You can identify frontend, backend pool, probe, rule and the NSG needed for the service.

If it does not work

For a failed backend, check service listening address/port, probe path/type, NSG probe allowance, NIC pool membership and route. Delete the temporary client and load balancer after the exercise.
[R20]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

3B · Run the supplied app and publish it

Planning estimate: 105 minutes. Finish with: Eight local tests pass and the production website answers /health.

3B.01 Download the complete project and open it in Cloud Shell

AZ-400 · CORE

Why this task: Using one complete set of files prevents missing-file and inconsistent-pipeline errors.

3B.01 · ACTION 1 OF 6

YOUR COMPUTER · This guide → Download project ZIP → Azure Cloud Shell upload

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER

Search: This guide

This guide > Download project ZIP > Azure Cloud Shell upload

This guide

Download project ZIP

Azure Cloud Shell upload

1 Archive

2 Extract into

3 Files at root

OneLab-GitHub-Project.zip

~/clouddrive/one-lab

app.py and requirements.txt

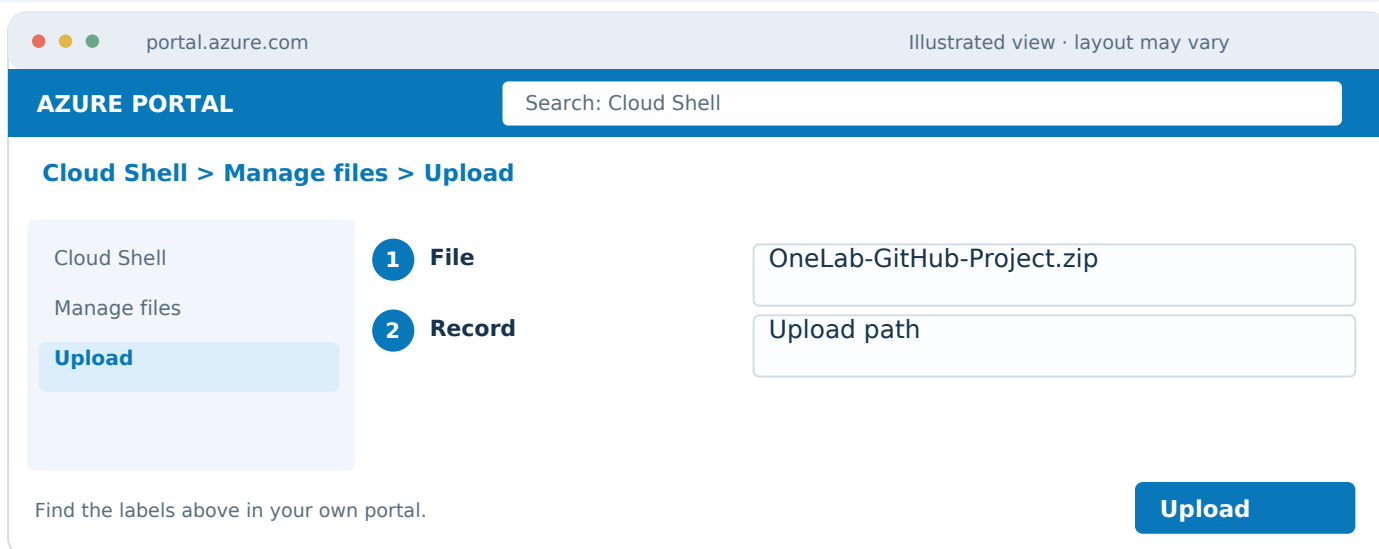
Find the labels above in your own portal.

Download project ZIP

Illustrated navigation view · use your own saved names and IDs

Why this action: Using one complete set of files prevents missing-file and inconsistent-pipeline errors.

1. In the HTML guide click Download project ZIP, or use the separate OneLab-GitHub-Project.zip supplied with this guide.
2. The same ZIP is attached inside the PDF: open the Attachments/paperclip pane in a desktop PDF reader and save it.

3B.01 · ACTION 2 OF 6**AZURE PORTAL · Cloud Shell → Manage files → Upload**


portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Cloud Shell

Cloud Shell > Manage files > Upload

Cloud Shell
Manage files
Upload

1 File
2 Record

OneLab-GitHub-Project.zip
Upload path

Find the labels above in your own portal. **Upload**

Illustrated navigation view · use your own saved names and IDs

Why this action: Using one complete set of files prevents missing-file and inconsistent-pipeline errors.

1. In Cloud Shell choose Manage files > Upload.
2. Upload OneLab-GitHub-Project.zip.
3. The upload message shows its path, usually your home directory.

3B.01 · ACTION 3 OF 6**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

The dark code panel below is the terminal/editor view for this action.

Why this action: Using one complete set of files prevents missing-file and inconsistent-pipeline errors.

1. Run the unzip block below from your existing one-lab directory.
2. It adds project files and does not replace lab.env.

USE IN: BASH · COPY ONLY THE CODE

```
cd ~/clouddrive/one-lab
unzip -n ~/OneLab-GitHub-Project.zip -d .
ls -a
# Only for manual file creation if you cannot use the ZIP:
mkdir -p infra scripts pipelines/templates .github/workflows
```

What it means: unzip -n keeps any existing file instead of overwriting it. If the upload message showed a different ZIP path, use that exact path.

unzip -n / -d . · Extract without overwriting existing files; put the files in the current folder.

mkdir -p · Create the folder and missing parent folders; keep an existing folder.

3B.01 · ACTION 4 OF 6

YOUR COMPUTER · This guide → Download project ZIP → Azure Cloud Shell upload

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER

Search: This guide

This guide > Download project ZIP > Azure Cloud Shell upload

This guide

Download project ZIP

Azure Cloud Shell upload

1 Locate

2 Action

Azure Cloud Shell upload

The ZIP is the recommended route.

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: Using one complete set of files prevents missing-file and inconsistent-pipeline errors.

1. The ZIP is the recommended route.
2. If attachments are hidden in your PDF reader, use the separate ZIP or the HTML download button.
3. The Source files appendix also contains every runtime and pipeline file for manual recovery.

3B.01 · ACTION 5 OF 6

YOUR COMPUTER · This guide → Download project ZIP → Azure Cloud Shell upload

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER

Search: This guide

This guide > Download project ZIP > Azure Cloud Shell upload

This guide

Download project ZIP

Azure Cloud Shell upload

1 Locate

2 Action

Azure Cloud Shell upload

Run ls -a and confirm app.py, requirements.txt, test_app.py, azure-pipelines.yml, infra, pipelines and .github are present.

Find the labels above in your own portal.

Run

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Run ls -a and confirm app.py, requirements.txt, test_app.py, azure-pipelines.yml, infra, pipelines and .github are present.

3B.01 · ACTION 6 OF 6**SEE THE IDEA · Download the complete project and open it in Cloud Shell → Explain the decision****A SMALL DECISION MAP****Download the complete project and open it in Cloud Shell > Explain the decision**

- | | | |
|----------|---------------|--|
| 1 | Locate | Explain the decision |
| 2 | Action | Do not upload or commit your private SSH key, lab.env, a downloaded publish profile, or real company data. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Using one complete set of files prevents missing-file and inconsistent-pipeline errors.

1. Do not upload or commit your private SSH key, lab.env, a downloaded publish profile, or real company data.

You should see

The project files are in ~/clouddrive/one-lab beside your existing private lab.env.

If it does not work

If the ZIP extracts into an extra folder, move into the folder containing app.py or copy its contents into one-lab. All later commands assume app.py is in the current directory.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

The website returns a small JSON response: service name, version and a feature flag. /health returns a health check; /secret-status reports only whether a secret was resolved; /fail deliberately returns HTTP 500 for monitoring. JSON is structured text with named fields. HTTP 200 means success, 404 means the path was not found, and 500 means the server failed.

3B.02 Install dependencies and run tests

AZ-400 · CORE

Why this task: A virtual environment keeps this project's Python packages separate. Automated tests detect mistakes before deployment.

3B.02 · ACTION 1 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Run the commands below.
2. Wait for installation to finish before running tests.

USE IN: BASH · COPY ONLY THE CODE

```
python -m venv /tmp/onelab-venv
source /tmp/onelab-venv/bin/activate
python -m pip install -r requirements-dev.txt
python -m pytest --junitxml=results.xml --cov=app --cov-report=xml
```

What it means: `-r` reads a dependency file. `pytest` executes `test_app.py`. JUnit XML is a portable test-result format; `coverage.xml` records executed source lines.

--junitxml · Write test results in the XML format a pipeline can publish.

--cov · Measure which parts of the named Python module were exercised.

--cov-report · Choose a coverage-report format; `xml` writes `coverage.xml`.

source FILE · Load settings or activate an environment in the current Bash shell.

python -m MODULE · Run a module using the currently selected Python interpreter.

-r requirements file · Read the dependency list from the named requirements file.

3B.02 · ACTION 2 OF 5**CLOUD SHELL · Cloud Shell → Bash → Python virtual environment**

The screenshot shows the Azure Cloud Shell interface. At the top, the browser address bar displays 'portal.azure.com' and a note 'Illustrated view · layout may vary'. The main header is 'CLOUD SHELL' with a search bar containing 'Cloud Shell'. Below the header, the breadcrumb path is 'Cloud Shell > Bash > Python virtual environment'. On the left, a sidebar lists 'Cloud Shell', 'Bash', and 'Python virtual environment' (highlighted in blue). To the right of the sidebar, two numbered steps are shown: '1 Locate' and '2 Action'. The '2 Action' step has a text input field containing 'Python virtual environment' and a confirmation box below it that says 'Confirm eight tests pass.'. At the bottom right, there is a green 'View' button. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Confirm eight tests pass.
2. Read the coverage percentage as a measurement, not proof that every behavior is correct.

3B.02 · ACTION 3 OF 5**CLOUD SHELL · Cloud Shell → Bash → Python virtual environment**

The screenshot shows the Azure Cloud Shell interface. At the top, the browser address bar displays 'portal.azure.com' and a note 'Illustrated view · layout may vary'. The main header is 'CLOUD SHELL' with a search bar containing 'Cloud Shell'. Below the header, the breadcrumb path is 'Cloud Shell > Bash > Python virtual environment'. On the left, a sidebar lists 'Cloud Shell', 'Bash', and 'Python virtual environment' (highlighted in blue). To the right of the sidebar, two numbered steps are shown: '1 Locate' and '2 Action'. The '2 Action' step has a text input field containing 'Python virtual environment' and a confirmation box below it that says 'Open app.py with Nano and find the /health function.'. At the bottom right, there is a green 'Save' button. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open app.py with Nano and find the /health function.
2. Change healthy to broken, save, and notice that the existing health test checks only the status code.

3B.02 · ACTION 4 OF 5**CLOUD SHELL · Cloud Shell → Bash → Python virtual environment**

portal.azure.com Illustrated view · layout may vary

CLOUD SHELL

Cloud Shell > Bash > Python virtual environment

Cloud Shell

Bash

Python virtual environment

1 **Locate**

2 **Action**

Find the labels above in your own portal. **Add**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Restore healthy.
2. Explain why a test can pass while missing a content mistake.
3. Add a content assertion to the health test if you want to strengthen it.

3B.02 · ACTION 5 OF 5**CLOUD SHELL · Cloud Shell → Bash → Python virtual environment**

portal.azure.com Illustrated view · layout may vary

CLOUD SHELL

Cloud Shell > Bash > Python virtual environment

Cloud Shell

Bash

Python virtual environment

1 **Environment**

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: A virtual environment keeps this project's Python packages separate. Automated tests detect mistakes before deployment.

1. Keep the source in clouddrive.
2. The Python environment is in /tmp/onelab-venv to avoid file-share executable/symlink problems.
3. Each new session uses the restart routine in Day 1, Block B to recreate and activate it.

You should see

The tests pass and you understand one limitation of the tests.

If it does not work

If pip cannot download a package, check network access and the exact file contents. The supplied versions are a reproducible teaching baseline, not a claim of permanent vulnerability-free status.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

3B.03 Start a local instance and make a package

AZ-400 · CORE

Why this task: First prove the application works before introducing Azure hosting.

3B.03 · ACTION 1 OF 4

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: First prove the application works before introducing Azure hosting.

1. Start gunicorn using the block below.
2. It runs in the background
3. LAB_APP_PID remembers that process for this shell.

USE IN: BASH · COPY ONLY THE CODE

```
gunicorn --bind 0.0.0.0:8000 app:app > /tmp/onelab.log 2>&1 &  
LAB_APP_PID=$!
```

What it means: & starts the app in the background. \$! is that process ID. The log stays in /tmp.

--bind · The address and port where gunicorn listens; 0.0.0.0 accepts connections on the host interfaces.

> log 2>&1 & · Send standard output and errors to the log, then run the process in the background.

\$! / kill "\$LAB_APP_PID" · Remember the last background process ID, then stop that exact process later.

3B.03 · ACTION 2 OF 4

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: First prove the application works before introducing Azure hosting.

1. curl the health URL and home URL.
2. You should see healthy and OneLab.
3. You may also use Cloud Shell Web preview on port 8000.

USE IN: BASH · COPY ONLY THE CODE

```
curl --retry 5 --retry-connrefused --retry-delay 2 http://127.0.0.1:8000/health
curl http://127.0.0.1:8000/
```

What it means: curl sends a web request. Retries cover the few seconds the process needs to start.

--retry · Retry the request this many times for the supported failure cases.

--retry-connrefused · Also retry connection-refused errors during application startup.

--retry-delay · Wait this many seconds between retries.

curl · Send an HTTP request. -f fails on HTTP error status; -s quiets the progress meter; -v shows connection details when used.

3B.03 · ACTION 3 OF 4**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

The dark code panel below is the terminal/editor view for this action.

Why this action: First prove the application works before introducing Azure hosting.

1. Stop the process using kill with that saved PID after the checks.

USE IN: BASH · COPY ONLY THE CODE

```
kill "$LAB_APP_PID"
```

What it means: Stop only the process you started in this shell.

\$NAME / \${NAME} · Read a variable from this Bash session. Source lab.env first when these are lab names.

3B.03 · ACTION 4 OF 4**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

The dark code panel below is the terminal/editor view for this action.

Why this action: First prove the application works before introducing Azure hosting.

1. Create app.zip containing only app.py and requirements.txt at the ZIP root.
2. Inspect the ZIP listing.
3. Do not ZIP the parent folder or include .venv.

USE IN: BASH · COPY ONLY THE CODE

```
python -m zipfile -c app.zip app.py requirements.txt
python -m zipfile -l app.zip
```

What it means: -c creates the archive. -l lists it. These two files must be at the archive root.

python -m MODULE · Run a module using the currently selected Python interpreter.

You should see

The app responds locally and the ZIP lists only the two runtime files.

If it does not work

A port-in-use error means another process is already using 8000. Stop your earlier lab process; do not terminate unrelated system processes.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

3B.04 Create an App Service plan and web app

AZ-104 + AZ-400 · CORE - ongoing plan cost

Why this task: The plan is the paid compute capacity. The app is the website running on that capacity.

3B.04 · ACTION 1 OF 7

AZURE PORTAL · App Services → Create → Web App

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL

App Services > Create > Web App

App Services	1 Resource group	rg-PREFIX-core
Create	2 Name	PREFIX-web
Web App	3 Publish	Code
	4 Runtime	Python 3.12 (if supported)
	5 Operating system	Linux
	6 Plan	asp-PREFIX, Standard S1

Find the labels above in your own portal. [Review + create](#)

Illustrated navigation view · use your own saved names and IDs

Why this action: The plan is the paid compute capacity. The app is the website running on that capacity.

1. Search App Services > Create > Web App.

2. Subscription: your practice subscription; group: rg-PREFIX-core; name: PREFIX-web.

3B.04 · ACTION 2 OF 7**AZURE PORTAL · Create Web App → Basics**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Create Web App

Create Web App > Basics

Create Web App

Basics

- 1 Publish** Code
- 2 Runtime** Python 3.12 if supported
- 3 OS** Linux
- 4 Region** Primary region

Find the labels above in your own portal. **Next**

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. Publish: Code; runtime: Python 3.12 or a currently supported matching Python 3 version; operating system: Linux; region: the primary region.

3B.04 · ACTION 3 OF 7**AZURE PORTAL · Create Web App → App Service plan**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Create Web App

Create Web App > App Service plan

Create Web App

App Service plan

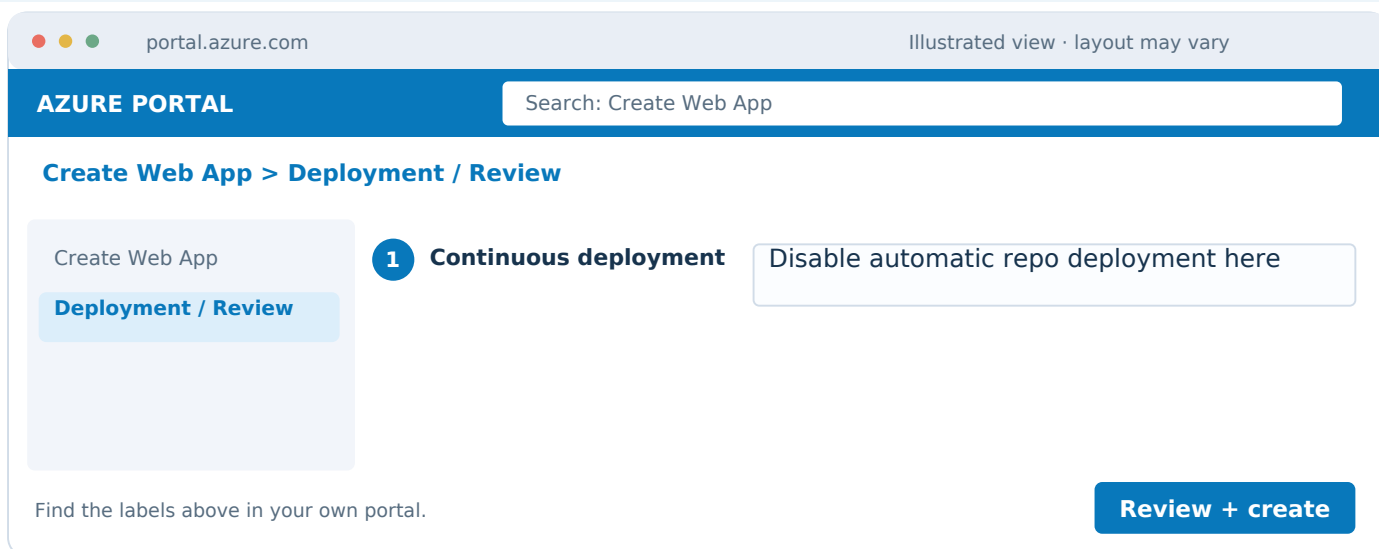
- 1 Name** asp-PREFIX
- 2 Pricing plan** Standard S1
- 3 Review** Displayed price

Find the labels above in your own portal. **Select**

Illustrated navigation view · use your own saved names and IDs

Why this action: The plan is the paid compute capacity. The app is the website running on that capacity.

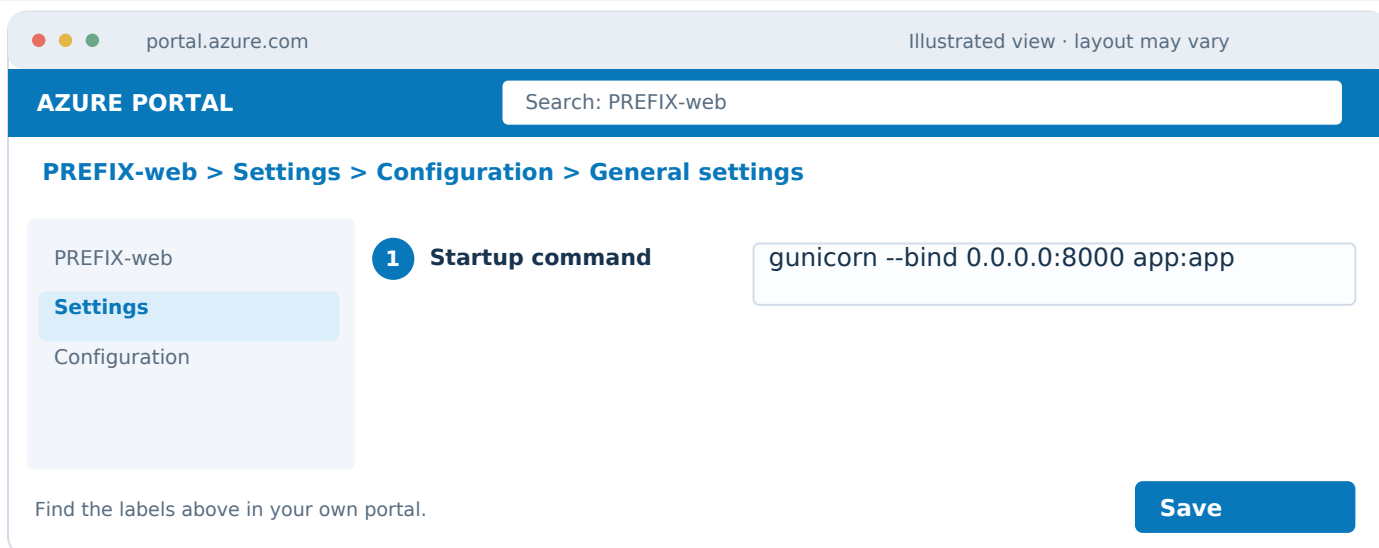
1. Create plan asp-PREFIX.
2. Choose Standard S1 for deployment slots and the scaling/backup exercises.
3. Check the price before selecting it.
4. The Free tier does not provide all these features.

3B.04 · ACTION 4 OF 7**AZURE PORTAL · Create Web App → Deployment / Review**

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. For this step leave automatic repository deployment off.
2. Create the app, wait for completion, and open the resource.

3B.04 · ACTION 5 OF 7**AZURE PORTAL · PREFIX-web → Settings → Configuration → General settings**

Illustrated navigation view · use your own saved names and IDs

Why this action: The plan is the paid compute capacity. The app is the website running on that capacity.

1. Open Settings > Configuration > General settings, or Configuration (preview)/Stack settings if the navigation has changed.
2. Set Startup command to unicorn --bind 0.0.0.0:8000 app:app.
3. Save.

3B.04 · ACTION 6 OF 7**AZURE PORTAL · PREFIX-web → Settings → Environment variables → App settings**

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page header includes 'AZURE PORTAL' and a search bar containing 'PREFIX-web'. The breadcrumb navigation path is 'PREFIX-web > Settings > Environment variables > App settings'. On the left sidebar, 'PREFIX-web' is selected, and under 'Settings', 'App settings' is highlighted. The main content area shows two numbered steps: '1 Name' with the value 'SCM_DO_BUILD_DURING_DEPLOYMENT' and '2 Value' with the value 'true'. At the bottom right, there is an 'Apply' button. A note at the bottom left states: 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: The plan is the paid compute capacity. The app is the website running on that capacity.

1. Open Environment variables > App settings.
2. Add SCM_DO_BUILD_DURING_DEPLOYMENT with value true.
3. Save/apply and accept the restart.

3B.04 · ACTION 7 OF 7**AZURE PORTAL · PREFIX-web → Configuration → General settings**

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page header includes 'AZURE PORTAL' and a search bar containing 'PREFIX-web'. The breadcrumb navigation path is 'PREFIX-web > Configuration > General settings'. On the left sidebar, 'PREFIX-web' is selected, and under 'Configuration', 'General settings' is highlighted. The main content area shows three numbered steps: '1 HTTPS Only' with the value 'On', '2 Minimum TLS' with the value '1.2 or stricter supported', and '3 Always On' with the value 'On'. At the bottom right, there is a 'Save' button. A note at the bottom left states: 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: The plan is the paid compute capacity. The app is the website running on that capacity.

1. Open Configuration/General settings.
2. Enable HTTPS Only, minimum TLS 1.2 or stricter supported policy, and Always On for the S1 plan.
3. Save.

You should see

The app exists on S1 with the expected Python runtime and startup command.

If it does not work

If a menu label changed, search within the resource menu for the setting name. These are exact setting names; the navigation grouping can change. [R21]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

3B.05 Deploy the package and read the actual hostname

AZ-104 + AZ-400 · CORE - ongoing plan cost

Why this task: App Service installs the Python requirements during ZIP build and starts the process you specified.

3B.05 · ACTION 1 OF 4

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: App Service installs the Python requirements during ZIP build and starts the process you specified.

1. Run the block below.
2. If deployment is still running, follow the deployment log/status before retrying.

USE IN: BASH · COPY ONLY THE CODE

```
az webapp deploy -g "$RG" -n "$APP" --src-path app.zip --type zip
HOST=$(az webapp show -g "$RG" -n "$APP" --query defaultHostName -o tsv)
echo "https://$HOST"
bash scripts/smoke.sh "https://$HOST"
```

What it means: `--type zip` uploads the package. `--query defaultHostName` reads the real service hostname. The smoke script retries short startup delays and fails if health never succeeds.

--src-path · The local file to upload as the deployment package.

--type · For `az webapp deploy`, `zip` means the upload is a ZIP package.

--query · Select or transform returned fields with a JMESPath expression; this does not itself change the resource.

-g · Short form of `--resource-group`.

-n · Short form of `--name` in these Azure commands.

-o · Azure CLI output format: `table`, `tsv` or `none`.

\$NAME / \${NAME} · Read a variable from this Bash session. Source `lab.env` first when these are lab names.

\$(command) · Run the inner command and use its output as the value.

3B.05 · ACTION 2 OF 4**YOUR BROWSER · Your browser → Exact defaultHostName URL**

Your own browser tab

Illustrated view · layout may vary

YOUR BROWSER

Your browser > Exact defaultHostName URL

Your browser

Exact defaultHostName URL

1 Address

2 Expected

Find the labels above in your own portal.

Open

Illustrated navigation view · use your own saved names and IDs

Why this action: App Service installs the Python requirements during ZIP build and starts the process you specified.

1. Copy the displayed URL into your browser.
2. You should see the OneLab JSON response.

3B.05 · ACTION 3 OF 4**YOUR BROWSER · Your app hostname → /health**

Your own browser tab

Illustrated view · layout may vary

YOUR BROWSER

Your app hostname > /health

Your app hostname

/health

1 Expected

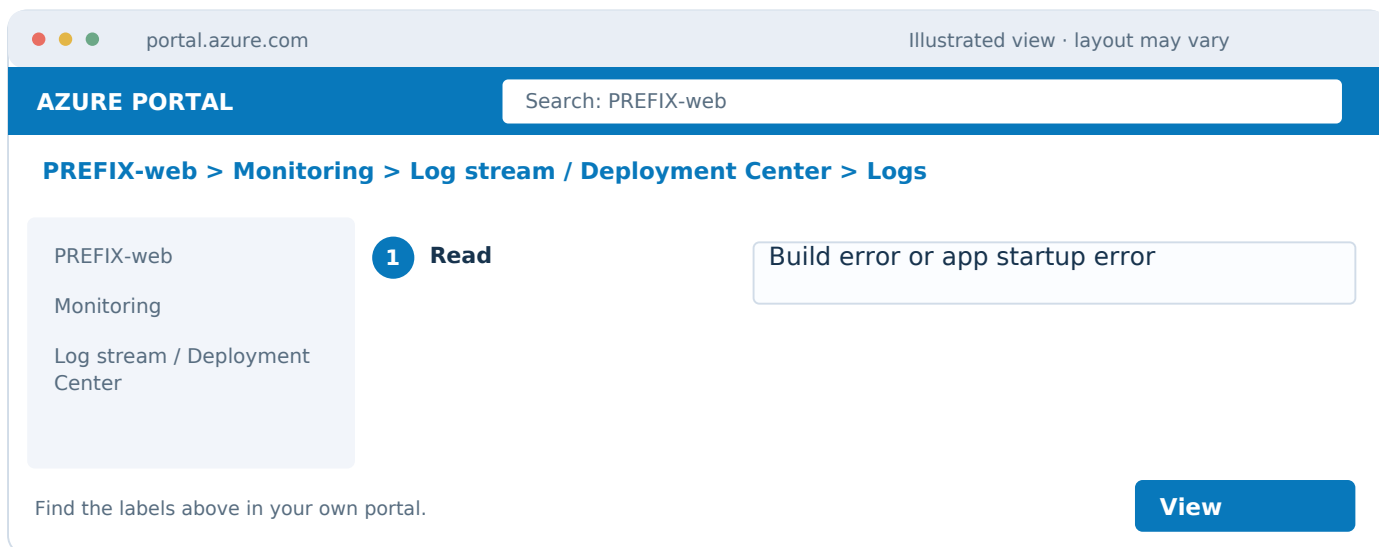
Find the labels above in your own portal.

Open

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Visit /health at that hostname.
2. Record the result and the deployment time.

3B.05 · ACTION 4 OF 4**AZURE PORTAL · PREFIX-web → Monitoring → Log stream / Deployment Center → Logs**

Illustrated navigation view · use your own saved names and IDs

Why this action: App Service installs the Python requirements during ZIP build and starts the process you specified.

1. If the app fails, use App Service > Monitoring > Log stream and Deployment Center > Logs.
2. Distinguish deployment build failure from startup failure.

You should see

The public HTTPS app URL returns OneLab and /health returns 200.

If it does not work

Do not guess the hostname by adding azurewebsites.net to the app name. Use defaultHostName because some app names include an additional generated hostname component. [R22]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

3C · Add staging and protected settings

Planning estimate: 100 minutes. Finish with: Two healthy slots with separately authorized secret access.

3C.01 Create the staging slot

AZ-104 + AZ-400 · CORE - ongoing plan cost

Why this task: A slot gives the same app a separate deployment endpoint for testing before promotion.

3C.01 · ACTION 1 OF 5

AZURE PORTAL · App Services → PREFIX-web → Deployment slots → Add slot

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: App Services

App Services > PREFIX-web > Deployment slots > Add slot

App Services
PREFIX-web
Deployment slots

1 Name staging

2 Clone settings from Production app

3 Automatic swap Off

Find the labels above in your own portal.

Add

Illustrated navigation view · use your own saved names and IDs

Why this action: A slot gives the same app a separate deployment endpoint for testing before promotion.

1. Open the web app > Deployment > Deployment slots > Add slot.
2. Name staging; clone settings from the production app.
3. Add.

3C.01 · ACTION 2 OF 5**AZURE PORTAL · PREFIX-web → Deployment slots → staging → Configuration**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL

PREFIX-web > Deployment slots > staging > Configuration

PREFIX-web
Deployment slots
staging

1 Runtime
2 Startup
3 Build setting

Matching Python version
gunicorn --bind 0.0.0.0:8000 app:app
SCM_DO_BUILD_DURING_DEPLOYMENT=true

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open the staging slot as its own resource.
2. Verify the Python runtime, startup command and SCM_DO_BUILD_DURING_DEPLOYMENT setting.
3. A slot has its own URL and identity.

3C.01 · ACTION 3 OF 5**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

The dark code panel below is the terminal/editor view for this action.

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. Run the deployment block below to publish the same initial version to staging.

USE IN: BASH · COPY ONLY THE CODE

```
az webapp deploy -g "$RG" -n "$APP" --slot staging \
  --src-path app.zip --type zip
STAGEHOST=$(az webapp show -g "$RG" -n "$APP" --slot staging \
  --query defaultHostName -o tsv)
bash scripts/smoke.sh "https://$STAGEHOST"
```

What it means: --slot staging directs the operation to the test slot rather than the live app.

--slot · Target the named deployment slot, such as staging.

--src-path · The local file to upload as the deployment package.

--type · For az webapp deploy, zip means the upload is a ZIP package.

--query · Select or transform returned fields with a JMESPath expression; this does not itself change the resource.

-g · Short form of --resource-group.

-n · Short form of --name in these Azure commands.

-o · Azure CLI output format: table, tsv or none.

\$NAME / \${NAME} · Read a variable from this Bash session. Source lab.env first when these are lab names.

\${command} · Run the inner command and use its output as the value.

Backslash at line end · Continue the same Bash command on the next line. No spaces may follow the backslash.

3C.01 · ACTION 4 OF 5**AZURE PORTAL · App Services → PREFIX-web → Deployment slots → Add slot**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: App Services

App Services > PREFIX-web > Deployment slots > Add slot

App Services
PREFIX-web
Deployment slots

1 Name staging

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Use the returned staging hostname to test /health.
2. Keep both slots healthy before pipeline work.

3C.01 · ACTION 5 OF 5**SEE THE IDEA · Create the staging slot → Explain the decision****A SMALL DECISION MAP****Create the staging slot > Explain the decision**

- | | |
|-----------------|------------------------------------|
| 1 Locate | Explain the decision |
| 2 Action | Do not enable automatic slot swap. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: A slot gives the same app a separate deployment endpoint for testing before promotion.

1. Do not enable automatic slot swap.
2. The pipeline will request approval and perform an explicit swap.

You should see

Both production and staging serve the initial working version.

If it does not work

If Add slot is unavailable, confirm the App Service plan tier supports slots. Stopping one slot does not remove the plan cost.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

3C.02 Inspect scale, TLS, domain, backup and networking

AZ-104 + AZ-400 · CORE - ongoing plan cost

Why this task: These administration settings determine capacity, connectivity and recovery.

3C.02 · ACTION 1 OF 7

AZURE PORTAL · App Service plans → asp-PREFIX → Scale up / Scale out

The screenshot shows the Azure Portal interface for configuring an App Service plan. The breadcrumb navigation is 'App Service plans > asp-PREFIX > Scale up / Scale out'. On the left, a sidebar lists 'Scale', 'Custom domains', and 'Certificates'. The main area contains two numbered steps: '1 Compare' and '2 Optional count'. Under '1 Compare', there is a text input field with 'Bigger instance'. Under '2 Optional count', there is a text input field with '2 briefly, then restore 1'. At the bottom right, there is a blue 'Apply' button. Below the main content area, a note says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: These administration settings determine capacity, connectivity and recovery.

1. Open the plan > Scale up.
2. Compare larger sizes without purchasing one.
3. Open Scale out and temporarily change instance count from 1 to 2 only if budget permits; restore 1 after observation.

3C.02 · ACTION 2 OF 7**AZURE PORTAL · asp-PREFIX → Scale out → Custom autoscale**

The screenshot shows the Azure Portal interface for configuring Custom autoscale for the resource 'asp-PREFIX'. The breadcrumb navigation is 'asp-PREFIX > Scale out > Custom autoscale'. On the left, a sidebar lists 'Scale' (selected), 'Custom domains', 'Certificates', 'Backups', and 'Networking'. The main area contains four numbered steps: 1. Metric (CPU Percentage), 2. Minimum (1), 3. Maximum (2), and 4. After observation (Remove the practice rule). A 'Save' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: asp-PREFIX

asp-PREFIX > Scale out > Custom autoscale

Scale

- Custom domains
- Certificates
- Backups
- Networking

1 Metric CPU Percentage

2 Minimum 1

3 Maximum 2

4 After observation Remove the practice rule

Find the labels above in your own portal.

Save

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. For metric autoscale, select a supported CPU rule, minimum 1, maximum 2 and a cooldown.
2. Record what triggers scale out and scale in; remove the test rule afterward.

3C.02 · ACTION 3 OF 7**AZURE PORTAL · PREFIX-web → Custom domains → Add custom domain**

The screenshot shows the Azure Portal interface for adding a custom domain for the resource 'PREFIX-web'. The breadcrumb navigation is 'PREFIX-web > Custom domains > Add custom domain'. On the left, a sidebar lists 'Scale', 'Custom domains' (selected), and 'Certificates'. The main area contains three numbered steps: 1. Domain (Your owned domain), 2. DNS TXT (Exact verification value shown), and 3. DNS CNAME (Exact target shown). A 'Validate' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: PREFIX-web

PREFIX-web > Custom domains > Add custom domain

Scale

- Custom domains**
- Certificates

1 Domain Your owned domain

2 DNS TXT Exact verification value shown

3 DNS CNAME Exact target shown

Find the labels above in your own portal.

Validate

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Open the app > Custom domains > Add custom domain.
2. If you own a domain, add the shown TXT verification record and CNAME at your DNS provider, validate, then add the binding.
3. Use the exact target hostname shown by Azure.

3C.02 · ACTION 4 OF 7**AZURE PORTAL · PREFIX-web → Certificates / Custom domains**

The screenshot shows the Azure Portal interface for the resource 'PREFIX-web'. The breadcrumb navigation is 'PREFIX-web > Certificates / Custom domains'. On the left, a sidebar contains 'Scale', 'Custom domains' (highlighted), and 'Certificates'. The main area has two numbered steps: '1 Certificate' with the description 'Eligible managed or uploaded certificate', and '2 TLS binding' with the description 'Your verified custom domain'. At the bottom right is an 'Add binding' button. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: These administration settings determine capacity, connectivity and recovery.

1. For an eligible custom domain, create a managed certificate or upload an appropriate certificate under Certificates, then add its TLS binding.
2. Without a domain, complete this as DESIGN; the default Azure hostname already has platform TLS.

3C.02 · ACTION 5 OF 7**AZURE PORTAL · PREFIX-web → Backups → Configure**

The screenshot shows the Azure Portal 'Configure' page for backups of 'PREFIX-web'. The breadcrumb navigation is 'PREFIX-web > Backups > Configure'. The left sidebar contains 'Scale', 'Custom domains', and 'Certificates'. The main area has three numbered steps: '1 Storage' with 'Private backup container', '2 Retention' with 'Short supported period', and '3 Run' with 'Backup now'. At the bottom right is a 'Save' button. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Open Backups and configure a supported backup to a private container, with a short retention and appropriate SAS/permissions if requested.
2. Run Backup now and inspect its result.
3. Back up application data separately where required.

3C.02 · ACTION 6 OF 7**AZURE PORTAL · PREFIX-web → Networking → VNet integration → Add**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: PREFIX-web

PREFIX-web > Networking > VNet integration > Add

Scale
Custom domains
Certificates

1 VNet vnet-main

2 Subnet integration

Find the labels above in your own portal.

Connect

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Open Networking > VNet integration and add vnet-main/integration.
2. This gives outbound access to the VNet; it does not turn the public website into a private inbound endpoint.

3C.02 · ACTION 7 OF 7**AZURE PORTAL · PREFIX-web → Networking → Access restrictions / Private endpoints**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: PREFIX-web

PREFIX-web > Networking > Access restrictions / Private endpoints

Scale
Custom domains
Certificates

1 Inspect App and SCM access

2 Hosted agent Must still reach deployment endpoint

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: These administration settings determine capacity, connectivity and recovery.

1. Inspect Access restrictions and Private endpoint options.
2. Do not restrict the app or SCM endpoint so your hosted deployment agent can no longer reach it.
3. A private app generally needs a reachable self-hosted agent or supported private connectivity.

You should see

You can explain scale up/out, DNS ownership, TLS binding, backup and inbound versus outbound networking.

If it does not work

Each slot can need its own networking and identity settings. A slot swap does not swap everything. Backups also have plan and content limitations. [R23, R24]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

3C.03 Create a Key Vault and a harmless secret

AZ-104 + AZ-400 · CORE

Why this task: A vault stores secrets separately from source code and access is granted to identities.

3C.03 · ACTION 1 OF 6

AZURE PORTAL · Key vaults → Create

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Key vaults

Key vaults > Create

Key vaults

Create

- 1 **Resource group** rg-PREFIX-core
- 2 **Name** PREFIX-kv
- 3 **Tier** Standard
- 4 **Permission model** Azure RBAC
- 5 **Secret name** demo-message

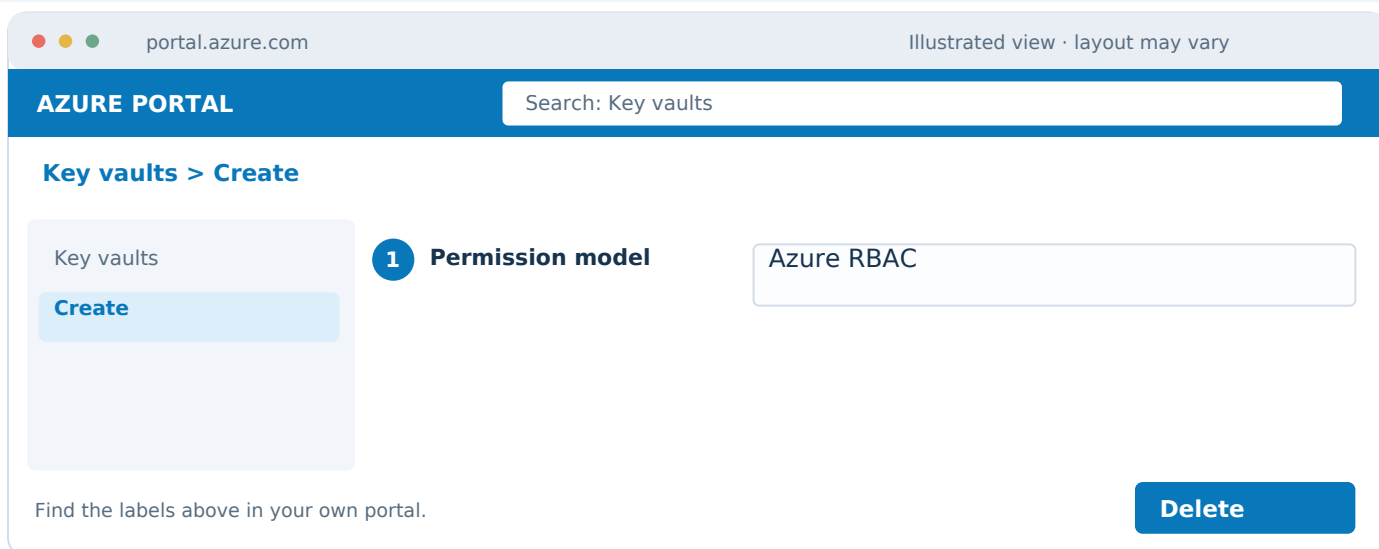
Find the labels above in your own portal.

Review + create

Illustrated navigation view · use your own saved names and IDs

Why this action: A vault stores secrets separately from source code and access is granted to identities.

1. Search Key vaults > Create.
2. Group: rg-PREFIX-core; name: PREFIX-kv; primary region
3. Standard tier.

3C.03 · ACTION 2 OF 6**AZURE PORTAL · Key vaults → Create**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Key vaults

Key vaults > Create

Key vaults

1 Permission model Azure RBAC

Create

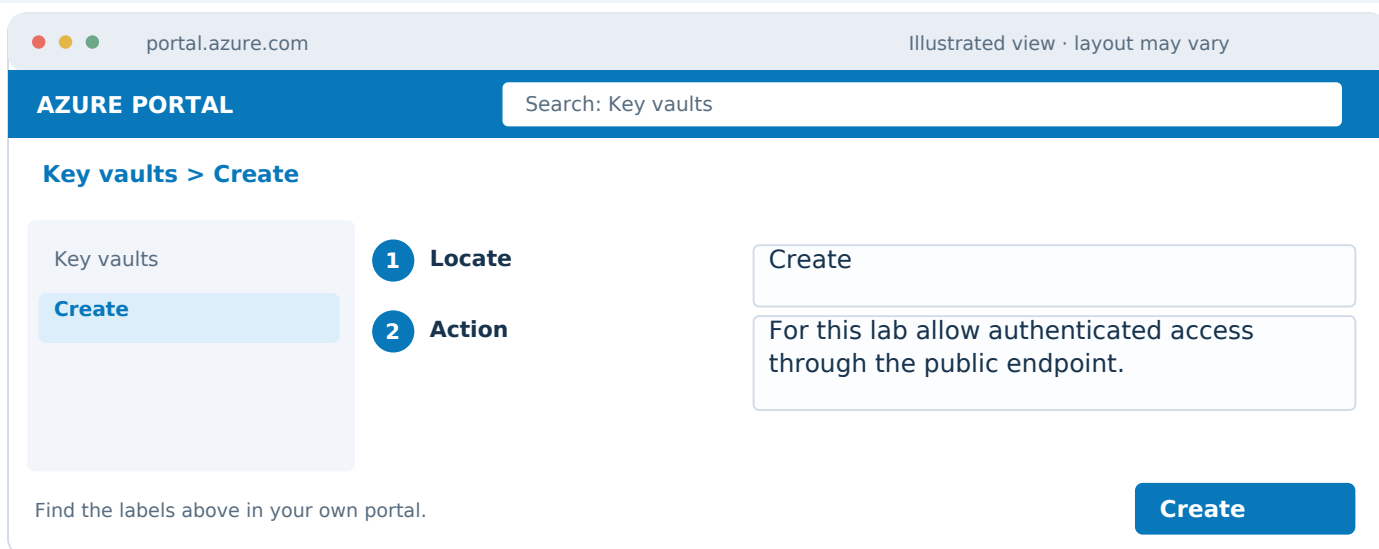
Find the labels above in your own portal.

Delete

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Use the Azure RBAC permission model.
2. Keep soft delete enabled.
3. Review purge protection before choosing it: once enabled it cannot simply be switched off for immediate cleanup.

3C.03 · ACTION 3 OF 6**AZURE PORTAL · Key vaults → Create**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Key vaults

Key vaults > Create

Key vaults

1 Locate Create

2 Action For this lab allow authenticated access through the public endpoint.

Create

Find the labels above in your own portal.

Create

Illustrated navigation view · use your own saved names and IDs

Why this action: A vault stores secrets separately from source code and access is granted to identities.

1. For this lab allow authenticated access through the public endpoint.
2. Create the vault.

3C.03 · ACTION 4 OF 6**AZURE PORTAL · PREFIX-kv → Access control (IAM) → Add role assignment**

The screenshot shows the Azure Portal interface for adding a role assignment. The breadcrumb path is PREFIX-kv > Access control (IAM) > Add role assignment. On the left, a sidebar shows the navigation path: PREFIX-kv, Access control (IAM), and Add role assignment (highlighted). The main area has three numbered steps: 1. Role (Key Vault Secrets Officer), 2. Member (Your own user), and 3. Scope (This vault). A 'Review + assign' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: PREFIX-kv

PREFIX-kv > Access control (IAM) > Add role assignment

PREFIX-kv

Access control (IAM)

Add role assignment

1 Role Key Vault Secrets Officer

2 Member Your own user

3 Scope This vault

Find the labels above in your own portal.

Review + assign

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Open IAM > Add role assignment.
2. Give your own user Key Vault Secrets Officer at this vault.
3. Wait for propagation.

3C.03 · ACTION 5 OF 6**AZURE PORTAL · PREFIX-kv → Objects → Secrets → Generate/Import**

The screenshot shows the Azure Portal interface for generating or importing a secret. The breadcrumb path is PREFIX-kv > Objects > Secrets > Generate/Import. On the left, a sidebar shows the navigation path: PREFIX-kv, Objects, Secrets, and Generate/Import (highlighted). The main area has two numbered steps: 1. Name (demo-message) and 2. Value (OneLab training only). A 'Create' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: PREFIX-kv

PREFIX-kv > Objects > Secrets > Generate/Import

PREFIX-kv

Objects

Secrets

1 Name demo-message

2 Value OneLab training only

Find the labels above in your own portal.

Create

Illustrated navigation view · use your own saved names and IDs

Why this action: A vault stores secrets separately from source code and access is granted to identities.

1. Open Objects > Secrets > Generate/Import.
2. Name demo-message; value a harmless phrase such as OneLab training only.
3. Create.

3C.03 · ACTION 6 OF 6**AZURE PORTAL · Key vaults → Create**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Key vaults

Key vaults > Create

Key vaults

Create

1 Name PREFIX-kv

2 Secret name demo-message

Find the labels above in your own portal. View

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. Open the secret and copy its versionless URI pattern:
`https://PREFIX-kv.vault.azure.net/secrets/demo-message`.
2. Never print the secret value into pipeline logs.

You should see

The vault contains one dummy secret and you know its URI.

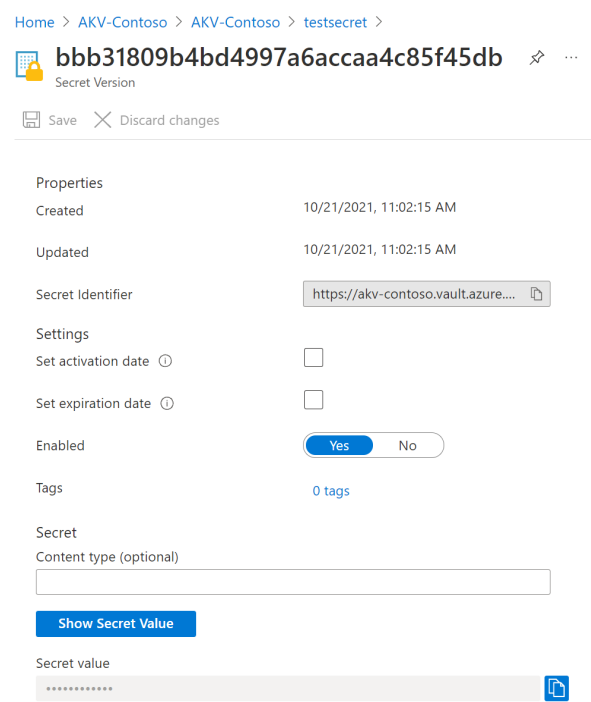
If it does not work

Key Vault Contributor manages the vault but does not automatically read/write secret values. This lab uses a separate data role. [R25]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Microsoft reference screenshot · Secret properties



[Microsoft reference source](#) · Sample values belong to that reference, not OneLab.

3C.04 Give each slot its own managed identity

AZ-104 + AZ-400 · CORE

Why this task: The running app can authenticate without keeping a client password.

3C.04 · ACTION 1 OF 6**AZURE PORTAL · App Services → PREFIX-web → Identity**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: App Services

App Services > PREFIX-web > Identity

App Services
PREFIX-web
Identity

- 1 System assigned** On
- 2 Repeat for** staging slot
- 3 Vault data role** Key Vault Secrets User
- 4 App setting** DEMO_MESSAGE
- 5 Slot setting** Selected

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Open production web app > Settings > Identity > System assigned.
2. Set On and Save.
3. Record its principal/object ID privately.

3C.04 · ACTION 2 OF 6**AZURE PORTAL · PREFIX-web → Deployment slots → staging → Identity**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: PREFIX-web

PREFIX-web > Deployment slots > staging > Identity

PREFIX-web
Deployment slots
staging

- 1 System assigned** On
- 2 Principal ID** Different from production

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Open the staging slot > Identity.
2. Enable its own system-assigned identity and record its separate principal ID.

3C.04 · ACTION 3 OF 6**AZURE PORTAL · PREFIX-kv → Access control (IAM) → Add role assignment**

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page title is 'AZURE PORTAL'. A search bar contains 'PREFIX-kv'. The breadcrumb navigation is 'PREFIX-kv > Access control (IAM) > Add role assignment'. On the left sidebar, 'PREFIX-kv' is selected, and 'Add role assignment' is highlighted. The main content area shows three numbered steps: 1. Role (Key Vault Secrets User), 2. Member type (Managed identity), and 3. Member (Production web app). A 'Review + assign' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Open the vault > IAM > Add role assignment > Key Vault Secrets User.
2. Under Members choose Managed identity > App Service, then select the production app.
3. Review + assign.

3C.04 · ACTION 4 OF 6**AZURE PORTAL · PREFIX-kv → Access control (IAM) → Add role assignment**

This screenshot is similar to the previous one, but the 'Member' field (step 3) is set to 'staging slot identity' instead of 'Production web app'. All other elements, including the breadcrumb navigation and sidebar, remain the same.

Illustrated navigation view · use your own saved names and IDs

Why this action: The staging slot has a different identity, so it needs its own permission.

1. Repeat the role assignment for the staging slot identity.
2. Verify both in Role assignments.

3C.04 · ACTION 5 OF 6

AZURE PORTAL · Each slot → Environment variables → App settings → Add

The screenshot shows the Azure Portal interface for adding a new environment variable. The breadcrumb trail is 'Each slot > Environment variables > App settings > Add'. On the left, a sidebar lists 'Each slot', 'Environment variables', and 'App settings'. The main area contains three numbered steps: 1. Name (set to DEMO_MESSAGE), 2. Value (set to 'The Key Vault reference shown in the action'), and 3. Deployment slot setting (set to 'Selected'). An 'Apply' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Each slot

Each slot > Environment variables > App settings > Add

Each slot
Environment variables
App settings

1 Name DEMO_MESSAGE

2 Value The Key Vault reference shown in the action

3 Deployment slot setting Selected

Find the labels above in your own portal. **Apply**

Illustrated navigation view · use your own saved names and IDs

Why this action: The running app can authenticate without keeping a client password.

1. On each slot, open Environment variables > App settings > Add.
2. Name DEMO_MESSAGE; value
@Microsoft.KeyVault(SecretUri=https://PREFIX-kv.vault.azure.net/secrets/demo-message).
3. Replace PREFIX.

3C.04 · ACTION 6 OF 6

AZURE PORTAL · App Services → PREFIX-web → Identity

The screenshot shows the Azure Portal interface for the 'Identity' settings of a web app. The breadcrumb trail is 'App Services > PREFIX-web > Identity'. On the left, a sidebar lists 'App Services', 'PREFIX-web', and 'Identity'. The main area shows a single step: 1. Slot setting (set to 'Selected'). A 'Save' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: App Services

App Services > PREFIX-web > Identity

App Services
PREFIX-web
Identity

1 Slot setting Selected

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Mark it as a Deployment slot setting and save.
2. Wait for the reference status to become resolved.
3. Test /secret-status on both URLs.

You should see

Each slot returns secret_resolved: true without revealing the phrase.

If it does not work

If unresolved, check the exact secret URI, slot-specific identity, data role, network access and propagation delay. A system identity belongs to one resource/slot; it is not exchanged by a slot swap. [R26]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

System-assigned identity belongs to one resource. User-assigned identity is separately created and reusable. A service principal represents an application in a tenant. Federation exchanges a trusted pipeline token for short-lived access; Day 4, Block B implements it.

3D · Turn on monitoring

Planning estimate: 60 minutes. Finish with: Real request, failure and dependency telemetry, or a tracked ingestion wait.

3D.01 Create a workspace and instrument the application

AZ-104 + AZ-400 · CORE

Why this task: The application must send telemetry before a dashboard can display it.

3D.01 · ACTION 1 OF 6

AZURE PORTAL · Log Analytics workspaces → Create

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Log Analytics workspaces

Log Analytics workspaces > Create

Log Analytics workspaces

Create

- 1 Group** rg-PREFIX-core
- 2 Workspace** law-PREFIX
- 3 Application Insights** ai-PREFIX
- 4 App setting** APPLICATIONINSIGHTS_CONNECTION_STRING

Find the labels above in your own portal.

Review + create

Illustrated navigation view · use your own saved names and IDs

Why this action: The application must send telemetry before a dashboard can display it.

1. Search Log Analytics workspaces > Create.
2. Use rg-PREFIX-core, name law-PREFIX and the primary region.
3. Review retention and usage settings after creation.

3D.01 · ACTION 2 OF 6**AZURE PORTAL · Application Insights → Create**

The screenshot shows the Azure Portal interface for creating a new Application Insights instance. The browser address bar shows 'portal.azure.com'. The page title is 'AZURE PORTAL' with a search bar containing 'Application Insights'. The breadcrumb navigation is 'Application Insights > Create'. On the left, a sidebar shows 'Application Insights' with a 'Create' button highlighted. The main content area has three numbered steps: 1. Name (ai-PREFIX), 2. Group (rg-PREFIX-core), and 3. Workspace (law-PREFIX). Each step has a corresponding text input field. At the bottom right, there is a 'Review + create' button. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Application Insights

Application Insights > Create

Application Insights

Create

1 Name ai-PREFIX

2 Group rg-PREFIX-core

3 Workspace law-PREFIX

Find the labels above in your own portal.

Review + create

Illustrated navigation view · use your own saved names and IDs

Why this action: The application must send telemetry before a dashboard can display it.

1. Search Application Insights > Create.
2. Name ai-PREFIX; same group/region; use the workspace you just created.
3. Create.

3D.01 · ACTION 3 OF 6**AZURE PORTAL · Application Insights → ai-PREFIX → Overview**

The screenshot shows the Azure Portal interface for the 'Overview' page of an Application Insights instance. The browser address bar shows 'portal.azure.com'. The page title is 'AZURE PORTAL' with a search bar containing 'Application Insights'. The breadcrumb navigation is 'Application Insights > ai-PREFIX > Overview'. On the left, a sidebar shows 'Application Insights' with 'ai-PREFIX' selected and the 'Overview' button highlighted. The main content area has two numbered steps: 1. Copy (Connection string) and 2. Store (App settings, not source). Each step has a corresponding text input field. At the bottom right, there is a 'Copy' button. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Application Insights

Application Insights > ai-PREFIX > Overview

Application Insights

ai-PREFIX

Overview

1 Copy Connection string

2 Store App settings, not source

Find the labels above in your own portal.

Copy

Illustrated navigation view · use your own saved names and IDs

Why this action: The application must send telemetry before a dashboard can display it.

1. Open Application Insights > Overview and copy the Connection string.
2. Treat it as configuration; do not paste it into the repository.

3D.01 · ACTION 4 OF 6**AZURE PORTAL · Each app slot → Environment variables → App settings**

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page title is 'AZURE PORTAL' with a search bar containing 'Each app slot'. The breadcrumb navigation is 'Each app slot > Environment variables > App settings'. On the left, a sidebar shows 'Each app slot', 'Environment variables', and 'App settings' (highlighted). The main area has two numbered steps: 1. Name: APPLICATIONINSIGHTS_CONNECTION_STRING; 2. Value: Your Insights connection string. An 'Apply' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: The application must send telemetry before a dashboard can display it.

1. Open production web app > Environment variables > App settings.
2. Add APPLICATIONINSIGHTS_CONNECTION_STRING with that value.
3. Save.
4. Repeat for staging, preferably with a separate telemetry resource if you want clean environment separation.

3D.01 · ACTION 5 OF 6**SEE THE IDEA · Create a workspace and instrument the application → Explain the decision****A SMALL DECISION MAP****Create a workspace and instrument the application > Explain the decision**

- | | |
|-----------------|--|
| 1 Locate | Explain the decision |
| 2 Action | The supplied app calls <code>configure_azure_monitor</code> only when this setting exists. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: The application must send telemetry before a dashboard can display it.

1. The supplied app calls `configure_azure_monitor` only when this setting exists.
2. This enables supported Flask/request dependency instrumentation.
3. Do not also enable a second overlapping auto-instrumentation method without checking for duplicate telemetry.

3D.01 · ACTION 6 OF 6**AZURE PORTAL · ai-PREFIX → Transaction search / Failures / Performance**

The illustration shows a browser window with the URL `portal.azure.com`. The page title is "AZURE PORTAL". A search bar contains "ai-PREFIX". The breadcrumb navigation path is "ai-PREFIX > Transaction search / Failures / Performance". On the left sidebar, "ai-PREFIX" is selected, and "Transaction search / Failures / Performance" is highlighted. A numbered list indicates the steps: "1 First" and "2 Then". To the right of the list are two input fields: "Browse app /dependency" and "Wait for ingestion and inspect request". At the bottom right is a blue "View" button. Below the illustration, a note says "Find the labels above in your own portal."

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Use the app in a browser and visit `/dependency` to generate an outgoing request.
2. Wait a few minutes for ingestion, then open Application Insights > Transaction search or Failures/Performance.

You should see

You see requests from the running application rather than an empty resource with no data.

If it does not work

Check the connection string, package installation, app restart, network egress, time range and resource scope if nothing appears. A successful deployment does not prove telemetry was configured. [R53]

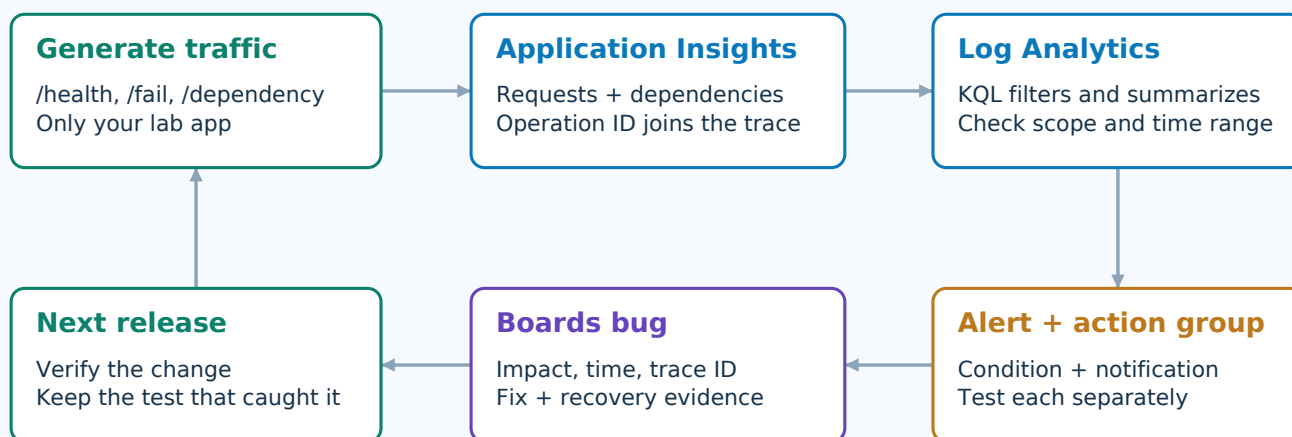
Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

A metric is a number over time, such as CPU percentage. A log is a recorded event with fields. A trace follows one request across components. An alert evaluates a condition; an action group decides whom or what to notify. An alert processing rule changes notification handling, for example during maintenance.

Turn one failed request into a useful fix



3D.02 Generate a small amount of controlled traffic

AZ-104 + AZ-400 · CORE

Why this task: You need a visible signal to practise interpreting telemetry.

3D.02 · ACTION 1 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: You need a visible signal to practise interpreting telemetry.

1. Read the actual production hostname again using the command below.
2. Run the short loop against only your own lab app.

USE IN: BASH · COPY ONLY THE CODE

```

HOST=$(az webapp show -g "$RG" -n "$APP" --query defaultHostName -o tsv)
for n in 1 2 3 4 5; do curl -s "https://$HOST/health"; done
curl -s -o /dev/null -w '%{http_code}\n' "https://$HOST/fail"
curl -s "https://$HOST/dependency"
# Optional tiny concurrent test of your own endpoint:
seq 1 20 | xargs -P 4 -I '{}' curl -s -o /dev/null "https://$HOST/health"
  
```

What it means: `-w` prints the HTTP status. `-o /dev/null` discards response bodies. `-P 4` runs at most four curl processes at once; it does not emulate thousands of users.

--query · Select or transform returned fields with a JMESPath expression; this does not itself change the resource.

-g · Short form of `--resource-group`.

-n · Short form of `--name` in these Azure commands.

-o · Azure CLI output format: table, tsv or none.

\$NAME / \${NAME} · Read a variable from this Bash session. Source `lab.env` first when these are lab names.

\$(command) · Run the inner command and use its output as the value.

curl · Send an HTTP request. `-f` fails on HTTP error status; `-s` quiets the progress meter; `-v` shows connection details when used.

curl -o /dev/null / -w · Discard the response body, then print only the requested status field.

3D.02 · ACTION 2 OF 5**AZURE PORTAL · ai-PREFIX → Failures / Performance**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: ai-PREFIX

ai-PREFIX > Failures / Performance

ai-PREFIX

Failures / Performance

1 Failure route /fail

2 Compare /health and /dependency

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: You need a visible signal to practise interpreting telemetry.

1. Open Application Insights > Failures and find the intentional /fail responses.
2. Open Performance and compare normal requests with /dependency.

3D.02 · ACTION 3 OF 5**AZURE PORTAL · ai-PREFIX → Transaction search → One request → End-to-end transaction**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: ai-PREFIX

ai-PREFIX > Transaction search > One request > End-to-end transaction

ai-PREFIX

Transaction search

One request

1 Inspect Operation ID and dependency timing

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: You need a visible signal to practise interpreting telemetry.

1. Select an individual request and inspect its operation/trace and dependency details.
2. This is distributed tracing evidence, not only a total request count.

3D.02 · ACTION 4 OF 5

CLOUD SHELL · Cloud Shell → Bash → Generate lab traffic

portal.azure.com

Illustrated view · layout may vary

CLOUD SHELL

Search: Cloud Shell

Cloud Shell > Bash > Generate lab traffic

Cloud Shell

Bash

Generate lab traffic

1 Locate

2 Action

Generate lab traffic

Record response time, failure percentage and time range.

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

- 1. Record response time, failure percentage and time range.
- 2. Do not confuse total requests with requests per second.

3D.02 · ACTION 5 OF 5

CLOUD SHELL · Cloud Shell → Bash → Generate lab traffic

portal.azure.com

Illustrated view · layout may vary

CLOUD SHELL

Search: Cloud Shell

Cloud Shell > Bash > Generate lab traffic

Cloud Shell

Bash

Generate lab traffic

1 Locate

2 Action

Generate lab traffic

For a light concurrency exercise, run the optional 20-request command.

Find the labels above in your own portal.

Run

Illustrated navigation view · use your own saved names and IDs

Why this action: You need a visible signal to practise interpreting telemetry.

- 1. For a light concurrency exercise, run the optional 20-request command.
- 2. It is a tiny controlled load test, not a capacity benchmark or a production stress test.

You should see

You can identify healthy requests, the intentional failure and one dependency call.

If it does not work

Application Insights sampling can mean not every request appears as an individual event. Read count/weight semantics before treating displayed event rows as exact traffic totals.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Finish Day 3

Save your place and check cost

Save both actual website URLs. Check /health and /secret-status on each. Restore the storage network settings described in Block A. Review running costs.

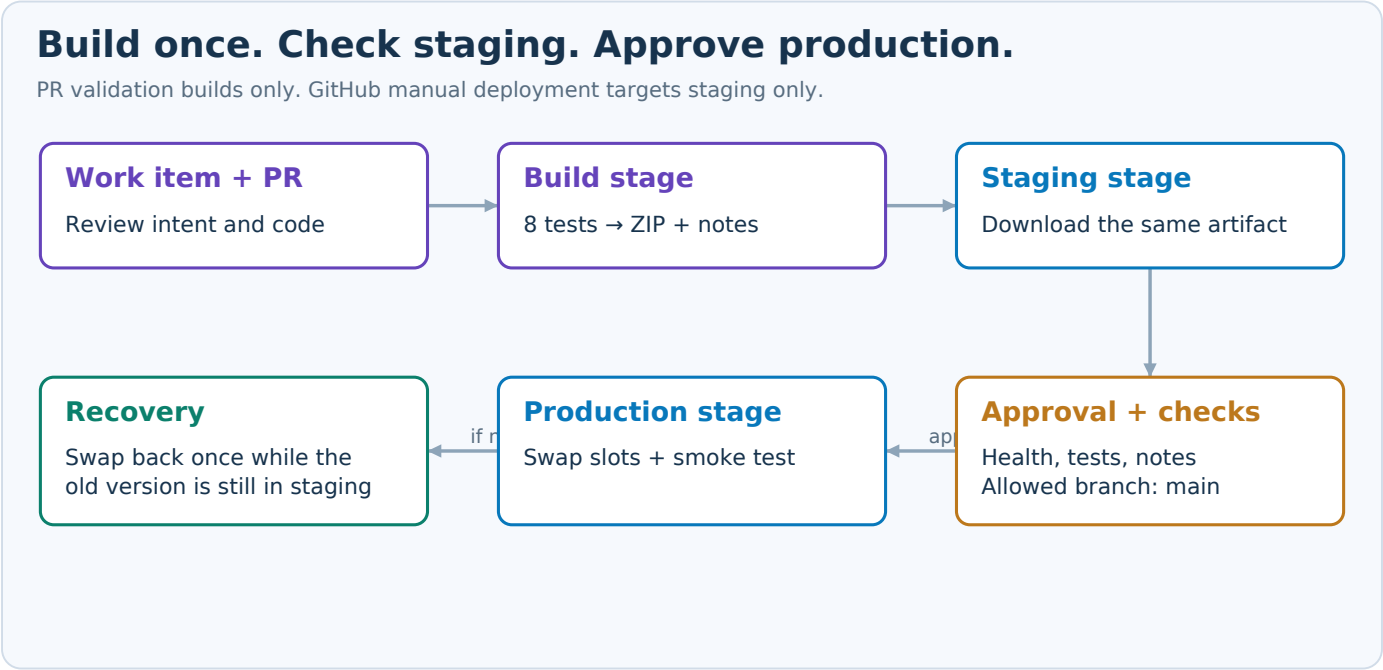
Say it in your own words: Why does the staging slot need its own Key Vault permission?

Check your explanation: Each slot has a separate system-assigned identity. Production permission does not automatically give the staging identity access.

DAY 4 OF 7

Let a pipeline release your changes

A change is linked to work, tested, deployed to staging and approved before production.



Start today

Run the restart routine. Confirm both website URLs still work. Check Pipelines parallel-job access before starting the first cloud run.

Start early and track the job

While a pipeline runs, inspect its logs and work-item links. Do not start another deployment to the same staging slot.

Block	Today's work	Minutes
A	Plan one change and save the code	90
B	Protect Git and connect the pipeline	100
C	Run the full release once	110
D	Keep a version you can reuse	60

4A · Plan one change and save the code

Planning estimate: 90 minutes. Finish with: A Boards item, Wiki page and the OneLab source in Azure Repos.

4A.01 Create one traceable feature and a bug

AZ-400 · CORE

Why this task: A work item explains why a code change exists and how you will judge it.

4A.01 · ACTION 1 OF 5

AZURE DEVOPS · OneLab → Boards → Work items → New work item

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: OneLab

OneLab > Boards > Work items > New work item

OneLab
Boards
Work items
New work item

1 Type User Story

2 Title Show OneLab website version

3 Assigned to Your own account

4 Acceptance Version visible

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. Open Boards > Work items > New work item > User Story.
2. Title: Show OneLab website version.
3. Description: As an operator I want a visible version so I can identify the deployed release.

4A.01 · ACTION 2 OF 5

AZURE DEVOPS · Boards → Work items → Your story

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS

Search: Boards

Boards > Work items > Your story

Boards

Work items

Your story

1 Acceptance

2 Assigned to

Version visible, health 200, tests block release

Yourself

Find the labels above in your own portal.

Save

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Set acceptance criteria: home returns OneLab and version; /health returns 200; failed tests block release.
2. Assign it to yourself and save.
3. Record its numeric ID.

4A.01 · ACTION 3 OF 5

AZURE DEVOPS · Your story → Links → Add link → New item

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS

Search: Your story

Your story > Links > Add link > New item

Your story

Links

Add link

1 Link type

2 Type

3 Titles

Child

Task

Code, test, deploy, monitor

Find the labels above in your own portal.

Add link

Illustrated navigation view · use your own saved names and IDs

Why this action: A work item explains why a code change exists and how you will judge it.

1. Add child Tasks for coding, testing, deployment and monitoring.
2. Open Boards > Boards and move the story through New, Active, Resolved and Closed only as real work reaches those states.

4A.01 · ACTION 4 OF 5**AZURE DEVOPS · Boards → Work items → New work item → Bug**

The screenshot shows the Azure DevOps web interface. At the top, the browser address bar displays 'dev.azure.com / YOUR_ORG / OneLab'. The page header includes the 'AZURE DEVOPS' logo and a search bar containing 'Boards'. The breadcrumb navigation path is 'Boards > Work items > New work item > Bug'. On the left, a sidebar lists 'Boards', 'Work items', and 'New work item'. The main content area features two numbered steps: '1 Title' with the input 'Health check failed', and '2 Describe' with the input 'Expected, actual, reproduce, harmless log'. A 'Save' button is located at the bottom right. A note at the bottom left states: 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: A work item explains why a code change exists and how you will judge it.

1. Create a Bug titled Health check failed, with expected behavior, actual result, steps to reproduce and a harmless log excerpt.

4A.01 · ACTION 5 OF 5**AZURE DEVOPS · Your bug → Links → Add link**

The screenshot shows the Azure DevOps web interface for adding links to a bug. The browser address bar displays 'dev.azure.com / YOUR_ORG / OneLab'. The page header includes the 'AZURE DEVOPS' logo and a search bar containing 'Your bug'. The breadcrumb navigation path is 'Your bug > Links > Add link'. On the left, a sidebar lists 'Your bug', 'Links', and 'Add link'. The main content area features two numbered steps: '1 Related item' with the input 'Your story', and '2 Later' with the input 'PR, build and release links'. A 'Save' button is located at the bottom right. A note at the bottom left states: 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. Link the bug to the story.
2. Later link the pull request, build and release so you can travel from requirement to running version.

You should see

The story has acceptance criteria, tasks and a related bug.

If it does not work

If your process uses Issue instead of User Story, your project was created with Basic. Use the equivalent work item or create the intended Agile project before substantial work.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

4A.02 Create useful queries and a dashboard

AZ-400 · CORE

Why this task: Metrics should answer a decision, not just make a colorful chart.

4A.02 · ACTION 1 OF 5

AZURE DEVOPS · OneLab → Boards → Queries → New query

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS

OneLab > Boards > Queries > New query

OneLab	1 Team Project	@Project
Boards	2 Work Item Type	Bug
Queries	3 State	<> Closed
New query	4 Query name	Open OneLab bugs

Find the labels above in your own portal. **Save query**

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Open Boards > Queries > New query.
2. Add Team Project = @Project, Work Item Type = Bug, and State <> Closed.
3. Save as Open OneLab bugs under Shared queries if you have permission.

4A.02 · ACTION 2 OF 5**AZURE DEVOPS · Overview → Dashboards → New dashboard**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: Overview

Overview > Dashboards > New dashboard

Overview
Dashboards
New dashboard

1 Widgets Query Results and Chart for Work Items

2 Query Open OneLab bugs

3 Group by State

Find the labels above in your own portal. **Create**

Illustrated navigation view · use your own saved names and IDs

Why this action: Metrics should answer a decision, not just make a colorful chart.

1. Open Overview > Dashboards > New dashboard.
2. Add Query Results and select that query.
3. Add a Chart for Work Items widget showing bugs by state.

4A.02 · ACTION 3 OF 5**AZURE DEVOPS · OneLab dashboard → Edit → Add widget**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: OneLab dashboard

OneLab dashboard > Edit > Add widget

OneLab dashboard
Edit
Add widget

1 Widgets Available Lead Time and Cycle Time

2 Scope OneLab team

Find the labels above in your own portal. **Save**

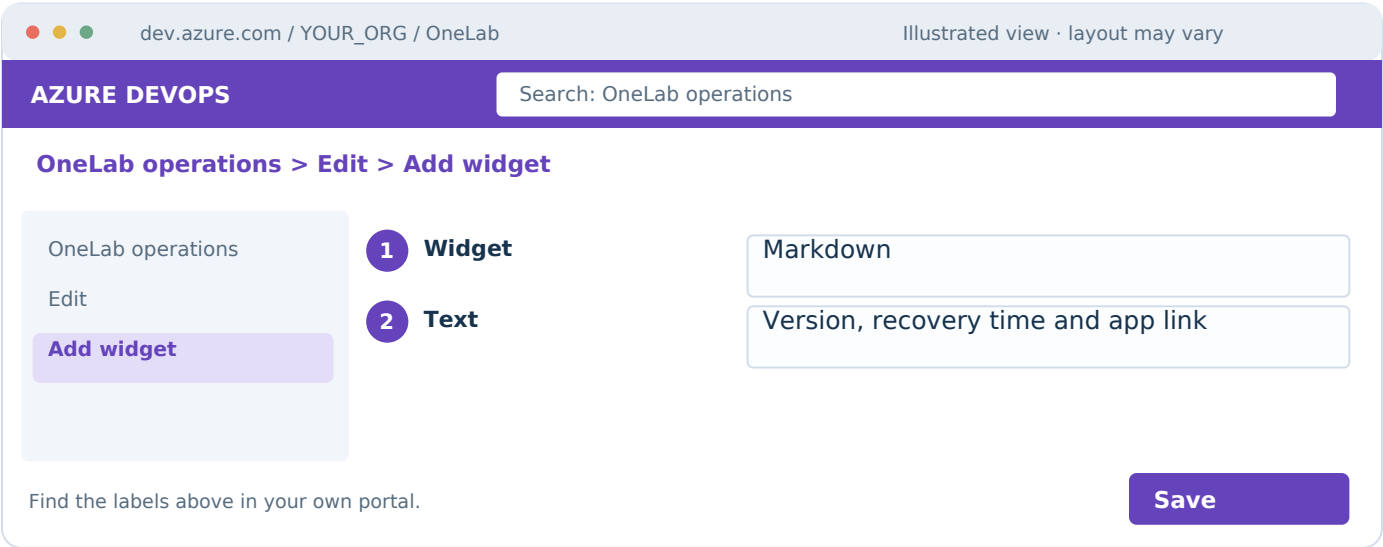
Illustrated navigation view · use your own saved names and IDs

Why this action: Metrics should answer a decision, not just make a colorful chart.

1. Add available Lead Time and Cycle Time widgets for your team.
2. A new empty project may have little data until work items move through states.

4A.02 · ACTION 4 OF 5

AZURE DEVOPS · OneLab operations → Edit → Add widget



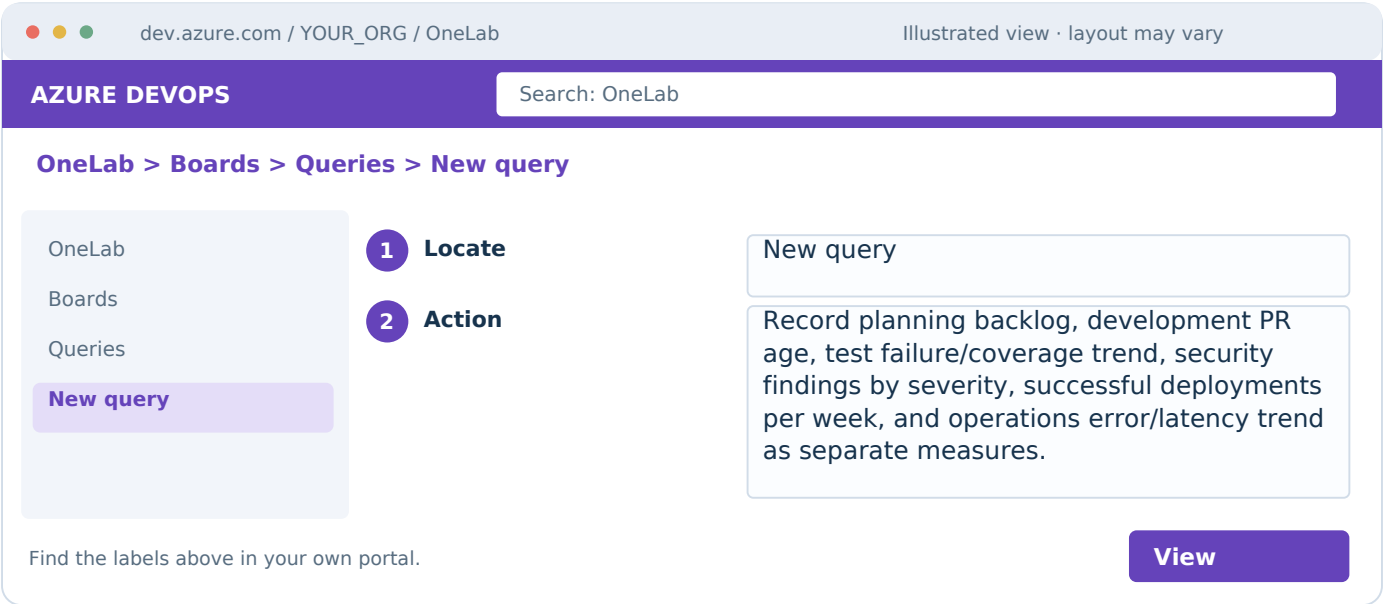
Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

- 1. Add a Markdown widget with release version, last recovery time and a link to the application.
- 2. Add available build/test trend widgets after your first pipeline run.

4A.02 · ACTION 5 OF 5

AZURE DEVOPS · OneLab → Boards → Queries → New query



Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

- 1. Record planning backlog, development PR age, test failure/coverage trend, security findings by severity, successful deployments per week, and operations error/latency trend as separate measures.

You should see

Your dashboard has at least one working query and a clear purpose for each planned metric.

If it does not work

Lead time usually starts when a request is recorded; cycle time starts when active work begins. Time to recovery measures restoring service after an incident. Define start/end events consistently. [R32]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED
Worked: ☐ Independently ☐ With the guide

4A.03 Write project documentation

AZ-400 · CORE

Why this task: The project should explain its own operation.

4A.03 · ACTION 1 OF 5

AZURE DEVOPS · OneLab → Overview → Wiki → Create project wiki

dev.azure.com / YOUR_ORG / OneLabIllustrated view · layout may vary

AZURE DEVOPS

Search: OneLab

OneLab > Overview > Wiki > Create project wiki

OneLab

Overview

Wiki

1 Page

2 Include

3 Do not include

Runbook

URL, health, release, rollback

Credentials

Find the labels above in your own portal.

Save

Illustrated navigation view · use your own saved names and IDs

Why this action: The project should explain its own operation.

- 1. Open Wiki > Create project wiki.
- 2. Add a page named Runbook.

4A.03 · ACTION 2 OF 5**AZURE DEVOPS · OneLab → Overview → Wiki → Create project wiki**

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS

OneLab > Overview > Wiki > Create project wiki

OneLab

Overview

Wiki

1 Include

2 Do not include

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: The project should explain its own operation.

1. Write the app URL, deployment route, health check, rollback steps and where to find monitoring.
2. Do not include credentials.

4A.03 · ACTION 3 OF 5**AZURE DEVOPS · OneLab → Overview → Wiki → Create project wiki**

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS

OneLab > Overview > Wiki > Create project wiki

OneLab

Overview

Wiki

1 Locate

2 Action

Find the labels above in your own portal. **Add**

Illustrated navigation view · use your own saved names and IDs

Why this action: The project should explain its own operation.

1. Add a Markdown table listing /, /health, /secret-status and /fail with purpose and expected HTTP status.
2. This is the application's small API reference.

4A.03 · ACTION 4 OF 5**FILE / WIKI EDITOR · Azure DevOps Wiki → Edit**

The dark code panel below is the terminal/editor view for this action.

Why this action: The project should explain its own operation.

1. Add the Mermaid example below using the wiki's ::: mermaid block syntax, then preview and save.

USE IN: WIKI MARKDOWN · COPY ONLY THE CODE

```

::: mermaid
flowchart TD
  A[Work item] --> B[Pull request]
  B --> C[Tests]
  C --> D[Staging]
  D --> E[Approval]
  E -->|Approve| F[Production]
  E -->|Reject| B
:::

```

What it means: The diagram connects the requirement, code review, automation and approval decision.

4A.03 · ACTION 5 OF 5**AZURE DEVOPS · OneLab → Overview → Wiki → Create project wiki**

The screenshot shows the Azure DevOps OneLab portal interface. At the top, the breadcrumb navigation is 'OneLab > Overview > Wiki > Create project wiki'. On the left, a sidebar lists 'OneLab', 'Overview', and 'Wiki' (which is highlighted). In the center, there are two numbered steps: '1 Locate' and '2 Action'. To the right of step 2, there is a text box containing the instruction: 'After each release, add the version, work-item/PR links, user-facing change, test evidence and rollback note.' Below this, there is a purple 'Add' button. At the bottom left, a note says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. After each release, add the version, work-item/PR links, user-facing change, test evidence and rollback note.
2. Day 4, Block C generates a Git-history starting point for release notes.

You should see

A new learner can locate the app, verify health and find rollback instructions from the wiki.

If it does not work

GitHub Markdown uses a fenced mermaid code block; Azure DevOps Wiki supports its ::: mermaid form. Do not assume every Markdown renderer has identical syntax.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

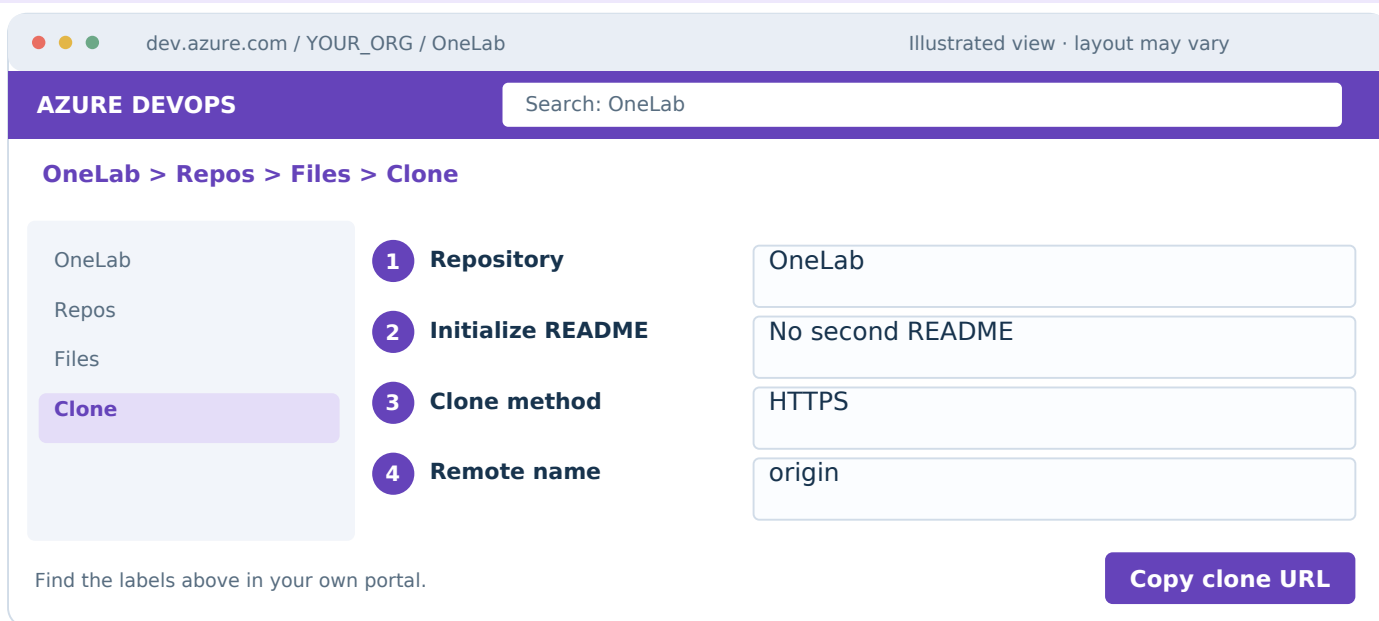
4A.04 Create an empty repository and push the project

AZ-400 · CORE

Why this task: Git records the content and history of the project; the remote repository stores the shared copy.

4A.04 · ACTION 1 OF 6

AZURE DEVOPS · OneLab → Repos → Files → Clone



The screenshot shows the Azure DevOps web interface for cloning a repository. The browser address bar displays 'dev.azure.com / YOUR_ORG / OneLab'. The page title is 'Illustrated view · layout may vary'. The main header is 'AZURE DEVOPS' with a search bar containing 'OneLab'. The breadcrumb navigation is 'OneLab > Repos > Files > Clone'. On the left, a sidebar lists 'OneLab', 'Repos', 'Files', and 'Clone' (highlighted). The main content area shows a numbered list of steps: 1. Repository, 2. Initialize README, 3. Clone method, and 4. Remote name. To the right of these steps are input fields: 'OneLab' for the repository name, 'No second README' for the README option, 'HTTPS' for the clone method, and 'origin' for the remote name. At the bottom left, it says 'Find the labels above in your own portal.' and at the bottom right, there is a 'Copy clone URL' button.

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: OneLab

OneLab > Repos > Files > Clone

OneLab
Repos
Files
Clone

1 Repository OneLab

2 Initialize README No second README

3 Clone method HTTPS

4 Remote name origin

Find the labels above in your own portal. **Copy clone URL**

Illustrated navigation view · use your own saved names and IDs

Why this action: Git records the content and history of the project; the remote repository stores the shared copy.

1. In OneLab open Repos > Files.
2. Use the empty default OneLab repository or create a Git repository named OneLab.
3. Do not initialize another README if you are going to push the complete local project.

4A.04 · ACTION 2 OF 6**AZURE DEVOPS · OneLab → Repos → Files → Clone**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS

Search: OneLab

OneLab > Repos > Files > Clone

OneLab

Repos

Files

1 Repository

2 Clone method

OneLab

HTTPS

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: Git records the content and history of the project; the remote repository stores the shared copy.

1. Copy the HTTPS clone URL.
2. It contains the organization, project and repository names.
3. A URL from another organization will not work even if its project has the same name.

4A.04 · ACTION 3 OF 6**AZURE DEVOPS · User settings → Personal access tokens → New Token**

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS

User settings > Personal access tokens > New Token

User settings
Personal access tokens
New Token

1 **Name**

2 **Organization**

3 **Expiry**

4 **Scope**

Find the labels above in your own portal. **Create**

Illustrated navigation view · use your own saved names and IDs

Why this action: Git records the content and history of the project; the remote repository stores the shared copy.

1. For this beginner HTTPS route, open User settings at the top right > Personal access tokens > New Token.
2. Name onelab-git; choose the correct organization, a short expiry, and only Code Read & write.
3. Create and copy it privately.

4A.04 · ACTION 4 OF 6**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

The dark code panel below is the terminal/editor view for this action.

Why this action: Git records the content and history of the project; the remote repository stores the shared copy.

1. In Cloud Shell run the Git setup below, replacing the author values and remote URL.
2. When prompted for a username use your account name; for password paste the PAT.
3. It will not appear while typing.

USE IN: BASH · COPY ONLY THE CODE

```
git init -b main
git config user.name "Your Name"
git config user.email "your-email@example.com"
git config credential.helper ""
git status --short
git add .
git diff --cached --stat
git commit -m "Create OneLab application and delivery files"
git remote add origin 'REPLACE_WITH_HTTPS_CLONE_URL'
git push -u origin main
```

What it means: git add stages files; commit creates a local snapshot; push sends it to the remote. -u remembers the tracking branch. Inspect the staged files before committing.

--short · Show compact Git status.

--cached · Inspect the staged changes that the next commit will include.

--stat · Show a per-file change summary.

git add / commit / push · Stage chosen files; record a local snapshot; then send commits to the named remote.

git push -u · Push and set this branch's upstream remote-tracking branch.

Empty credential helper · Do not configure Git to store the token through a credential helper for this folder.

4A.04 · ACTION 5 OF 6

SEE THE IDEA · A Git credential stays private

A SMALL DECISION MAP

A Git credential stays private

- 1 **Enter** At the password prompt
- 2 **Never include** In the remote URL or repository

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Never place the token in the remote URL, source file or command text.
2. Revoke it when finished.
3. If PAT creation is blocked by policy, use an organization-approved Git credential method instead.

4A.04 · ACTION 6 OF 6

AZURE DEVOPS · OneLab → Repos → Files

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: OneLab

OneLab > Repos > Files

OneLab	1 Present	app.py and pipeline files
Repos	2 Absent	lab.env and private keys
Files		

Find the labels above in your own portal.

Refresh

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Refresh Repos > Files.
2. Confirm app.py and the pipeline files are present, and lab.env is absent.

You should see

The main branch in Azure Repos contains the complete safe project.

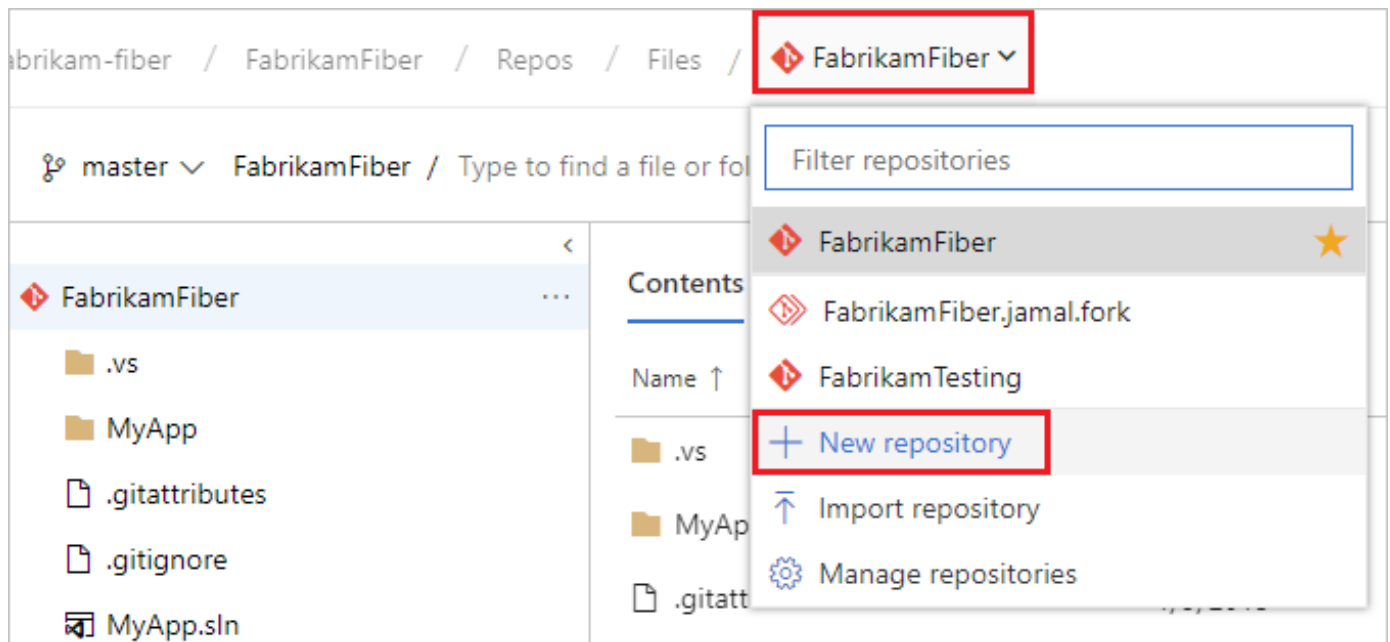
If it does not work

If rejected because the remote already contains commits, clone it into a separate folder, copy only project files into that clone, commit and push. Do not force-push over existing work. [R33]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

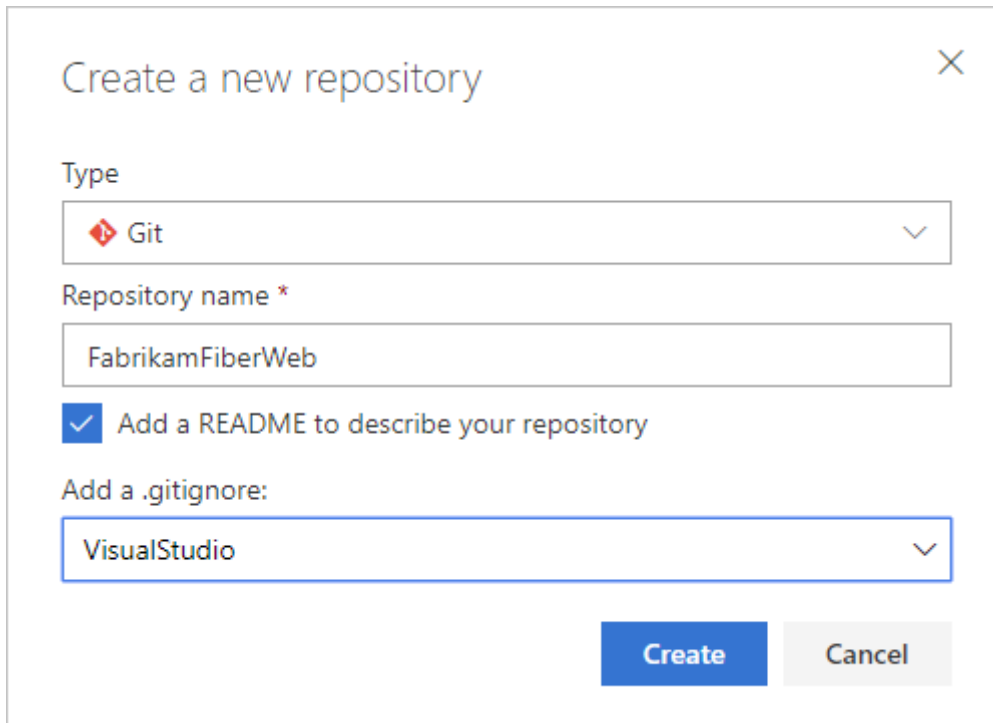
Worked: ☐ Independently ☐ With the guide

Microsoft reference screenshot · Screenshot of repository menu, select New repository.



[Microsoft reference source](#) · Sample values belong to that reference, not OneLab.

Microsoft reference screenshot · Screenshot of Create a new repository dialog.



Create a new repository

Type

Git

Repository name *

FabrikamFiberWeb

☒ Add a README to describe your repository

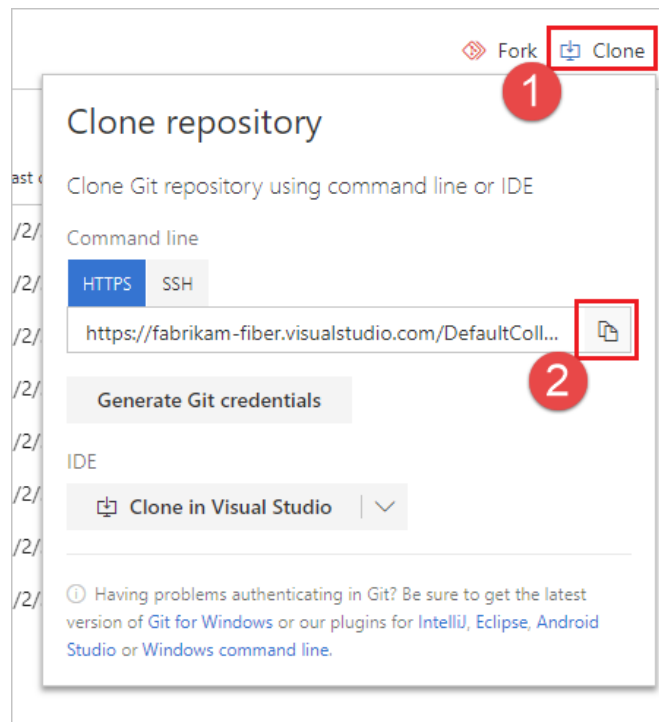
Add a .gitignore:

VisualStudio

Create Cancel

[Microsoft reference source](#) · Sample values belong to that reference, not OneLab.

Microsoft reference screenshot · Clone repository dialog.



Clone repository

Clone Git repository using command line or IDE

Command line

HTTPS SSH

https://fabrikam-fiber.visualstudio.com/DefaultColl...

Generate Git credentials

IDE

Clone in Visual Studio

Having problems authenticating in Git? Be sure to get the latest version of Git for Windows or our plugins for IntelliJ, Eclipse, Android Studio or Windows command line.

[Microsoft reference source](#) · Sample values belong to that reference, not OneLab.

4B · Protect Git and connect the pipeline

Planning estimate: 100 minutes. Finish with: A review path, federated connection, variable group and protected environments.

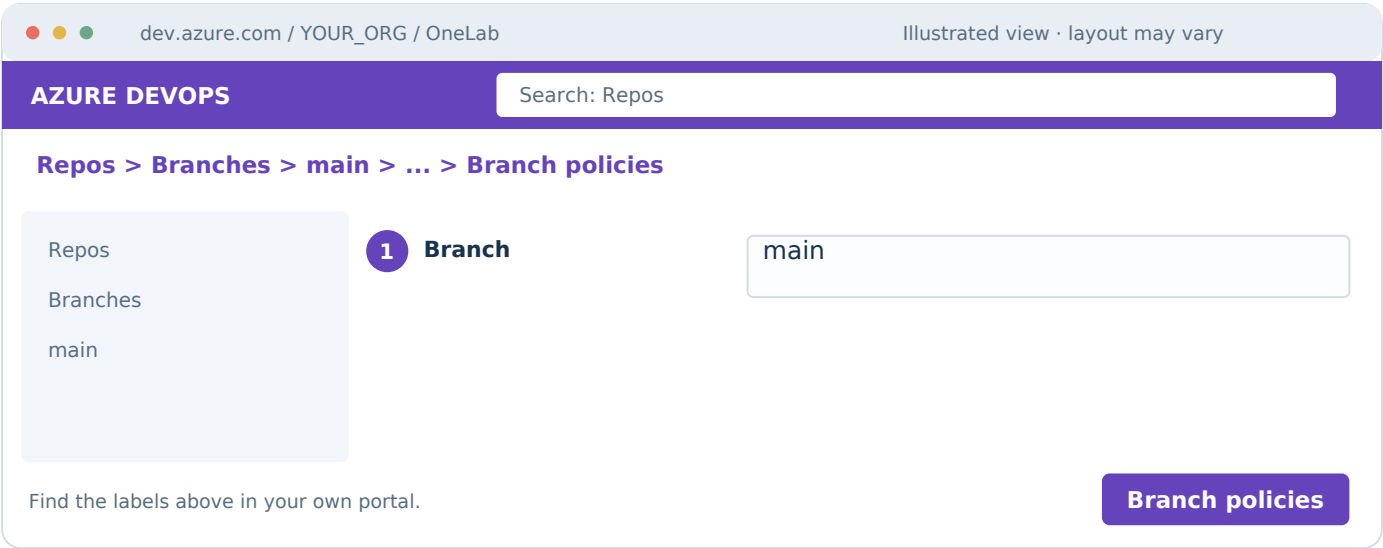
4B.01 Protect main and make a feature branch

AZ-400 · CORE

Why this task: A pull request is a proposed change that can be reviewed and tested before entering main.

4B.01 · ACTION 1 OF 6

AZURE DEVOPS · Repos → Branches → main → ... → Branch policies



Illustrated navigation view · use your own saved names and IDs

Why this action: A pull request is a proposed change that can be reviewed and tested before entering main.

1. Open Repos > Branches > main > More options > Branch policies.

4B.01 · ACTION 2 OF 6**AZURE DEVOPS · main → Branch policies → Minimum reviewers**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: main

main > Branch policies > Minimum reviewers

main

Branch policies

Minimum reviewers

1 Minimum 1

2 Solo only Allow requestor approval

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: A pull request is a proposed change that can be reviewed and tested before entering main.

1. Enable a minimum reviewer count of one if a second reviewer exists.
2. For a solo exercise, allow the requestor's vote only as a labelled lab simplification; a real team should require independent review.

4B.01 · ACTION 3 OF 6**AZURE DEVOPS · main → Branch policies**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: main

main > Branch policies

main

Branch policies

1 Linked work items Required

2 Comment resolution Required

3 Build validation Add OneLab-Web when created

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: A pull request is a proposed change that can be reviewed and tested before entering main.

1. Enable linked work items and comment resolution.
2. After the pipeline exists in Day 4, Block C, add it under Build validation: automatic and required.

4B.01 · ACTION 4 OF 6**AZURE DEVOPS · main → Branch policies → Limit merge types**

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS

main > Branch policies > Limit merge types

main

Branch policies

Limit merge types

1 Allowed type

2 Security ☒ Do not grant broad bypass

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: A pull request is a proposed change that can be reviewed and tested before entering main.

1. Inspect allowed merge types.
2. Choose Squash merge for short feature branches in this lab.
3. Inspect repository security under Project settings > Repositories and avoid granting bypass rights broadly.

4B.01 · ACTION 5 OF 6**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

The dark code panel below is the terminal/editor view for this action.

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Run `git switch -c feature/version-note`.
2. Edit `app.py` to change version 1.0.0 to 1.0.1; keep the tests working.
3. Commit with a message containing `#YOUR_STORY_ID` and push the branch.

USE IN: BASH · COPY ONLY THE CODE

```
git switch -c feature/version-note
nano app.py
python -m pytest
git add app.py
git commit -m "Show version 1.0.1 for #REPLACE_STORY_ID"
git push -u origin feature/version-note
```

What it means: A feature branch is a short-lived line of work. The story ID links the reason for the change; replace the placeholder before committing.

python -m MODULE · Run a module using the currently selected Python interpreter.

git add / commit / push · Stage chosen files; record a local snapshot; then send commits to the named remote.

git switch -c · Create and switch to a new branch.

git push -u · Push and set this branch's upstream remote-tracking branch.

Nano keys · Ctrl+O then Enter saves. Ctrl+X exits. Paste only the intended file content.

4B.01 · ACTION 6 OF 6**AZURE DEVOPS · Repos → Pull requests → New pull request**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: Repos

Repos > Pull requests > New pull request

Repos

Pull requests

New pull request

1 Source feature/version-note

2 Target main

3 Work item Your story ID

Find the labels above in your own portal.

Create

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. In Repos > Pull requests > New pull request, source feature/version-note, target main.
2. Add a meaningful description and link the story.
3. Keep it open until the pipeline validation is configured.

You should see

There is a protected main branch and a linked pull request.

If it does not work

For Azure Repos, PR build validation is a branch policy. Writing pr: in YAML does not configure Azure Repos PR validation. [R34]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

4B.02 Practise a conflict and safe recovery in a local copy

AZ-400 · CORE

Why this task: Recovering Git history should not require guessing or overwriting shared work.

4B.02 · ACTION 1 OF 6

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: Recovering Git history should not require guessing or overwriting shared work.

1. Use the block below to create a disposable local repository next to the project.
2. It does not modify the Azure Repos remote.

USE IN: BASH · COPY ONLY THE CODE

```
mkdir -p ../git-practice
cd ../git-practice
git init -b main
git config user.name "OneLab"
git config user.email "lab@example.com"
printf 'Original\n' > note.txt
git add note.txt
git commit -m initial
git switch -c branch-a
printf 'Version A\n' > note.txt
git commit -am A
git switch main
printf 'Version B\n' > note.txt
git commit -am B
git merge branch-a
```

What it means: The final merge intentionally produces a conflict because both branches changed the same line. This is a controlled exercise, not an unexpected failure.

git add / commit / push · Stage chosen files; record a local snapshot; then send commits to the named remote.

git switch -c · Create and switch to a new branch.

mkdir -p · Create the folder and missing parent folders; keep an existing folder.

4B.02 · ACTION 2 OF 6

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Observe the conflict markers in note.txt after the merge.
2. Edit the file to the intended combined sentence; remove <<<<<<, ===== and >>>>>> markers.

USE IN: BASH · COPY ONLY THE CODE

```
nano note.txt
```

What it means: Resolve the displayed conflict in the disposable git-practice folder. Keep one intended sentence.

Nano keys · Ctrl+O then Enter saves. Ctrl+X exits. Paste only the intended file content.

4B.02 · ACTION 3 OF 6

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Run `git add note.txt` and `git commit` to complete the merge.
2. Or run `git merge --abort` before committing to return to the pre-merge state.

USE IN: BASH · COPY ONLY THE CODE

```
git add note.txt
git commit -m "Resolve training conflict"
```

What it means: Stage the resolved content and finish the merge.

git add / commit / push · Stage chosen files; record a local snapshot; then send commits to the named remote.

4B.02 · ACTION 4 OF 6

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: Recovering Git history should not require guessing or overwriting shared work.

1. Use `git log --oneline` to inspect history.
2. In a shared branch use `git revert COMMIT_ID` to create a new reversing commit rather than resetting everyone's history.

USE IN: BASH · COPY ONLY THE CODE

```
git log --oneline -6
```

What it means: Read only: show the last six commits. Use a real commit ID if practising a revert.

--oneline · Show one short line per Git commit.

4B.02 · ACTION 5 OF 6**CLOUD SHELL · Cloud Shell → Bash → ../git-practice**

portal.azure.com Illustrated view · layout may vary

CLOUD SHELL Search: Cloud Shell

Cloud Shell > Bash > ../git-practice

Cloud Shell

Bash

../git-practice

1 Locate

2 Action

../git-practice

For a deleted uncommitted file, git restore PATH retrieves the committed version and discards that file's local changes. git reflog can help locate a recent local commit you moved away from.

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. For a deleted uncommitted file, git restore PATH retrieves the committed version and discards that file's local changes. git reflog can help locate a recent local commit you moved away from.

4B.02 · ACTION 6 OF 6**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

The dark code panel below is the terminal/editor view for this action.

Why this action: Recovering Git history should not require guessing or overwriting shared work.

1. Return to the one-lab directory afterward.
2. Do not run practice reset or history-removal operations against the shared main branch.

USE IN: BASH · COPY ONLY THE CODE

```
cd ~/clouddrive/one-lab
pwd
```

What it means: Return to the real project before the next task.

You should see

You resolve or abort a conflict and can explain revert versus restore versus reset.

If it does not work

A real leaked secret must be revoked first. Removing it from the latest file does not remove it from Git history, forks, build artifacts or logs.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

Strategy/tool	Decision for OneLab
Trunk-based	Use short branches and frequent integration into main; protect main with tests.
Release branches	Use only when maintaining several released versions; merge hotfixes back to avoid losing fixes.
Git LFS	Store large binaries using pointer files; track the pattern before adding the binary.
git-fat	An alternative external large-file approach; every collaborator and runner needs the required tooling. LFS is the practical choice here.
Scalar / sparse checkout	Improve work with large repos or only needed folders; they do not replace permissions or review.
Cross-repo sharing	Publish a versioned package or shared pipeline template; avoid copying code that silently diverges.

DESIGN removal exercise: make a disposable copy, identify the unwanted path/history, use an approved history-rewrite tool such as git-filter-repo, verify all relevant refs, coordinate force updates and fresh clones, and remove cached artifacts. Never treat history rewriting as secret revocation. For LFS on an eligible workstation: git lfs install; git lfs track "*.bin"; commit .gitattributes before the large file.

4B.03 Create a federated service connection

AZ-400 · CORE

Why this task: A pipeline needs its own identity; it does not inherit your browser login.

4B.03 · ACTION 1 OF 7**AZURE DEVOPS · OneLab → Project settings → Service connections**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: OneLab

OneLab > Project settings > Service connections

OneLab
Project settings
Service connections

1 Action New service connection

Find the labels above in your own portal. **New service connection**

Illustrated navigation view · use your own saved names and IDs

Why this action: A pipeline needs its own identity; it does not inherit your browser login.

1. Open Project settings > Service connections > New service connection > Azure Resource Manager > Next.

4B.03 · ACTION 2 OF 7**AZURE DEVOPS · New service connection → Azure Resource Manager**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: New service connection

New service connection > Azure Resource Manager

New service connection
Azure Resource Manager

1 Identity type App registration (automatic)
2 Credential Workload identity federation

Find the labels above in your own portal. **Next**

Illustrated navigation view · use your own saved names and IDs

Why this action: A pipeline needs its own identity; it does not inherit your browser login.

1. Choose App registration (automatic) and Workload identity federation.
2. Select subscription scope, the correct subscription, and resource group rg-PREFIX-core.

4B.03 · ACTION 3 OF 7

AZURE DEVOPS · New Azure service connection → Scope

dev.azure.com / YOUR_ORG / OneLabIllustrated view · layout may vary

AZURE DEVOPS

Search: New Azure service connection

New Azure service connection > Scope

New Azure service connection

Scope

1 Subscription

2 Resource group

3 Connection name

4 Grant access to all pipelines

Your practice subscription

rg-PREFIX-core

sc-onelab

Unchecked

Find the labels above in your own portal.

Save

Illustrated navigation view · use your own saved names and IDs

Why this action: The connection will be trusted by only the pipelines you explicitly allow.

1. Name the connection sc-onelab.
2. Leave Grant access permission to all pipelines unchecked.
3. Save.

4B.03 · ACTION 4 OF 7

MICROSOFT ENTRA · sc-onelab → Manage service principal → Overview

entra.microsoft.comIllustrated view · layout may vary

MICROSOFT ENTRA

Search: sc-onelab

sc-onelab > Manage service principal > Overview

sc-onelab

Manage service principal

Overview

1 Application (client) ID

2 Object ID

Record separately

Record the service-principal object ID

Find the labels above in your own portal.

View

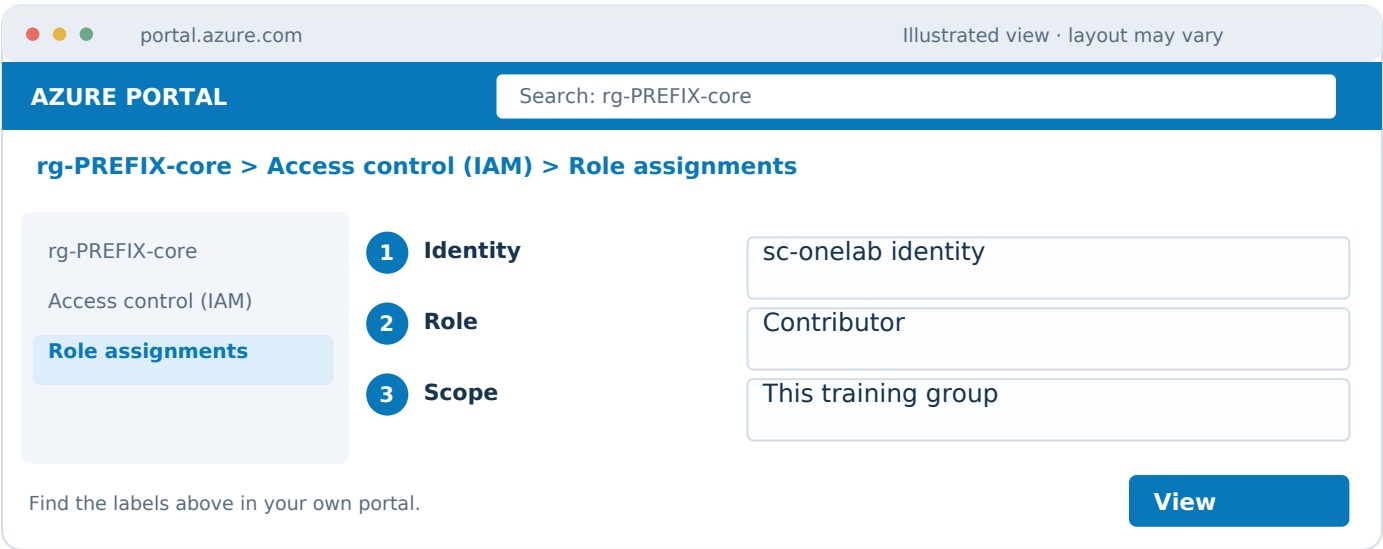
Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Open the connection and use Manage service principal to inspect its identity.
2. Note its application/client ID and service-principal object ID as different fields.

4B.03 · ACTION 5 OF 7

AZURE PORTAL · rg-PREFIX-core → Access control (IAM) → Role assignments



Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. In Azure Portal > rg-PREFIX-core > IAM, verify the connection identity has the resource permissions required for deployment.
2. Contributor scoped to this training group is sufficient for the shown web-app deployment operations, but does not grant secret data access.

4B.03 · ACTION 6 OF 7

SEE THE IDEA · If automatic federation creation fails

A SMALL DECISION MAP

If automatic federation creation fails

1 Azure scope	Role-assignment rights
2 Directory	App-registration rights
3 DevOps	Service-connection permissions

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. If automatic creation fails, confirm your Azure subscription role, permission to create application registrations, directory and DevOps service-connection permissions.
2. A tenant admin may need to create an approved identity and a manual federation connection.

4B.03 · ACTION 7 OF 7

AZURE DEVOPS · New service connection → Manual workload identity federation

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS

New service connection > Manual workload identity federation

New service connection

Manual workload identity federation

1 Issuer

2 Subject

3 Audience

Copy the CURRENT wizard value

Copy the CURRENT wizard value

Copy the CURRENT wizard value

Find the labels above in your own portal.

Verify and save

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. For a manual connection, copy the issuer, subject and audience presented by the current wizard exactly.
2. Do not hard-code issuer strings from old tutorials
3. Microsoft is changing issuer behavior for new connections.
4. In Entra open the selected identity > Federated credentials > Add credential, enter those three values as supplied, then return to the DevOps wizard to Verify and save.
5. If your tenant blocks identity creation, mark BLOCKED and continue with local tests until an authorized administrator supplies the identity.

You should see

sc-onelab verifies and is scoped to the practice group.

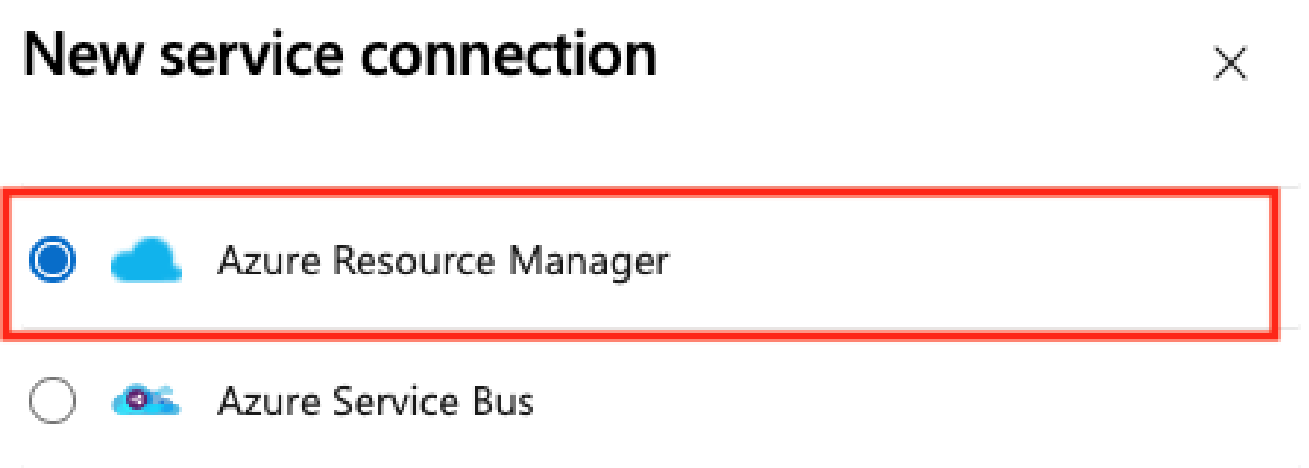
If it does not work

A federated token trust and an Azure role assignment are both necessary. A client ID is not the service-principal object ID used for role assignments. [R35]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

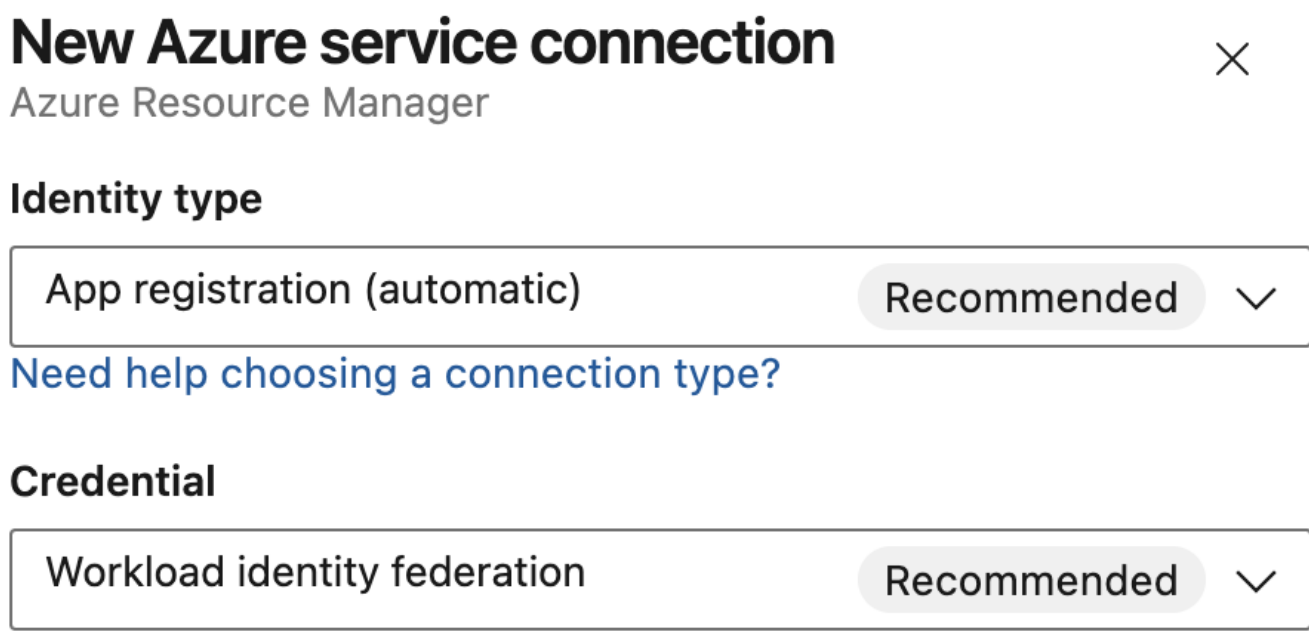
Worked: ☐ Independently ☐ With the guide

Microsoft reference screenshot · Screenshot of New service connection dialog with Azure Resource Manager option selected and Azure Service Bus option below.



[Microsoft reference source](#) · Sample values belong to that reference, not OneLab.

Microsoft reference screenshot · Screenshot of app registration (automatic) authentication method selection with workload identity selected.



[Microsoft reference source](#) · Sample values belong to that reference, not OneLab.

Microsoft reference screenshot · Screenshot that shows the app registration client ID and tenant ID.

Display name

federatedidentitydevops

Application (client) ID

00000000-0000-0000-0000-000000000000

Object ID

00000000-0000-0000-0000-000000000000

Directory (tenant) ID

00000000-0000-0000-0000-000000000000

Microsoft reference source · Sample values belong to that reference, not OneLab.

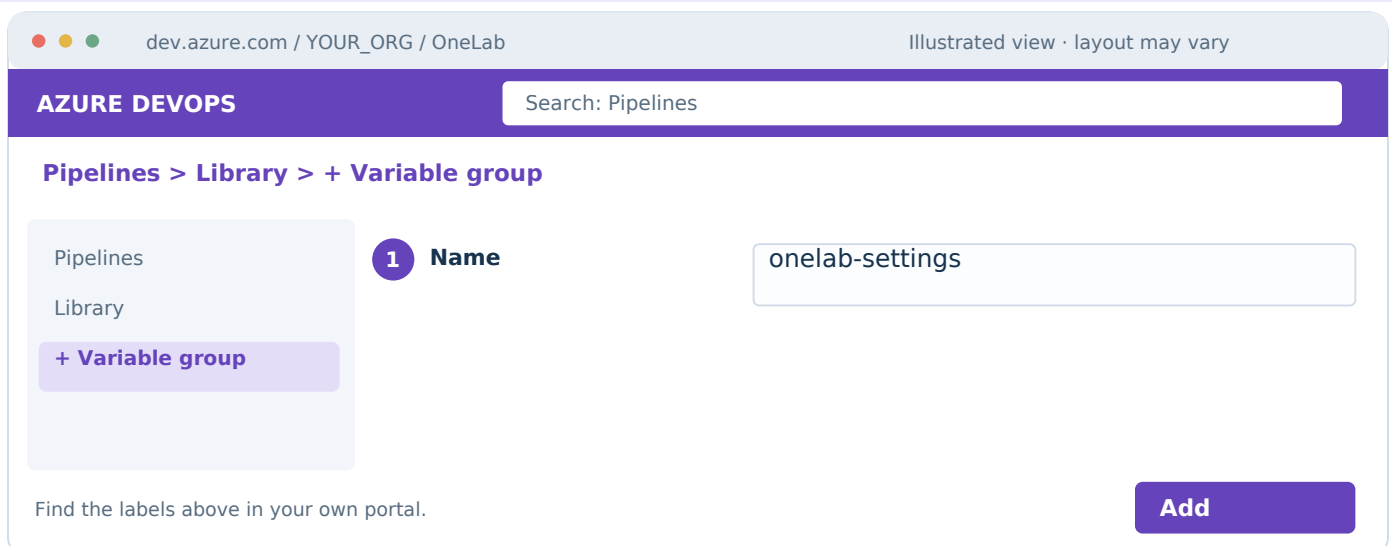
4B.04 Create the nonsecret variable group

AZ-400 · CORE

Why this task: Central settings reduce name mismatch mistakes.

4B.04 · ACTION 1 OF 4

AZURE DEVOPS · Pipelines → Library → + Variable group



Illustrated navigation view · use your own saved names and IDs

Why this action: Central settings reduce name mismatch mistakes.

1. Choose + Variable group.
2. Name onelab-settings.

4B.04 · ACTION 2 OF 4**AZURE DEVOPS · onelab-settings → Variables**

The screenshot shows the Azure DevOps web interface. At the top, the breadcrumb is 'dev.azure.com / YOUR_ORG / OneLab'. The page title is 'AZURE DEVOPS' with a search bar containing 'onelab-settings'. The main heading is 'onelab-settings > Variables'. On the left, a sidebar shows 'onelab-settings' and 'Variables' (highlighted). The main area lists three variables: 1. 'subscriptionId' with value 'Your subscription GUID', 2. 'resourceGroup' with value 'rg-PREFIX-core', and 3. 'appName' with value 'PREFIX-web'. At the bottom, there is a 'Save' button and a note: 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Central settings reduce name mismatch mistakes.

1. Add subscriptionId with your real subscription GUID, resourceGroup with rg-PREFIX-core, and appName with PREFIX-web.
2. These are identifiers, not secrets.

4B.04 · ACTION 3 OF 4**AZURE DEVOPS · onelab-settings → Pipeline permissions**

The screenshot shows the Azure DevOps web interface. At the top, the breadcrumb is 'dev.azure.com / YOUR_ORG / OneLab'. The page title is 'AZURE DEVOPS' with a search bar containing 'onelab-settings'. The main heading is 'onelab-settings > Pipeline permissions'. On the left, a sidebar shows 'onelab-settings' and 'Pipeline permissions' (highlighted). The main area lists two permissions: 1. 'Allow' with value 'OneLab-Web after it exists', and 2. 'Open access' with value 'Off where offered'. At the bottom, there is an 'Add pipeline' button and a note: 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Save.
2. Open Pipeline permissions and authorize only your OneLab pipeline after it is created.
3. If your UI offers Open access, leave it off for protected groups.
4. Nonsecret variables are not a security boundary.
5. Protect real secrets using the secret-resource permissions and Key Vault steps.

4B.04 · ACTION 4 OF 4**SEE THE IDEA · Runtime secret versus pipeline setting****A SMALL DECISION MAP****Runtime secret versus pipeline setting**

- | | |
|-------------------------|-----------------------------|
| 1 Pipeline group | Resource names and IDs |
| 2 Key Vault | Dummy secret |
| 3 App identity | Reads the secret at runtime |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Central settings reduce name mismatch mistakes.

1. Do not store the Key Vault dummy secret here.
2. The runtime resolves its own Key Vault reference through managed identity.

You should see

The variable names exactly match the YAML, including capitalization.

If it does not work

A YAML variable reference that does not exist may remain as literal text and cause a confusing downstream CLI error.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

4B.05 Create environments and an approval check

AZ-400 · CORE

Why this task: A protected environment can require approval independently of editable YAML.

4B.05 · ACTION 1 OF 5

AZURE DEVOPS · OneLab → Pipelines → Environments → New environment

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS

Search: OneLab

OneLab > Pipelines > Environments > New environment

OneLab

Pipelines

Environments

New environment

1 Staging

2 Production

3 Resource

4 Production approval

onelab-staging

onelab-production

None

Your test reviewer

Find the labels above in your own portal.

Create

Illustrated navigation view · use your own saved names and IDs

Why this action: A protected environment can require approval independently of editable YAML.

1. Create environment onelab-staging with resource type None.
2. Create onelab-production the same way.

4B.05 · ACTION 2 OF 5**AZURE DEVOPS · onelab-production → ... → Approvals and checks → +**

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS Search: onelab-production

onelab-production > ... > Approvals and checks > +

onelab-production

...

Approvals and checks

1 Check type Approvals

2 Approver Your controlled reviewer account

Find the labels above in your own portal. **Create**

Illustrated navigation view · use your own saved names and IDs

Why this action: A protected environment can require approval independently of editable YAML.

1. Open onelab-production > Approvals and checks > + > Approvals.
2. Choose your reviewer; for a solo lab choose yourself and explicitly allow approvers to approve their own runs.

4B.05 · ACTION 3 OF 5**AZURE DEVOPS · OneLab → Pipelines → Environments → New environment**

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS Search: OneLab

OneLab > Pipelines > Environments > New environment

OneLab

Pipelines

Environments

1 Staging onelab-staging

Find the labels above in your own portal. **Create**

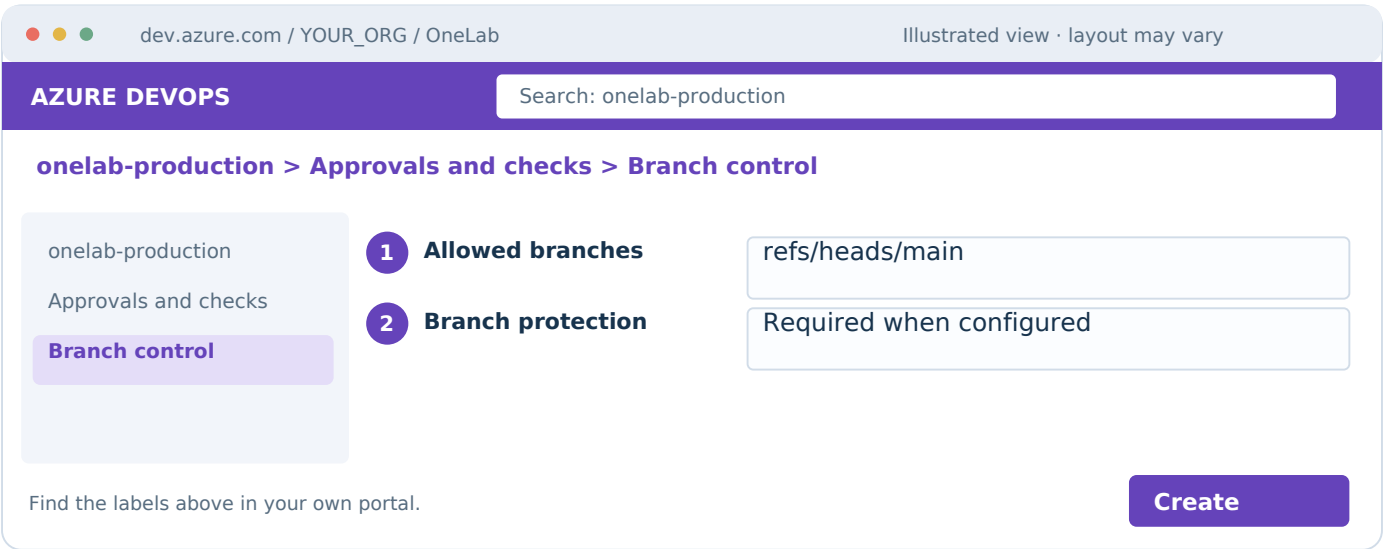
Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Add instructions: check staging health, tests, release notes and rollback plan.
2. Set a reasonable timeout and create.

4B.05 · ACTION 4 OF 5

AZURE DEVOPS · onelab-production → Approvals and checks → Branch control



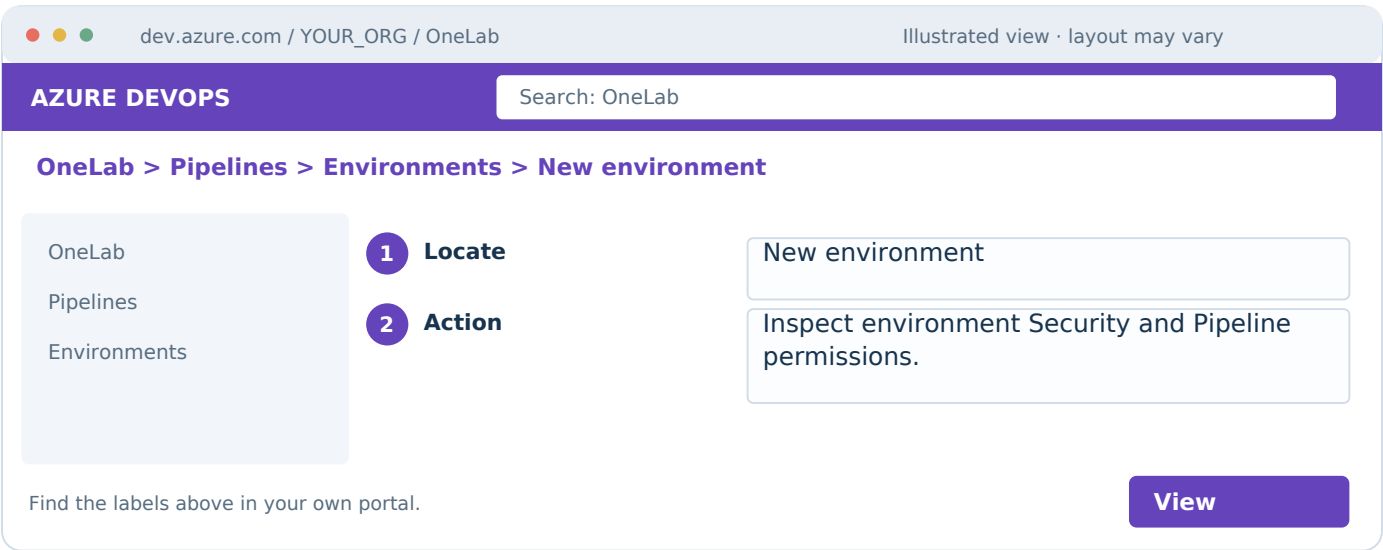
Illustrated navigation view · use your own saved names and IDs

Why this action: A protected environment can require approval independently of editable YAML.

1. Add a Branch control check allowing refs/heads/main, and require branch protection after Day 4, Block A, Day 4, Block B policies are configured.

4B.05 · ACTION 5 OF 5

AZURE DEVOPS · OneLab → Pipelines → Environments → New environment



Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Inspect environment Security and Pipeline permissions.
2. Authorize your intended pipeline only when it exists.

You should see

Production requires approval and only the allowed branch can deploy.

If it does not work

Approvals/checks belong to the protected resource in the portal. A stage condition in YAML is useful but is not the same independent control. [R36]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

4C · Run the full release once

Planning estimate: 110 minutes. Finish with: Build, staging, approval and production evidence; a failure caught before release.

4C.01 Understand the supplied YAML

AZ-400 · CORE

Why this task: You should know what each part does before running it.

4C.01 · ACTION 1 OF 6

AZURE DEVOPS · OneLab → Repos → Files → azure-pipelines.yml

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS

Search: OneLab

OneLab > Repos > Files > azure-pipelines.yml

OneLab

Repos

Files

1 Build

2 Staging

3 Production

Test and make one ZIP

Deploy ZIP + smoke test

Approve + swap + smoke test

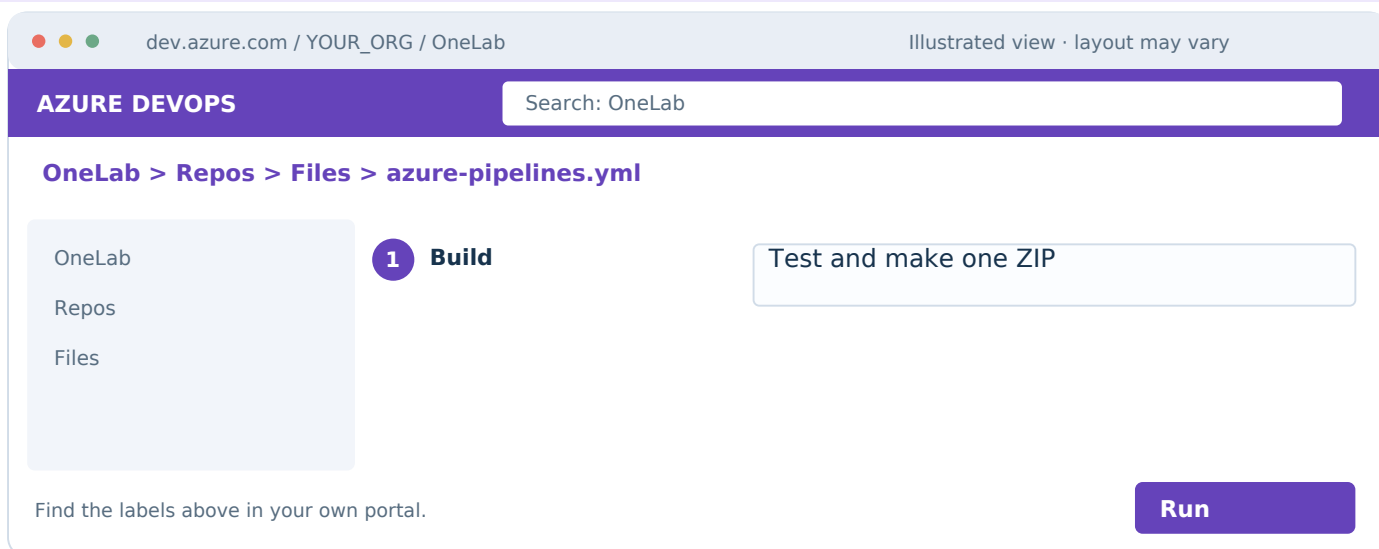
Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: You should know what each part does before running it.

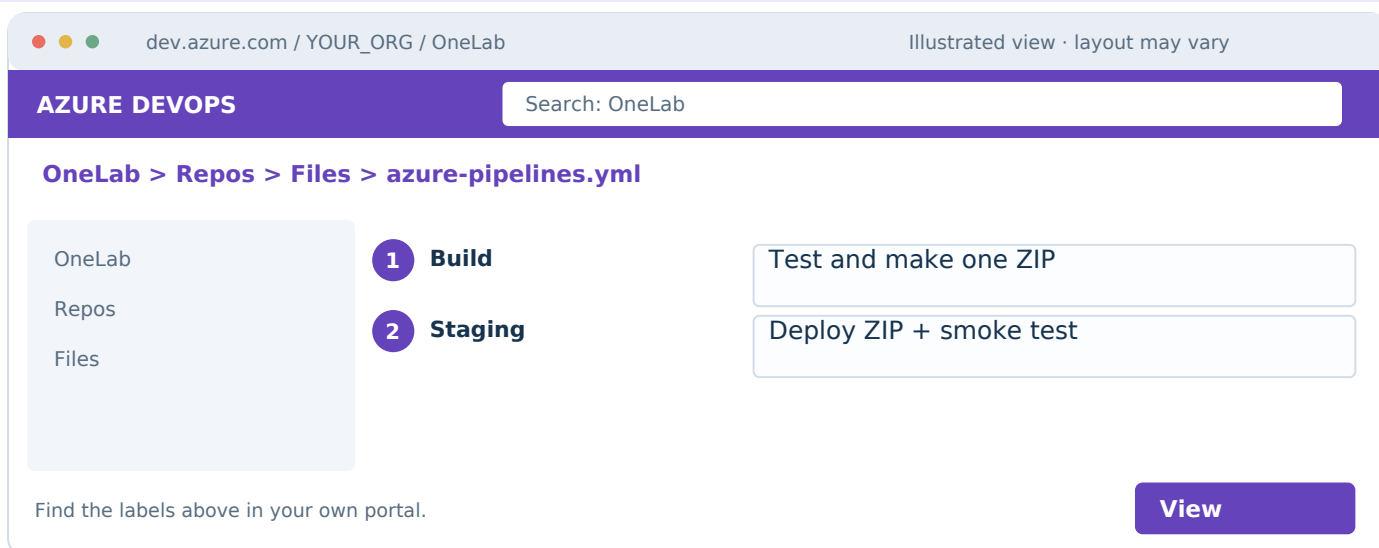
1. Open azure-pipelines.yml in Repos > Files. trigger includes main, so changes merged into main start CI.
2. Azure Repos PR validation comes from the branch policy.

4C.01 · ACTION 2 OF 6**AZURE DEVOPS · OneLab → Repos → Files → azure-pipelines.yml**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Read the Build stage.
2. It checks out code, runs the test template, creates app.zip, and publishes a run-specific artifact plus Git-derived notes.

4C.01 · ACTION 3 OF 6**AZURE DEVOPS · OneLab → Repos → Files → azure-pipelines.yml**

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. Read Staging.
2. It runs only for a successful main-branch build, downloads that same artifact, deploys it to staging and executes a health check.

4C.01 · ACTION 4 OF 6**AZURE DEVOPS · OneLab → Repos → Files → azure-pipelines.yml**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: OneLab

OneLab > Repos > Files > azure-pipelines.yml

OneLab
Repos
Files

1 Staging Deploy ZIP + smoke test

2 Production Approve + swap + smoke test

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Read Production.
2. It waits for Staging and the environment approval, swaps staging with production, and checks health again.

4C.01 · ACTION 5 OF 6**AZURE DEVOPS · OneLab → Repos → Files → azure-pipelines.yml**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: OneLab

OneLab > Repos > Files > azure-pipelines.yml

OneLab
Repos
Files

1 Locate azure-pipelines.yml

2 Action Find the literal sc-onelab service connection name and the three variable-group settings.

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: You should know what each part does before running it.

1. Find the literal sc-onelab service connection name and the three variable-group settings.
2. Every name must match what you created in Day 4, Block B.

4C.01 · ACTION 6 OF 6

AZURE DEVOPS · OneLab → Repos → Files → azure-pipelines.yml

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: OneLab

OneLab > Repos > Files > azure-pipelines.yml

OneLab
Repos
Files

1 Locate azure-pipelines.yml

2 Action Read pipelines/templates/test.yml.

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: You should know what each part does before running it.

1. Read pipelines/templates/test.yml.
2. The template installs the chosen Python, runs tests and an idempotent local migration demonstration, and publishes test/coverage results.

You should see

You can trace source commit to artifact to staging to production without a second rebuild.

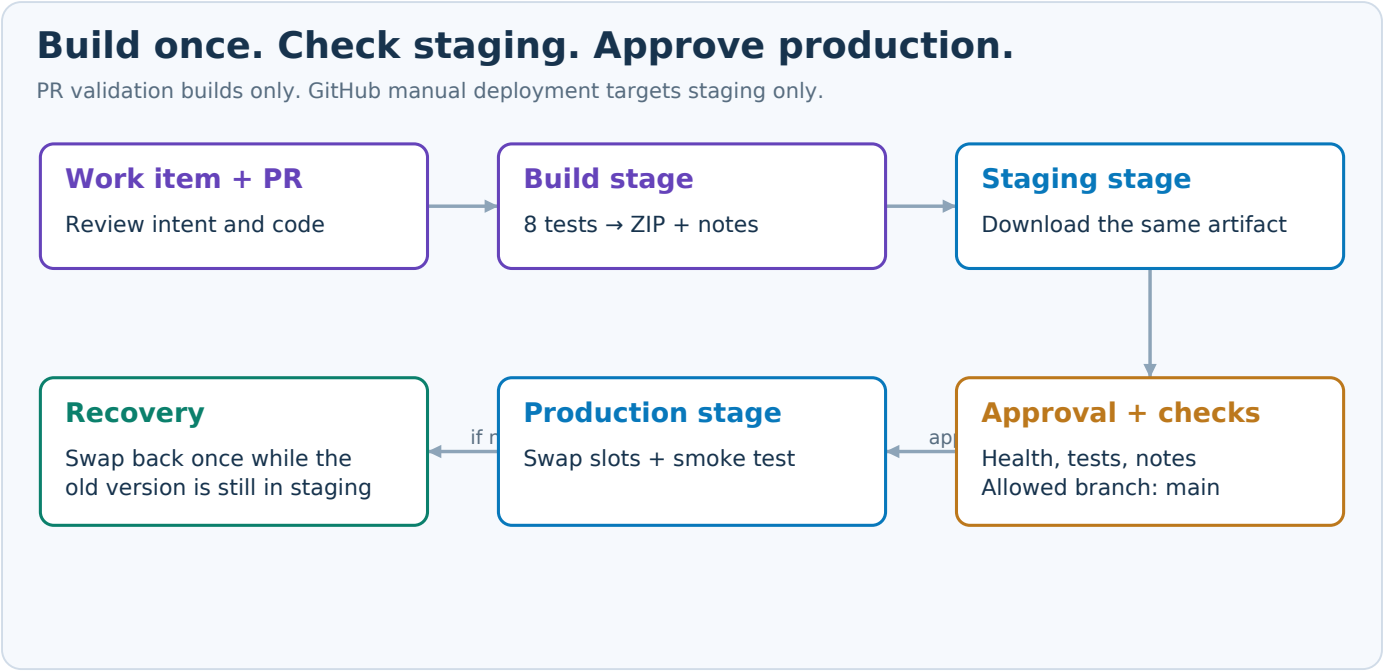
If it does not work

The SQLite migration is a disposable pipeline demonstration. It is not a deployed production database. Day 5, Block B explains how a real database changes release ordering.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result



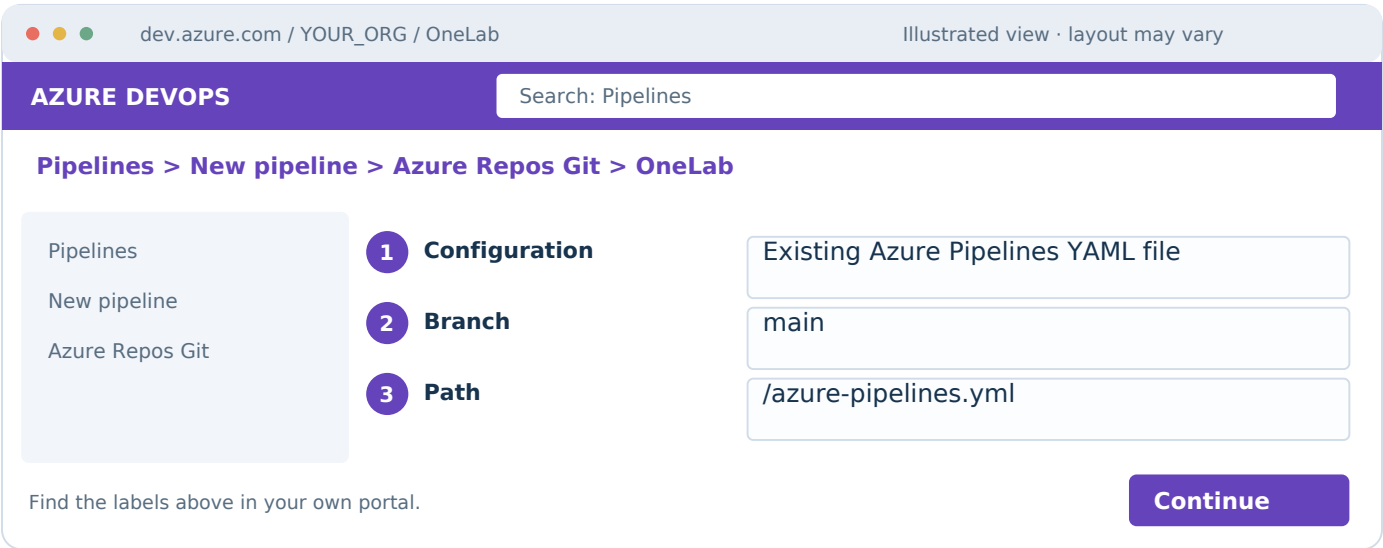
4C.02 Create and authorize the pipeline

AZ-400 · CORE

Why this task: The first run needs resource authorization in addition to valid YAML.

4C.02 · ACTION 1 OF 6

AZURE DEVOPS · Pipelines → New pipeline → Azure Repos Git → OneLab



Illustrated navigation view · use your own saved names and IDs

Why this action: The first run needs resource authorization in addition to valid YAML.

- 1. Choose Pipelines > New pipeline > Azure Repos Git > OneLab > Existing Azure Pipelines YAML file.
- 2. Choose branch main and path /azure-pipelines.yml.

4C.02 · ACTION 2 OF 6**AZURE DEVOPS · Review your pipeline YAML**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: Review your pipeline YAML

Review your pipeline YAML

Review your pipeline YAML

1 YAML /azure-pipelines.yml

2 Name OneLab-Web

Find the labels above in your own portal. **Run**

Illustrated navigation view · use your own saved names and IDs

Why this action: The first run needs resource authorization in addition to valid YAML.

1. Review the YAML and select Run.
2. Name the pipeline OneLab-Web if the portal offers a name edit.

4C.02 · ACTION 3 OF 6**AZURE DEVOPS · Pipeline run → Permission needed → View**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: Pipeline run

Pipeline run > Permission needed > View

Pipeline run

Permission needed

View

1 Resources sc-onelab, onelab-settings, environments

2 Permission This pipeline only

Find the labels above in your own portal. **Permit**

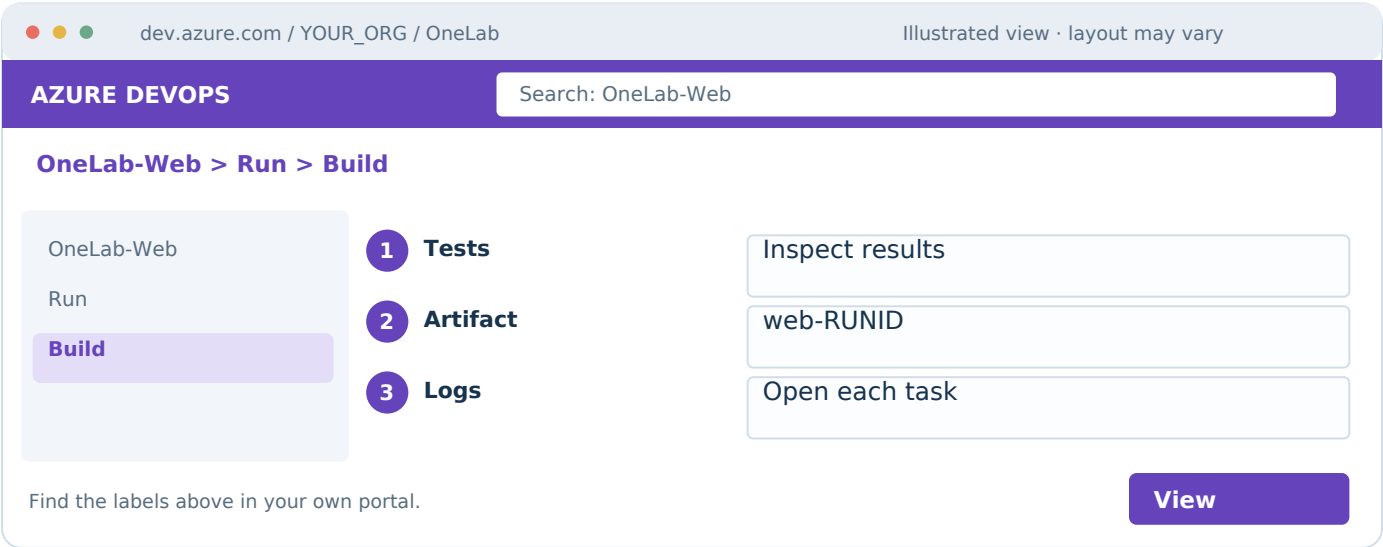
Illustrated navigation view · use your own saved names and IDs

Why this action: Azure DevOps resource permission is separate from the identity role in Azure.

1. If a banner says the run needs permission, open View/Permit and authorize sc-onelab, onelab-settings and the environments only for this pipeline.
2. Alternative path: Project settings > Service connections > sc-onelab > Pipeline permissions > + Add pipeline > OneLab-Web > Save.
3. Keep the list restricted.
4. Do this only after OneLab-Web exists.

4C.02 · ACTION 4 OF 6

AZURE DEVOPS · OneLab-Web → Run → Build



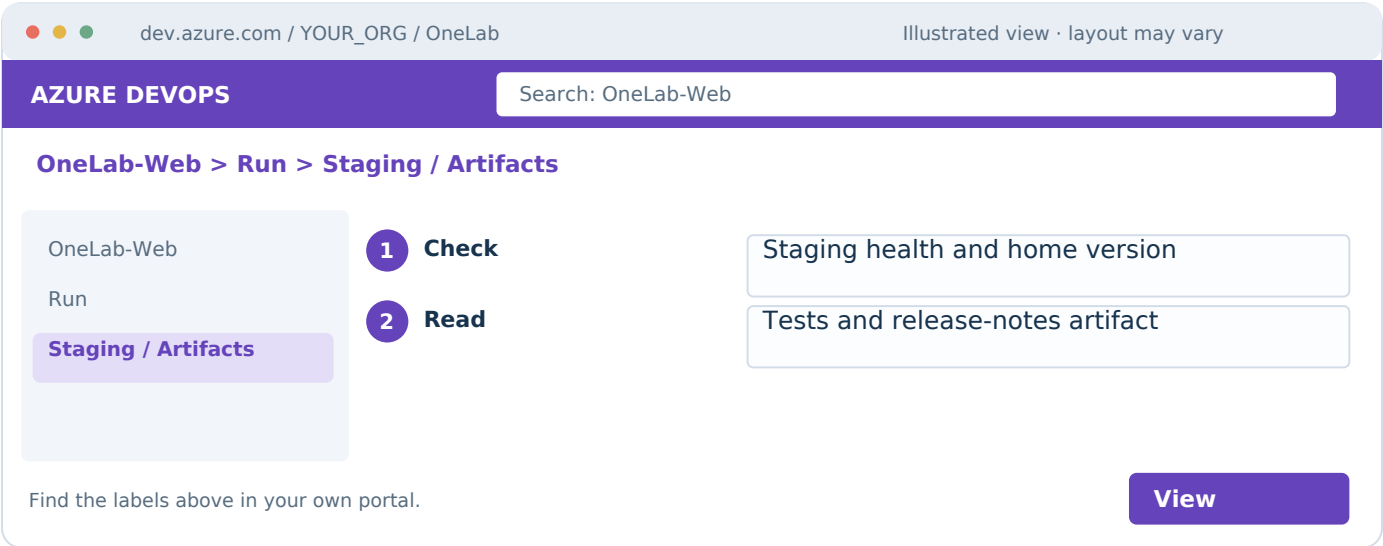
Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

- 1. Open the run, select the Build stage and each task.
- 2. Confirm test results and artifact web-RUNID appear.

4C.02 · ACTION 5 OF 6

AZURE DEVOPS · OneLab-Web → Run → Staging / Artifacts



Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

- 1. When Staging finishes, open its hostname and /health.
- 2. Review release-notes artifact and tests.

4C.02 · ACTION 6 OF 6

AZURE DEVOPS · OneLab-Web → Run → Production → Review

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: OneLab-Web

OneLab-Web > Run > Production > Review

OneLab-Web	1 Staging	Succeeded
Run	2 Approver	Your authorized reviewer
Production	3 Decision	Review evidence first

Find the labels above in your own portal.

Approve

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open the pending Production approval.
2. Approve only after those checks.
3. Confirm Production succeeds and the production endpoint is healthy.
4. In onelab-production > Approvals and checks add an Exclusive lock for the shared slot workflow.
5. For this solo lab run only one deployment at a time, including GitHub runs; a lock on one platform does not lock the other.

You should see

A complete run shows Build, Staging and Production succeeded, with a recorded approval.

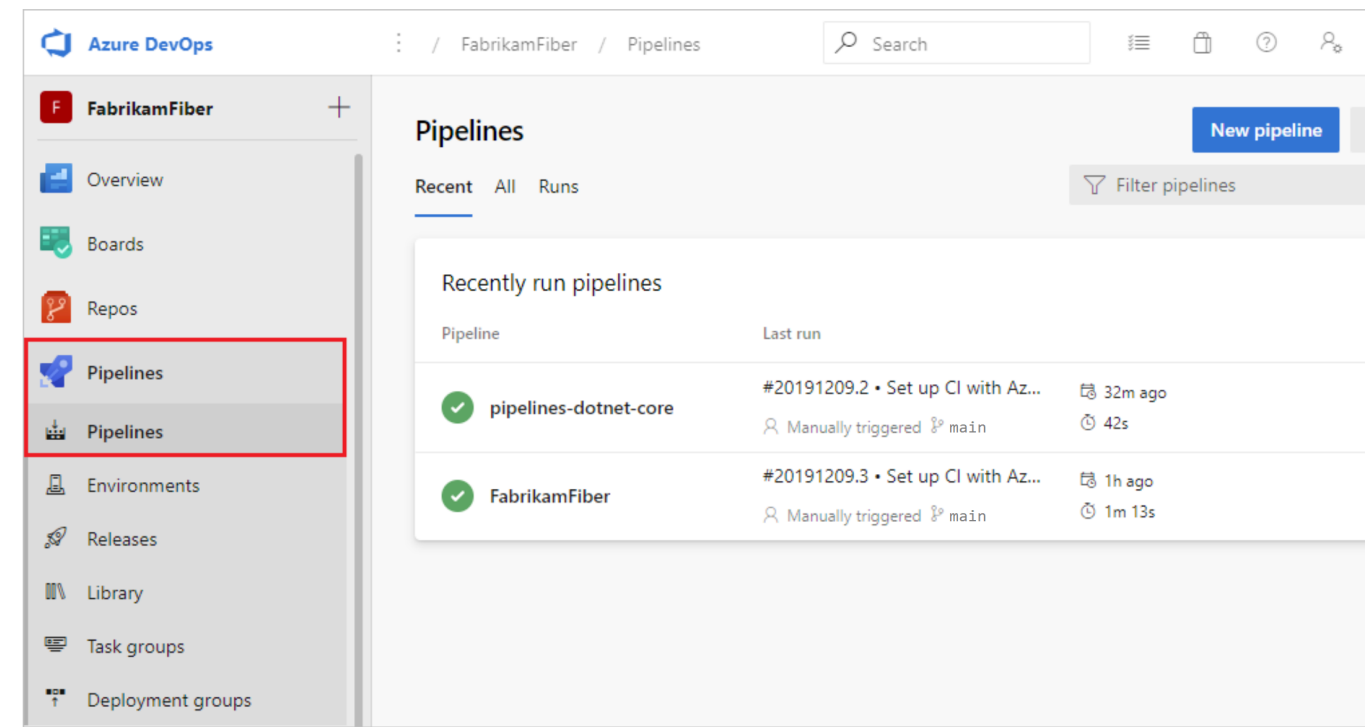
If it does not work

A queued job can mean no parallelism; Waiting for approval means a check is pending; Unauthorized means a connection/resource permission problem. They require different fixes.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Microsoft reference screenshot · Screenshot of pipelines landing page.



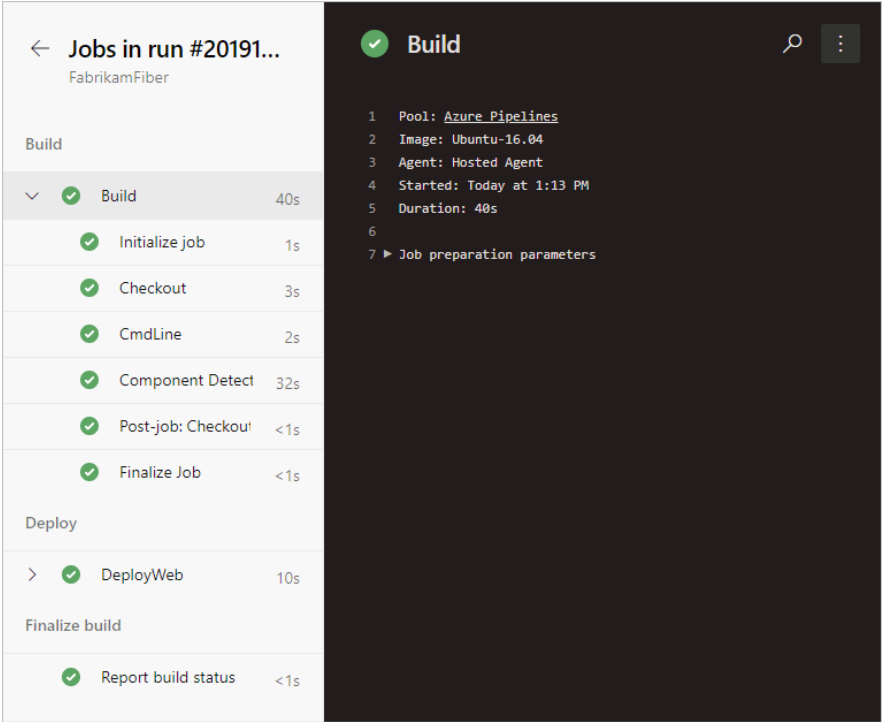
Microsoft reference source · Sample values belong to that reference, not OneLab.

Microsoft reference screenshot · Screenshot of pipeline jobs and stages.

Stages <u>Jobs</u>			
Name	Status	Stage	Duration
✓ Build	Success	Build	⌚ 40s
✓ DeployWeb	Success	Deploy	⌚ 10s

Microsoft reference source · Sample values belong to that reference, not OneLab.

Microsoft reference screenshot · Screenshot of pipeline tasks.



Microsoft reference source · Sample values belong to that reference, not OneLab.

4C.03 Turn on PR validation and complete the feature

AZ-400 · CORE

Why this task: A broken change should fail before it reaches main.

4C.03 · ACTION 1 OF 5**AZURE DEVOPS · OneLab → Repos → Branches → main → Branch policies**

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS Search: OneLab

OneLab > Repos > Branches > main > Branch policies

OneLab
Repos
Branches
main
Branch policies

1 Build validation OneLab-Web

2 Trigger Automatic

3 Policy Required

4 PR Your feature branch -> main

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: A broken change should fail before it reaches main.

1. Open main branch policies > Build validation > Add build policy.
2. Choose OneLab-Web; trigger Automatic; policy Required.
3. Save.

4C.03 · ACTION 2 OF 5**AZURE DEVOPS · Repos → Pull requests → feature/version-note → Checks**

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS Search: Repos

Repos > Pull requests > feature/version-note > Checks

Repos
Pull requests
feature/version-note

1 Build validation OneLab-Web

2 Expected Build runs

Find the labels above in your own portal. **Queue build**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open the pending feature/version-note PR.
2. Queue/requeue validation if needed.
3. Confirm Build runs while Staging and Production are skipped for the PR.

4C.03 · ACTION 3 OF 5**AZURE DEVOPS · feature/version-note PR → Overview**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: feature/version-note PR

feature/version-note PR > Overview

feature/version-note PR

Overview

1 Review Diff, work item, tests and comments

2 Merge type Squash

Find the labels above in your own portal.

Complete

Illustrated navigation view · use your own saved names and IDs

Why this action: A broken change should fail before it reaches main.

1. Review the diff, linked work item, tests and comments.
2. Approve with the configured reviewer and complete using Squash merge.

4C.03 · ACTION 4 OF 5**AZURE DEVOPS · Pipelines → OneLab-Web → Main-branch run**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: Pipelines

Pipelines > OneLab-Web > Main-branch run

Pipelines

OneLab-Web

Main-branch run

1 Staging Version 1.0.1

2 Then Approve production after verification

Find the labels above in your own portal.

Review

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Watch the main-branch pipeline.
2. Verify staging reports 1.0.1, approve promotion and confirm production reports 1.0.1.

4C.03 · ACTION 5 OF 5

AZURE DEVOPS · Boards → Work items → Your story

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS

Search: Boards

Boards > Work items > Your story

Boards

Work items

Your story

1 Acceptance criteria

2 State

All verified in production

Closed

Find the labels above in your own portal.

Save

Illustrated navigation view · use your own saved names and IDs

Why this action: A broken change should fail before it reaches main.

1. Close the story only after acceptance criteria and production verification are satisfied.

You should see

The work item, PR, test run, artifact and deployment can be linked together.

If it does not work

A PR validation run may test a temporary merge commit. Do not assume its commit identifier always equals the source branch head.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED
Worked: ☐ Independently ☐ With the guide

4C.04 Prove the quality gate catches a failure

AZ-400 · CORE

Why this task: Seeing an intentional failure teaches where a release stops.

4C.04 · ACTION 1 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: Seeing an intentional failure teaches where a release stops.

1. Return to main locally with `git switch main` and `git pull --ff-only`.
2. Create feature/broken-test.

USE IN: BASH · COPY ONLY THE CODE

```
git switch main
git pull --ff-only origin main
git switch -c feature/broken-test
```

What it means: Start from the latest accepted Azure Repos version. `--ff-only` stops instead of inventing a merge.

--ff-only · Pull only if Git can fast-forward; stop instead of creating an unexpected merge.

git switch -c · Create and switch to a new branch.

4C.04 · ACTION 2 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: Seeing an intentional failure teaches where a release stops.

1. In `test_app.py` change the expected service value from OneLab to `WrongValue`.
2. Commit and push this branch, then open a PR.

USE IN: BASH · COPY ONLY THE CODE

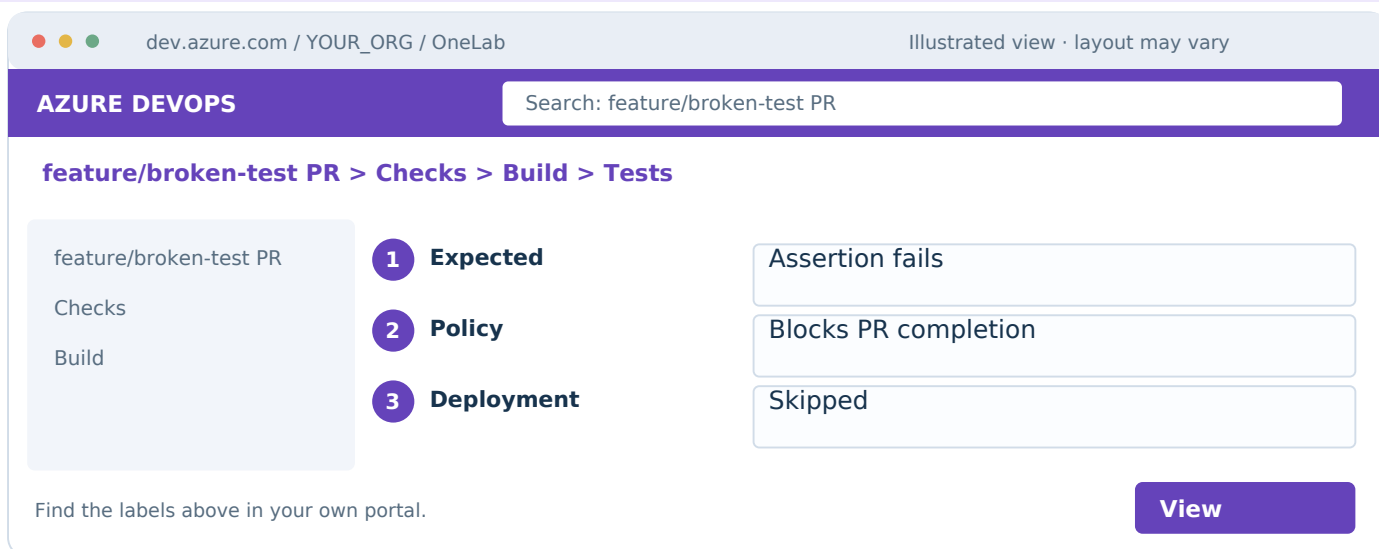
```
nano test_app.py
git add test_app.py
git commit -m "Demonstrate a failing service assertion"
git push -u origin feature/broken-test
```

What it means: Change the expected OneLab value to `WrongValue`. Then open a PR in Azure DevOps.

git add / commit / push · Stage chosen files; record a local snapshot; then send commits to the named remote.

git push -u · Push and set this branch's upstream remote-tracking branch.

Nano keys · `Ctrl+O` then `Enter` saves. `Ctrl+X` exits. Paste only the intended file content.

4C.04 · ACTION 3 OF 5**AZURE DEVOPS · feature/broken-test PR → Checks → Build → Tests**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Read the failing assertion in the Tests tab.
2. Confirm the required build policy blocks completion and no deployment occurs.

4C.04 · ACTION 4 OF 5**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

The dark code panel below is the terminal/editor view for this action.

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Restore the assertion to OneLab, commit and push.
2. Confirm a new validation run passes.
3. Abandon the exercise PR if it contains no useful final change.

USE IN: BASH · COPY ONLY THE CODE

```
nano test_app.py
python -m pytest
git add test_app.py
git commit -m "Restore the expected service value"
git push origin feature/broken-test
```

What it means: Restore OneLab in the assertion before committing. The new push reruns PR validation.

python -m MODULE · Run a module using the currently selected Python interpreter.

git add / commit / push · Stage chosen files; record a local snapshot; then send commits to the named remote.

Nano keys · Ctrl+O then Enter saves. Ctrl+X exits. Paste only the intended file content.

4C.04 · ACTION 5 OF 5

AZURE DEVOPS · Boards → Work items → Health check failed

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS Search: Boards

Boards > Work items > Health check failed

Boards
Work items
Health check failed

1 Record

Error, failed test, correction and successful rerun

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Record the error, the check that caught it, the correction and the proof of recovery in the bug work item.

You should see

The same PR changes from failed to passing after a concrete fix.

If it does not work

Do not disable tests or make their task `continueOnError` just to get a green release.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

4D · Keep a version you can reuse

Planning estimate: 60 minutes. Finish with: A package/feed version and a retained release artifact.

4D.01 Create a feed and understand its views

AZ-400 · CORE + package extension

Why this task: A package feed stores reusable versioned components; a pipeline artifact belongs to a particular run.

4D.01 · ACTION 1 OF 5

AZURE DEVOPS · OneLab → Artifacts → Create feed

dev.azure.com / YOUR_ORG / OneLabIllustrated view · layout may vary

AZURE DEVOPS

Search: OneLab

OneLab > Artifacts > Create feed

OneLab

Artifacts

Create feed

1 Name

2 Scope

3 Visibility

onelab

Project

Your project

Find the labels above in your own portal.

Create

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. Open Artifacts > Create feed.
2. Name onelab; visibility limited to your project/team; project scope.
3. Create.

4D.01 · ACTION 2 OF 5**AZURE DEVOPS · onelab → Feed settings → Upstream sources**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: onelab

onelab > Feed settings > Upstream sources

onelab
Feed settings
Upstream sources

1 Source

Only an approved public Python source

Find the labels above in your own portal.

Add

Illustrated navigation view · use your own saved names and IDs

Why this action: A package feed stores reusable versioned components; a pipeline artifact belongs to a particular run.

1. Open feed settings > Upstream sources.
2. Add an allowed public Python source if you want to study caching of external dependencies.
3. Keep internal and public sources governed to reduce dependency-confusion risk.

4D.01 · ACTION 3 OF 5**AZURE DEVOPS · onelab → Feed settings → Permissions**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: onelab

onelab > Feed settings > Permissions

onelab
Feed settings
Permissions

1 Identity

OneLab Build Service (YOUR_ORGANIZATION)

2 Role

Feed Publisher (Contributor)

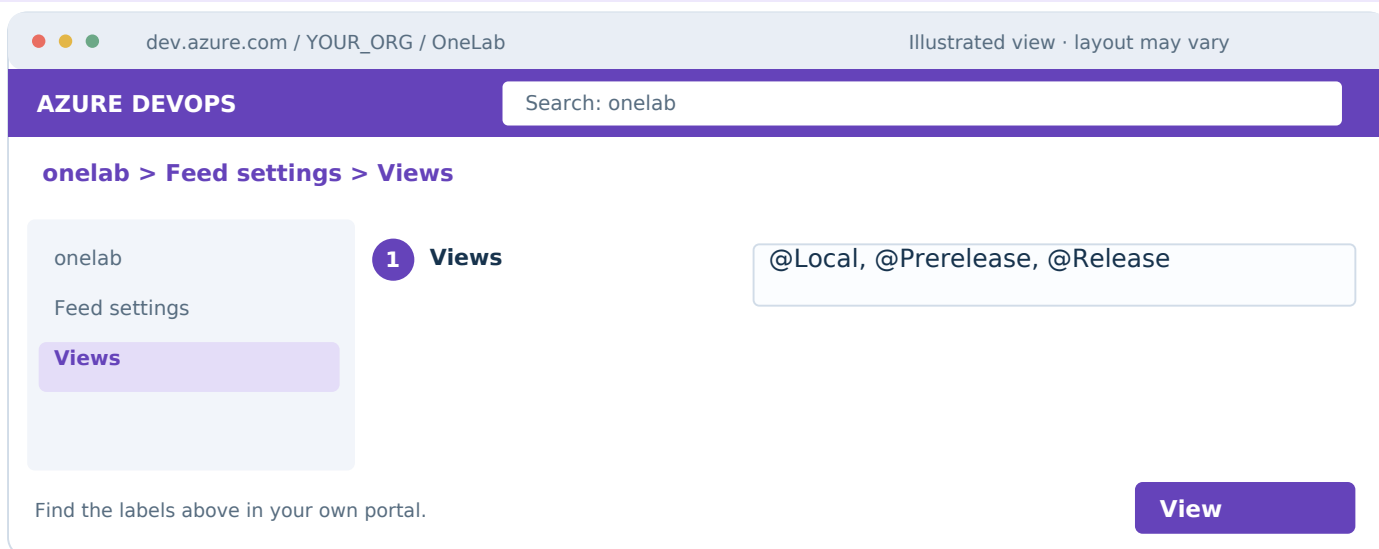
Find the labels above in your own portal.

Save

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

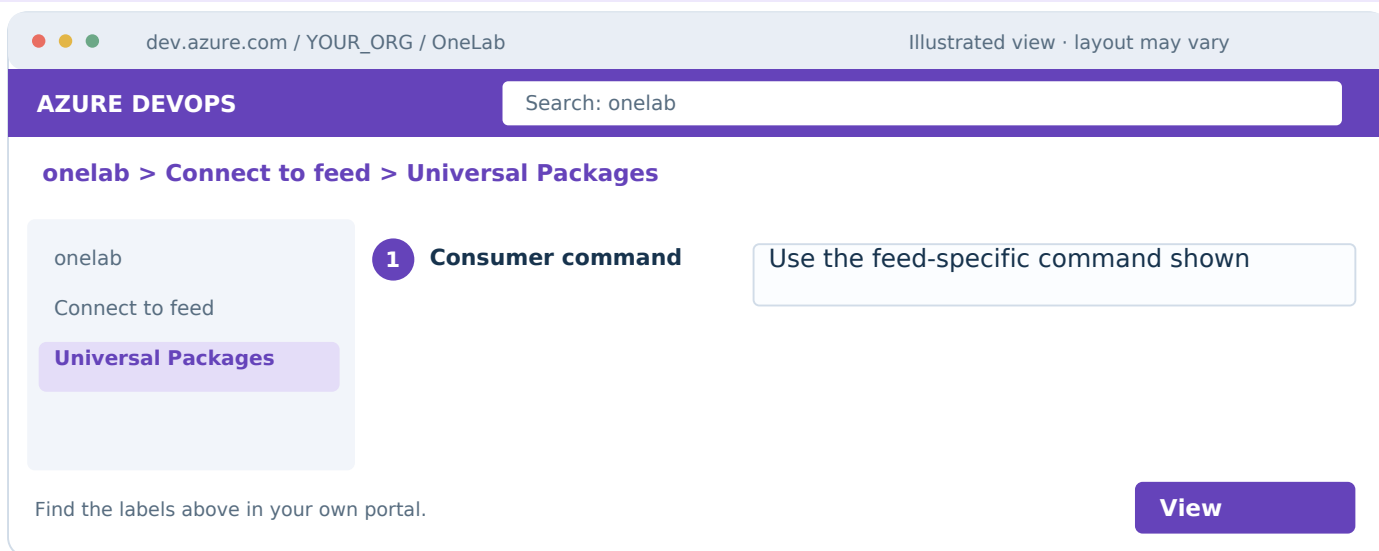
1. Open Permissions.
2. Add OneLab Build Service (YOUR_ORGANIZATION) as Feed Publisher (Contributor) if it will publish.
3. Grant readers only the role they need.

4D.01 · ACTION 4 OF 5**AZURE DEVOPS · onelab → Feed settings → Views**

Illustrated navigation view · use your own saved names and IDs

Why this action: A package feed stores reusable versioned components; a pipeline artifact belongs to a particular run.

1. Inspect Views such as @Local, @Prerelease and @Release.
2. A view lets consumers choose packages at a maturity level; it is not a separate source repository.

4D.01 · ACTION 5 OF 5**AZURE DEVOPS · onelab → Connect to feed → Universal Packages**

Illustrated navigation view · use your own saved names and IDs

Why this action: A package feed stores reusable versioned components; a pipeline artifact belongs to a particular run.

1. Open Connect to feed and select Universal Packages to see the feed-specific instructions.
2. Use the pipeline task below to publish a small package from this project.

You should see

The feed exists and its publisher identity is scoped appropriately.

If it does not work

A pipeline's Azure service connection does not automatically grant Azure Artifacts feed rights. Feed permissions belong to Azure DevOps identities. [R37]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

4D.02 Publish and promote a package

AZ-400 · CORE + package extension

Why this task: A version should identify immutable release content.

4D.02 · ACTION 1 OF 5

AZURE DEVOPS · Repos → Files → pipelines/packages.yml

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS

Search: Repos

Repos > Files > pipelines/packages.yml

Repos

Files

pipelines/packages.yml

1 Existing file

2 Feed

Use the supplied file

OneLab/onelab

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: A version should identify immutable release content.

1. Open the supplied pipelines/packages.yml from the source appendix.
2. It is already included in the ZIP.
3. If you edit it, commit that change through a branch and PR; do not create a duplicate file.

4D.02 · ACTION 2 OF 5**AZURE DEVOPS · Pipelines → New pipeline → Existing YAML**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: Pipelines

Pipelines > New pipeline > Existing YAML

Pipelines

New pipeline

Existing YAML

1 File /pipelines/packages.yml

2 Version 1.0.BuildId

Find the labels above in your own portal.

Run

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. Create a second YAML pipeline selecting that file.
2. Run it manually.
3. It packages only app.py and requirements.txt and publishes onelab-web with version 1.0.BUILDID.

4D.02 · ACTION 3 OF 5**AZURE DEVOPS · Artifacts → onelab → onelab-web**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: Artifacts

Artifacts > onelab > onelab-web

Artifacts

onelab

onelab-web

1 Version Your actual build-based version

Find the labels above in your own portal.

Open

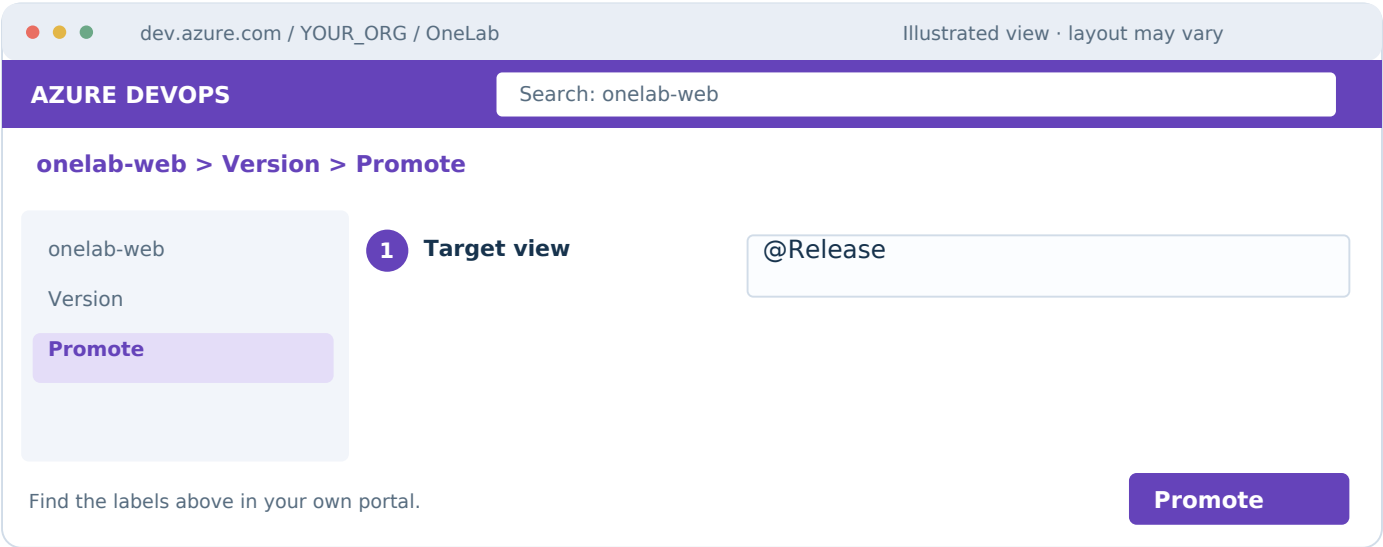
Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open Artifacts > onelab > onelab-web.
2. Confirm that version exists and inspect its files.

4D.02 · ACTION 4 OF 5

AZURE DEVOPS · onelab-web → Version → Promote



Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. Promote the tested version to @Release using the package's Promote action.
2. Download it using Connect to feed instructions and compare it to the published files.

4D.02 · ACTION 5 OF 5

SEE THE IDEA · A package version means one content set

A SMALL DECISION MAP

A package version means one content set

1	Keep	Known-good version
2	New content	New version number

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. Keep the release version used for rollback.
2. Do not reuse a published version number for different content.

You should see

A versioned package appears in the feed and its release view.

If it does not work

If permission is denied, check the project Build Service feed role. Universal package versions need their supported numeric format; the build ID makes each sample version unique.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED
Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

Version / storage choice	Meaning
SemVer 2.4.1	Major for incompatible changes, minor for compatible features, patch for compatible fixes. Pre-release labels identify unfinished releases where the package format supports them.
CalVer 2026.09.22	Version based on release date; add a revision when several releases share a date.
Pipeline artifact web-123	Output associated with one build run; retain the run if this output is needed.
Container digest	Use the digest for precise image content; do not rely on a mutable latest tag for rollback.
GitHub Packages	Useful when distributing packages with GitHub workflows and permissions. Choose the registry supported by the package ecosystem.
Dependency lock	Pin direct dependencies and generate/review a full resolved lock for reproducibility. A requirements file pinning only direct dependencies does not freeze every transitive package.

Decision: OneLab promotes the same tested ZIP to staging and production. A shared library would use a package feed with its own version. A container release would store its digest alongside the commit and run ID. Retain the current production release and a known-good previous release, with retention consistent with cost and audit needs.

Finish Day 4

Save your place and check cost

Record one successful release, one deliberately failed test and the artifact version. Keep the known-good run for rollback. Check costs.

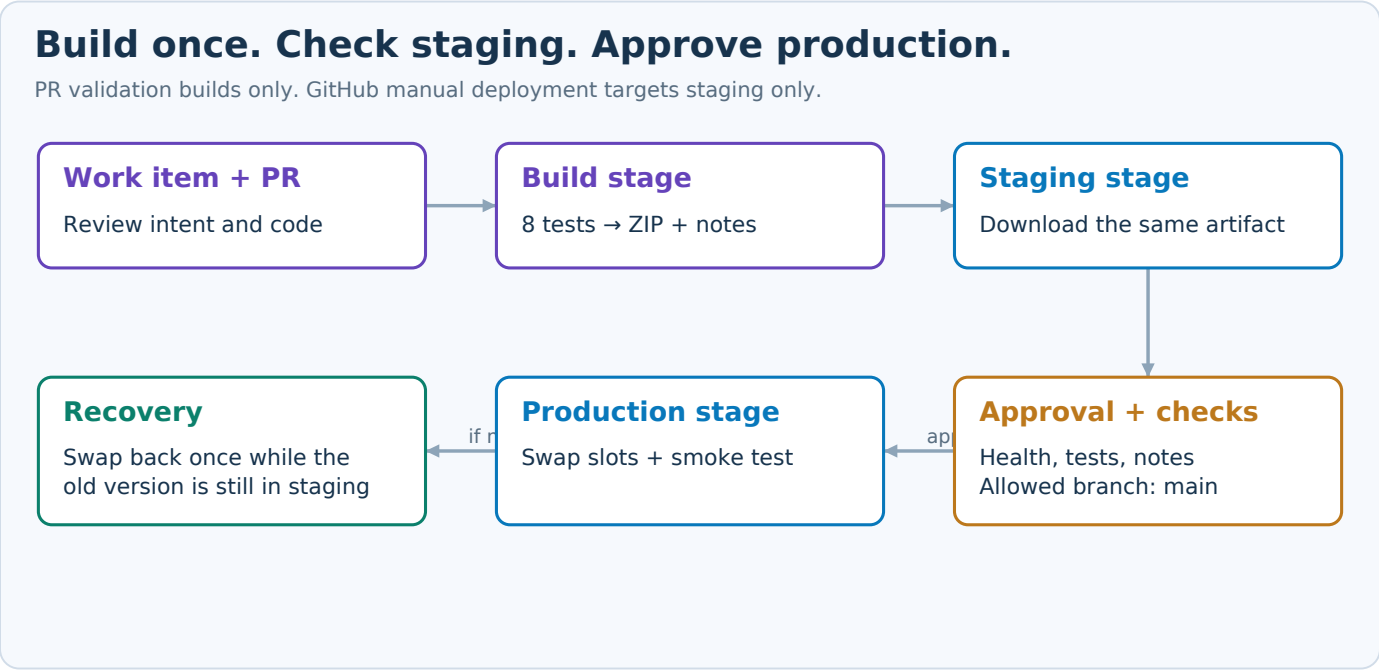
Say it in your own words: Why should production receive the same artifact that staging tested?

Check your explanation: Rebuilding can produce different content. Promoting the same version lets you trace exactly what was tested and released.

DAY 5 OF 7

Practise safe changes in both platforms

You can preview infrastructure, undo a release and run the same application through GitHub.



Start today
Run the restart routine and confirm Azure Repos main is current. Keep Azure Repos as the main source; GitHub is the deliberate second route.

Start early and track the job
Run the GitHub deployment only after Azure Pipelines is idle. It targets staging, so production stays under the existing release process.

Block	Today's work	Minutes
A	Preview and apply a small infrastructure file	90
B	Rollback, flags and database order	90
C	Use GitHub with the same application	125
D	Read the pipeline controls	45

5A · Preview and apply a small infrastructure file

Planning estimate: 90 minutes. Finish with: Bicep/ARM and PowerShell evidence plus an intentional infrastructure pipeline run.

5A.01 Read the Bicep file before deploying

AZ-104 + AZ-400 · CORE

Why this task: Infrastructure code makes settings reviewable and repeatable.

5A.01 · ACTION 1 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: Infrastructure code makes settings reviewable and repeatable.

1. Open infra/storage.bicep.
2. It declares a separate small account, not the main storage account already used in the portal.

USE IN: BASH · COPY ONLY THE CODE

```
nano infra/storage.bicep
```

What it means: Read the file. Ctrl+X exits. Save only if you deliberately change it.

Nano keys · Ctrl+O then Enter saves. Ctrl+X exits. Paste only the intended file content.

5A.01 · ACTION 2 OF 5

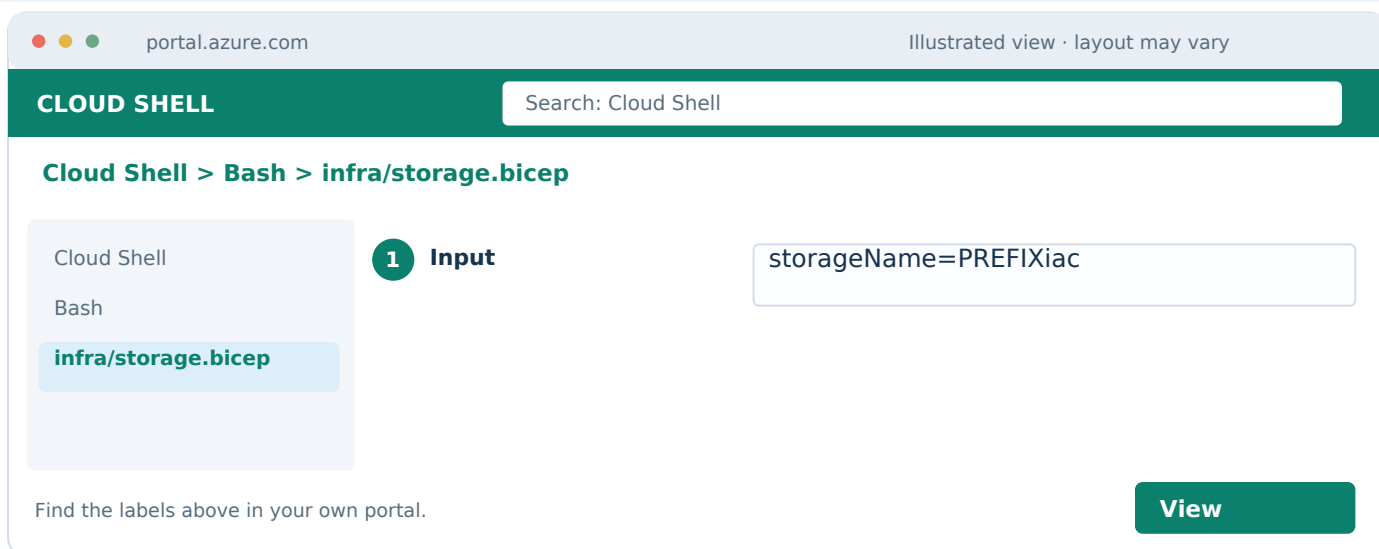
CLOUD SHELL · Cloud Shell → Bash → infra/storage.bicep

The image shows a simulated Azure Cloud Shell interface. At the top, there's a header bar with 'portal.azure.com' on the left and 'Illustrated view · layout may vary' on the right. Below this is a dark green bar with 'CLOUD SHELL' on the left and a search bar containing 'Search: Cloud Shell'. The main area has a breadcrumb 'Cloud Shell > Bash > infra/storage.bicep'. On the left, a navigation pane shows 'Cloud Shell', 'Bash', and 'infra/storage.bicep' (highlighted in blue). To the right of this pane, a green circle with the number '1' is followed by the word 'Input'. Further right is a text input field containing 'storageName=PREFIXiac'. At the bottom left, a note says 'Find the labels above in your own portal.' and at the bottom right is a green 'View' button.

Illustrated navigation view · use your own saved names and IDs

Why this action: Infrastructure code makes settings reviewable and repeatable.

1. Find param storageName and the default location.
2. A parameter is input; a resource block is a desired Azure object; properties are its settings; output returns a result.

5A.01 · ACTION 3 OF 5**CLOUD SHELL · Cloud Shell → Bash → infra/storage.bicep**

Illustrated navigation view · use your own saved names and IDs

Why this action: Infrastructure code makes settings reviewable and repeatable.

1. Notice HTTPS, TLS 1.2 and anonymous-access settings.
2. The allowed SKU list limits accepted input to LRS and ZRS.

5A.01 · ACTION 4 OF 5**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

The dark code panel below is the terminal/editor view for this action.

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Run the what-if command below.
2. Read every proposed change.
3. It should create PREFIXiac only, not delete your main account.

USE IN: BASH · COPY ONLY THE CODE

```
az bicep install
az bicep version
az deployment group what-if -g "$RG" \
  \
  --template-file infra/storage.bicep --parameters storageName=${P}iac
```

What it means: what-if previews changes. Stop here and read the preview before the next action.

--template-file · The Bicep or ARM file describing the intended resource.

--parameters · Supply input values required by the infrastructure template.

-g · Short form of --resource-group.

\$NAME / \${NAME} · Read a variable from this Bash session. Source lab.env first when these are lab names.

Backslash at line end · Continue the same Bash command on the next line. No spaces may follow the backslash.

5A.01 · ACTION 5 OF 5**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

The dark code panel below is the terminal/editor view for this action.

Why this action: Infrastructure code makes settings reviewable and repeatable.

1. Run create only when the preview is understood.
2. Open the resource group > Deployments in the portal and inspect template, inputs and outputs.

USE IN: BASH · COPY ONLY THE CODE

```
az deployment group create -g "$RG" -n onelab-storage \  
  \--template-file infra/storage.bicep --parameters storageName=${P}iac
```

What it means: create applies the desired state and records a named deployment.

--template-file · The Bicep or ARM file describing the intended resource.

--parameters · Supply input values required by the infrastructure template.

-g · Short form of --resource-group.

-n · Short form of --name in these Azure commands.

\$NAME / \${NAME} · Read a variable from this Bash session. Source lab.env first when these are lab names.

Backslash at line end · Continue the same Bash command on the next line. No spaces may follow the backslash.

You should see

A code-created PREFIXiac storage account exists and you can trace it to a deployment.

If it does not work

Bicep operates through Azure Resource Manager. The resource API version is different from the Azure CLI version. What-if is a preview, not a guarantee that deployment will pass quota and policy. [R27]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

5A.02 Modify, compile, export and compare

AZ-104 + AZ-400 · CORE

Why this task: You should be able to read both Bicep and ARM JSON, not just run a template.

5A.02 · ACTION 1 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: You should be able to read both Bicep and ARM JSON, not just run a template.

1. In `infra/storage.bicep` change the Purpose tag value to `laC-reviewed`.
2. Run what-if again, then deploy and inspect the changed tag.

USE IN: BASH · COPY ONLY THE CODE

```
nano infra/storage.bicep
az deployment group what-if -g "$RG" \
  --template-file infra/storage.bicep --parameters storageName=${P}iac
```

What it means: Change Purpose to `laC-reviewed`, save, exit and inspect the tag update preview.

--template-file · The Bicep or ARM file describing the intended resource.

--parameters · Supply input values required by the infrastructure template.

-g · Short form of `--resource-group`.

\$NAME / \${NAME} · Read a variable from this Bash session. Source `lab.env` first when these are lab names.

Backslash at line end · Continue the same Bash command on the next line. No spaces may follow the backslash.

Nano keys · `Ctrl+O` then `Enter` saves. `Ctrl+X` exits. Paste only the intended file content.

USE IN: BASH · COPY ONLY THE CODE

```
az deployment group create -g "$RG" -n onelab-storage \
  --template-file infra/storage.bicep --parameters storageName=${P}iac
```

What it means: Run after reading the preview to apply this one tag change.

--template-file · The Bicep or ARM file describing the intended resource.

--parameters · Supply input values required by the infrastructure template.

-g · Short form of `--resource-group`.

-n · Short form of `--name` in these Azure commands.

\$NAME / \${NAME} · Read a variable from this Bash session. Source `lab.env` first when these are lab names.

Backslash at line end · Continue the same Bash command on the next line. No spaces may follow the backslash.

5A.02 · ACTION 2 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: You should be able to read both Bicep and ARM JSON, not just run a template.

1. Compile to ARM JSON with the command below.
2. Open the JSON and find parameters, resources and outputs.

USE IN: BASH · COPY ONLY THE CODE

```
az bicep build --file infra/storage.bicep --outfile infra/storage.json
nano infra/storage.json
```

What it means: Compile to ARM JSON, then read the output. Exit Nano with Ctrl+X.

--file · The source file for the Bicep operation.

--outfile · The path where the compiled output is written.

Nano keys · Ctrl+O then Enter saves. Ctrl+X exits. Paste only the intended file content.

5A.02 · ACTION 3 OF 5

CLOUD SHELL · On the resource group → Deployments → onelab-storage, use Template → Download or the available export action to inspect the deployed ARM template

The image shows a screenshot of the Azure portal's Cloud Shell interface. At the top, there's a search bar with the text "Search: On the resource group". Below the search bar, a navigation path is highlighted: "On the resource group > Deployments > onelab-storage, use Template > Download or the available export action to inspect the deployed ARM template". This path is broken down into two steps: 1. Locate (On the resource group > Deployments) and 2. Action (onelab-storage, use Template > Download or the available export action to inspect the deployed ARM template). A "View" button is located at the bottom right of the navigation view.

Illustrated navigation view · use your own saved names and IDs

Why this action: You should be able to read both Bicep and ARM JSON, not just run a template.

1. On the resource group > Deployments > onelab-storage, use Template > Download or the available export action to inspect the deployed ARM template.

5A.02 · ACTION 4 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. For conversion practice, copy storage.json to imported.json, then run decompile below.
2. Read warnings and review imported.bicep; a generated template can need cleanup.

USE IN: BASH · COPY ONLY THE CODE

```
cp infra/storage.json infra/imported.json
az bicep decompile --file infra/imported.json
```

What it means: cp makes a practice copy. Decompile produces an approximation to review.

--file · The source file for the Bicep operation.

5A.02 · ACTION 5 OF 5**SEE THE IDEA · Modify, compile, export and compare → Explain the decision****A SMALL DECISION MAP****Modify, compile, export and compare > Explain the decision**

- | | | |
|----------|---------------|--|
| 1 | Locate | Explain the decision |
| 2 | Action | Explain idempotence: applying the same desired configuration again should not create a second randomly named copy. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: You should be able to read both Bicep and ARM JSON, not just run a template.

1. Explain idempotence: applying the same desired configuration again should not create a second randomly named copy.
2. Explain drift: a portal edit makes actual settings differ from the code.

You should see

You can map one Bicep resource to its ARM JSON equivalent.

If it does not work

Exported templates are a starting point, not automatically production-ready. Never keep real secrets in plain template parameters or deployment outputs.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

5A.03 See the same inventory through PowerShell

AZ-104 + AZ-400 · CORE

Why this task: AZ-104 expects familiarity with PowerShell as well as Azure CLI.

5A.03 · ACTION 1 OF 4

CLOUD SHELL · Cloud Shell → Switch to PowerShell

The image shows a browser window with the URL 'portal.azure.com'. The page title is 'CLOUD SHELL'. A search bar contains 'Search: Cloud Shell'. Below the search bar, the breadcrumb 'Cloud Shell > Switch to PowerShell' is displayed. On the left, a sidebar shows 'Cloud Shell' and a button 'Switch to PowerShell'. To the right, a numbered list indicates the steps: 1. Subscription, 2. Resource group, and 3. Then return to. Each step has a corresponding input field: 'Your practice subscription' for Subscription, 'rg-PREFIX-core' for Resource group, and 'Bash and restart routine' for Then return to. At the bottom right, there is an 'Open' button. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Record the subscription ID and group name from lab.env.
2. In Cloud Shell choose Switch to PowerShell.

5A.03 · ACTION 2 OF 4

CLOUD SHELL · Cloud Shell → PowerShell → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: AZ-104 expects familiarity with PowerShell as well as Azure CLI.

1. Run the block below after replacing both values.
2. Do not paste Bash export statements into this shell.

USE IN: POWERSHELL · COPY ONLY THE CODE

```
Set-AzContext -SubscriptionId 'REPLACE_SUBSCRIPTION_ID'
Get-AzResourceGroup -Name 'rg-REPLACE_PREFIX-core'
Get-AzResource -ResourceGroupName 'rg-REPLACE_PREFIX-core' |
  Select-Object Name, ResourceType, Location
```

What it means: Set-AzContext chooses the subscription. Select-Object chooses displayed properties; it does not modify resources.

source FILE · Load settings or activate an environment in the current Bash shell.

5A.03 · ACTION 3 OF 4

CLOUD SHELL · Cloud Shell → Switch to PowerShell

The screenshot shows the Azure Cloud Shell interface. At the top, there's a header bar with 'portal.azure.com' on the left and 'Illustrated view · layout may vary' on the right. Below this is a dark green bar with 'CLOUD SHELL' on the left and a search bar containing 'Search: Cloud Shell' on the right. The main content area has a breadcrumb 'Cloud Shell > Switch to PowerShell'. On the left, there's a sidebar with 'Cloud Shell' and 'Switch to PowerShell' (highlighted in blue). To the right of the sidebar, there are two numbered steps: '1 Locate' and '2 Action'. Step 1 points to the 'Switch to PowerShell' button. Step 2 points to a text box that says 'Compare the listed resources with the Azure Portal group.' At the bottom right, there's a green button labeled 'View'.

Illustrated navigation view · use your own saved names and IDs

Why this action: AZ-104 expects familiarity with PowerShell as well as Azure CLI.

1. Compare the listed resources with the Azure Portal group.

5A.03 · ACTION 4 OF 4

CLOUD SHELL · Cloud Shell → Switch to PowerShell

The screenshot shows the Azure Cloud Shell interface. At the top, there's a header bar with 'portal.azure.com' on the left and 'Illustrated view · layout may vary' on the right. Below this is a dark green bar with 'CLOUD SHELL' on the left and a search bar containing 'Search: Cloud Shell' on the right. The main content area has a breadcrumb 'Cloud Shell > Switch to PowerShell'. On the left, there's a sidebar with 'Cloud Shell' and 'Switch to PowerShell' (highlighted in blue). To the right of the sidebar, there are two numbered steps: '1 Locate' and '2 Action'. Step 1 points to the 'Switch to PowerShell' button. Step 2 points to a text box that says 'Switch back to Bash and run the session restart routine from Day 1, Block B.' At the bottom right, there's a green button labeled 'Run'.

Illustrated navigation view · use your own saved names and IDs

Why this action: AZ-104 expects familiarity with PowerShell as well as Azure CLI.

1. Switch back to Bash and run the session restart routine from Day 1, Block B.

You should see

You recognize Get-AzResourceGroup and Get-AzResource as read operations.

If it does not work

A PowerShell pipeline passes objects to the next command; Bash commands commonly pass text.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

5A.04 Create a separate infrastructure pipeline

AZ-400 · CORE + platform extensions

Why this task: Infrastructure changes deserve preview and controlled deployment just like application changes.

5A.04 · ACTION 1 OF 6

AZURE DEVOPS · Pipelines → Library → onelab-settings

The screenshot shows the Azure DevOps web interface. At the top, the browser address bar displays 'dev.azure.com / YOUR_ORG / OneLab'. The page header includes 'AZURE DEVOPS' and a search bar containing 'Pipelines'. The breadcrumb navigation is 'Pipelines > Library > onelab-settings'. On the left, a sidebar lists 'Pipelines', 'Library', and 'onelab-settings' (which is highlighted). The main content area shows a configuration for a variable. It has two numbered steps: '1 Variable' and '2 Value'. Under '1 Variable', there is a text input field containing 'iacStorageName'. Under '2 Value', there is a text input field containing 'PREFIXiac'. At the bottom left, there is a note: 'Find the labels above in your own portal.' At the bottom right, there is a purple 'Save' button.

Illustrated navigation view · use your own saved names and IDs

Why this action: Infrastructure changes deserve preview and controlled deployment just like application changes.

1. In Library > onelab-settings add iacStorageName with value PREFIXiac.
2. Save.

5A.04 · ACTION 2 OF 6

AZURE DEVOPS · Pipelines → New pipeline → Existing YAML

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS Search: Pipelines

Pipelines > New pipeline > Existing YAML

Pipelines
New pipeline
Existing YAML

1 File /pipelines/iac.yml

2 Run parameter deploy false

Find the labels above in your own portal. **Run**

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Create a YAML pipeline using /pipelines/iac.yml.
2. Run it with deploy unchecked/false.
3. Name it OneLab-Infrastructure.
4. At any resource-permission prompt, permit this pipeline specifically to use sc-onelab and onelab-settings.

5A.04 · ACTION 3 OF 6

AZURE DEVOPS · Infrastructure pipeline → Run → Logs

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS Search: Infrastructure pipeline

Infrastructure pipeline > Run > Logs

Infrastructure pipeline
Run
Logs

1 Read Compile, validate and what-if

2 Expected target PREFIXiac

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: Infrastructure changes deserve preview and controlled deployment just like application changes.

1. Inspect Bicep compilation, Azure validation and what-if output.
2. These are separate checks.

5A.04 · ACTION 4 OF 6

AZURE DEVOPS · Infrastructure pipeline → Run pipeline

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: Infrastructure pipeline

Infrastructure pipeline > Run pipeline

Infrastructure pipeline

Run pipeline

1 deploy true only after reading preview

2 Permission Permit sc-onelab and variable group for this pipeline

Find the labels above in your own portal.

Run

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. For the lab, run again with deploy true only after reading the preview.
2. Confirm the deployment record and storage settings.

5A.04 · ACTION 5 OF 6

SEE THE IDEA · A team approves infrastructure changes

A SMALL DECISION MAP

A team approves infrastructure changes

- | | |
|-------------------|--------------------------|
| 1 Review | Infrastructure PR |
| 2 Gate | Protected environment |
| 3 Identity | Limited deployment scope |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Infrastructure changes deserve preview and controlled deployment just like application changes.

1. For a team workflow, place the applying job in a protected deployment environment, require an infrastructure PR and approval, and keep the deploying identity scoped to its target.

5A.04 · ACTION 6 OF 6

SEE THE IDEA · Incremental deployment

A SMALL DECISION MAP

Incremental deployment

- | | | |
|---|--------------------------|---------------------------------|
| 1 | Creates / updates | Declared resources |
| 2 | Deletion | Needs an explicit reviewed plan |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Do not make a CI run that automatically deletes resources missing from a template.
2. Understand incremental versus complete deployment behavior and deletion controls before using them.

You should see

A pipeline can preview and intentionally apply the infrastructure file.

If it does not work

The boolean parameter in the example controls the action; it is not a substitute for a production resource-owner approval.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

Technology	What it manages in this project
ARM / Bicep	Azure resource declarations: accounts, networks, plans and their properties.
Terraform	A similar IaC choice using providers and state; protect remote state, locking and saved plan/apply workflow. It is not required for this Bicep route.
Configuration management	Packages, files and services inside a VM; repeated runs should converge on the same desired state.
Azure Machine Configuration	Audit or apply supported machine configuration through policy/guest configuration, including Arc-connected machines.
Automation State Configuration	Legacy DSC configuration compilation, pull and reporting. Still understand its model, but it is scheduled to retire on 30 September 2027.
Azure Deployment Environments	A catalog-driven self-service environment platform. It is scheduled to retire on 22 February 2027; learn the objective and migration implications.

5B · Rollback, flags and database order

Planning estimate: 90 minutes. Finish with: A tested slot rollback, a flag change and a repeatable local migration.

5B.01 Make an immediate rollback while the old slot is available

AZ-400 · CORE + design exercises

Why this task: After a successful swap, the previous production content is normally in the other slot.

5B.01 · ACTION 1 OF 5

AZURE PORTAL · App Services → PREFIX-web → Deployment slots → Swap

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page title is 'AZURE PORTAL'. A search bar contains 'App Services'. The breadcrumb navigation is 'App Services > PREFIX-web > Deployment slots > Swap'. On the left, a sidebar lists 'App Services', 'PREFIX-web', and 'Deployment slots'. The main content area shows a 'Swap' action with three steps: 1. Source (staging), 2. Target (production), and 3. Check before swap (Which version is in each slot). A 'Swap' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Before changing staging again, record production and staging versions.
2. Open App Service > Deployment slots > Swap.

5B.01 · ACTION 2 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: After a successful swap, the previous production content is normally in the other slot.

1. Review source staging and target production, along with settings that change or remain slot-specific.
2. Swap back only for this intentional rollback drill.
3. Choose one route: click Swap in the portal OR run the equivalent command shown here.
4. Do not do both for the same rollback.

USE IN: BASH · COPY ONLY THE CODE

```
az webapp deployment slot swap -g "$RG" -n "$APP" \
  --slot staging --target-slot production
```

What it means: This exchanges eligible content/settings between the two slots. It is an action, not a preview; run it only for the planned exercise. Choose the portal Swap OR this equivalent command once. Running both swaps twice and cancels your rollback.

--slot · Target the named deployment slot, such as staging.

--target-slot · The other slot in a swap; production is the live slot.

-g · Short form of --resource-group.

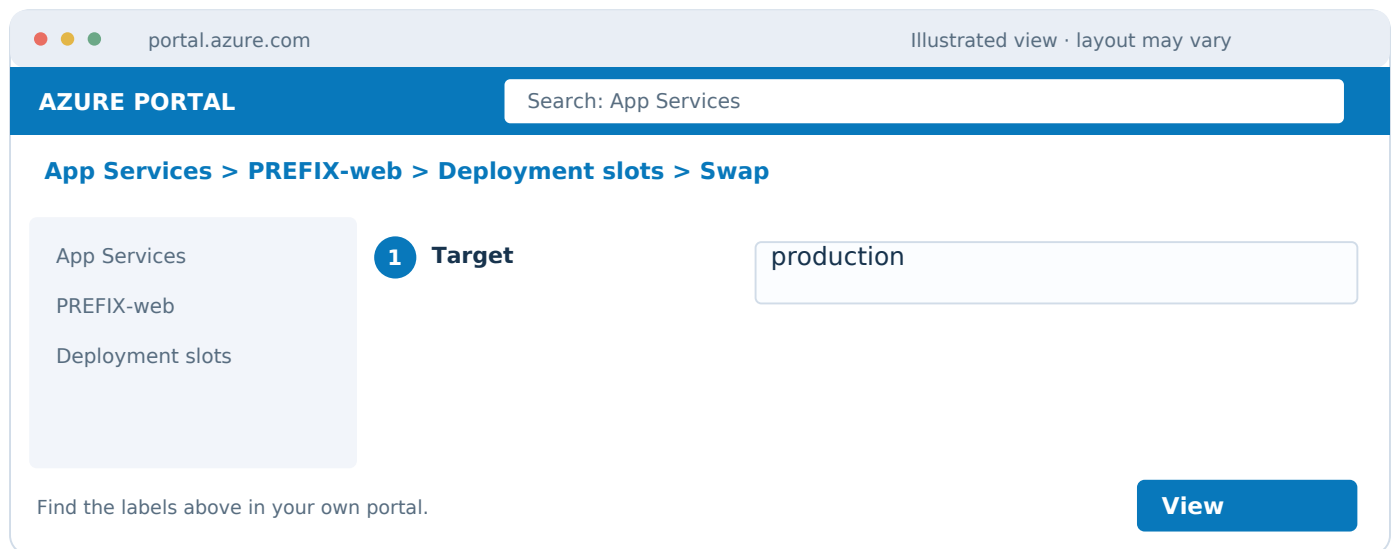
-n · Short form of --name in these Azure commands.

\$NAME / \${NAME} · Read a variable from this Bash session. Source lab.env first when these are lab names.

Backslash at line end · Continue the same Bash command on the next line. No spaces may follow the backslash.

5B.01 · ACTION 3 OF 5

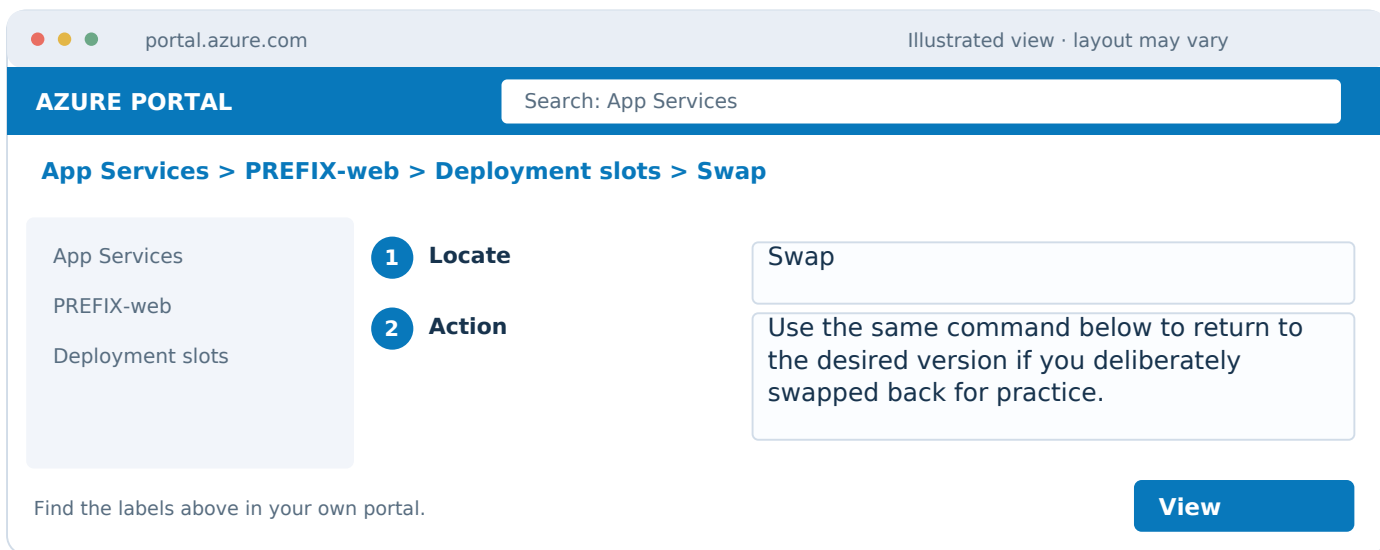
AZURE PORTAL · App Services → PREFIX-web → Deployment slots → Swap



Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Check /health and version on production.
2. Record time from starting the drill to restored healthy version.

5B.01 · ACTION 4 OF 5**AZURE PORTAL · App Services → PREFIX-web → Deployment slots → Swap**

The screenshot shows the Azure Portal interface. At the top, there's a search bar with 'App Services' entered. Below the search bar, the breadcrumb navigation is 'App Services > PREFIX-web > Deployment slots > Swap'. On the left, a sidebar lists 'App Services', 'PREFIX-web', and 'Deployment slots'. In the main area, there are two numbered steps: '1 Locate' and '2 Action'. The 'Swap' button is highlighted. A text box explains: 'Use the same command below to return to the desired version if you deliberately swapped back for practice.' At the bottom right, there is a 'View' button.

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: App Services

App Services > PREFIX-web > Deployment slots > Swap

App Services
PREFIX-web
Deployment slots

1 Locate

2 Action

Swap

Use the same command below to return to the desired version if you deliberately swapped back for practice.

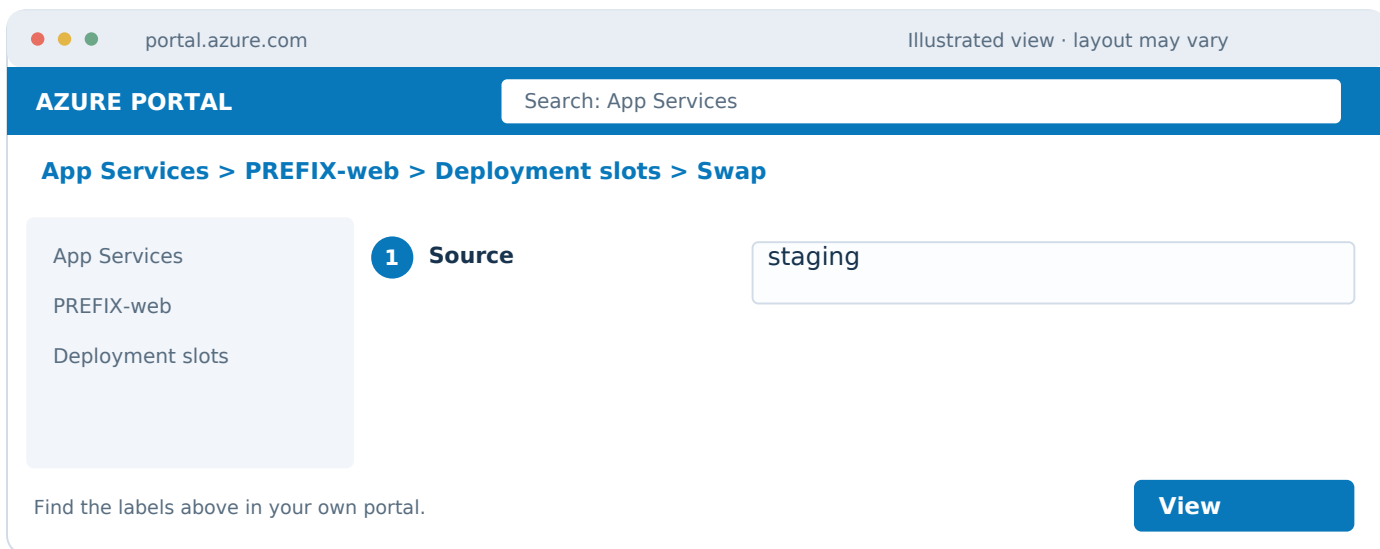
Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. Use the same command below to return to the desired version if you deliberately swapped back for practice.
2. Check before each swap; the command exchanges the slots.

5B.01 · ACTION 5 OF 5**AZURE PORTAL · App Services → PREFIX-web → Deployment slots → Swap**

The screenshot shows the Azure Portal interface. At the top, there's a search bar with 'App Services' entered. Below the search bar, the breadcrumb navigation is 'App Services > PREFIX-web > Deployment slots > Swap'. On the left, a sidebar lists 'App Services', 'PREFIX-web', and 'Deployment slots'. In the main area, there is a numbered step '1 Source'. A text box contains the word 'staging'. At the bottom right, there is a 'View' button.

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: App Services

App Services > PREFIX-web > Deployment slots > Swap

App Services
PREFIX-web
Deployment slots

1 Source

staging

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. If staging has since been overwritten, redeploy a retained known-good artifact and test it before promotion.
2. Do not assume the old version is still there.

You should see

You demonstrate rollback and identify exactly which version is live.

If it does not work

A slot swap does not reverse a database migration or external side effect. Rollback needs an application and data compatibility plan. [R23]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

5B.02 Add a feature flag without rebuilding code

AZ-400 · CORE + design exercises

Why this task: A feature flag controls a behavior separately from deploying the application.

5B.02 · ACTION 1 OF 6

AZURE PORTAL · App Configuration → Create

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: App Configuration

App Configuration > Create

App Configuration

Create

1 Group rg-PREFIX-core

2 Name PREFIX-config

3 Tier Free if eligible, otherwise review cost

Find the labels above in your own portal. **Review + create**

Illustrated navigation view · use your own saved names and IDs

Why this action: A feature flag controls a behavior separately from deploying the application.

1. Search App Configuration > Create.
2. Use rg-PREFIX-core, name PREFIX-config and an available Free tier if eligible; otherwise inspect the charge before creating.

5B.02 · ACTION 2 OF 6**AZURE PORTAL · PREFIX-config → IAM, then Feature manager → Create**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL

PREFIX-config > IAM, then Feature manager > Create

PREFIX-config
IAM, then Feature manager
Create

1 User role
2 Feature
3 Enabled

App Configuration Data Owner
NewBanner
Yes

Find the labels above in your own portal. **Create**

Illustrated navigation view · use your own saved names and IDs

Why this action: A feature flag controls a behavior separately from deploying the application.

1. Open IAM and give your user App Configuration Data Owner for the exercise.
2. Open Feature manager > Create.
3. Flag name NewBanner; enable it and save.

5B.02 · ACTION 3 OF 6**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

The dark code panel below is the terminal/editor view for this action.

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Run the block below to read the flag and set NEW_BANNER on staging.
2. Open staging home and confirm new_banner is true.

USE IN: BASH · COPY ONLY THE CODE

```
FLAG=$(az appconfig feature show -n "$CONFIG" --feature NewBanner \
  --auth-mode login --query enabled -o tsv)
test "$FLAG" = true || test "$FLAG" = false
az webapp config appsettings set -g "$RG" -n "$APP" --slot staging \
  --slot-settings NEW_BANNER="$FLAG" -o none
```

What it means: The explicit true/false check avoids silently using an empty or malformed result. --slot-settings keeps the feature value with this slot.

--feature · The feature flag name stored in App Configuration.

--auth-mode · login uses your signed-in Entra identity for this data operation.

--query · Select or transform returned fields with a JMESPath expression; this does not itself change the resource.

--slot · Target the named deployment slot, such as staging.

--slot-settings · Set an app setting and mark it as sticky to this slot.

-g · Short form of --resource-group.

-n · Short form of --name in these Azure commands.

-o · Azure CLI output format: table, tsv or none.

\$NAME / \${NAME} · Read a variable from this Bash session. Source lab.env first when these are lab names.

\$(command) · Run the inner command and use its output as the value.

Backslash at line end · Continue the same Bash command on the next line. No spaces may follow the backslash.

5B.02 · ACTION 4 OF 6

AZURE PORTAL · PREFIX-config → Feature manager → NewBanner

The screenshot shows the Azure Portal interface. At the top, there's a search bar with 'PREFIX-config' entered. Below the search bar, the breadcrumb navigation is 'PREFIX-config > Feature manager > NewBanner'. On the left, a sidebar shows a list of items: 'PREFIX-config', 'Feature manager', and 'NewBanner' (which is highlighted). To the right of the sidebar, there are two numbered steps: '1 Enabled' and '2 Then'. The 'Enabled' step has a dropdown menu set to 'Off'. The 'Then' step has a text input field containing 'Repeat the bridge command from Action 3'. At the bottom right, there is a blue 'Apply' button. Below the main content area, there is a note: 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Disable NewBanner in Feature manager and run the same block again.
2. Confirm false without rebuilding or changing app.py.

5B.02 · ACTION 5 OF 6

SEE THE IDEA · This lab refreshes the flag when you run the bridge

A SMALL DECISION MAP

This lab refreshes the flag when you run the bridge

- | | |
|----------------------------|------------------|
| 1 App Configuration | Stores the flag |
| 2 Bridge command | Copies its value |
| 3 App setting | NEW_BANNER |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. This simple lab synchronizes the flag into an app setting when you run the command.
2. It does not implement continuous runtime refresh.
3. A real-time design uses the App Configuration provider/feature-management SDK and a refresh strategy.

5B.02 · ACTION 6 OF 6

SEE THE IDEA · Add a feature flag without rebuilding code → Explain the decision

A SMALL DECISION MAP

Add a feature flag without rebuilding code > Explain the decision

- | | | |
|----------|---------------|---|
| 1 | Locate | Explain the decision |
| 2 | Action | DESIGN targeted rollout: enable for internal staff first, then a small cohort, then all users, while watching errors. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. DESIGN targeted rollout: enable for internal staff first, then a small cohort, then all users, while watching errors.
2. Record how you would identify the cohort and turn it off quickly.

You should see

The application behavior changes from a centrally stored feature setting.

If it does not work

Key-value reader access and App Configuration resource management are separate permissions. Wait for role propagation. [R38]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

Approach	OneLab explanation and trade-off
Blue-green	Keep two complete versions; switch traffic when the new one is healthy. Fast reversal, extra capacity. Slots approximate this deployment pattern.
Canary / progressive exposure	Send a small percentage to the new version, compare health, then increase. Stop if error or latency limits are exceeded.
Ring rollout	Release to named groups such as operators, pilot users, then everyone. Membership matters more than a random percentage.
Rolling deployment	Replace batches of instances while others keep serving. Mixed versions must remain compatible.
A/B test	Compare user outcomes for two variants; it is an experiment, not automatically a safe-release check.
Hotfix	Branch from the deployed release if necessary, make the smallest fix, run required checks, release with recorded approval and merge the fix into ongoing work.

5B.03 Practise safe database change ordering

AZ-400 · CORE + design exercises

Why this task: Application rollback is easier when schema changes preserve compatibility.

5B.03 · ACTION 1 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: Application rollback is easier when schema changes preserve compatibility.

1. Run scripts/migrate.py twice.
2. It creates a local SQLite tickets table and adds a status column only if missing.
3. Both runs should finish successfully.

USE IN: BASH · COPY ONLY THE CODE

```
python scripts/migrate.py
python scripts/migrate.py
```

What it means: Expected columns include id and status on both runs. The with block commits successful changes and closes the database connection.

5B.03 · ACTION 2 OF 5

CLOUD SHELL · Cloud Shell → Bash → scripts/migrate.py

The image shows a simulated Azure Cloud Shell interface. At the top, the browser address bar shows 'portal.azure.com' and a note 'Illustrated view · layout may vary'. Below this is a green header bar with 'CLOUD SHELL' and a search bar containing 'Cloud Shell'. The main content area shows a breadcrumb path: 'Cloud Shell > Bash > scripts/migrate.py'. On the left, a file explorer shows 'Cloud Shell', 'Bash', and 'scripts/migrate.py' (highlighted). To the right of the explorer, two numbered steps are listed: '1 Locate' and '2 Action'. The 'scripts/migrate.py' file is shown in a text box. Below the text box, a description reads: 'Inspect the file: the migration checks existing columns before adding one.' At the bottom right, there is a green 'Run' button. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Application rollback is easier when schema changes preserve compatibility.

1. Inspect the file: the migration checks existing columns before adding one.
2. This is why the second run is safe.

5B.03 · ACTION 3 OF 5

SEE THE IDEA · Practise safe database change ordering → Explain the decision

A SMALL DECISION MAP

Practise safe database change ordering > Explain the decision

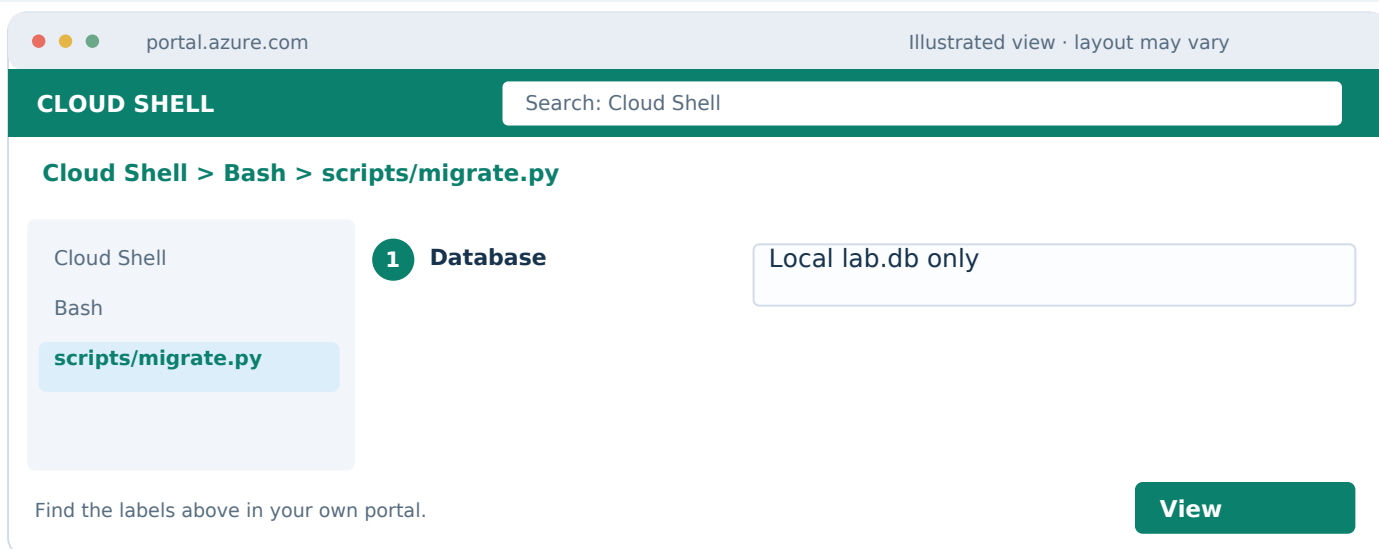
- 1 **Locate** Explain the decision
- 2 **Action** DESIGN real deployment order: verify backup/recovery, apply a backward-compatible schema expansion, deploy compatible app code, migrate data, and remove old schema only after old code is no longer needed.

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

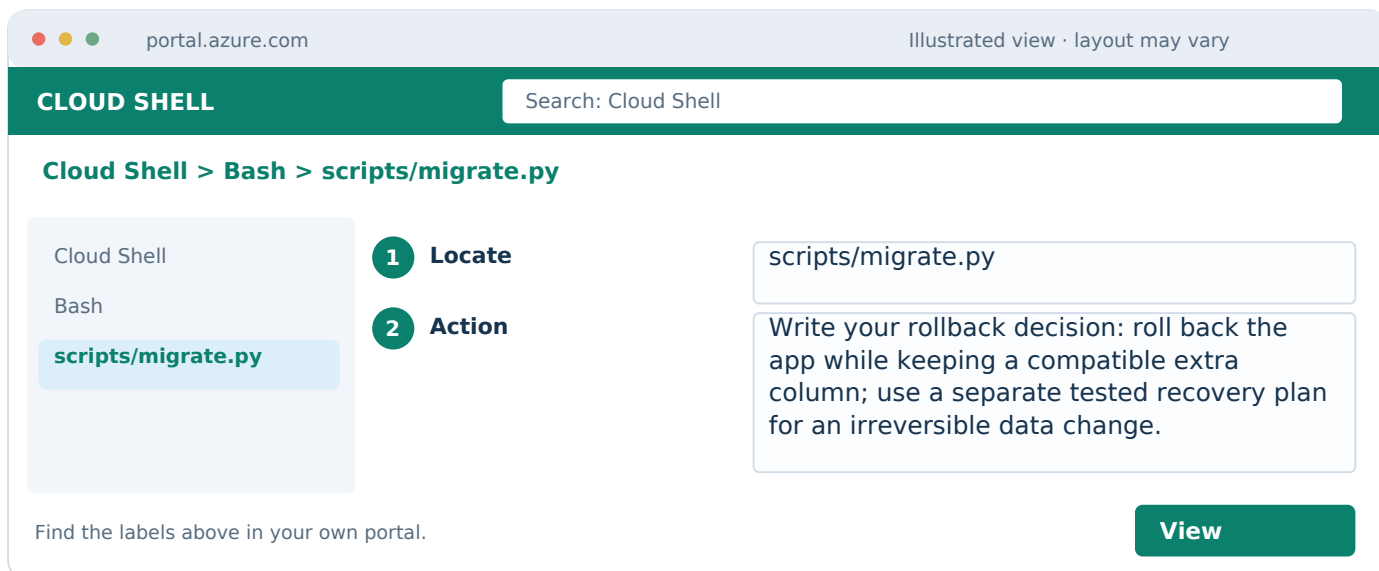
1. DESIGN real deployment order: verify backup/recovery, apply a backward-compatible schema expansion, deploy compatible app code, migrate data, and remove old schema only after old code is no longer needed.

5B.03 · ACTION 4 OF 5**CLOUD SHELL · Cloud Shell → Bash → scripts/migrate.py**

Illustrated navigation view · use your own saved names and IDs

Why this action: Application rollback is easier when schema changes preserve compatibility.

1. Place a real database migration in a controlled deployment job with private database connectivity and a least-privilege identity.
2. Avoid running competing migration jobs concurrently.

5B.03 · ACTION 5 OF 5**CLOUD SHELL · Cloud Shell → Bash → scripts/migrate.py**

Illustrated navigation view · use your own saved names and IDs

Why this action: Application rollback is easier when schema changes preserve compatibility.

1. Write your rollback decision: roll back the app while keeping a compatible extra column; use a separate tested recovery plan for an irreversible data change.

You should see

The local migration is repeatable and you can explain why data recovery differs from app rollback.

If it does not work

lab.db is disposable local demonstration data and is ignored by Git. This exercise does not claim a production SQL database was provisioned.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

5C · Use GitHub with the same application

Planning estimate: 125 minutes. Finish with: GitHub CI, a manual federated staging deployment and traceable work/notifications.

5C.01 Create the second remote and run GitHub CI

AZ-400 · CORE + account-dependent features

Why this task: AZ-400 includes GitHub as well as Azure DevOps. Reuse the same application so you can compare the tools.

5C.01 · ACTION 1 OF 6

GITHUB · Your profile → + → New repository

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB Search: Your profile

Your profile > + > New repository

Your profile
+
New repository

1 Name onelab

2 Visibility Private by default

3 Initialize README Off

4 Second remote github

Find the labels above in your own portal.

Create repository

Illustrated navigation view · use your own saved names and IDs

Why this action: AZ-400 includes GitHub as well as Azure DevOps. Reuse the same application so you can compare the tools.

1. On GitHub select + > New repository.
2. Name onelab.
3. Choose Private, or Public only if you intentionally want to publish this dummy training code.
4. Do not initialize another README.

5C.01 · ACTION 2 OF 6

GITHUB · For HTTPS push, open your profile → Settings → Developer settings → Personal access tokens → Fine-grained tokens

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB Search: For HTTPS push, open your profile

For HTTPS push, open your profile > Settings > Developer settings > Personal access tokens > Fine-grained tokens

For HTTPS push, open your profile

Settings

Developer settings

1 Locate

2 Action

Fine-grained tokens

For HTTPS push, open your profile > Settings > Developer settings > Personal access tokens > Fine-grained tokens.

Find the labels above in your own portal.

Create

Illustrated navigation view · use your own saved names and IDs

Why this action: AZ-400 includes GitHub as well as Azure DevOps. Reuse the same application so you can compare the tools.

1. For HTTPS push, open your profile > Settings > Developer settings > Personal access tokens > Fine-grained tokens.
2. Create a short-lived token for this repository only with Contents Read and write and Workflows Read and write.

5C.01 · ACTION 3 OF 6

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: AZ-400 includes GitHub as well as Azure DevOps. Reuse the same application so you can compare the tools.

1. In Cloud Shell switch to main and pull the completed Azure Repos changes.
2. Add a second remote named github with the repository HTTPS URL.
3. Push main; use the token as the password when prompted, never as part of the URL.
4. Do not change main to track github: keep origin/main as your regular pull source.
5. The script below sets that explicitly.

USE IN: BASH · COPY ONLY THE CODE

```
git switch main
git pull --ff-only origin main
git remote add github 'https://github.com/REPLACE_OWNER/onelab.git'
git push github main
git branch --set-upstream-to=origin/main main
```

What it means: origin and github are local names for two remote URLs. Explicitly name the desired remote when pushing to avoid confusion.

--ff-only · Pull only if Git can fast-forward; stop instead of creating an unexpected merge.

--set-upstream-to · Set the remote-tracking branch used by a later ordinary pull.

5C.01 · ACTION 4 OF 6

GITHUB · GitHub → repository → Actions → OneLab CI



The screenshot shows the GitHub Actions configuration page for a repository named 'onelab'. The breadcrumb navigation is 'GitHub > repository > Actions > OneLab CI'. On the left, there is a sidebar with 'GitHub', 'repository', and 'Actions' links. The main area has two numbered steps: '1 Name' with a text input containing 'onelab', and '2 Second remote' with a text input containing 'github'. At the bottom right is a 'Run' button. A note at the bottom left says 'Find the labels above in your own portal.'

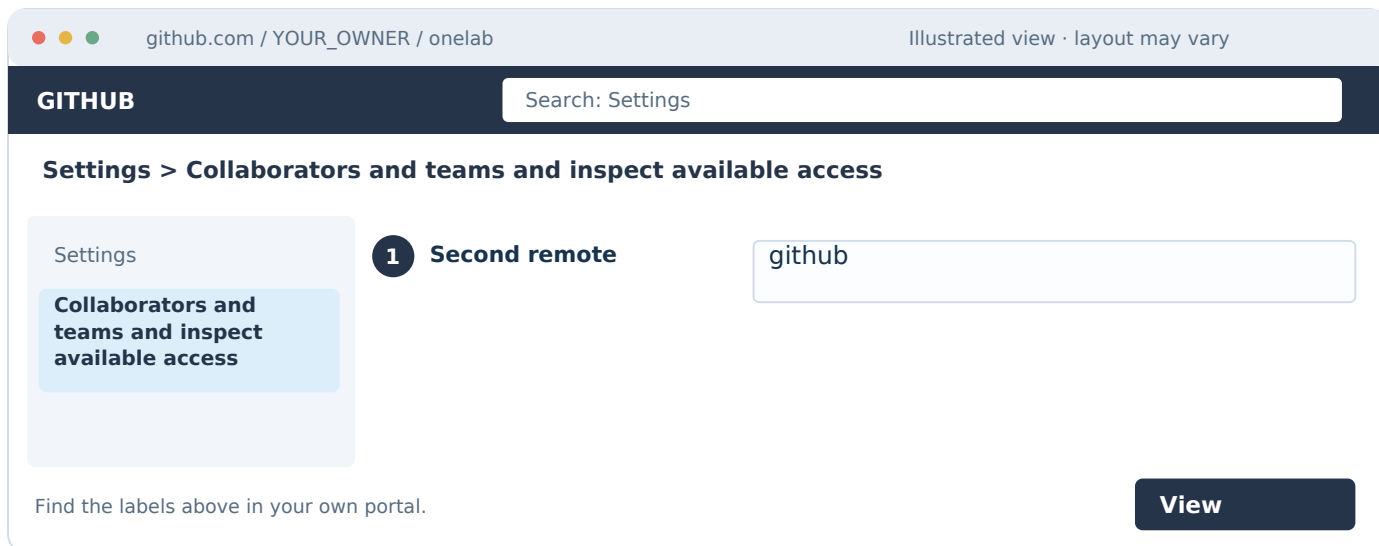
Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open GitHub > repository > Actions > OneLab CI.
2. Inspect checkout, Python setup, test run and uploaded test-results artifact.

5C.01 · ACTION 5 OF 6

GITHUB · Settings → Collaborators and teams and inspect available access



The screenshot shows the GitHub Settings page for a repository named 'onelab'. The breadcrumb navigation is 'Settings > Collaborators and teams and inspect available access'. On the left, there is a sidebar with 'Settings' and 'Collaborators and teams and inspect available access' links. The main area has one numbered step: '1 Second remote' with a text input containing 'github'. At the bottom right is a 'View' button. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Open Settings > Collaborators and teams and inspect available access.
2. A GitHub organization can manage teams and outside collaborators; a personal repository has simpler collaborator permissions.

5C.01 · ACTION 6 OF 6**SEE THE IDEA · Create the second remote and run GitHub CI → Explain the decision****A SMALL DECISION MAP****Create the second remote and run GitHub CI > Explain the decision**

- | | | |
|----------|---------------|---|
| 1 | Locate | Explain the decision |
| 2 | Action | Keep a clear source-of-truth rule: use Azure Repos for the main lab workflow and deliberately push accepted main changes to GitHub. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: AZ-400 includes GitHub as well as Azure DevOps. Reuse the same application so you can compare the tools.

1. Keep a clear source-of-truth rule: use Azure Repos for the main lab workflow and deliberately push accepted main changes to GitHub.
2. Two remotes do not automatically stay synchronized.

You should see

The same source code passes a GitHub Actions workflow.

If it does not work

The CodeQL workflow may fail or be unavailable in a private repository without the required product/plan. Mark that exercise as unavailable or use an intentionally public, sanitized training repo; do not expose personal code merely to get a green check. [R39]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

5C.02 Protect a branch and track work in GitHub

AZ-400 · CORE + account-dependent features

Why this task: A branch policy has a similar purpose in both platforms but different settings.

5C.02 · ACTION 1 OF 5

GITHUB · onelab → Settings → Rules → Rulesets

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB

Search: onelab

onelab > Settings > Rules > Rulesets

onelab

Settings

Rules

Rulesets

1 Target

2 Require

3 Status check

4 Bypass

main

Pull request

OneLab CI test job

Only the intended policy

Find the labels above in your own portal.

Save changes

Illustrated navigation view · use your own saved names and IDs

Why this action: A branch policy has a similar purpose in both platforms but different settings.

1. Open Settings > Rules > Rulesets, or Branches > Branch protection rules.
2. Target main, require pull requests and the CI status check, and restrict bypass according to your available plan.

5C.02 · ACTION 2 OF 5**GITHUB · onelab → Issues → New issue**

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB Search: onelab

onelab > Issues > New issue

onelab

Issues

New issue

1 Title Show a release note

2 Include Description and acceptance criteria

Find the labels above in your own portal.

Create

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Open Issues > New issue.
2. Title Show a release note; add a description and acceptance criteria.
3. Add labels such as enhancement or bug.

5C.02 · ACTION 3 OF 5**GITHUB · Your profile or organization → Projects → New project**

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB Search: Your profile or organization

Your profile or organization > Projects > New project

Your profile or organization

Projects

New project

1 Project OneLab work

2 Item Your new issue

3 Statuses Todo, In progress, Done

Find the labels above in your own portal.

Create

Illustrated navigation view · use your own saved names and IDs

Why this action: A branch policy has a similar purpose in both platforms but different settings.

1. Create a GitHub Project from your profile or organization > Projects > New project.
2. Add this issue, with Todo, In progress and Done statuses.

5C.02 · ACTION 4 OF 5**GITHUB · onelab → README.md → Edit → Commit changes**

github.com / YOUR_OWNER / onelab Illustrated view · layout may vary

GITHUB Search: onelab

onelab > README.md > Edit > Commit changes

onelab

README.md

Edit

1 Branch New docs feature branch

2 PR text Closes #YOUR_ISSUE_NUMBER

Find the labels above in your own portal. **Propose changes**

Illustrated navigation view · use your own saved names and IDs

Why this action: A branch policy has a similar purpose in both platforms but different settings.

1. Create a feature branch, make a small documentation change, open a PR referencing the issue, and inspect the required checks before merging.
2. Use the GitHub web editor for a docs-only change: open README.md > pencil > make one sentence clearer > Commit changes > Create a new branch.
3. Open Compare & pull request, write Closes #ISSUE_NUMBER, and submit.
4. After this GitHub-only exercise, carry the same documentation change back through an Azure Repos PR before the next synchronization.

5C.02 · ACTION 5 OF 5**GITHUB · Your OneLab Project → Insights → New chart**

github.com / YOUR_OWNER / onelab Illustrated view · layout may vary

GITHUB Search: Your OneLab Project

Your OneLab Project > Insights > New chart

Your OneLab Project

Insights

New chart

1 Group by Status

2 Feature access Depends on your plan

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Open the Project > Insights and create an available chart grouping items by status.
2. Some chart/history features are plan-dependent; record the limitation if the option is absent.

You should see

You can connect an issue, PR, CI result and project status.

If it does not work

GitHub Flow means a short branch, PR/review, merge and deploy. Protected-branch features for private repositories depend on the plan.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

5C.03 Use OIDC to deploy only to staging

AZ-400 · CORE + account-dependent features

Why this task: Federation ties one GitHub environment to one Azure identity without a long-lived Azure password.

5C.03 · ACTION 1 OF 8

AZURE PORTAL · Managed identities → Create

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Managed identities

Managed identities > Create

Managed identities

Create

- 1 Group** rg-PREFIX-core
- 2 Name** github-onelab
- 3 Record** Client ID and Tenant ID

Find the labels above in your own portal.

Review + create

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. In Azure Portal search Managed identities > Create.
2. Use rg-PREFIX-core; name github-onelab.
3. Open it and record Client ID and Tenant ID.

5C.03 · ACTION 2 OF 8

AZURE PORTAL · rg-PREFIX-core → Access control (IAM) → Add role assignment

The screenshot shows the Azure Portal interface. At the top, the breadcrumb navigation is 'rg-PREFIX-core > Access control (IAM) > Add role assignment'. On the left sidebar, 'rg-PREFIX-core' is selected, and 'Add role assignment' is highlighted. The main area has two numbered steps: 1. Role, with 'Contributor' selected in a dropdown; 2. Member, with 'github-onelab managed identity' selected in a dropdown. A 'Review + assign' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. On rg-PREFIX-core > IAM, grant this managed identity Contributor at this lab group.
2. For a real environment narrow the role and resource scope further.

5C.03 · ACTION 3 OF 8

AZURE PORTAL · github-onelab → Federated credentials → Add credential

The screenshot shows the Azure Portal interface. At the top, the breadcrumb navigation is 'github-onelab > Federated credentials > Add credential'. On the left sidebar, 'github-onelab' is selected, and 'Add credential' is highlighted. The main area has three numbered steps: 1. Scenario, with 'GitHub Actions deploying Azure resources' selected; 2. Organization, with 'Your exact GitHub owner' selected; 3. Repository, with 'onelab' selected. A 'Next' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Federation ties one GitHub environment to one Azure identity without a long-lived Azure password.

1. Open the identity > Federated credentials > Add credential.
2. Choose GitHub Actions deploying Azure resources.
3. Fill in your exact GitHub organization/owner and repository onelab.

5C.03 · ACTION 4 OF 8

AZURE PORTAL · Add federated credential → Entity

The screenshot shows the Azure Portal interface for adding a federated credential. The browser address bar displays 'portal.azure.com'. The page title is 'AZURE PORTAL'. A search bar contains the text 'Add federated credential'. The main heading is 'Add federated credential > Entity'. On the left, a sidebar shows 'Add federated credential' and 'Entity' (selected). The main content area has three numbered steps: 1. Entity type (Environment), 2. Environment (github-staging), and 3. Audience (Use the offered Azure token-exchange value). An 'Add' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Add federated credential

Add federated credential > Entity

Add federated credential

Entity

1 Entity type Environment

2 Environment github-staging

3 Audience Use the offered Azure token-exchange value

Find the labels above in your own portal.

Add

Illustrated navigation view · use your own saved names and IDs

Why this action: Federation ties one GitHub environment to one Azure identity without a long-lived Azure password.

1. Choose entity type Environment and name github-staging.
2. Use the audience offered for Azure token exchange and save.
3. The subject must match this environment, not the main branch subject.

5C.03 · ACTION 5 OF 8

GITHUB · onelab → Settings → Environments → New environment

The screenshot shows the GitHub 'New environment' page. The browser address bar displays 'github.com / YOUR_OWNER / onelab'. The page title is 'GITHUB'. A search bar contains the text 'onelab'. The main heading is 'onelab > Settings > Environments > New environment'. On the left, a sidebar shows 'onelab', 'Settings', and 'Environments' (selected). The main content area has three numbered steps: 1. Name (github-staging), 2. Deployment branches (main), and 3. Reviewer (If your plan supports it). A 'Configure environment' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB Search: onelab

onelab > Settings > Environments > New environment

onelab

Settings

Environments

1 Name github-staging

2 Deployment branches main

3 Reviewer If your plan supports it

Find the labels above in your own portal.

Configure environment

Illustrated navigation view · use your own saved names and IDs

Why this action: Federation ties one GitHub environment to one Azure identity without a long-lived Azure password.

1. In GitHub repository > Settings > Environments > New environment, create github-staging.
2. Restrict deployment branches to main.
3. Add a reviewer if your plan supports it.

5C.03 · ACTION 6 OF 8

GITHUB · github-staging → Environment variables

github.com / YOUR_OWNER / onelab Illustrated view · layout may vary

GITHUB

github-staging > Environment variables

github-staging

Environment variables

1	AZURE_CLIENT_ID	github-onelab client ID
2	AZURE_TENANT_ID	Your tenant ID
3	AZURE_SUBSCRIPTION_ID	Your subscription ID
4	AZURE_RESOURCE_GROUP	rg-PREFIX-core
5	AZURE_APP_NAME	PREFIX-web

Find the labels above in your own portal.

Add variable

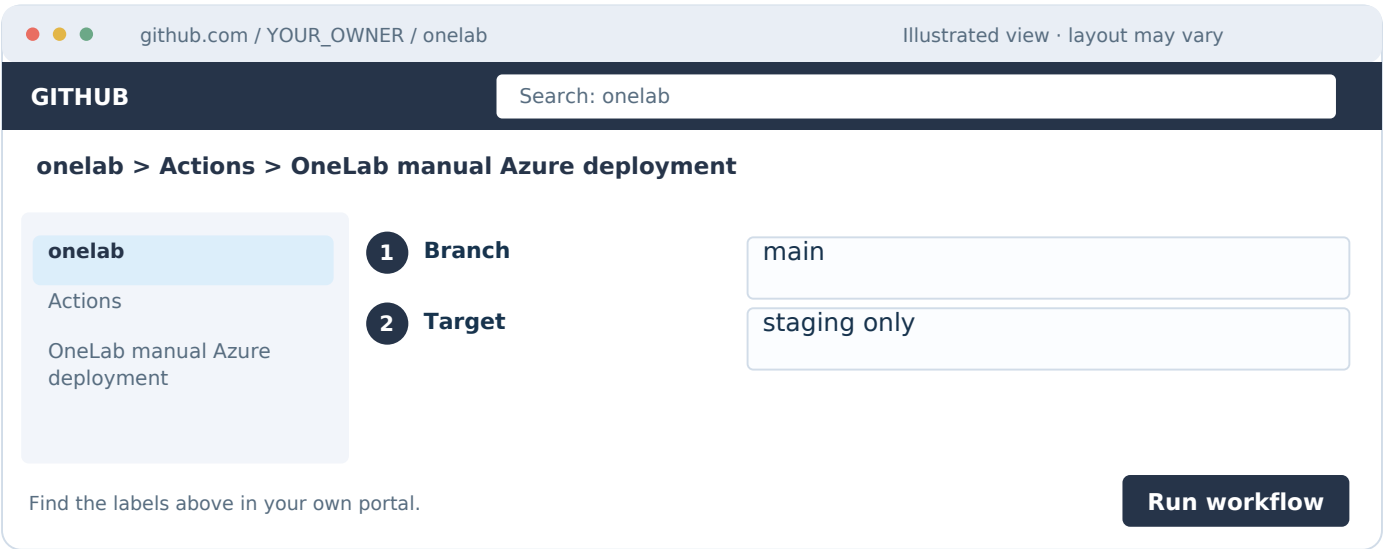
Illustrated navigation view · use your own saved names and IDs

Why this action: Federation ties one GitHub environment to one Azure identity without a long-lived Azure password.

1. Under that environment > Variables add AZURE_CLIENT_ID, AZURE_TENANT_ID, AZURE_SUBSCRIPTION_ID, AZURE_RESOURCE_GROUP and AZURE_APP_NAME with the exact values.

5C.03 · ACTION 7 OF 8

GITHUB · onelab → Actions → OneLab manual Azure deployment



Illustrated navigation view · use your own saved names and IDs

Why this action: Federation ties one GitHub environment to one Azure identity without a long-lived Azure password.

1. Open Actions > OneLab manual Azure deployment > Run workflow on main.
2. Inspect azure/login and the staging deployment.
3. Check staging /health.
4. This workflow never swaps production.

5C.03 · ACTION 8 OF 8

SEE THE IDEA · Use OIDC to deploy only to staging → Explain the decision

A SMALL DECISION MAP

Use OIDC to deploy only to staging > Explain the decision

1 Locate	Explain the decision
2 Action	Do not run the GitHub and Azure DevOps deployment workflows at the same time; they target the same staging slot.

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Federation ties one GitHub environment to one Azure identity without a long-lived Azure password.

1. Do not run the GitHub and Azure DevOps deployment workflows at the same time; they target the same staging slot.

You should see

A manually triggered GitHub workflow deploys to staging using a short-lived identity token.

If it does not work

AADSTS federation errors usually mean a mismatched owner, repo, branch/environment, issuer or audience. id-token: write allows requesting the OIDC token; Azure RBAC still decides resource access. [R40]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

5C.04 Connect Boards, GitHub, notifications and hooks

AZ-400 · CORE + account-dependent features

Why this task: Feedback should reach the people who can act on it.

5C.04 · ACTION 1 OF 7

AZURE DEVOPS · OneLab → Project settings → Boards → GitHub connections

The screenshot shows the Azure DevOps web interface. At the top, the browser address bar displays 'dev.azure.com / YOUR_ORG / OneLab'. The page header includes 'AZURE DEVOPS' and a search bar containing 'OneLab'. The breadcrumb navigation path is 'OneLab > Project settings > Boards > GitHub connections'. On the left, a sidebar lists 'OneLab', 'Project settings', and 'Boards'. The main content area is divided into three numbered steps: 1. Connection, 2. Repository, and 3. Link text. Step 1 has a text input field for 'Your GitHub account'. Step 2 has a dropdown menu currently showing 'onelab only'. Step 3 has a text input field for 'AB#YOUR_WORK_ITEM_ID'. At the bottom left, a note says 'Find the labels above in your own portal.' At the bottom right, there is a prominent purple 'Connect' button.

Illustrated navigation view · use your own saved names and IDs

Why this action: Feedback should reach the people who can act on it.

1. In Azure DevOps open Project settings > Boards > GitHub connections > Connect your GitHub account.
2. Authorize the Azure Boards app only for the intended repository.

5C.04 · ACTION 2 OF 7**GITHUB · onelab → Pull requests → Your PR → Description**

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB Search: onelab

onelab > Pull requests > Your PR > Description

onelab
Pull requests
Your PR

1 Link text AB#YOUR_WORK_ITEM_ID

2 Then Verify Azure Boards Development link

Find the labels above in your own portal.

Save

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Create a GitHub commit or PR message containing AB#YOUR_WORK_ITEM_ID.
2. Open the Azure Boards work item and confirm the Development link.

5C.04 · ACTION 3 OF 7**AZURE DEVOPS · User settings → Notifications → New subscription**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: User settings

User settings > Notifications > New subscription

User settings
Notifications
New subscription

1 Category Build

2 Event Build fails

3 Project OneLab

4 Delivery Your own email

Find the labels above in your own portal.

Finish

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. In Azure DevOps use your notification settings to subscribe to failed OneLab builds and assigned work-item updates.
2. Send notifications only to your own test account.

5C.04 · ACTION 4 OF 7

YOUR BROWSER · Microsoft Teams → Your test channel → Apps

Your own browser tab

Illustrated view · layout may vary

YOUR BROWSER

Search: Microsoft Teams

Microsoft Teams > Your test channel > Apps

Microsoft Teams

Your test channel

Apps

1 App

2 Subscription

3 Plan

Azure Pipelines or Azure Boards

Your OneLab pipeline or area

DESIGN if Teams is unavailable

Find the labels above in your own portal.

Add

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. For a Teams extension, add the Azure Pipelines or Azure Boards app to a test channel that you control, sign in, and configure a subscription to the OneLab pipeline or area.
2. Generate one test event and verify it; then remove unwanted subscriptions.

5C.04 · ACTION 5 OF 7

AZURE DEVOPS · Project settings → Service hooks → Create subscription → Web Hooks

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS

Search: Project settings

Project settings > Service hooks > Create subscription > Web Hooks

Project settings

Service hooks

Create subscription

Web Hooks

1 Event

2 Build

3 Status

4 Receiver

Build completed

OneLab-Web

Failed

HTTPS endpoint YOU own

Find the labels above in your own portal.

Next

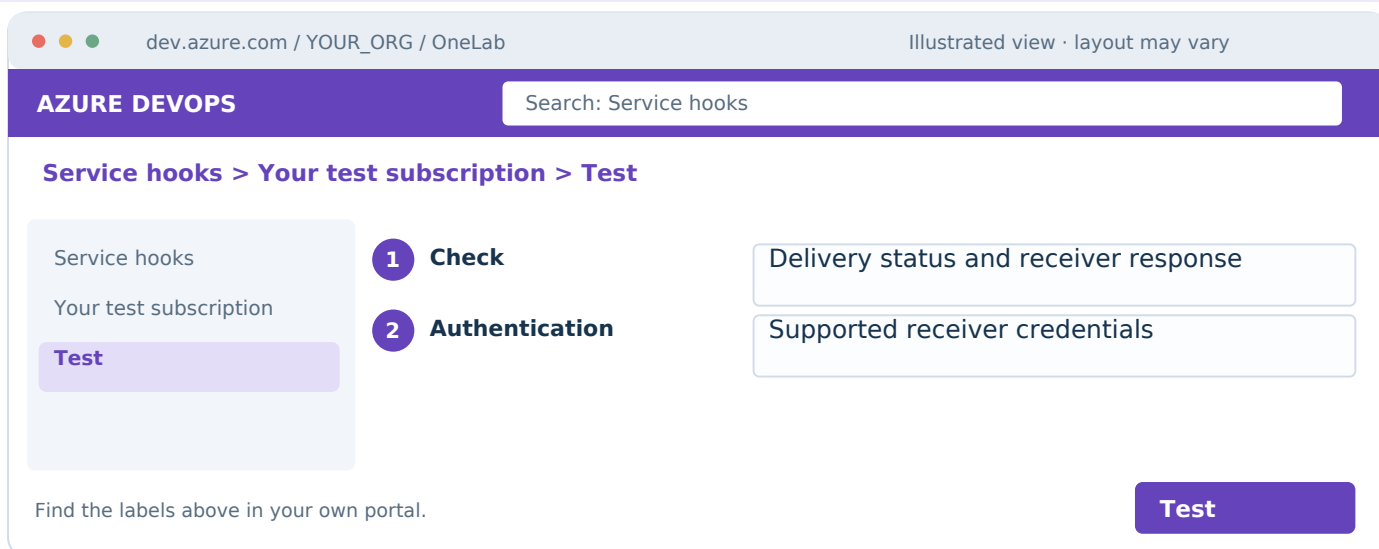
Illustrated navigation view · use your own saved names and IDs

Why this action: Feedback should reach the people who can act on it.

1. For a webhook extension, open Project settings > Service hooks > Create subscription > Web Hooks.
2. Choose Build completed, filter OneLab-Web and Failed, and use an HTTPS receiving endpoint you own.

5C.04 · ACTION 6 OF 7

AZURE DEVOPS · Service hooks → Your test subscription → Test



Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Use Test to inspect delivery status and response.
2. Protect the receiver with its supported authentication and minimize payload data.
3. Never send project events to an arbitrary public request-capture service.

5C.04 · ACTION 7 OF 7

SEE THE IDEA · No Teams account or webhook receiver?

A SMALL DECISION MAP

No Teams account or webhook receiver?

- | | | |
|---|------------------------|-------------------------------------|
| 1 | Record | Event, filter, destination and auth |
| 2 | Delivery design | Retries and duplicate handling |
| 3 | Status | DESIGN |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Without Teams or a receiver, record the event, filter, destination, authentication and retry/duplicate-handling design.
2. The app itself has no webhook receiver endpoint.

You should see

At least one relevant notification works; optional integrations are recorded accurately.

If it does not work

Webhooks deliver event messages; receivers should handle duplicates and failures safely. A Boards connection does not automatically create an Azure Pipelines connection to GitHub. [R41, R42]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

5C.05 Run Azure Pipelines directly from GitHub

AZ-400 · CORE + account-dependent features

Why this task: Repository hosting and pipeline execution do not have to be in the same product.

5C.05 · ACTION 1 OF 4

FILE / WIKI EDITOR · Project file → ci-only.yml

The dark code panel below is the terminal/editor view for this action.

Why this action: Repository hosting and pipeline execution do not have to be in the same product.

1. Open the supplied ci-only.yml in the GitHub repo and compare it with the test-only YAML below.
2. It is already included in the source ZIP.
3. If absent, add it and commit through the repository review process.

USE IN: FILE: CI-ONLY.YML · COPY ONLY THE CODE

```
trigger: none
pr:
  branches:
    include: [main]
pool:
  vmImage: ubuntu-24.04
steps:
- checkout: self
- template: pipelines/templates/test.yml
```

What it means: This reuses the same test template while omitting Azure deployment stages.

5C.05 · ACTION 2 OF 4

AZURE DEVOPS · Pipelines → New pipeline → GitHub → Existing YAML

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS Search: Pipelines

Pipelines > New pipeline > GitHub > Existing YAML

Pipelines
New pipeline
GitHub

1 Repository Your owner/onelab

2 Path /ci-only.yml

3 GitHub App access This repository only

Find the labels above in your own portal. **Continue**

Illustrated navigation view · use your own saved names and IDs

Why this action: Repository hosting and pipeline execution do not have to be in the same product.

1. In Azure DevOps > Pipelines > New pipeline choose GitHub, authorize the Azure Pipelines GitHub app for this repository, and select Existing YAML file > /ci-only.yml.

5C.05 · ACTION 3 OF 4

AZURE DEVOPS · Your GitHub-connected Azure Pipeline → Run

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS Search: Your GitHub-connected Azure Pipeline

Your GitHub-connected Azure Pipeline > Run

Your GitHub-connected Azure Pipeline
Run

1 Scope Tests only

2 Deployment None

Find the labels above in your own portal. **Run**

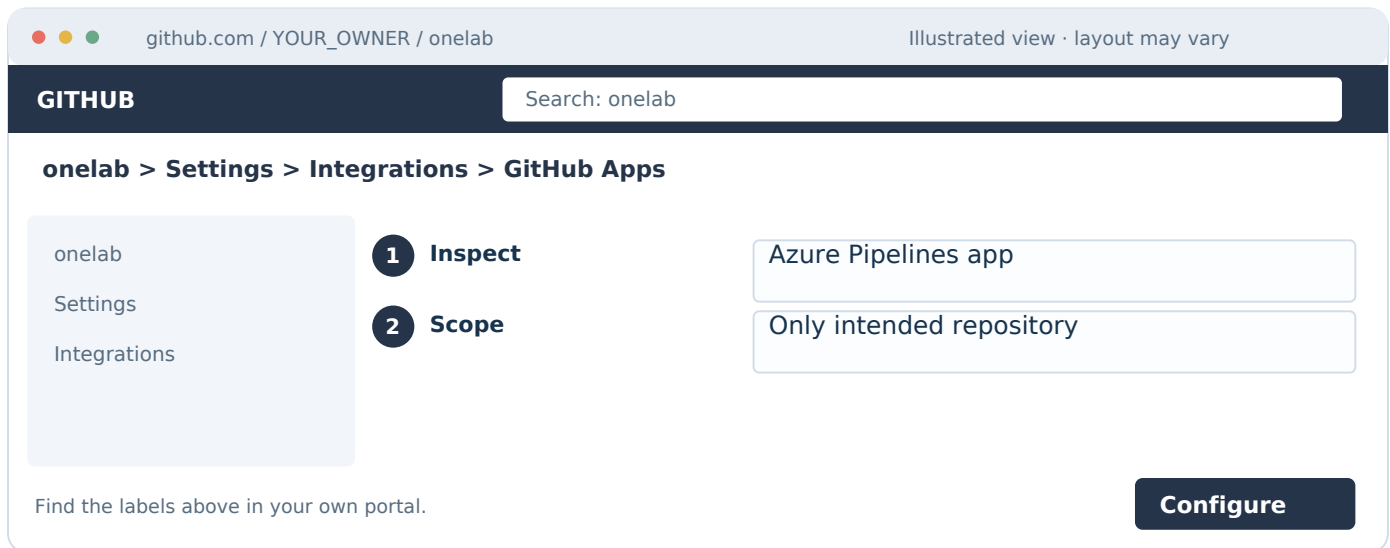
Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Run it and inspect the test result.
2. Keep this integration test-only so it does not compete with the main deployment pipeline.

5C.05 · ACTION 4 OF 4

GITHUB · onelab → Settings → Integrations → GitHub Apps



Illustrated navigation view · use your own saved names and IDs

Why this action: Repository hosting and pipeline execution do not have to be in the same product.

1. Open the repository's GitHub settings to inspect the installed app's repository access.
2. Remove unused connections when finished.

You should see

Azure Pipelines checks out and tests the GitHub repository.

If it does not work

For GitHub repositories, YAML pr triggers are supported. This differs from Azure Repos branch-policy validation. [R43]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

5D · Read the pipeline controls

Planning estimate: 45 minutes. Finish with: Retention, trigger, template and migration decisions recorded simply.

5D.01 Optimize without hiding failures

AZ-400 · CORE + self-hosted extension

Why this task: Use measurements to improve speed and reliability.

5D.01 · ACTION 1 OF 6

AZURE DEVOPS · OneLab → Pipelines → OneLab-Web → Analytics

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS

Search: OneLab

OneLab > Pipelines > OneLab-Web > Analytics

OneLab

Pipelines

OneLab-Web

1 Measure

2 Retain

3 Cache

Duration and real failures

Current and previous known-good run

Disposable downloaded dependencies

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Open OneLab-Web > Analytics or run history.
2. Record success/failure count, duration by stage and any repeatedly failing/flaky test.

5D.01 · ACTION 2 OF 6**AZURE DEVOPS · OneLab → Pipelines → OneLab-Web → Analytics**

The screenshot shows the Azure DevOps OneLab interface. At the top, the breadcrumb navigation is **OneLab > Pipelines > OneLab-Web > Analytics**. On the left, a sidebar lists the navigation path: OneLab, Pipelines, and OneLab-Web. The main content area shows the 'Action' step (2) with the title 'Analytics' and the description 'Separate deterministic failures from flaky tests caused by timing, shared state or external services.' A 'View' button is at the bottom right.

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: OneLab

OneLab > Pipelines > OneLab-Web > Analytics

OneLab
Pipelines
OneLab-Web

1 Locate

2 Action

Analytics

Separate deterministic failures from flaky tests caused by timing, shared state or external services.

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Separate deterministic failures from flaky tests caused by timing, shared state or external services.
2. Fix the cause; do not use unlimited retries as a test strategy.

5D.01 · ACTION 3 OF 6**AZURE DEVOPS · OneLab → Pipelines → OneLab-Web → Analytics**

The screenshot shows the Azure DevOps OneLab interface. At the top, the breadcrumb navigation is **OneLab > Pipelines > OneLab-Web > Analytics**. On the left, a sidebar lists the navigation path: OneLab, Pipelines, and OneLab-Web. The main content area shows the 'Action' step (2) with the title 'Analytics' and the description 'Use parallel jobs only for independent work.' A 'View' button is at the bottom right.

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: OneLab

OneLab > Pipelines > OneLab-Web > Analytics

OneLab
Pipelines
OneLab-Web

1 Locate

2 Action

Analytics

Use parallel jobs only for independent work.

Find the labels above in your own portal. **View**

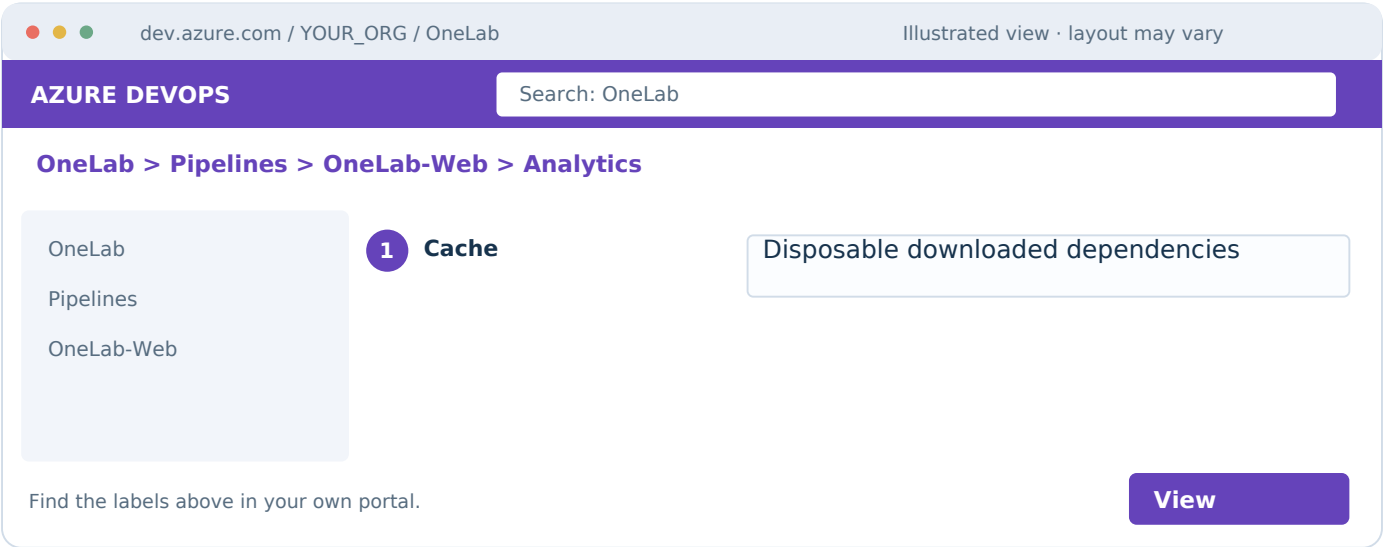
Illustrated navigation view · use your own saved names and IDs

Why this action: Use measurements to improve speed and reliability.

1. Use parallel jobs only for independent work.
2. Keep database migration before dependent app deployment and Staging before Production.

5D.01 · ACTION 4 OF 6

AZURE DEVOPS · OneLab → Pipelines → OneLab-Web → Analytics



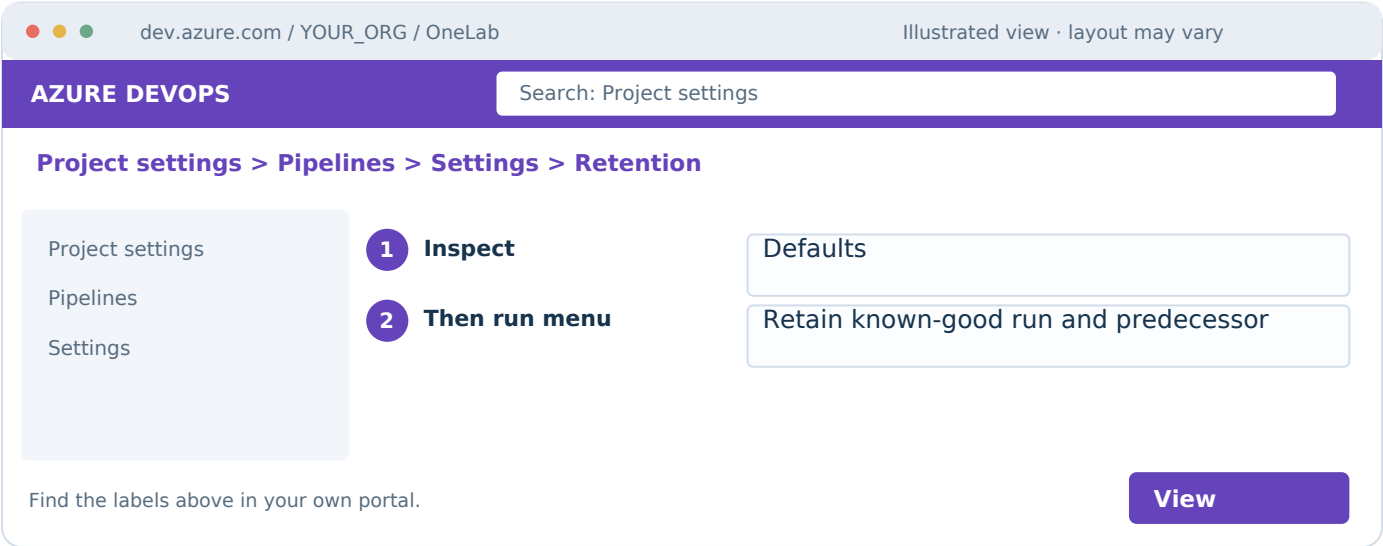
Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

- 1. Cache downloaded dependencies using a key that includes OS and dependency-file content.
- 2. A cache is disposable acceleration; an artifact is an output you rely on.

5D.01 · ACTION 5 OF 6

AZURE DEVOPS · Project settings → Pipelines → Settings → Retention



Illustrated navigation view · use your own saved names and IDs

Why this action: Use measurements to improve speed and reliability.

- 1. Open Project settings > Pipelines > Settings/Retention.
- 2. Inspect default retention, then retain the known-good production run and its predecessor using the run menu.

5D.01 · ACTION 6 OF 6

AZURE DEVOPS · Artifacts → onelab → Feed settings → Retention

The screenshot shows the Azure DevOps web interface. At the top, the browser address bar displays 'dev.azure.com / YOUR_ORG / OneLab'. The page title is 'AZURE DEVOPS' with a search bar containing 'Search: Artifacts'. The breadcrumb navigation is 'Artifacts > onelab > Feed settings > Retention'. On the left, a sidebar lists 'Artifacts', 'onelab', and 'Feed settings'. The main content area is titled '1 Protect' and contains a box labeled 'Release-view versions and rollback package'. At the bottom right, there is a 'View' button. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. In Artifacts feed settings inspect retention for packages and protect versions used by a release view.
2. Avoid deleting the rollback package to save a small amount of storage.

You should see

You can justify an optimization using duration, reliability and cost data.

If it does not work

A cancelled or expired run may lose its artifacts unless retention is configured. Keep a separate record of commit, package version/digest and deployment. [R52]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

5D.02 Understand trigger and template choices

AZ-400 · CORE + self-hosted extension

Why this task: Small syntax differences can change when and how automation runs.

5D.02 · ACTION 1 OF 6

SEE THE IDEA · Pipeline triggers, templates and migration

A SMALL DECISION MAP

Pipeline triggers, templates and migration

- | | | |
|---|----------|-----------------------------|
| 1 | CI | Push validation |
| 2 | PR | Policy or PR event |
| 3 | Schedule | Slower periodic checks |
| 4 | Template | Reusable reviewed structure |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Small syntax differences can change when and how automation runs.

1. Compare CI push triggers, PR validation, manual runs, scheduled runs and pipeline-completion triggers.
2. Use a nightly schedule for slower integration/security checks if needed.

5D.02 · ACTION 2 OF 6

SEE THE IDEA · Pipeline triggers, templates and migration

A SMALL DECISION MAP

Pipeline triggers, templates and migration

- | | | |
|---|----|-----------------|
| 1 | CI | Push validation |
|---|----|-----------------|

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Small syntax differences can change when and how automation runs.

1. Read dependsOn as execution ordering and condition as the decision to run.
2. Independent jobs can run in parallel if capacity exists.

5D.02 · ACTION 3 OF 6**SEE THE IDEA · Pipeline triggers, templates and migration****A SMALL DECISION MAP****Pipeline triggers, templates and migration**

- | | | |
|----------|-----------------|-----------------------------|
| 1 | Template | Reusable reviewed structure |
|----------|-----------------|-----------------------------|

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Small syntax differences can change when and how automation runs.

1. Variables hold values, variable groups share settings, parameters choose typed compile-time inputs, and templates reuse structure.
2. Do not pass a secret as a template parameter.

5D.02 · ACTION 4 OF 6**SEE THE IDEA · Pipeline triggers, templates and migration****A SMALL DECISION MAP****Pipeline triggers, templates and migration**

- | | | |
|----------|-----------|--------------------|
| 1 | CI | Push validation |
| 2 | PR | Policy or PR event |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. For a classic-to-YAML migration, inventory tasks, agent demands, variables, service connections, artifacts, approvals and triggers.
2. Export View YAML for build jobs where offered, then rebuild release stages/checks explicitly.

5D.02 · ACTION 5 OF 6**SEE THE IDEA · Pipeline triggers, templates and migration****A SMALL DECISION MAP****Pipeline triggers, templates and migration**

- 1 PR** Policy or PR event

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Small syntax differences can change when and how automation runs.

1. Run the YAML pipeline against a nonproduction target, compare outputs and approvals, then disable the old triggers before enabling the new release path.
2. Do not let both pipelines deploy simultaneously.

5D.02 · ACTION 6 OF 6**SEE THE IDEA · Pipeline triggers, templates and migration****A SMALL DECISION MAP****Pipeline triggers, templates and migration**

- 1 PR** Policy or PR event
- 2 Template** Reusable reviewed structure

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Small syntax differences can change when and how automation runs.

1. Classic task groups correspond to reusable task collections
2. YAML step/job/stage templates provide reviewed reusable structure.
3. A YAML extends template can enforce a common entry structure.

You should see

You can explain how to move an existing pipeline without losing its control points.

If it does not work

Classic release pipelines do not convert every gate/approval through a single View YAML action. Environment checks must be reviewed separately.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Finish Day 5

Save your place and check cost

Save the what-if result, rollback time and both CI results. Check that no competing deploy job is running. Record the GitHub remote explicitly in your notes.

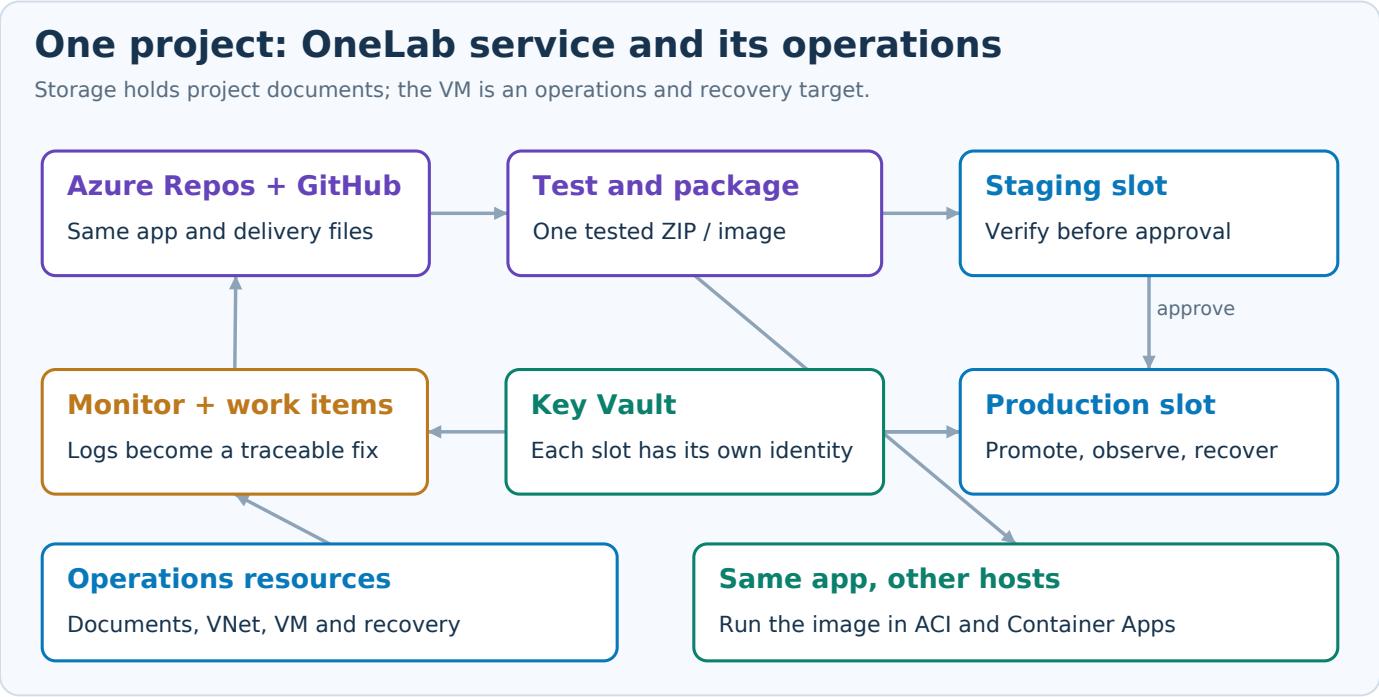
Say it in your own words: Does rolling back application code automatically undo a database change?

Check your explanation: No. Keep schema changes compatible and plan their order. Destructive data changes need their own recovery plan.

DAY 6 OF 7

Try other hosts and secure the delivery

You can compare VM/container choices, read security findings and observe a self-hosted agent.



Start today

Run the restart routine and start vm-ops if stopped. Review replication health before changing VM size; check supported sizes and target quota. Keep the protected VM in its existing group.

Start early and track the job

Start scans while working on the next independent check. A failed security scan is evidence to review, not a reason to hide the finding.

Block	Today's work	Minutes
A	Compare VM choices and repair drift	95
B	Run OneLab in two container hosts	95
C	Read and act on security findings	120
D	See how an agent runs a job	60

6A · Compare VM choices and repair drift

Planning estimate: 95 minutes. Finish with: Resize/availability observations, a repaired service and an environment-design decision.

6A.01 Resize, stop and compare availability choices

AZ-104 · CORE + availability extension

Why this task: Compute size, number of instances, and failure isolation solve different problems.

6A.01 · ACTION 1 OF 5

AZURE PORTAL · vm-ops → Size

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: vm-ops

vm-ops > Size

Overview
Size
Disks

1 Current size
2 New size
3 After Stop

Record before changing
Supported and quota-approved
Wait for Stopped (deallocated)

Find the labels above in your own portal.

Resize

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open vm-ops > Size.
2. Note current vCPU and memory.
3. Choose a quota-supported alternative, review cost, resize, and confirm the new size.
4. A resize can restart the VM.

6A.01 · ACTION 2 OF 5

AZURE PORTAL · Virtual machines → vm-ops → Overview

The screenshot shows the Azure Portal interface. At the top, there's a search bar with 'Virtual machines' entered. Below the search bar, the breadcrumb navigation reads 'Virtual machines > vm-ops > Overview'. On the left, a sidebar lists 'Overview', 'Size', and 'Disks'. The main content area shows a sequence of steps: 1. Action (Stop) and 2. Wait for (Stopped (deallocated)). A blue 'Stop' button is located at the bottom right of the main content area. Below the button, a note says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Use Stop in the portal and wait for Stopped (deallocated).
2. Compare this with simply shutting down inside Linux, which can leave compute allocated.

6A.01 · ACTION 3 OF 5

AZURE PORTAL · Virtual machines → vm-ops → Overview

The screenshot shows the Azure Portal interface. At the top, there's a search bar with 'Virtual machines' entered. Below the search bar, the breadcrumb navigation reads 'Virtual machines > vm-ops > Overview'. On the left, a sidebar lists 'Overview', 'Size', and 'Disks'. The main content area shows a sequence of steps: 1. Action (Start) and 2. Then (Run command to check nginx). A blue 'Start' button is located at the bottom right of the main content area. Below the button, a note says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Start the VM before network, monitoring or backup tasks.
2. Confirm its health page still exists.

6A.01 · ACTION 4 OF 5**SEE THE IDEA · Resize, stop and compare availability choices → Explain the decision****A SMALL DECISION MAP****Resize, stop and compare availability choices > Explain the decision**

- | | | |
|----------|---------------|--|
| 1 | Locate | Explain the decision |
| 2 | Action | DESIGN availability set: place two VMs in different fault and update domains to reduce shared hardware/maintenance risk. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Compute size, number of instances, and failure isolation solve different problems.

1. DESIGN availability set: place two VMs in different fault and update domains to reduce shared hardware/maintenance risk.
2. A zone separates data-center locations within a region.

6A.01 · ACTION 5 OF 5**SEE THE IDEA · Resize, stop and compare availability choices → Explain the decision****A SMALL DECISION MAP****Resize, stop and compare availability choices > Explain the decision**

- | | | |
|----------|---------------|--|
| 1 | Locate | Explain the decision |
| 2 | Action | DESIGN scale set: a managed fleet of similar VMs can increase/decrease instance count. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Compute size, number of instances, and failure isolation solve different problems.

1. DESIGN scale set: a managed fleet of similar VMs can increase/decrease instance count.
2. A bigger single VM is scale up; more VMs are scale out.

You should see

You can distinguish scaling, deallocation and availability placement.

If it does not work

Changing an existing VM's zone or availability set is not a simple size edit. Plan recreation or a supported move/replication method. [R16]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

6A.02 Practise a scale set and resource move

AZ-104 · CORE + availability extension

Why this task: The operations health service could need multiple instances or a new organizational location.

6A.02 · ACTION 1 OF 5

AZURE PORTAL · Virtual machine scale sets → Create

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Virtual machine scale sets

Virtual machine scale sets > Create

Virtual machine scale sets

Create

- 1 Group** rg-PREFIX-ops
- 2 Image** Supported Ubuntu image
- 3 Initial instances** 2 if quota permits
- 4 After observation** Scale to 1 and delete test set

Find the labels above in your own portal.

Review + create

Illustrated navigation view · use your own saved names and IDs

Why this action: The operations health service could need multiple instances or a new organizational location.

1. For a short paid exercise, search Virtual machine scale sets > Create.
2. Use rg-PREFIX-ops, a supported Ubuntu image and size, two instances, and vnet-main.
3. Inspect Flexible versus Uniform orchestration and use the wizard's supported defaults.

6A.02 · ACTION 2 OF 5

AZURE PORTAL · Your practice scale set → Scaling

The image shows a simulated Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com' and a note says 'Illustrated view · layout may vary'. The main header is 'AZURE PORTAL' with a search bar containing 'Your practice scale set'. Below the header, the breadcrumb 'Your practice scale set > Scaling' is visible. On the left, a sidebar shows 'Your practice scale set' with 'Scaling' selected. The main content area has two numbered steps: '1 Manual count' and '2 Optional autoscale'. Step 1 has a text input field with '1 after observing 2'. Step 2 has a text input field with 'Minimum 1 / maximum 2'. At the bottom right is a blue 'Save' button. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Your practice scale set

Your practice scale set > Scaling

Your practice scale set

Scaling

1 Manual count 1 after observing 2

2 Optional autoscale Minimum 1 / maximum 2

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. On Scaling configure a manual count of one after observing two instances.
2. For autoscale, create a rule on a supported CPU metric, minimum 1 and maximum 2; choose a cooldown.
3. Restore manual count 1 and delete this practice scale set after inspection.

6A.02 · ACTION 3 OF 5

AZURE PORTAL · rg-PREFIX-ops → Move → Move to another resource group



Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. For the group-move exercise, create an empty test resource group.
2. In rg-PREFIX-ops > Move > Move to another resource group, select the VM and required dependencies and inspect validation.
3. This VM is now protected: record any protection/dependency restriction and cancel.
4. Keep it in its current group for Day 7.
5. This is a validation/design exercise, not a completed move.

6A.02 · ACTION 4 OF 5

SEE THE IDEA · Practise a scale set and resource move → Explain the decision

A SMALL DECISION MAP

Practise a scale set and resource move > Explain the decision

1

Locate

Explain the decision

2

Action

DESIGN subscription move: verify both subscriptions, same tenant where required, provider registration, quota, permissions, dependent resources and role reassignment.

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. DESIGN subscription move: verify both subscriptions, same tenant where required, provider registration, quota, permissions, dependent resources and role reassignment.
2. A move does not copy a resource.

6A.02 · ACTION 5 OF 5**SEE THE IDEA · Practise a scale set and resource move → Explain the decision****A SMALL DECISION MAP****Practise a scale set and resource move > Explain the decision**

- | | | |
|----------|---------------|--|
| 1 | Locate | Explain the decision |
| 2 | Action | DESIGN region move: use a supported Resource Mover or replication/recreation workflow. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: The operations health service could need multiple instances or a new organizational location.

1. DESIGN region move: use a supported Resource Mover or replication/recreation workflow.
2. Editing the resource-group location does not relocate the VM.
3. Day 2, Block D, Day 7, Block C shows regional recovery.

You should see

You record the difference between group/subscription movement and region relocation.

If it does not work

If scale-set quota is unavailable, complete the sizing/scale policy design and mark deployment unperformed rather than claiming a fleet was built.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

6A.03 Observe configuration drift on the operations VM

AZ-400 · CORE + platform extensions

Why this task: Creating a VM does not ensure its internal services stay configured.

6A.03 · ACTION 1 OF 5

VM TERMINAL IN AZURE PORTAL · Azure Portal → vm-ops → Run command → RunShellScript

The dark code panel below is the terminal/editor view for this action.

Why this action: Creating a VM does not ensure its internal services stay configured.

1. Run `sudo systemctl stop nginx` on vm-ops.
2. Then run `curl http://127.0.0.1` and observe failure.

USE IN: VM RUNSHELLSCRIPT · COPY ONLY THE CODE

```
sudo systemctl stop nginx
curl --max-time 5 http://127.0.0.1
```

What it means: The failed curl is intentional. Continue to the repair action.

--max-time · Stop a curl request after this many seconds.

curl · Send an HTTP request. -f fails on HTTP error status; -s quiets the progress meter; -v shows connection details when used.

systemctl enable --now · Set the service to start at boot and start it now; stop/start act on the running service.

sudo · Run that Linux operation with administrator privileges inside the named VM.

6A.03 · ACTION 2 OF 5

VM TERMINAL IN AZURE PORTAL · Azure Portal → vm-ops → Run command → RunShellScript

The dark code panel below is the terminal/editor view for this action.

Why this action: Creating a VM does not ensure its internal services stay configured.

1. Run the desired-configuration command below.
2. It ensures the package is present and the service is enabled and running.

USE IN: TERMINAL: INSIDE VM · COPY ONLY THE CODE

```
sudo apt-get install -y nginx
sudo systemctl enable --now nginx
curl -f http://127.0.0.1
```

What it means: The install converges on an installed package and the service command converges on running/enabled state. It is intentionally a small demonstration.

curl · Send an HTTP request. -f fails on HTTP error status; -s quiets the progress meter; -v shows connection details when used.

systemctl enable --now · Set the service to start at boot and start it now; stop/start act on the running service.

sudo · Run that Linux operation with administrator privileges inside the named VM.

6A.03 · ACTION 3 OF 5**AZURE PORTAL · vm-ops → Operations → Run command → RunShellScript**

The image shows a browser window with the URL `portal.azure.com`. The page title is "AZURE PORTAL". A search bar contains the text "vm-ops". Below the search bar, the navigation path is displayed: **vm-ops > Operations > Run command > RunShellScript**. On the left, a sidebar lists the navigation items: "vm-ops", "Operations", and "Run command". To the right of the sidebar, two numbered steps are shown: **1 Locate** and **2 Action**. The "Locate" step has a text input field containing "RunShellScript". The "Action" step has a text input field containing "Run it a second time.". At the bottom right, there is a blue button labeled "Run". Below the navigation path, a note says "Find the labels above in your own portal."

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Run it a second time.
2. Confirm the service remains healthy instead of creating duplicate services.

6A.03 · ACTION 4 OF 5**SEE THE IDEA · Observe configuration drift on the operations VM → Explain the decision****A SMALL DECISION MAP****Observe configuration drift on the operations VM > Explain the decision**

- | | |
|-----------------|---|
| 1 Locate | Explain the decision |
| 2 Action | DESIGN large fleet: keep configuration definitions in Git, test on a small group, use a configuration tool/extension or Machine Configuration policy, and alert on drift. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. DESIGN large fleet: keep configuration definitions in Git, test on a small group, use a configuration tool/extension or Machine Configuration policy, and alert on drift.
2. An ad-hoc command is not ongoing enforcement.

6A.03 · ACTION 5 OF 5

AZURE PORTAL · vm-ops → Operations → Run command → RunShellScript

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: vm-ops

vm-ops > Operations > Run command > RunShellScript

vm-ops
Operations
Run command
RunShellScript

1 Locate
2 Action

RunShellScript

For a Machine Configuration extension, enable the required managed identity and guest configuration extension for a supported VM, assign a suitable built-in machine-configuration audit policy at lab scope, then inspect its compliance details.

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: Creating a VM does not ensure its internal services stay configured.

1. For a Machine Configuration extension, enable the required managed identity and guest configuration extension for a supported VM, assign a suitable built-in machine-configuration audit policy at lab scope, then inspect its compliance details.
2. Use an Audit policy first.

You should see

You demonstrate drift and restoration and can distinguish a one-time repair from continuous desired state.

If it does not work

The Machine Configuration and legacy DSC extensions have platform/OS prerequisites. Do not claim compliance until the guest reports its evaluated state. [R49]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

6A.04 Design or deploy an approved self-service environment

AZ-400 · CORE + platform extensions

Why this task: A platform team can give developers approved templates instead of unrestricted subscription control.

6A.04 · ACTION 1 OF 6

SEE THE IDEA · Approved developer environment design

A SMALL DECISION MAP

Approved developer environment design

- | | | |
|---|------------|----------------------------------|
| 1 | Catalog | Reviewed templates |
| 2 | Scope | Approved subscription / group |
| 3 | Permission | Developer can request a template |
| 4 | Lifetime | Owner and expiry |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Because Azure Deployment Environments is retiring, use DESIGN unless you specifically need a short-lived implementation before retirement.
2. Record this limitation in your coverage checklist.

6A.04 · ACTION 2 OF 6**AZURE PORTAL · Azure Deployment Environments → Dev centers → Create**

The screenshot shows the Azure Portal interface for creating a Dev center. The breadcrumb navigation is 'Azure Deployment Environments > Dev centers > Create'. On the left, a sidebar lists 'Azure Deployment Environments', 'Dev centers', and 'Create' (highlighted). The main area contains three numbered steps: 1. Eligibility (Only a permitted pre-retirement test), 2. Name (dc-PREFIX), and 3. Catalog (Offered quick-start catalog). A 'Review + create' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Azure Deployment Environments

Azure Deployment Environments > Dev centers > Create

Azure Deployment Environments

Dev centers

Create

1 Eligibility Only a permitted pre-retirement test

2 Name dc-PREFIX

3 Catalog Offered quick-start catalog

Find the labels above in your own portal.

Review + create

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. For an eligible test setup, search Azure Deployment Environments > Dev centers > Create.
2. Use the lab group, supported region, name dc-PREFIX and the offered quick-start catalog.

6A.04 · ACTION 3 OF 6**AZURE PORTAL · dc-PREFIX → Identity → System assigned**

The screenshot shows the Azure Portal interface for configuring system-assigned identity. The breadcrumb navigation is 'dc-PREFIX > Identity > System assigned'. On the left, a sidebar lists 'dc-PREFIX', 'Identity', and 'System assigned' (highlighted). The main area contains two numbered steps: 1. Enable (Managed identity) and 2. Separate subscription IAM (Contributor + User Access Administrator as required). A 'Save' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: dc-PREFIX

dc-PREFIX > Identity > System assigned

dc-PREFIX

Identity

System assigned

1 Enable Managed identity

2 Separate subscription IAM Contributor + User Access Administrator as required

Find the labels above in your own portal.

Save

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Open the dev center > Identity and enable a managed identity.
2. Grant the documented Contributor and User Access Administrator permissions only on a dedicated practice deployment subscription; these are broad rights.

6A.04 · ACTION 4 OF 6

AZURE PORTAL · dc-PREFIX → Environment types / Projects

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL

Search: dc-PREFIX

dc-PREFIX > Environment types / Projects

dc-PREFIX

Environment types / Projects

1 Environment type

2 Project

3 Target

Dev

Your lab project

Dedicated deployment subscription

Find the labels above in your own portal.

Add

Illustrated navigation view · use your own saved names and IDs

Why this action: A platform team can give developers approved templates instead of unrestricted subscription control.

1. Add an environment type Dev.
2. Create a project associated with the dev center and enable its Dev environment type with the target subscription and deployment identity.

6A.04 · ACTION 5 OF 6

YOUR BROWSER · Azure developer portal → Your project → Environments

Your own browser tab

Illustrated view · layout may vary

YOUR BROWSER

Search: Azure developer portal

Azure developer portal > Your project > Environments

Azure developer portal

Your project

Environments

1 Role

2 Definition

3 After test

Deployment Environments User

Approved small catalog item

Delete environment

Find the labels above in your own portal.

Create

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Grant your test developer Deployment Environments User at project scope.
2. In the developer portal create a small environment from an approved catalog definition, review its resources and delete it after testing.

6A.04 · ACTION 6 OF 6**SEE THE IDEA · Design or deploy an approved self-service environment → Explain the decision****A SMALL DECISION MAP****Design or deploy an approved self-service environment > Explain the decision**

- | | |
|-----------------|--|
| 1 Locate | Explain the decision |
| 2 Action | DESIGN for OneLab: keep the application's template in a governed catalog, separate Dev and Production permissions, enforce tags/region/expiry, and migrate this workflow to a supported successor before retirement. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. DESIGN for OneLab: keep the application's template in a governed catalog, separate Dev and Production permissions, enforce tags/region/expiry, and migrate this workflow to a supported successor before retirement.

You should see

You can explain catalog, project, environment type, deployment identity and developer permission.

If it does not work

Do not confuse an Azure Pipelines environment used for checks with an Azure Deployment Environments self-service environment. They are different products. [R50]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

6B · Run OneLab in two container hosts

Planning estimate: 95 minutes. Finish with: An ACR digest and ACI/Container Apps health checks before deleting the temporary hosts.

6B.01 Build an image in Azure Container Registry

AZ-104 + AZ-400 · EXTENSION - registry and runtime costs

Why this task: A container image packages the application and runtime together.

6B.01 · ACTION 1 OF 5

AZURE PORTAL · Container registries → Create

The screenshot shows the 'Create Container Registry' page in the Azure Portal. The browser address bar shows 'portal.azure.com'. The page title is 'AZURE PORTAL' with a search bar containing 'Container registries'. The breadcrumb is 'Container registries > Create'. On the left, there's a sidebar with 'Container registries' and a 'Create' button. The main form has five numbered steps: 1. Resource group (rg-PREFIX-core), 2. Name (PREFIXacr), 3. SKU (Basic), 4. Admin user (Disabled), and 5. Permissions (Ordinary RBAC). A 'Review + create' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

Step	Field	Value
1	Resource group	rg-PREFIX-core
2	Name	PREFIXacr
3	SKU	Basic
4	Admin user	Disabled
5	Permissions	Ordinary RBAC

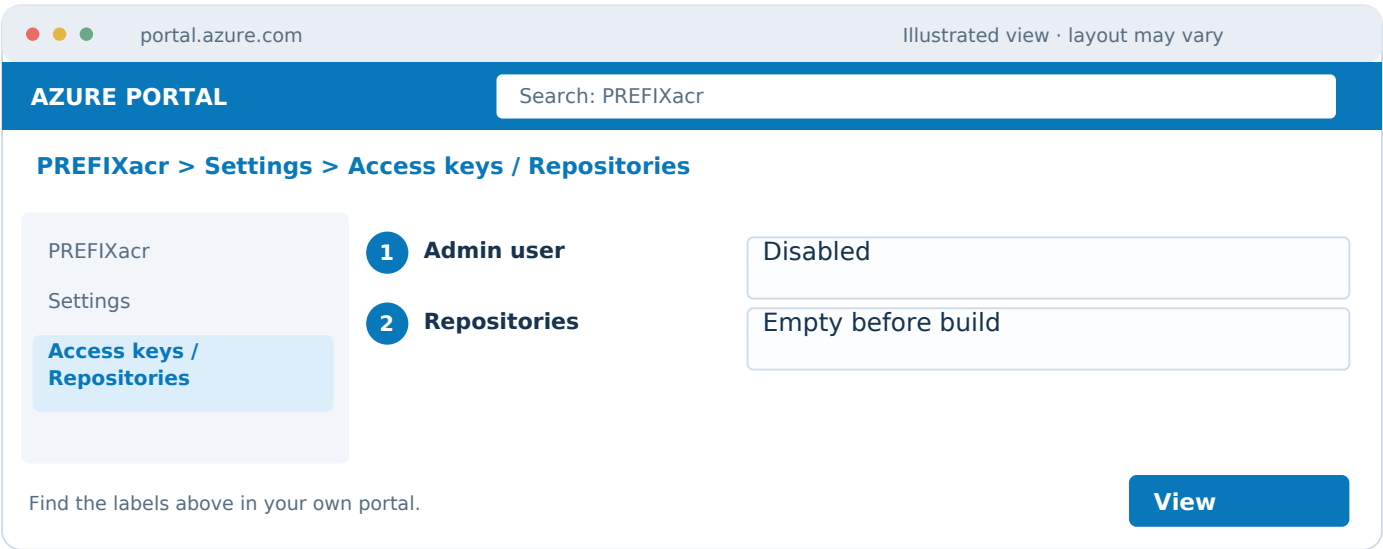
Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Search Container registries > Create.
2. Use rg-PREFIX-core; name PREFIXacr; primary region
3. Basic SKU.
4. Select ordinary RBAC registry permissions for this exercise rather than the repository ABAC mode, if the choice is shown.

6B.01 · ACTION 2 OF 5

AZURE PORTAL · PREFIXacr → Settings → Access keys / Repositories



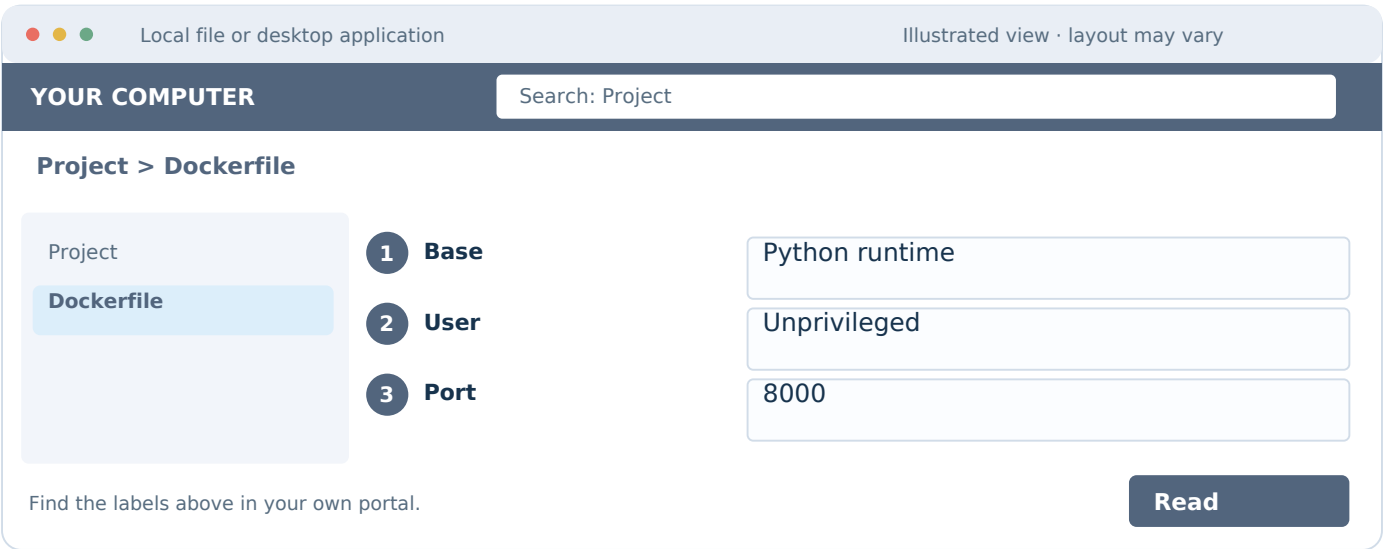
Illustrated navigation view · use your own saved names and IDs

Why this action: A container image packages the application and runtime together.

1. Leave the registry admin user disabled.
2. Create and open the registry.
3. Inspect Repositories; it is initially empty.

6B.01 · ACTION 3 OF 5

YOUR COMPUTER · Project → Dockerfile



Illustrated navigation view · use your own saved names and IDs

Why this action: A container image packages the application and runtime together.

1. Read Dockerfile.
2. It starts from Python, installs requirements, copies app.py, switches to an unprivileged user and starts gunicorn on port 8000.

6B.01 · ACTION 4 OF 5**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

The dark code panel below is the terminal/editor view for this action.

Why this action: A container image packages the application and runtime together.

1. Run the ACR build command below.
2. It uses a cloud build, so Cloud Shell does not need a local Docker daemon.

USE IN: BASH · COPY ONLY THE CODE

```
az acr build -r "$ACR" -t onelab:1.0.0 .
LOGIN=$(az acr show -n "$ACR" --query loginServer -o tsv)
echo "$LOGIN/oneLab:1.0.0"
```

What it means: The final dot is the current build context. .dockerignore excludes local settings, Git history and temporary files from that context.

--query · Select or transform returned fields with a JMESPath expression; this does not itself change the resource.

-n · Short form of --name in these Azure commands.

-o · Azure CLI output format: table, tsv or none.

az acr build -r / -t · -r selects the registry; -t names the image and tag. The final dot uses this folder as the build context.

\$NAME / \${NAME} · Read a variable from this Bash session. Source lab.env first when these are lab names.

\$(command) · Run the inner command and use its output as the value.

6B.01 · ACTION 5 OF 5**AZURE PORTAL · PREFIXacr → Repositories → onelab → 1.0.0**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: PREFIXacr

PREFIXacr > Repositories > onelab > 1.0.0

PREFIXacr
Repositories
onelab

1 Digest

Record exact sha256 value

2 Tag

1.0.0

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Open Repositories > onelab > 1.0.0.
2. Record the image digest.
3. A tag is a movable label; a digest identifies content.

You should see

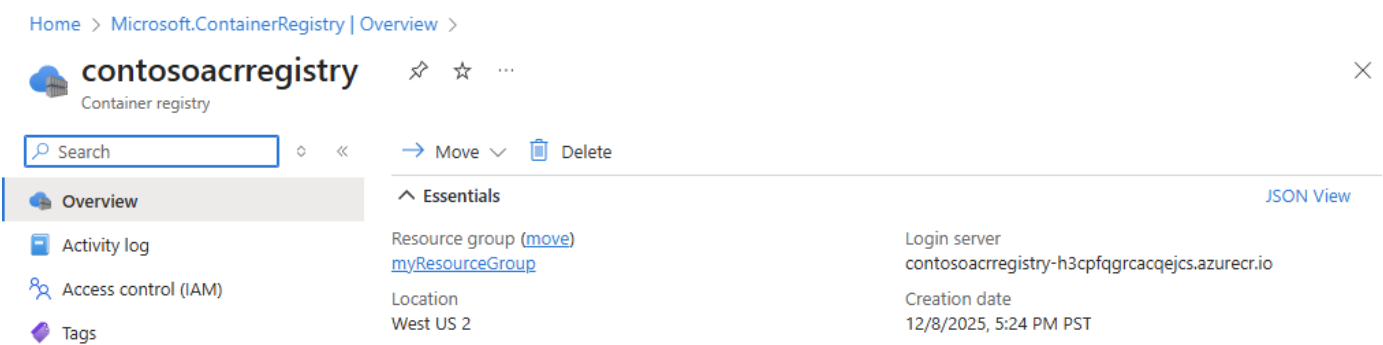
The registry contains an onelab:1.0.0 image and a digest.

If it does not work

If ACR Tasks/build is restricted for your subscription, use an approved build agent with Docker to build and push. Do not enable the admin account as a universal workaround. [R28]

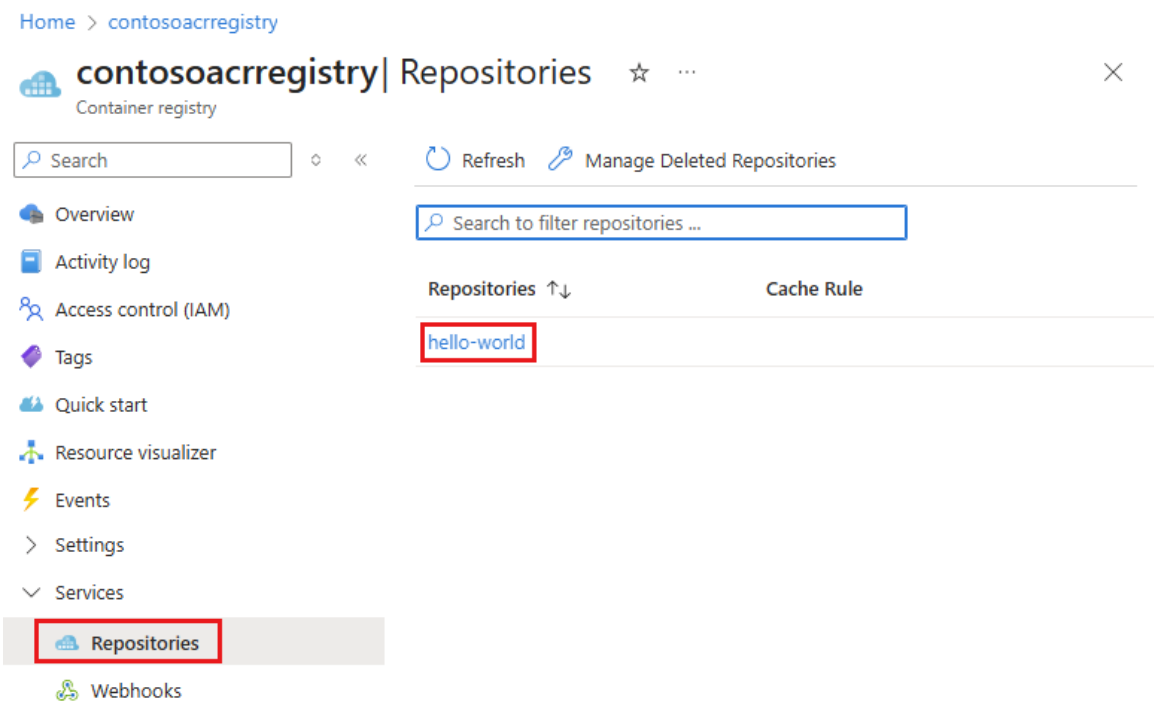
Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED
Worked: ☐ Independently ☐ With the guide

Microsoft reference screenshot · Screenshot of the overview page of a container registry in the Azure portal.



Microsoft reference source · Sample values belong to that reference, not OneLab.

Microsoft reference screenshot · Screenshot showing the container images for a container registry in the Azure portal.



Microsoft reference source · Sample values belong to that reference, not OneLab.

6B.02 Create an identity that may pull the image

AZ-104 + AZ-400 · EXTENSION - registry and runtime costs

Why this task: The runtime needs image-pull access, not permission to administer the whole subscription.

6B.02 · ACTION 1 OF 3

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: The runtime needs image-pull access, not permission to administer the whole subscription.

1. Run the identity creation and AcrPull assignment below.

USE IN: BASH · COPY ONLY THE CODE

```
az identity create -g "$RG" -n onelab-pull
PULL_ID=$(az identity show -g "$RG" -n onelab-pull --query id -o tsv)
PULL_OID=$(az identity show -g "$RG" -n onelab-pull \
  --query principalId -o tsv)
ACR_ID=$(az acr show -n "$ACR" --query id -o tsv)
az role assignment create --assignee-object-id "$PULL_OID" \
  --assignee-principal-type ServicePrincipal --role AcrPull --scope "$ACR_ID"
```

What it means: The resource ID selects the identity resource. principalId selects its Entra principal for a role assignment; these are not interchangeable.

--query · Select or transform returned fields with a JMESPath expression; this does not itself change the resource.

--assignee-object-id · The Entra principal object ID that receives the role.

--assignee-principal-type · ServicePrincipal identifies an application or managed-identity principal.

--role · The exact RBAC role name to assign.

--scope · The Azure resource ID where the role assignment applies.

-g · Short form of --resource-group.

-n · Short form of --name in these Azure commands.

-o · Azure CLI output format: table, tsv or none.

\$NAME / \${NAME} · Read a variable from this Bash session. Source lab.env first when these are lab names.

\$(command) · Run the inner command and use its output as the value.

Backslash at line end · Continue the same Bash command on the next line. No spaces may follow the backslash.

6B.02 · ACTION 2 OF 3**CLOUD SHELL · the registry → IAM and verify the assignment**

portal.azure.com Illustrated view · layout may vary

CLOUD SHELL Search: the registry

the registry > IAM and verify the assignment

the registry

1 Locate

2 Action

IAM and verify the assignment

IAM and verify the assignment

Open the registry > IAM and verify the assignment.

Find the labels above in your own portal. [View](#)

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Open the registry > IAM and verify the assignment.
2. Wait for propagation before creating a container.

6B.02 · ACTION 3 OF 3**CLOUD SHELL · Cloud Shell → Bash → Managed image-pull identity**

portal.azure.com Illustrated view · layout may vary

CLOUD SHELL Search: Cloud Shell

Cloud Shell > Bash > Managed image-pull identity

Cloud Shell

Bash

1 Identity

2 Role

Managed image-pull identity

onelab-pull

AcrPull

Find the labels above in your own portal. [View](#)

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Keep this identity for both the ACI and Container Apps exercises.
2. The role shown here applies to the registry's ordinary RBAC mode.

You should see

The onelab-pull managed identity has AcrPull on this registry only.

If it does not work

For a registry using repository ABAC mode, use the applicable repository-reader role/condition instead; the legacy AcrPull assignment is not the same authorization model.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

6B.03 Run the image with Azure Container Instances

AZ-104 + AZ-400 · EXTENSION - registry and runtime costs

Why this task: ACI is useful for a simple standalone container without managing a server fleet.

6B.03 · ACTION 1 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: ACI is useful for a simple standalone container without managing a server fleet.

1. Run the ACI command below.
2. It supplies explicit CPU, memory, OS type, port and managed identity for the private image.

USE IN: BASH · COPY ONLY THE CODE

```
az container create -g "$RG" -n ci-$P --location "$LOC" \
  --image "$LOGIN/onelab:1.0.0" --os-type Linux --cpu 1 --memory 1.5 \
  --ports 8000 --ip-address Public --restart-policy Always \
  --assign-identity "$PULL_ID" --acr-identity "$PULL_ID"
az container show -g "$RG" -n ci-$P --query ipAddress.ip -o tsv
```

What it means: --assign-identity attaches the identity to the container group; --acr-identity uses it for the registry pull.

--location · The Azure region to use.

--image · The full registry/image:tag to run.

--os-type · The container operating-system family; this image uses Linux.

--cpu · The requested number of CPU cores.

--memory · The requested memory quantity; use the unit format the command expects.

--ports · Ports that the container group exposes.

--ip-address · Public asks ACI for a publicly reachable group IP.

--restart-policy · Always restarts a stopped process in this container group.

--assign-identity · Attach this managed identity to the container group.

--acr-identity · Use this attached identity when ACI pulls the private ACR image.

--query · Select or transform returned fields with a JMESPath expression; this does not itself change the resource.

-g · Short form of --resource-group.

-n · Short form of --name in these Azure commands.

-o · Azure CLI output format: table, tsv or none.

\$NAME / \${NAME} · Read a variable from this Bash session. Source lab.env first when these are lab names.

Backslash at line end · Continue the same Bash command on the next line. No spaces may follow the backslash.

6B.03 · ACTION 2 OF 5**AZURE PORTAL · Container instances → ci-PREFIX → Containers → Logs**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Container instances

Container instances > ci-PREFIX > Containers > Logs

Container instances
ci-PREFIX
Containers

1 Process gunicorn

2 Listening port 8000

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. In the portal search Container instances > ci-PREFIX.
2. Open Containers > Logs and confirm gunicorn is listening.

6B.03 · ACTION 3 OF 5**YOUR BROWSER · http://YOUR_ACI_IP:8000/health**

Your own browser tab

Illustrated view · layout may vary

YOUR BROWSER Search: http://YOUR_ACI_IP:8000/health

http://YOUR_ACI_IP:8000/health

1 Expected healthy

2 Use Your ACI public IP

Find the labels above in your own portal.

Open

Illustrated navigation view · use your own saved names and IDs

Why this action: ACI is useful for a simple standalone container without managing a server fleet.

1. Read the public IP from Overview and browse http://IP:8000/health.
2. This small ACI exercise uses plain HTTP and dummy data; the main website continues using HTTPS.

6B.03 · ACTION 4 OF 5**AZURE PORTAL · ci-PREFIX → Overview / Containers**

The screenshot shows the Azure Portal interface for configuring a container instance. The browser address bar displays 'portal.azure.com'. The page title is 'AZURE PORTAL'. A search bar contains 'ci-PREFIX'. The breadcrumb navigation is 'ci-PREFIX > Overview / Containers'. On the left, a sidebar shows 'ci-PREFIX' with 'Overview / Containers' selected. The main content area has three numbered steps: 1. CPU (value: 1), 2. Memory (value: 1.5 GiB), and 3. Restart policy (value: Always). A 'View' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: ci-PREFIX

ci-PREFIX > Overview / Containers

ci-PREFIX

Overview / Containers

1 CPU 1

2 Memory 1.5 GiB

3 Restart policy Always

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: ACI is useful for a simple standalone container without managing a server fleet.

1. Inspect Restart policy and CPU/memory.
2. Scaling this example means changing/recreating resources or using other orchestration; it is not an App Service autoscale rule.

6B.03 · ACTION 5 OF 5**AZURE PORTAL · Container instances → ci-PREFIX → Overview**

The screenshot shows the Azure Portal interface for the 'Remove' action. The browser address bar displays 'portal.azure.com'. The page title is 'AZURE PORTAL'. A search bar contains 'Container instances'. The breadcrumb navigation is 'Container instances > ci-PREFIX > Overview'. On the left, a sidebar shows 'Container instances' with 'ci-PREFIX' and 'Overview' selected. The main content area has one numbered step: 1. Remove (value: Only this temporary container group). A 'Delete' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Container instances

Container instances > ci-PREFIX > Overview

Container instances

ci-PREFIX

Overview

1 Remove Only this temporary container group

Find the labels above in your own portal. **Delete**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Delete this ACI after the test to stop its compute charge.

You should see

The same application returns healthy from the ACI container.

If it does not work

ImagePull failures usually mean image name/tag, registry access or identity role issues. A running container can still fail HTTP if the configured port is wrong. [R29]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

6B.04 Run and scale with Azure Container Apps

AZ-104 + AZ-400 · EXTENSION - registry and runtime costs

Why this task: Container Apps adds managed HTTPS ingress, revisions and event/HTTP-based scaling.

6B.04 · ACTION 1 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: Container Apps adds managed HTTPS ingress, revisions and event/HTTP-based scaling.

1. Run the block below to install/update the extension and create consumption environment env-PREFIX and app ca-PREFIX.

USE IN: BASH · COPY ONLY THE CODE

```
az extension add --name containerapp --upgrade
az containerapp env create -g "$RG" -n env-$P --location "$LOC"
az containerapp create -g "$RG" -n ca-$P --environment env-$P \
  --image "$LOGIN/onelab:1.0.0" --ingress external --target-port 8000 \
  --cpu 0.5 --memory 1Gi --min-replicas 0 --max-replicas 2 \
  --user-assigned "$PULL_ID" --registry-server "$LOGIN" \
  --registry-identity "$PULL_ID"
```

What it means: Ingress reaches port 8000. The identity authorizes the image pull.

--name · The name of this resource or object.

--upgrade · Install a newer available CLI extension when one is present.

--location · The Azure region to use.

--environment · The Container Apps environment that hosts this application.

--image · The full registry/image:tag to run.

--ingress · external enables an externally reachable application endpoint.

--target-port · The internal port where the application process listens.

--cpu · The requested number of CPU cores.

--memory · The requested memory quantity; use the unit format the command expects.

--min-replicas · The minimum running replicas; zero permits scale-to-zero.

--max-replicas · The maximum replicas allowed by this app's scaling configuration.

--user-assigned · Attach the named user-assigned identity to the app.

--registry-server · The private registry login hostname.

--registry-identity · The identity used to pull this app's image.

-g · Short form of --resource-group.

-n · Short form of --name in these Azure commands.

\$NAME / \${NAME} · Read a variable from this Bash session. Source lab.env first when these are lab names.

Backslash at line end · Continue the same Bash command on the next line. No spaces may follow the backslash.

6B.04 · ACTION 2 OF 5

AZURE PORTAL · Container Apps → ca-PREFIX → Overview

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Container Apps

Container Apps > ca-PREFIX > Overview

Container Apps

ca-PREFIX

Overview

1 Application URL Copy exact HTTPS URL

Find the labels above in your own portal. **Open**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. In Container Apps > ca-PREFIX > Overview, copy Application URL and test /health over HTTPS.

6B.04 · ACTION 3 OF 5

AZURE PORTAL · ca-PREFIX → Scale and replicas

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: ca-PREFIX

ca-PREFIX > Scale and replicas

ca-PREFIX

Scale and replicas

1 Minimum 0

2 Maximum 2

3 CPU / memory 0.5 CPU / 1 GiB

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: Container Apps adds managed HTTPS ingress, revisions and event/HTTP-based scaling.

1. Open Containers to check CPU/memory.

2. Open Scale and replicas: minimum 0, maximum 2.

3. Expect a cold start after zero replicas.

6B.04 · ACTION 4 OF 5

AZURE PORTAL · ca-PREFIX → Revisions and replicas

The screenshot shows the Azure Portal interface for the resource 'ca-PREFIX'. The breadcrumb navigation is 'ca-PREFIX > Revisions and replicas'. The left sidebar shows 'ca-PREFIX' and 'Revisions and replicas' (highlighted). The main content area has two numbered steps: '1 New revision' and '2 Inspect'. Under '1 New revision' is a text box 'Change a harmless environment variable'. Under '2 Inspect' is a text box 'Traffic splitting and revision mode'. At the bottom right is a blue button 'Create new revision'. The top of the page has a search bar with 'ca-PREFIX' and a note 'Illustrated view · layout may vary'.

portal.azure.com

AZURE PORTAL

Search: ca-PREFIX

ca-PREFIX > Revisions and replicas

ca-PREFIX

Revisions and replicas

1 New revision

Change a harmless environment variable

2 Inspect

Traffic splitting and revision mode

Find the labels above in your own portal.

Create new revision

Illustrated navigation view · use your own saved names and IDs

Why this action: Container Apps adds managed HTTPS ingress, revisions and event/HTTP-based scaling.

1. Open Revisions and replicas.
2. Create a new revision by changing a harmless environment variable.
3. Inspect single-revision versus multiple-revision mode and traffic splitting.

6B.04 · ACTION 5 OF 5

AZURE PORTAL · ca-PREFIX → Overview → Delete

The screenshot shows the Azure Portal interface for the resource 'ca-PREFIX'. The breadcrumb navigation is 'ca-PREFIX > Overview > Delete'. The left sidebar shows 'ca-PREFIX', 'Overview', and 'Delete' (highlighted). The main content area has two numbered steps: '1 Then delete' and '2 Also inspect'. Under '1 Then delete' is a text box 'env-PREFIX'. Under '2 Also inspect' is a text box 'Automatically created monitoring resources'. At the bottom right is a blue button 'Delete'. The top of the page has a search bar with 'ca-PREFIX' and a note 'Illustrated view · layout may vary'.

portal.azure.com

AZURE PORTAL

Search: ca-PREFIX

ca-PREFIX > Overview > Delete

ca-PREFIX

Overview

Delete

1 Then delete

env-PREFIX

2 Also inspect

Automatically created monitoring resources

Find the labels above in your own portal.

Delete

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Delete the Container App and its environment after this comparison if you will not use them.
2. Check for its automatically created Log Analytics workspace separately.

You should see

You can compare ACI, Container Apps and App Service using the same application.

If it does not work

If ingress is healthy but returns an error, confirm target port 8000 and image startup. A consumption app scaling to zero does not mean every associated resource is free. [R30]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

6C · Read and act on security findings

Planning estimate: 120 minutes. Finish with: Scan reports, a secure-file exercise and a credential-incident response plan.

6C.01 Inspect the four kinds of security findings

AZ-400 · CORE + paid security extensions

Why this task: A green unit test does not prove an application is secure.

6C.01 · ACTION 1 OF 6

GITHUB · onelab → Settings → Code security

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB

Search: onelab

onelab > Settings > Code security

onelab

Settings

Code security

1 Dependency graph

2 Dependabot alerts

3 Secret scanning

4 Push protection

Enable if available

Enable if available

Plan-dependent

Plan-dependent

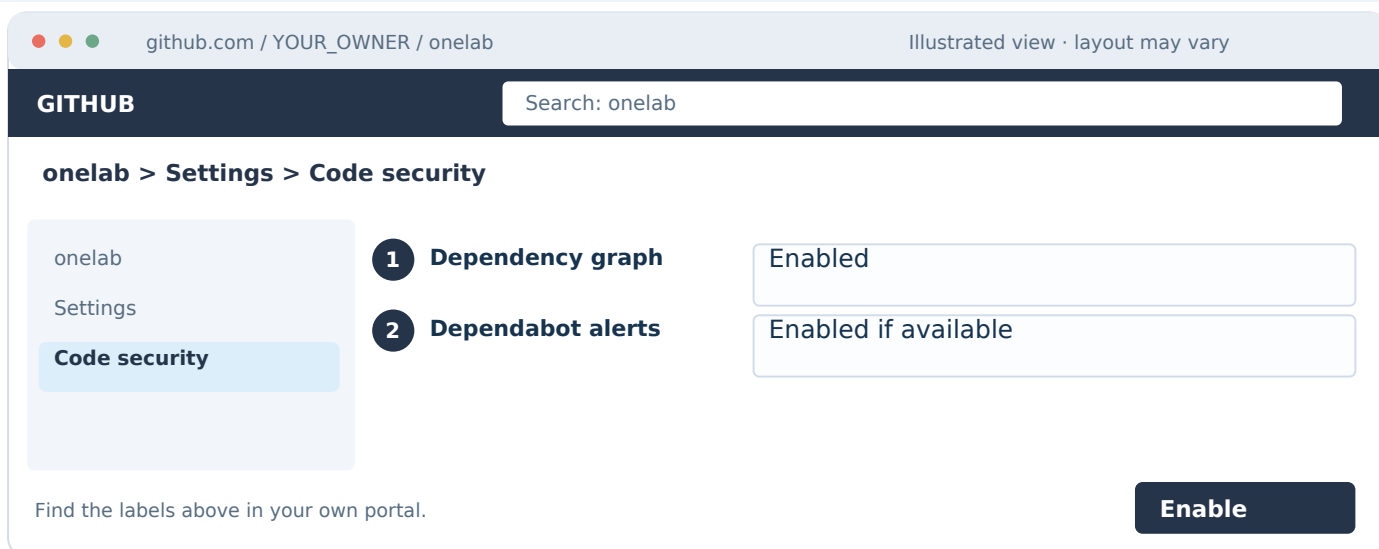
Find the labels above in your own portal.

Enable

Illustrated navigation view · use your own saved names and IDs

Why this action: A green unit test does not prove an application is secure.

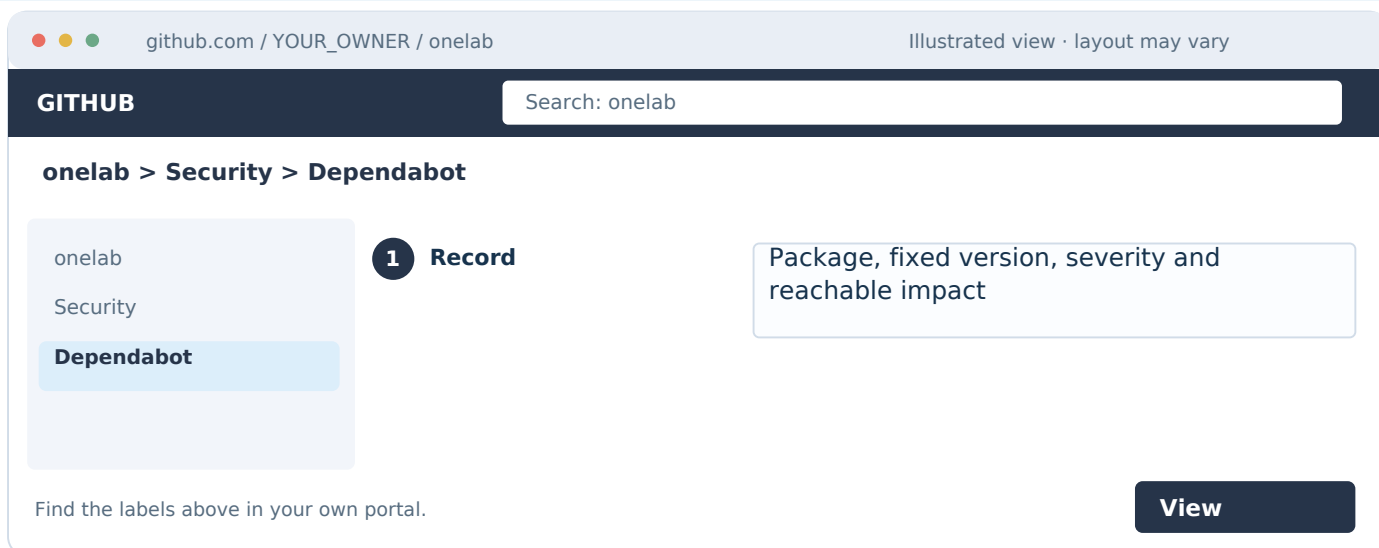
1. Open the GitHub repository > Settings > Code security and analysis, or the current Code security/Security configuration page.

6C.01 · ACTION 2 OF 6**GITHUB · onelab → Settings → Code security**

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. Enable the dependency graph and Dependabot alerts where available.
2. The supplied `.github/dependabot.yml` also schedules dependency-version update PRs.

6C.01 · ACTION 3 OF 6**GITHUB · onelab → Security → Dependabot**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open Security > Dependabot and inspect any alerts.
2. For each, record affected package, fixed version, severity, reachable impact and the update/test plan.
3. An empty alert list is not proof that all risks are absent.

6C.01 · ACTION 4 OF 6**GITHUB · onelab → Settings → Code security**

The screenshot shows the GitHub interface for the repository 'onelab'. The breadcrumb navigation is 'onelab > Settings > Code security'. On the left, a sidebar lists 'onelab', 'Settings', and 'Code security' (which is highlighted). In the center, there are three numbered steps: 1. Secret scanning, 2. Push protection, and 3. Data. To the right of these steps are three input fields: 'If licensed / available' for Secret scanning, 'If available' for Push protection, and 'Approved dummy formats only' for Data. At the bottom right, there is a 'Review' button. A note at the bottom left says 'Find the labels above in your own portal.'

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB Search: onelab

onelab > Settings > Code security

onelab
Settings
Code security

1 Secret scanning If licensed / available

2 Push protection If available

3 Data Approved dummy formats only

Find the labels above in your own portal.

Review

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Enable secret scanning and push protection if available for the repository.
2. Use only the platform's approved dummy test format if testing detection; never commit a real token.

6C.01 · ACTION 5 OF 6**GITHUB · onelab → Settings → Code security**

The screenshot shows the GitHub interface for the repository 'onelab'. The breadcrumb navigation is 'onelab > Settings > Code security'. On the left, a sidebar lists 'onelab', 'Settings', and 'Code security' (which is highlighted). In the center, there are two numbered steps: 1. Locate and 2. Action. To the right of these steps are two input fields: 'Code security' for the 'Locate' step and 'For licensing, generate a dependency inventory using pip list and review licences against your company's allowed list.' for the 'Action' step. At the bottom right, there is a 'View' button. A note at the bottom left says 'Find the labels above in your own portal.'

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB Search: onelab

onelab > Settings > Code security

onelab
Settings
Code security

1 Locate Code security

2 Action For licensing, generate a dependency inventory using pip list and review licences against your company's allowed list.

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: A green unit test does not prove an application is secure.

1. For licensing, generate a dependency inventory using pip list and review licences against your company's allowed list.
2. Vulnerability severity and licence compliance are different checks.

6C.01 · ACTION 6 OF 6**SEE THE IDEA · Inspect the four kinds of security findings → Explain the decision****A SMALL DECISION MAP****Inspect the four kinds of security findings > Explain the decision**

- | | | |
|----------|---------------|---|
| 1 | Locate | Explain the decision |
| 2 | Action | Explain the four scans: code analysis finds risky source patterns; dependency analysis checks third-party packages; secret scanning looks for credentials; licence review checks reuse obligations. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: A green unit test does not prove an application is secure.

1. Explain the four scans: code analysis finds risky source patterns; dependency analysis checks third-party packages; secret scanning looks for credentials; licence review checks reuse obligations.

You should see

You can name the finding type and the corresponding remediation.

If it does not work

Do not merely dismiss a real finding to pass a build. Record justified exceptions with an owner and expiry. Pinning a dependency makes it repeatable, not automatically secure. [R44]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

6C.02 Run CodeQL inside a container

AZ-400 · CORE + paid security extensions

Why this task: This demonstrates a code-analysis job whose tools and source live in the same container.

6C.02 · ACTION 1 OF 5

GITHUB · onelab → Actions → CodeQL in a container

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB

Search: onelab

onelab > Actions > CodeQL in a container

onelab

Actions

CodeQL in a container

1 Workflow

2 Language

3 Build mode

4 Container

.github/workflows/codeql.yml

Python

none

Supported glibc Python image

Run workflow

Find the labels above in your own portal.

Illustrated navigation view · use your own saved names and IDs

Why this action: This demonstrates a code-analysis job whose tools and source live in the same container.

1. Read `.github/workflows/codeql.yml`.
2. The job uses a glibc-based Python Debian container, checkout, CodeQL init for Python, and CodeQL analyze.

6C.02 · ACTION 2 OF 5**GITHUB · onelab → Actions → CodeQL in a container → Run workflow**

The screenshot shows the GitHub Actions interface for the 'CodeQL in a container' workflow. The breadcrumb path is 'onelab > Actions > CodeQL in a container > Run workflow'. On the left, a sidebar lists 'onelab', 'Actions', and 'CodeQL in a container'. The main area contains two numbered steps: '1 Branch' with a dropdown set to 'main', and '2 Then inspect' with a dropdown set to 'Security > Code scanning'. A 'Run workflow' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB Search: onelab

onelab > Actions > CodeQL in a container > Run workflow

onelab

Actions

CodeQL in a container

1 Branch main

2 Then inspect Security > Code scanning

Find the labels above in your own portal.

Run workflow

Illustrated navigation view · use your own saved names and IDs

Why this action: This demonstrates a code-analysis job whose tools and source live in the same container.

1. Run CodeQL in a container from Actions.
2. Open its logs and the Security > Code scanning results.

6C.02 · ACTION 3 OF 5**GITHUB · onelab → Actions → CodeQL in a container**

The screenshot shows the GitHub Actions interface for the 'CodeQL in a container' workflow. The breadcrumb path is 'onelab > Actions > CodeQL in a container'. On the left, a sidebar lists 'onelab', 'Actions', and 'CodeQL in a container'. The main area contains three numbered steps: '1 Language' with a dropdown set to 'Python', '2 Build mode' with a dropdown set to 'none', and '3 Container' with a dropdown set to 'Supported glibc Python image'. A 'View' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB Search: onelab

onelab > Actions > CodeQL in a container

onelab

Actions

CodeQL in a container

1 Language Python

2 Build mode none

3 Container Supported glibc Python image

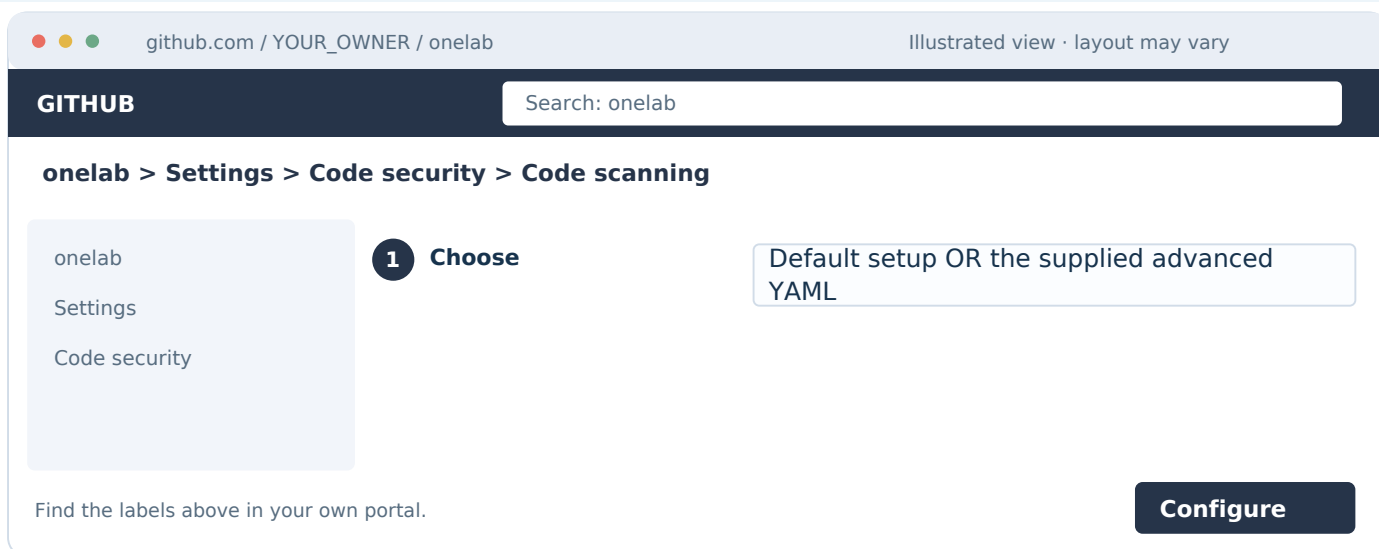
Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: This demonstrates a code-analysis job whose tools and source live in the same container.

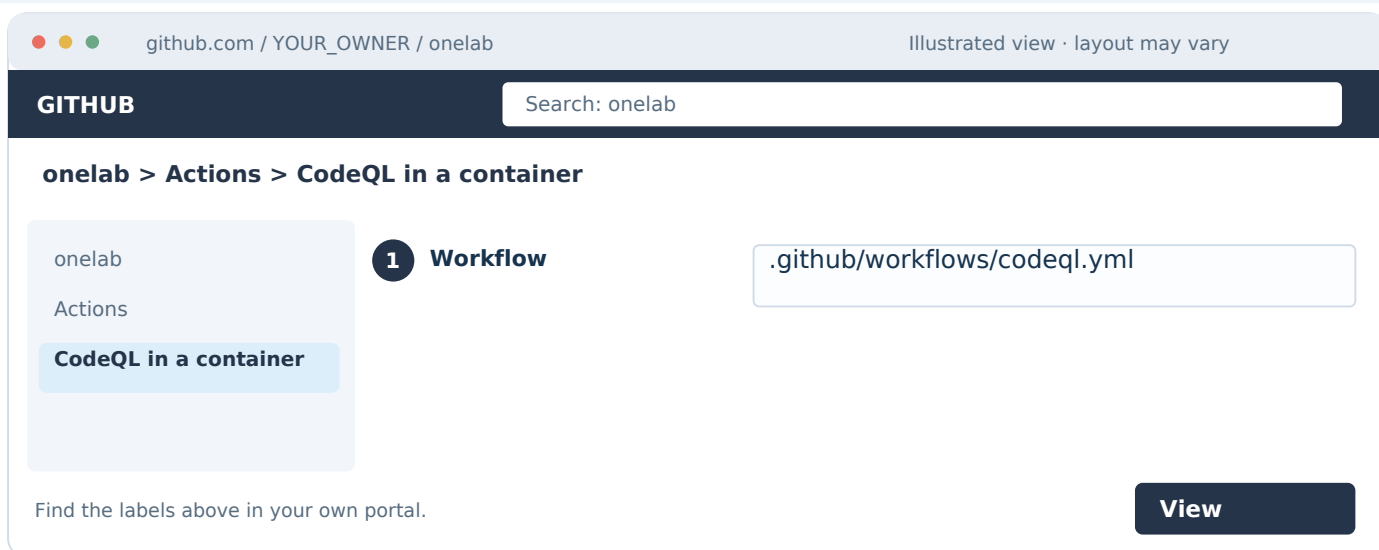
1. Python uses build-mode none here.
2. For compiled languages the build must occur where CodeQL can observe it; building secretly in another container can produce no-source errors.

6C.02 · ACTION 4 OF 5**GITHUB · onelab → Settings → Code security → Code scanning**

Illustrated navigation view · use your own saved names and IDs

Why this action: This demonstrates a code-analysis job whose tools and source live in the same container.

1. If default CodeQL setup is also enabled, choose either that managed setup or this advanced YAML exercise so you do not create confusing duplicate scans.

6C.02 · ACTION 5 OF 5**GITHUB · onelab → Actions → CodeQL in a container**

Illustrated navigation view · use your own saved names and IDs

Why this action: A recorded version lets you connect a change to the exact files tested and released.

1. The examples use readable action major-version tags.
2. For a hardened workflow, open the action's trusted release on GitHub, copy its exact commit SHA, replace the tag with that SHA and retain a comment naming the release.
3. Review Dependabot action update PRs.

You should see

The workflow completes and reports scan results, or you record a precise licence/plan limitation.

If it does not work

The CodeQL CLI is not supported on musl-only Alpine images. Private-repository code scanning needs the appropriate GitHub product/plan. [R45]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

6C.03 Add dependency and container scanning gates

AZ-400 · CORE + paid security extensions

Why this task: Different scanners inspect different layers of the release.

6C.03 · ACTION 1 OF 6

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: Different scanners inspect different layers of the release.

1. In the project virtual environment install pip-audit, run the command below and read its findings.
2. Update affected packages through a branch, rerun tests, and rescan.

USE IN: BASH · COPY ONLY THE CODE

```
python -m pip install pip-audit
python -m pip_audit -r requirements.txt
```

What it means: The scanner resolves dependencies and checks vulnerability advisories. A nonzero result should trigger review and an update, not an automatic claim that the whole application is exploitable.

python -m MODULE · Run a module using the currently selected Python interpreter.

-r requirements file · Read the dependency list from the named requirements file.

6C.03 · ACTION 2 OF 6

GITHUB · onelab → Actions → OneLab image scan

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB

Search: onelab

onelab > Actions > OneLab image scan

onelab

Actions

OneLab image scan

1 Branch

main

2 Operation

Build this repo image and scan it

Find the labels above in your own portal.

Run workflow

Illustrated navigation view · use your own saved names and IDs

Why this action: Different scanners inspect different layers of the release.

1. The supplied .github/workflows/image-scan.yml builds this project's image on a GitHub runner, then runs Anchore Grype through its official scan action.
2. Open Actions > OneLab image scan > Run workflow on main.

6C.03 · ACTION 3 OF 6

GITHUB · OneLab image scan → Run → Artifacts / Logs

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB

Search: OneLab image scan

OneLab image scan > Run > Artifacts / Logs

OneLab image scan

Run

Artifacts / Logs

1 Report

SARIF artifact

2 Gate

High or Critical fails

3 Fix

Update, test, rescan

Find the labels above in your own portal.

Download

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Read the scan log and download its SARIF report artifact.
2. The job fails on High or Critical findings.
3. Update the base image or dependencies through a PR, test and rescan; do not change fail-build to false just to clear a real finding.

6C.03 · ACTION 4 OF 6

GITHUB · onelab → Settings → Rules → Rulesets / Branch protection

github.com / YOUR_OWNER / onelab

Illustrated view · layout may vary

GITHUB

Search: onelab

onelab > Settings > Rules > Rulesets / Branch protection

onelab

Settings

Rules

1 Branch

2 Required check

main

Actual image scan job name after it runs

Find the labels above in your own portal.

Save

Illustrated navigation view · use your own saved names and IDs

Why this action: Different scanners inspect different layers of the release.

1. In GitHub branch protection/rulesets add the scan job as a required check after it has run.
2. In a release design, scan the exact image/digest being released, then push and deploy that same image.
3. This demo scan does not automatically assess the earlier ACR image if it was built from different content.

6C.03 · ACTION 5 OF 6

SEE THE IDEA · Add dependency and container scanning gates → Explain the decision

A SMALL DECISION MAP

Add dependency and container scanning gates > Explain the decision

1 Locate

2 Action

Explain the decision

DESIGN the gate explicitly: block fixable Critical findings; review High findings with a defined owner and deadline; expire exceptions; rerun scanning when the vulnerability database changes.

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Different scanners inspect different layers of the release.

1. DESIGN the gate explicitly: block fixable Critical findings; review High findings with a defined owner and deadline; expire exceptions; rerun scanning when the vulnerability database changes.

6C.03 · ACTION 6 OF 6

AZURE PORTAL · Defender for Cloud → Environment settings → Your subscription → Defender plans

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page header includes 'AZURE PORTAL' and a search bar containing 'Defender for Cloud'. The breadcrumb navigation path is 'Defender for Cloud > Environment settings > Your subscription > Defender plans'. On the left, a sidebar lists 'Defender for Cloud', 'Environment settings', and 'Your subscription'. The main content area shows two numbered steps: '1 Inspect' and '2 Cost'. Under '1 Inspect', there are two boxes: 'Container / registry assessment coverage' and 'Review before enabling'. At the bottom right, there is a blue 'View' button. Below the screenshot, a note says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. For Microsoft Defender for Containers/registry assessment, inspect the relevant Defender plan and enabled registry coverage before enabling a paid feature.
2. Confirm that the registry/image is actually assessed; enabling a plan is not evidence of a clean image.

You should see

You have dependency and container reports and understand the blocking severity gate.

If it does not work

CodeQL source analysis does not replace operating-system image scanning. This guide does not claim that an unrun scanner found zero vulnerabilities. [R64]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

6C.04 Inspect Azure DevOps Advanced Security and Defender integration

AZ-400 · CORE + paid security extensions

Why this task: Central security views combine findings across repositories.

6C.04 · ACTION 1 OF 6

AZURE DEVOPS · OneLab → Project settings → Repositories → OneLab → Settings

The screenshot shows the Azure DevOps web interface. The browser address bar displays 'dev.azure.com / YOUR_ORG / OneLab'. The page title is 'AZURE DEVOPS'. A search bar contains 'Search: OneLab'. The breadcrumb navigation is 'OneLab > Project settings > Repositories > OneLab > Settings'. On the left, a sidebar lists 'OneLab', 'Project settings', and 'Repositories'. The main content area has three numbered steps: 1. Secret Protection (with a toggle 'Check billing first'), 2. Code Security (with a toggle 'Check billing first'), and 3. Defender integration (with a toggle 'Authorize intended repositories only'). At the bottom, there is a text prompt 'Find the labels above in your own portal.' and a purple 'Inspect' button.

Illustrated navigation view · use your own saved names and IDs

Why this action: Central security views combine findings across repositories.

1. In Azure DevOps open Project settings > Repositories > OneLab > Settings.
2. Inspect Secret Protection and Code Security toggles and the billing notice before enabling either.

6C.04 · ACTION 2 OF 6

AZURE DEVOPS · Project settings → Repositories → OneLab → Settings

The screenshot shows the Azure DevOps web interface. The browser address bar displays 'dev.azure.com / YOUR_ORG / OneLab'. The page title is 'AZURE DEVOPS'. A search bar contains 'Search: Project settings'. The breadcrumb navigation is 'Project settings > Repositories > OneLab > Settings'. On the left, a sidebar lists 'Project settings', 'Repositories', and 'OneLab'. The main content area has two numbered steps: 1. Paid products (with a toggle 'Relevant Code Security / Secret Protection feature') and 2. After enabling (with a toggle 'Run actual scan'). At the bottom, there is a text prompt 'Find the labels above in your own portal.' and a purple 'Review' button.

Illustrated navigation view · use your own saved names and IDs

Why this action: Central security views combine findings across repositories.

1. If you choose the paid exercise, enable the relevant product, configure available dependency scanning and CodeQL setup, then run a pipeline scan and open the repository security results.

6C.04 · ACTION 3 OF 6**AZURE DEVOPS · OneLab → Project settings → Repositories → OneLab → Settings**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: OneLab

OneLab > Project settings > Repositories > OneLab > Settings

OneLab
Project settings
Repositories
OneLab
Settings

1 Locate

2 Action

Settings

For YAML task-based setup, use AdvancedSecurity-Codeql-Init@1 with languages python before analysis, AdvancedSecurity-Dependency-Scanning@1 after dependencies are restored, and AdvancedSecurity-Codeql-Analyze@1 at the analysis step.

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: Central security views combine findings across repositories.

1. For YAML task-based setup, use AdvancedSecurity-Codeql-Init@1 with languages python before analysis, AdvancedSecurity-Dependency-Scanning@1 after dependencies are restored, and AdvancedSecurity-Codeql-Analyze@1 at the analysis step.
2. These tasks require the service feature to be enabled.

6C.04 · ACTION 4 OF 6**AZURE PORTAL · Defender for Cloud → Environment settings → Add environment**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Defender for Cloud

Defender for Cloud > Environment settings > Add environment

Defender for Cloud
Environment settings
Add environment

1 Connector

2 Authorization

Azure DevOps or GitHub

Intended repositories only

Find the labels above in your own portal. **Connect**

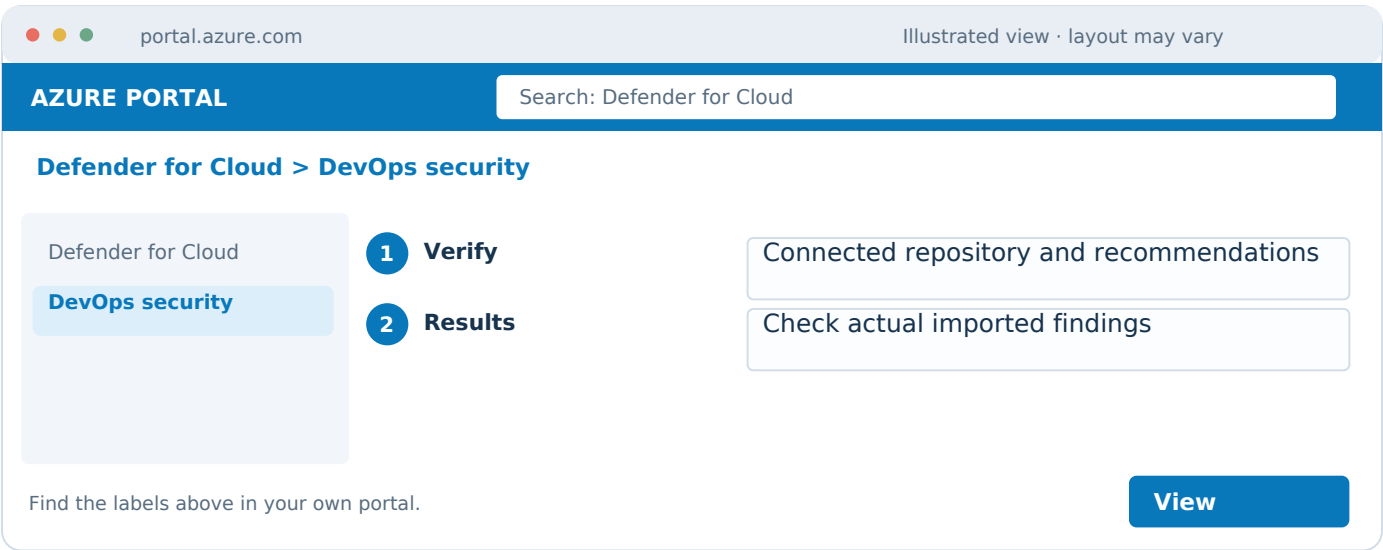
Illustrated navigation view · use your own saved names and IDs

Why this action: Central security views combine findings across repositories.

1. In Azure Portal > Microsoft Defender for Cloud > Environment settings, add the supported Azure DevOps or GitHub connection.
2. Authorize only the intended organization/repositories.

6C.04 · ACTION 5 OF 6

AZURE PORTAL · Defender for Cloud → DevOps security



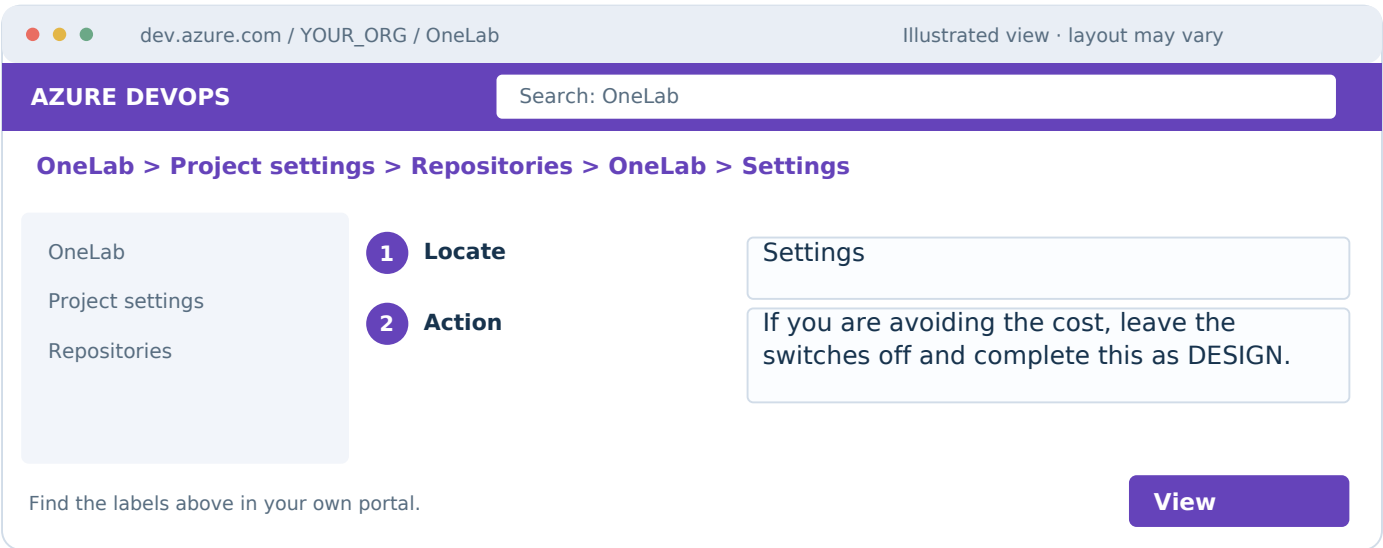
Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open Defender for Cloud > DevOps security and verify the connected repository and available recommendations.
2. Follow the connection's permission requirements for importing Advanced Security results.

6C.04 · ACTION 6 OF 6

AZURE DEVOPS · OneLab → Project settings → Repositories → OneLab → Settings



Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. If you are avoiding the cost, leave the switches off and complete this as DESIGN.
2. Record the required connection, product and permissions rather than marking it implemented.

You should see

Your record distinguishes a configured integration from actual findings imported from scans.

If it does not work

Azure DevOps currently offers Secret Protection and Code Security as separate billable products. Check costs and disable trial-only features after testing. [R46, R47]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

6C.05 Handle secure files and credential incidents

AZ-400 · CORE + paid security extensions

Why this task: Secrets can leak through files, logs, artifacts and shell debugging.

6C.05 · ACTION 1 OF 5

CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab

The dark code panel below is the terminal/editor view for this action.

Why this action: Secrets can leak through files, logs, artifacts and shell debugging.

1. Create a harmless file named training-certificate.txt containing dummy training data only.
2. Open Library > Secure files > + Secure file and upload it.

USE IN: BASH · COPY ONLY THE CODE

```
printf 'OneLab dummy certificate data\n' > /tmp/training-certificate.txt
```

What it means: Create harmless data in Cloud Shell. Manage files > Download /tmp/training-certificate.txt; upload that download in Azure DevOps.

6C.05 · ACTION 2 OF 5**AZURE DEVOPS · Pipelines → Library → Secure files → training-certificate.txt**

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS

Pipelines > Library > Secure files > training-certificate.txt

Pipelines
Library
Secure files

1 Pipeline permissions
2 Checks

Only the secure-file test pipeline
Inspect approval options

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open its security and pipeline permissions.
2. Authorize only the test pipeline; inspect approval/check options.

6C.05 · ACTION 3 OF 5**FILE / WIKI EDITOR · Project file → pipelines/secure-file-demo.yml**

The dark code panel below is the terminal/editor view for this action.

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. In a test YAML job use DownloadSecureFile@1 with secureFile: training-certificate.txt and a task name such as demoFile.
2. Reference \$(demoFile.secureFilePath) without printing the contents.

USE IN: FILE: PIPELINES/SECURE-FILE-DEMO.YML · COPY ONLY THE CODE

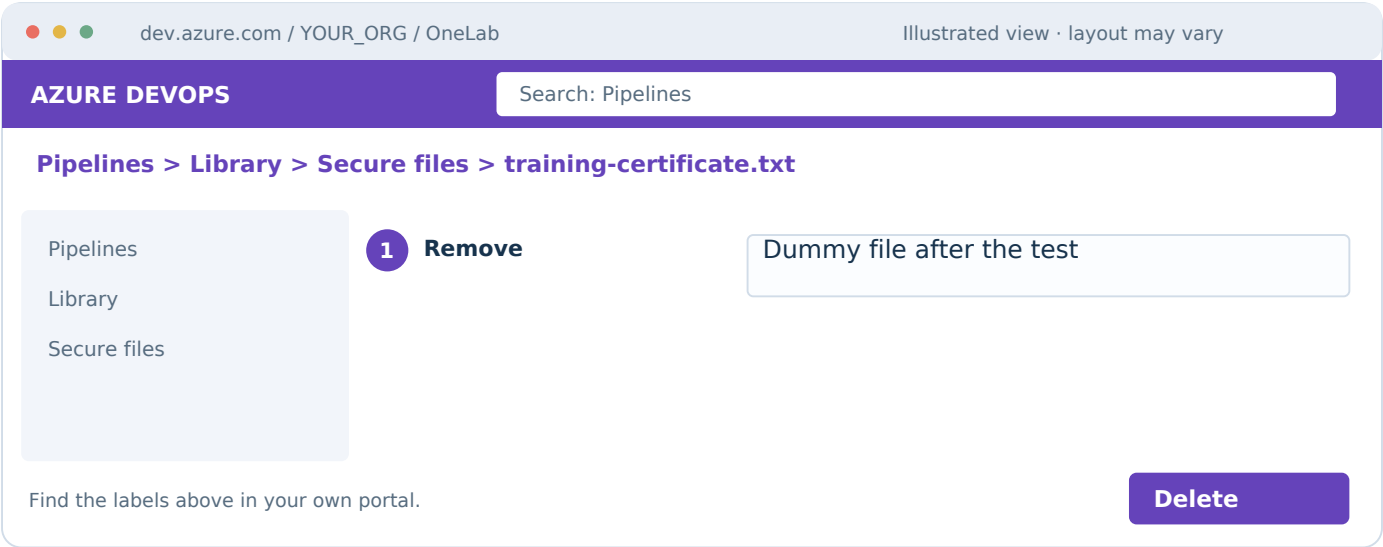
```
trigger: none
pr: none
pool:
  vmImage: ubuntu-24.04
steps:
- task: DownloadSecureFile@1
  name: demoFile
  inputs:
    secureFile: training-certificate.txt
- bash: |
  test -s "$DEMO_FILE"
  echo 'Dummy secure file is present; content was not printed.'
env:
  DEMO_FILE: $(demoFile.secureFilePath)
```

What it means: In Repos > Files create pipelines/secure-file-demo.yml on a branch, paste this YAML, commit and merge the PR. Create an Existing YAML pipeline for that path; permit only it to use the uploaded dummy file.

\$NAME / \${NAME} · Read a variable from this Bash session. Source lab.env first when these are lab names.

6C.05 · ACTION 4 OF 5

AZURE DEVOPS · Pipelines → Library → Secure files → training-certificate.txt



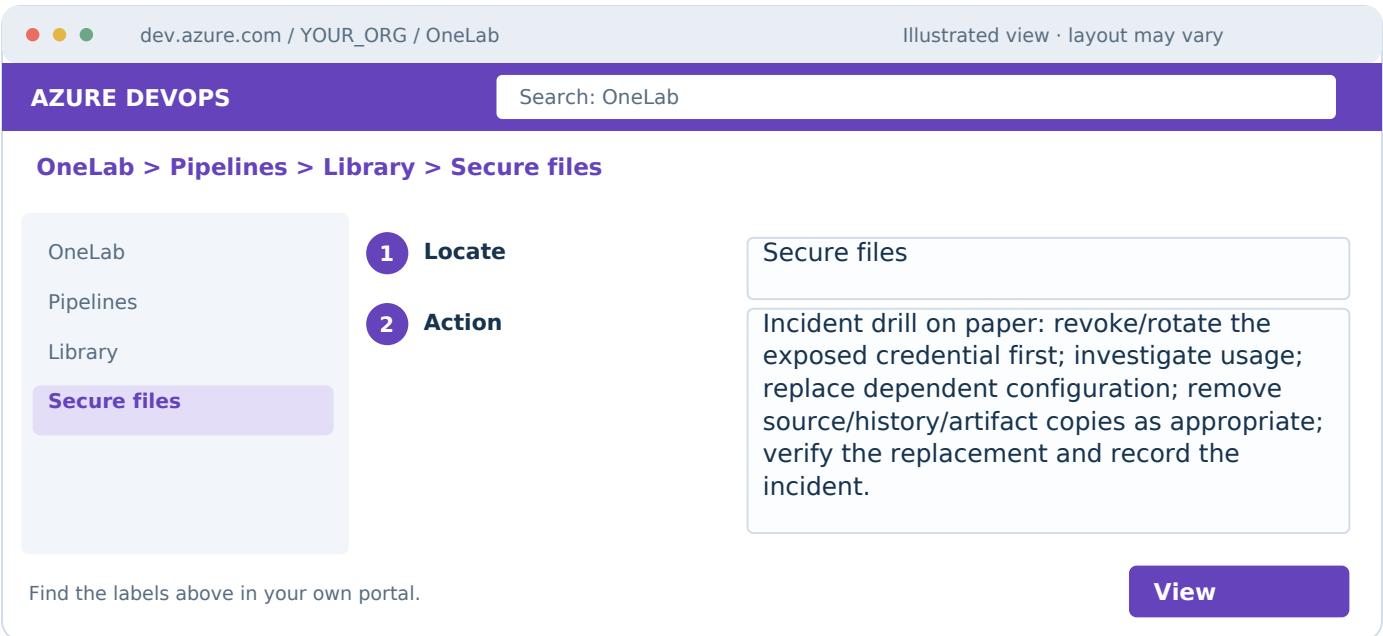
Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. After the exercise delete the dummy secure file.
2. For a real signing certificate, store its password separately, restrict the agent and remove local copies after use.

6C.05 · ACTION 5 OF 5

AZURE DEVOPS · OneLab → Pipelines → Library → Secure files



Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Incident drill on paper: revoke/rotate the exposed credential first; investigate usage; replace dependent configuration; remove source/history/artifact copies as appropriate; verify the replacement and record the incident.

You should see

You understand that log masking is not a complete secret-leak prevention mechanism.

If it does not work

Avoid set -x, echoing secrets, verbose HTTP headers and serializing whole environment-variable dictionaries. Substrings or transformed values may not be masked. [R48]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

6D · See how an agent runs a job

Planning estimate: 60 minutes. Finish with: Hosted-agent comparison and a one-command self-hosted agent test, if accessible.

6D.01 Choose the correct runner or agent

AZ-400 · CORE + self-hosted extension

Why this task: The machine that runs a pipeline needs tools, network access and maintenance.

6D.01 · ACTION 1 OF 5

AZURE DEVOPS · OneLab → Project settings → Agent pools

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS

Search: OneLab

OneLab > Project settings > Agent pools

OneLab

Project settings

Agent pools

1 Main application pipeline

2 Private network job

3 Untrusted PR

Microsoft-hosted ubuntu-24.04

Reachable self-hosted agent

Do not give privileged private access

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: The machine that runs a pipeline needs tools, network access and maintenance.

1. For the core lab use Microsoft-hosted ubuntu-24.04.
2. Each job gets a fresh machine, so files must be published/downloaded to survive across jobs.

6D.01 · ACTION 2 OF 5

AZURE DEVOPS · OneLab → Project settings → Agent pools

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS

Search: OneLab

OneLab > Project settings > Agent pools

OneLab

Project settings

Agent pools

1 Locate

2 Action

Agent pools

Choose a self-hosted agent when a job must reach private resources, needs special tools or hardware, or benefits from controlled caching.

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: The machine that runs a pipeline needs tools, network access and maintenance.

1. Choose a self-hosted agent when a job must reach private resources, needs special tools or hardware, or benefits from controlled caching.
2. You become responsible for patching, isolation and cleaning secrets.

6D.01 · ACTION 3 OF 5

AZURE DEVOPS · OneLab → Project settings → Agent pools

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS

Search: OneLab

OneLab > Project settings > Agent pools

OneLab

Project settings

Agent pools

1 Locate

2 Action

Agent pools

For a hybrid pipeline, run public build/tests on hosted agents, then deploy using a private self-hosted pool that can reach the private app/database.

Find the labels above in your own portal.

Run

Illustrated navigation view · use your own saved names and IDs

Why this action: The machine that runs a pipeline needs tools, network access and maintenance.

1. For a hybrid pipeline, run public build/tests on hosted agents, then deploy using a private self-hosted pool that can reach the private app/database.

6D.01 · ACTION 4 OF 5

AZURE DEVOPS · OneLab → Project settings → Agent pools

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS

Search: OneLab

OneLab > Project settings > Agent pools

OneLab

Project settings

Agent pools

1

Locate

2

Action

Agent pools

Windows builds may require Windows tooling or licensing.

Find the labels above in your own portal.

Add

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Windows builds may require Windows tooling or licensing.
2. A Linux container job needs a compatible host and container runtime; a container does not magically add private-network connectivity.

6D.01 · ACTION 5 OF 5

AZURE DEVOPS · OneLab → Project settings → Agent pools

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS

Search: OneLab

OneLab > Project settings > Agent pools

OneLab

Project settings

Agent pools

1

Locate

2

Action

Agent pools

Parallel jobs are capacity slots, not the same as the number of configured agents.

Find the labels above in your own portal.

Run

Illustrated navigation view · use your own saved names and IDs

Why this action: The machine that runs a pipeline needs tools, network access and maintenance.

1. Parallel jobs are capacity slots, not the same as the number of configured agents.
2. Two ready agents cannot run two jobs concurrently if the organization has only one permitted parallel job.

You should see

You can choose an agent from connectivity, tools, licence, cost and maintenance requirements.

If it does not work

Do not run untrusted fork PR code on a persistent agent that holds access to private systems. [R51]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

6D.02 Register a test self-hosted Linux agent

AZ-400 · CORE + self-hosted extension

Why this task: This exercise proves the difference between an agent record and a running worker.

6D.02 · ACTION 1 OF 6

AZURE DEVOPS · Organization settings → Agent pools → OneLabAgents

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: Organization settings

Organization settings > Agent pools > OneLabAgents

Organization settings

Agent pools

OneLabAgents

1 **Pool type** Self-hosted

2 **OS** Linux

3 **Agent name** vm-ops

4 **Registration token** Short-lived

5 **Test job** uname -a

Find the labels above in your own portal.

New agent

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. In Organization settings > Agent pools > Add pool, create a self-hosted pool OneLabAgents.
2. Grant the project appropriate use permission.

6D.02 · ACTION 2 OF 6**AZURE DEVOPS · Organization settings → Agent pools → OneLabAgents → Agents → New agent**

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS Search: Organization settings

Organization settings > Agent pools > OneLabAgents > Agents > New agent

Organization settings
Agent pools
OneLabAgents

1 OS
2 Architecture
3 Download

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: This exercise proves the difference between an agent record and a running worker.

1. Open the pool > Agents > New agent > Linux x64.
2. Copy the portal's current download/extract commands.
3. Use a separate agent directory on vm-ops while connected interactively through Bastion or an approved SSH path.

6D.02 · ACTION 3 OF 6**AZURE DEVOPS · User settings → Personal access tokens → New Token**

dev.azure.com / YOUR_ORG / OneLab Illustrated view · layout may vary

AZURE DEVOPS Search: User settings

User settings > Personal access tokens > New Token

User settings
Personal access tokens
New Token

1 Scope
2 Expiry
3 Input

Find the labels above in your own portal. **Create**

Illustrated navigation view · use your own saved names and IDs

Why this action: This exercise proves the difference between an agent record and a running worker.

1. Create a short-lived Azure DevOps PAT with only Agent Pools Read & manage for registration.
2. Do not paste it into command text; supply it at the interactive prompt.

6D.02 · ACTION 4 OF 6**CLOUD SHELL · vm-ops → Interactive SSH/Bastion session → Separate agent folder**

portal.azure.com Illustrated view · layout may vary

CLOUD SHELL

vm-ops > Interactive SSH/Bastion session > Separate agent folder

vm-ops

Interactive SSH/Bastion session

Separate agent folder

1 Run

2 Pool

3 Agent

4 Then

Find the labels above in your own portal.

Enter

Illustrated navigation view · use your own saved names and IDs

Why this action: This exercise proves the difference between an agent record and a running worker.

1. Run `./config.sh` and answer organization URL, PAT authentication, token, pool OneLabAgents, agent name vm-ops, and work folder `_work`.
2. Then run `./run.sh` as the ordinary agent user.

6D.02 · ACTION 5 OF 6**FILE / WIKI EDITOR · Project file → agent test**

The dark code panel below is the terminal/editor view for this action.

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Confirm the agent appears Online.
2. Create a test pipeline with pool name OneLabAgents and one Bash step `uname -a`.
3. Run it and verify it executed on this VM.

USE IN: YAML: AGENT TEST · COPY ONLY THE CODE

```
trigger: none
pool:
  name: OneLabAgents
steps:
- bash: uname -a
  displayName: Identify the self-hosted machine
```

What it means: `pool.name` selects your pool. `pool.vmImage` would select a Microsoft-hosted image instead.

6D.02 · ACTION 6 OF 6

CLOUD SHELL · vm-ops → Agent terminal

portal.azure.com Illustrated view · layout may vary

CLOUD SHELL Search: vm-ops

vm-ops > Agent terminal

vm-ops

Agent terminal

- 1 Stop** Ctrl+C
- 2 Remove** ./config.sh remove
- 3 Then** Revoke registration PAT

Find the labels above in your own portal. **Enter**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Stop `./run.sh` with Ctrl+C after the test.
2. Remove the agent registration using `./config.sh remove` if finished, revoke the registration PAT, and delete the empty pool if no longer needed.

You should see

You have observed Online, job execution and Offline states.

If it does not work

Do not run the full application pipeline on this pool until its Python/Azure CLI/Docker needs are installed. The one-line test is only agent connectivity validation.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Finish Day 6

Save your place and check cost

Delete the short-lived ACI, Container App/environment and practice scale set after their checks. Stop/remove the test agent and revoke its registration token when finished. Keep vm-ops for recovery.

Say it in your own words: Does passing a code scan prove the container image has no vulnerable operating-system packages?

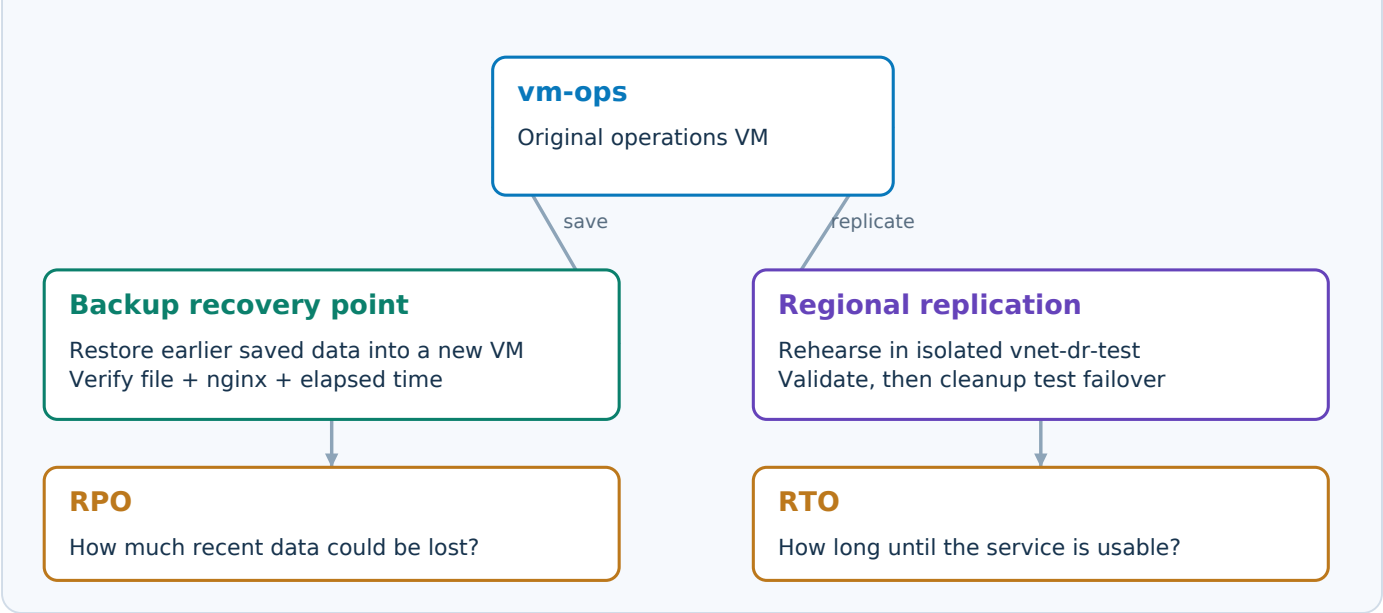
Check your explanation: No. Source, dependency, secret and image checks inspect different things. Review the findings for the exact release you intend to use.

DAY 7 OF 7

Recover, explain and clean up

You prove what works, recover a saved service and remove the resources you no longer need.

Backup and replication solve different problems



Start today

Run the restart routine and start vm-ops. Check recovery points and replication health first. Start the VM and disk restore jobs early, then return to monitoring while they run.

Start early and track the job

Run the isolated test failover as soon as replication is healthy. Finish its cleanup before disabling protection. Do the practical checks before deleting the project.

Block	Today's work	Minutes
A	Find the evidence and trigger an alert	90
B	Restore a copy of the service	100
C	Rehearse recovery and explain your work	140
D	Save your work and remove the lab	60

7A · Find the evidence and trigger an alert

Planning estimate: 90 minutes. Finish with: Fresh KQL results, VM/storage logs, an alert and a linked incident.

7A.01 Query the telemetry with KQL

AZ-104 + AZ-400 · CORE

Why this task: KQL filters, groups and summarizes recorded events.

7A.01 · ACTION 1 OF 4

AZURE PORTAL · Log Analytics workspaces → law-PREFIX → Logs

The screenshot shows the Azure Portal interface for configuring a Log Analytics query. The browser address bar shows 'portal.azure.com'. The page title is 'AZURE PORTAL'. A search bar contains 'Log Analytics workspaces'. The breadcrumb navigation is 'Log Analytics workspaces > law-PREFIX > Logs'. On the left, a sidebar shows 'Log Analytics workspaces', 'law-PREFIX', and 'Logs' (selected). The main area has three numbered steps: 1. Scope (set to 'law-PREFIX'), 2. Time range (set to 'Last 30-60 minutes'), and 3. Table (set to 'AppRequests / AppDependencies'). A 'Run' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: KQL filters, groups and summarizes recorded events.

1. Before querying, repeat the small traffic loop from Day 3, Block D against your own app.
2. Open law-PREFIX > Logs.
3. Select KQL mode if the editor opens in Simple mode.
4. Close example queries if needed.
5. Set Last 30 minutes.
6. Old traffic from Day 3 will not be in this window.

7A.01 · ACTION 2 OF 4

AZURE LOGS · KQL · Azure Portal → law-PREFIX → Logs → KQL mode

The dark code panel below is the terminal/editor view for this action.

Why this action: KQL filters, groups and summarizes recorded events.

1. Run each query below separately.
2. Read the columns and compare the results to the traffic you generated.

USE IN: KQL · COPY ONLY THE CODE

```
AppRequests
| where TimeGenerated > ago(30m)
| project TimeGenerated, Name, ResultCode, DurationMs, OperationId
| order by TimeGenerated desc
```

What it means: where filters rows, project selects fields, and order by sorts them.

where · Keep rows matching the condition.

project · Choose the columns to display.

ago · Use a time offset back from now.

OperationId · Correlate a request with dependencies from the same operation.

USE IN: KQL · COPY ONLY THE CODE

```
AppRequests
| where TimeGenerated > ago(1h)
| summarize Requests=sum(ItemCount), Failures=sumif(ItemCount, Success == false),
    P95ms=percentile(DurationMs, 95) by bin(TimeGenerated, 5m)
| order by TimeGenerated asc
```

What it means: ItemCount accounts for represented requests when telemetry is sampled. P95 is the response duration under which 95% of the sampled durations fall; it is not the average. bin groups timestamps into five-minute windows.

where · Keep rows matching the condition.

summarize · Calculate grouped totals or statistics.

bin · Group timestamps into equal-sized time windows.

ago · Use a time offset back from now.

percentile · Estimate the duration below which the stated percentage falls.

USE IN: KQL · COPY ONLY THE CODE

```
AppDependencies
| where TimeGenerated > ago(1h)
| project TimeGenerated, Name, Target, Success, DurationMs, OperationId
| order by TimeGenerated desc
```

What it means: OperationId lets you correlate a dependency with the request that initiated it. Trace collection depends on instrumented components.

where · Keep rows matching the condition.

project · Choose the columns to display.

ago · Use a time offset back from now.

OperationId · Correlate a request with dependencies from the same operation.

7A.01 · ACTION 3 OF 4**AZURE PORTAL · Log Analytics workspaces → law-PREFIX → Logs**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Log Analytics workspaces

Log Analytics workspaces > law-PREFIX > Logs

Log Analytics workspaces
law-PREFIX
Logs

1 Table AppRequests / AppDependencies

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: KQL filters, groups and summarizes recorded events.

1. AppRequests is the workspace table.
2. If you query from an Application Insights resource with classic-style aliases, the equivalent name can be requests and its field names differ.

7A.01 · ACTION 4 OF 4**AZURE PORTAL · Log Analytics workspaces → law-PREFIX → Logs**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Log Analytics workspaces

Log Analytics workspaces > law-PREFIX > Logs

Log Analytics workspaces
law-PREFIX
Logs

1 Locate
2 Action

Logs
Save the useful query.

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: KQL filters, groups and summarizes recorded events.

1. Save the useful query.
2. Pin a result to a dashboard or add it to a workbook so the next operator can find it.

You should see

You can explain where, project, summarize, count, percentile and bin in your own words.

If it does not work

A table-not-found error often means wrong scope or no data collection for that table, not invalid permissions to every Azure resource.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

7A.02 Collect VM guest logs and inspect resource Insights

AZ-104 + AZ-400 · CORE

Why this task: Platform metrics do not automatically contain guest operating-system logs or memory measurements.

7A.02 · ACTION 1 OF 7

AZURE PORTAL · Monitor → Data collection rules → Create

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Monitor

Monitor > Data collection rules > Create

Monitor

Data collection rules

Create

- 1 Name** dcr-onelab-syslog
- 2 Platform** Linux
- 3 Resource** vm-ops
- 4 Data source** Linux Syslog: user, Warning+
- 5 Destination** law-PREFIX

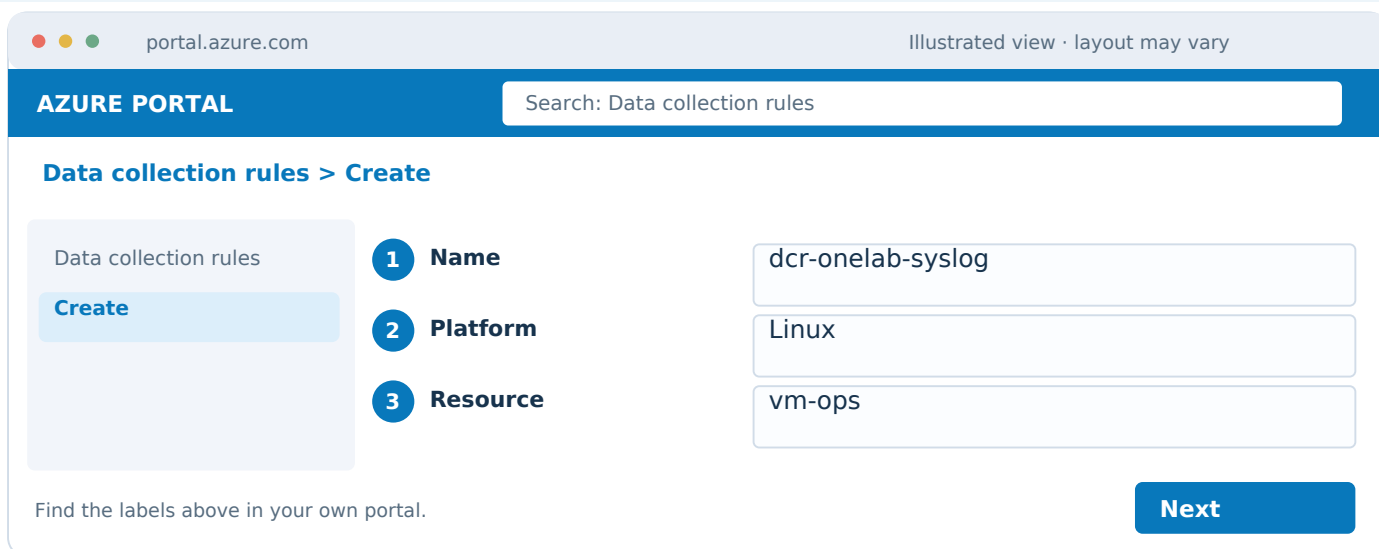
Find the labels above in your own portal.

Review + create

Illustrated navigation view · use your own saved names and IDs

Why this action: Platform metrics do not automatically contain guest operating-system logs or memory measurements.

1. Open vm-ops > Monitoring > Insights or the current Monitor experience > Enable.
2. Select Azure Monitor Agent and the intended workspace/data collection rule where prompted.

7A.02 · ACTION 2 OF 7**AZURE PORTAL · Data collection rules → Create**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Data collection rules

Data collection rules > Create

Data collection rules

Create

1 Name dcr-onelab-syslog

2 Platform Linux

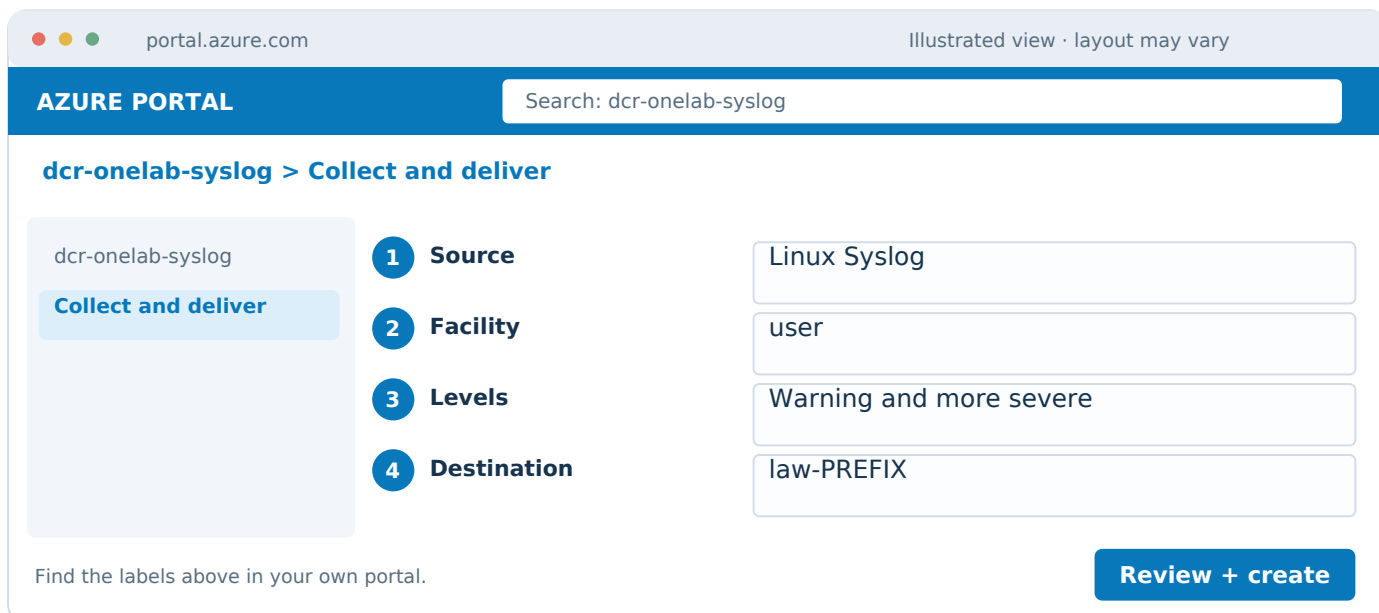
3 Resource vm-ops

Find the labels above in your own portal. **Next**

Illustrated navigation view · use your own saved names and IDs

Why this action: Platform metrics do not automatically contain guest operating-system logs or memory measurements.

1. Search Data collection rules > Create.
2. Name dcr-onelab-syslog; use a compatible region/platform Linux.
3. Add vm-ops as a resource.

7A.02 · ACTION 3 OF 7**AZURE PORTAL · dcr-onelab-syslog → Collect and deliver**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: dcr-onelab-syslog

dcr-onelab-syslog > Collect and deliver

dcr-onelab-syslog

Collect and deliver

1 Source Linux Syslog

2 Facility user

3 Levels Warning and more severe

4 Destination law-PREFIX

Find the labels above in your own portal. **Review + create**

Illustrated navigation view · use your own saved names and IDs

Why this action: Platform metrics do not automatically contain guest operating-system logs or memory measurements.

1. Add Linux Syslog as a data source, selecting user facility and Warning or more severe levels.
2. Select law-PREFIX as the destination; review and create.

7A.02 · ACTION 4 OF 7**AZURE LOGS · KQL · Azure Portal → law-PREFIX → Logs → KQL mode**

The dark code panel below is the terminal/editor view for this action.

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Inside vm-ops Run command execute logger -p user.warning "OneLab test event".
2. Wait for ingestion, then run the Syslog query below.

USE IN: KQL · COPY ONLY THE CODE

```
Syslog
| where TimeGenerated > ago(1h)
| where SyslogMessage has "OneLab"
| project TimeGenerated, Computer, Facility, SeverityLevel, SyslogMessage
```

What it means: Run this from the workspace that receives the VM's DCR output.

where · Keep rows matching the condition.

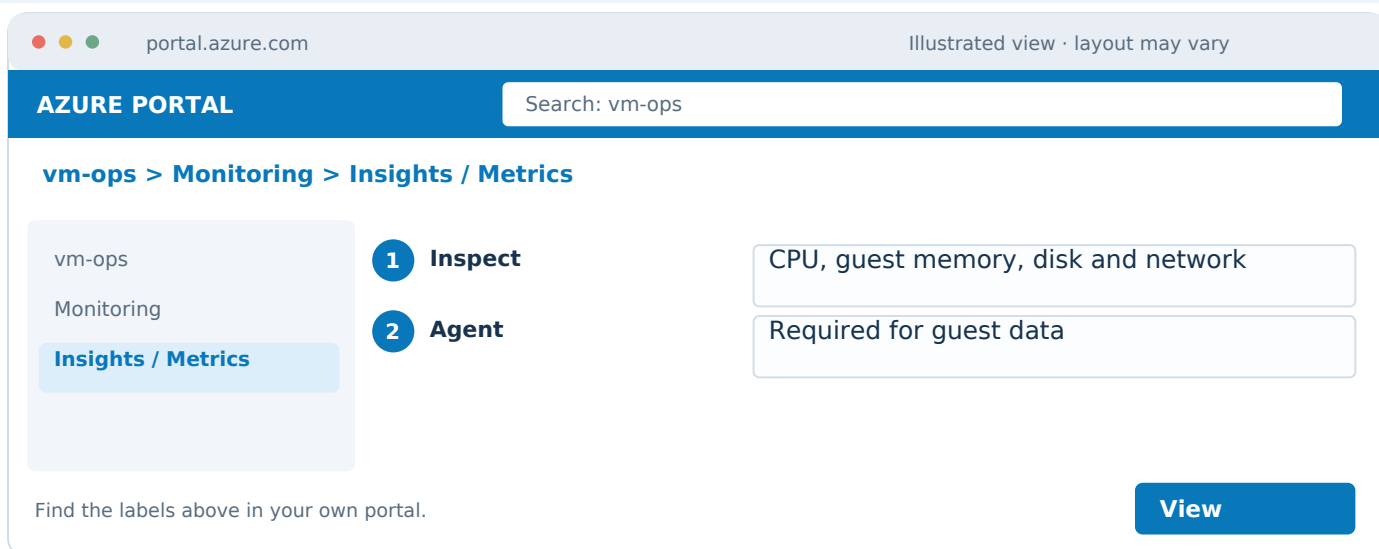
project · Choose the columns to display.

ago · Use a time offset back from now.

USE IN: VM RUNSHELLSCRIPT · COPY ONLY THE CODE

```
logger -p user.warning "OneLab test event"
```

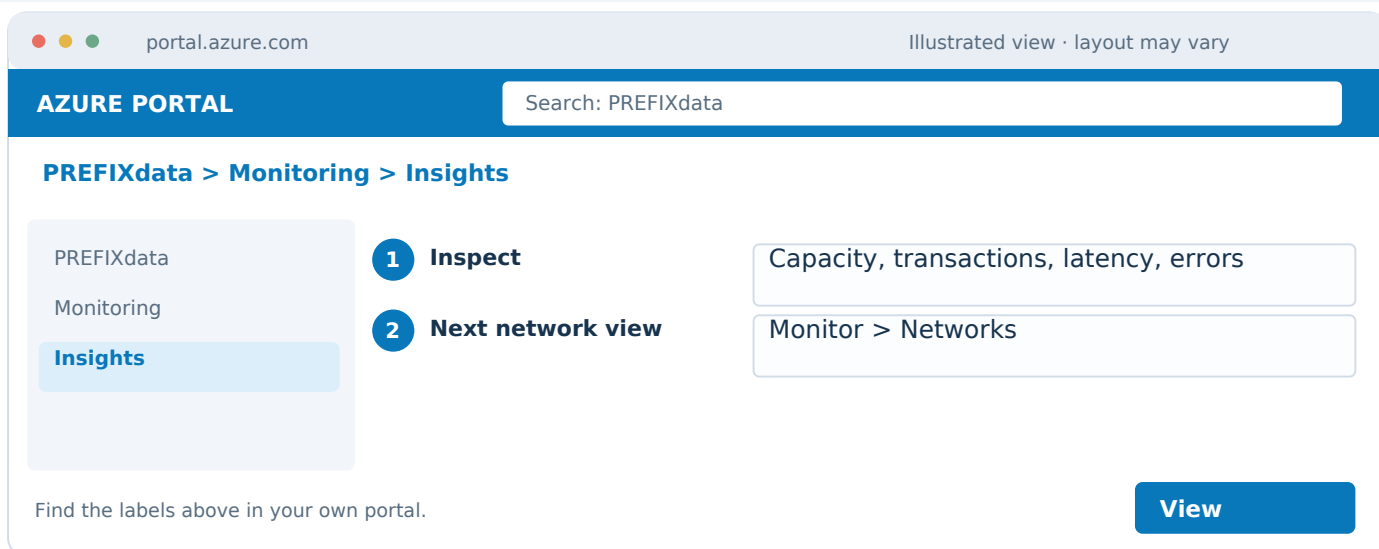
What it means: This creates one Linux user-facility warning for the data collection rule.

7A.02 · ACTION 5 OF 7**AZURE PORTAL · vm-ops → Monitoring → Insights / Metrics**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

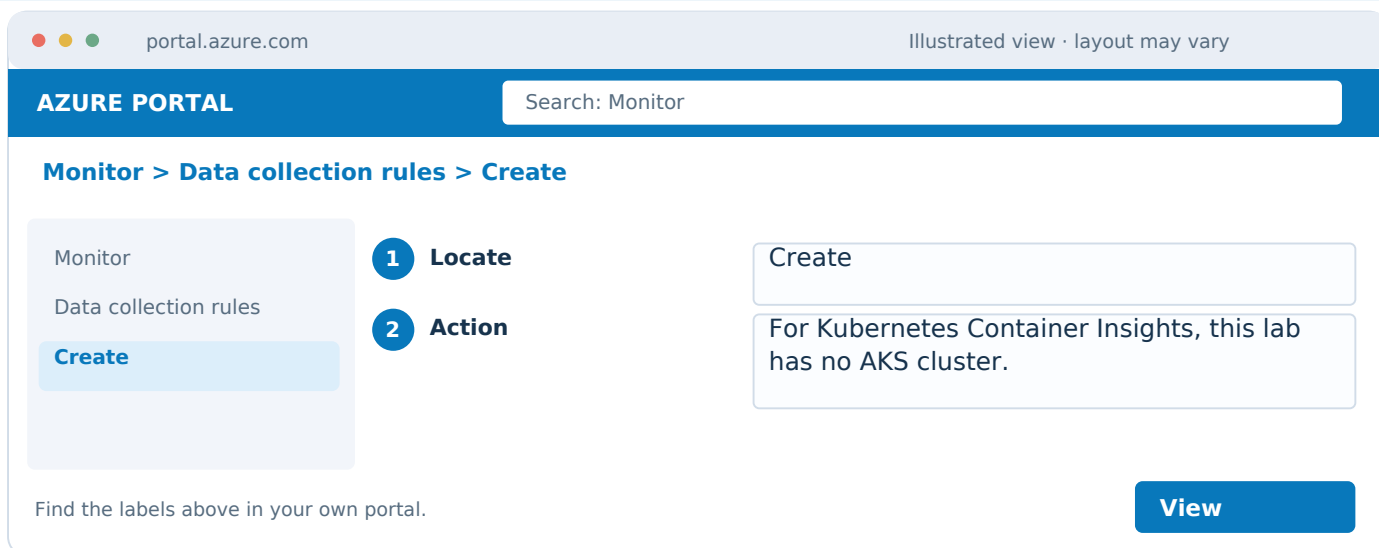
1. Inspect VM CPU, available memory, disk read/write latency/throughput and network traffic using enabled guest/Insights metrics.
2. Note which are host metrics and which require an agent.

7A.02 · ACTION 6 OF 7**AZURE PORTAL · PREFIXdata → Monitoring → Insights**

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Open PREFIXdata > Monitoring > Insights to inspect capacity, transactions, latency and errors.
2. Open Monitor > Networks or Network Watcher for topology/connectivity views.

7A.02 · ACTION 7 OF 7**AZURE PORTAL · Monitor → Data collection rules → Create**

Illustrated navigation view · use your own saved names and IDs

Why this action: Platform metrics do not automatically contain guest operating-system logs or memory measurements.

1. For Kubernetes Container Insights, this lab has no AKS cluster.
2. DESIGN: enable supported container monitoring on an eligible cluster with a workspace/DCR and inspect node/pod/container signals.
3. Container Apps logs/metrics are not evidence that AKS Container Insights was deployed.

You should see

A real OneLab Syslog event appears and you can identify the telemetry source.

If it does not work

A DCR states what to collect and where to send it; installing an agent alone does not select every log. Confirm resource association and supported region if data is absent. [R54]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

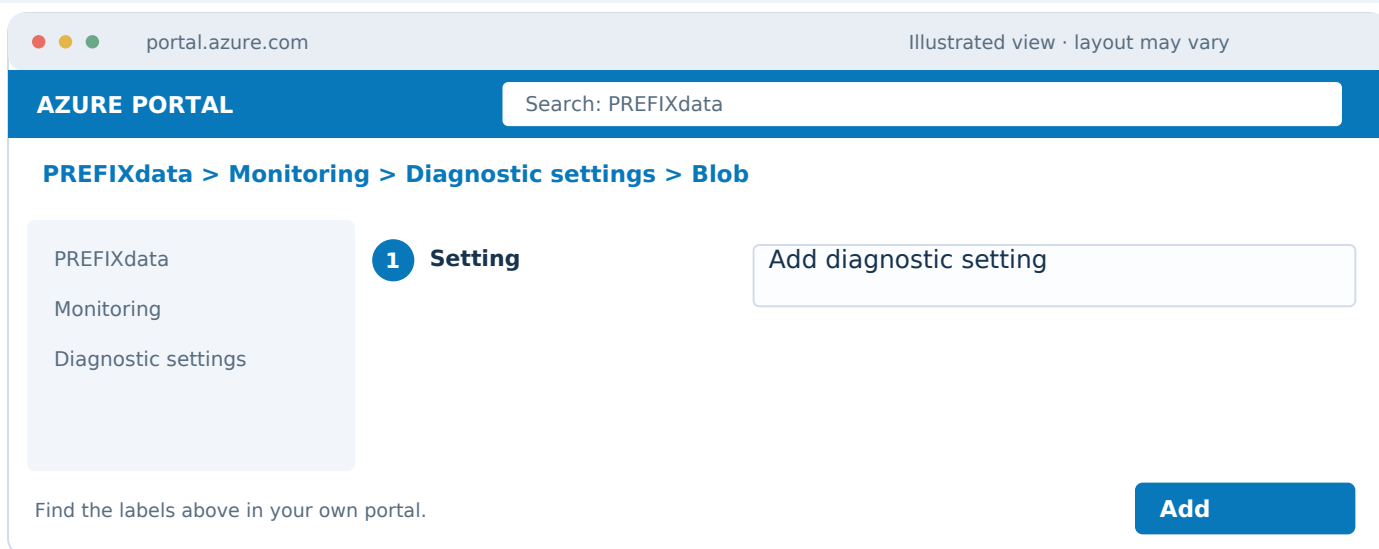
7A.03 Send storage and activity logs to the workspace

AZ-104 + AZ-400 · CORE

Why this task: Diagnostic settings connect resource logs to a destination.

7A.03 · ACTION 1 OF 5

AZURE PORTAL · PREFIXdata → Monitoring → Diagnostic settings → Blob



Illustrated navigation view · use your own saved names and IDs

Why this action: Diagnostic settings connect resource logs to a destination.

1. Open PREFIXdata > Monitoring > Diagnostic settings.
2. Select the Blob service diagnostic scope if needed, then Add diagnostic setting.

7A.03 · ACTION 2 OF 5

AZURE PORTAL · Blob → Diagnostic setting

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Blob

Blob > Diagnostic setting

Blob

Diagnostic setting

- 1 Name** blob-logs
- 2 Logs** Read / write / delete
- 3 Destination** law-PREFIX
- 4 Tables** Resource-specific

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Name blob-logs.
2. Select read/write/delete log categories available for that service, send to law-PREFIX, and choose resource-specific tables when offered.
3. Save.

7A.03 · ACTION 3 OF 5

AZURE LOGS · KQL · Azure Portal → law-PREFIX → Logs → KQL mode

The dark code panel below is the terminal/editor view for this action.

Why this action: Diagnostic settings connect resource logs to a destination.

1. Read or upload a lab blob to generate an event.
2. Query StorageBlobLogs in the workspace after ingestion.

USE IN: KQL · COPY ONLY THE CODE

```
StorageBlobLogs
| where TimeGenerated > ago(30m)
| project TimeGenerated, OperationName, StatusCode, AuthenticationType
| order by TimeGenerated desc
```

What it means: Run in law-PREFIX > Logs after uploading or reading a blob.

where · Keep rows matching the condition.

project · Choose the columns to display.

ago · Use a time offset back from now.

7A.03 · ACTION 4 OF 5

AZURE PORTAL · Subscriptions → Your subscription → Activity log → Export Activity Logs

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL

Search: Subscriptions

Subscriptions > Your subscription > Activity log > Export Activity Logs

Subscriptions

Your subscription

Activity log

1 Destination

2 Categories

law-PREFIX

Administrative lab events

Find the labels above in your own portal.

Save

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Open the subscription > Activity log > Export Activity Logs/Diagnostic settings.
2. Send selected administrative events to the workspace if your permissions allow.

7A.03 · ACTION 5 OF 5

SEE THE IDEA · Two log sources

A SMALL DECISION MAP

Two log sources

1 Activity log

2 StorageBlobLogs

Resource setting changes

Blob data operations

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Diagnostic settings connect resource logs to a destination.

1. Compare Activity log with data-plane logs: changing the storage account settings is an administrative action; reading a blob is a storage data operation.

You should see

You can locate both administrative and data-operation evidence.

If it does not work

Log ingestion and retention cost money. Collect the categories needed for the exercise and remove unnecessary diagnostic settings during cleanup.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

7A.04 Create and test an alert with a maintenance rule

AZ-104 + AZ-400 · CORE

Why this task: An alert is useful only if its evaluation and notification path work.

7A.04 · ACTION 1 OF 6

AZURE PORTAL · Monitor → Alerts → Action groups → Create

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL

Monitor > Alerts > Action groups > Create

Monitor	1 Name	ag-onelab
Alerts	2 Short name	onelab
Action groups	3 Notification	Your own email
Create	4 Lab alert	Requests total > 0

Find the labels above in your own portal. **Review + create**

Illustrated navigation view · use your own saved names and IDs

Why this action: An alert is useful only if its evaluation and notification path work.

1. Open Monitor > Alerts > Action groups > Create.
2. Group: rg-PREFIX-core; name ag-onelab; short name onelab.
3. Add an email notification to your own email and create.

7A.04 · ACTION 2 OF 6**AZURE PORTAL · Monitor → Alerts → Action groups → ag-onelab → Test**

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page header includes 'AZURE PORTAL' and a search bar containing 'Monitor'. The breadcrumb navigation path is 'Monitor > Alerts > Action groups > ag-onelab > Test'. On the left, a sidebar lists 'Monitor', 'Alerts', and 'Action groups'. The main content area features two numbered steps: '1 Notification' with the input 'Your own email', and '2 Expected' with the input 'Receive test email'. A 'Test' button is located at the bottom right. A note at the bottom left states: 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Use Test action group and confirm receipt.
2. This proves notification delivery, not yet the metric condition.

7A.04 · ACTION 3 OF 6**AZURE PORTAL · PREFIX-web → Alerts → Create alert rule → Condition**

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page header includes 'AZURE PORTAL' and a search bar containing 'PREFIX-web'. The breadcrumb navigation path is 'PREFIX-web > Alerts > Create alert rule > Condition'. On the left, a sidebar lists 'PREFIX-web', 'Alerts', 'Create alert rule', and 'Condition' (which is highlighted). The main content area features four numbered steps: '1 Signal' with the input 'Requests or supported request metric', '2 Aggregation' with the input 'Total', '3 Threshold' with the input 'Greater than 0', and '4 Window' with the input '5 minutes for this lab'. A 'Next' button is located at the bottom right. A note at the bottom left states: 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: An alert is useful only if its evaluation and notification path work.

1. Open the web app > Alerts > Create alert rule.
2. Scope the app; choose Requests or another available request-count metric; condition Total greater than 0 in a five-minute window for this short exercise.

7A.04 · ACTION 4 OF 6**AZURE PORTAL · Create alert rule → Actions / Details**

The screenshot shows the Azure Portal interface for creating an alert rule. The browser address bar displays 'portal.azure.com'. The page title is 'AZURE PORTAL' with a search bar containing 'Create alert rule'. The breadcrumb navigation is 'Create alert rule > Actions / Details'. On the left, a sidebar shows 'Create alert rule' and 'Actions / Details' (selected). The main content area has three numbered steps: 1. Action group (value: ag-onelab), 2. Name (value: lab-request-test), and 3. Severity (value: Informational / low). A 'Create' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Attach ag-onelab.
2. Name lab-request-test, choose a low/informational severity, review evaluation interval and create.
3. Generate one request and wait for the alert to fire.

7A.04 · ACTION 5 OF 6**AZURE PORTAL · Monitor → Alerts → Alert processing rules → Create**

The screenshot shows the Azure Portal interface for creating an alert processing rule. The browser address bar displays 'portal.azure.com'. The page title is 'AZURE PORTAL' with a search bar containing 'Monitor'. The breadcrumb navigation is 'Monitor > Alerts > Alert processing rules > Create'. On the left, a sidebar shows 'Monitor', 'Alerts', and 'Alert processing rules' (selected). The main content area has three numbered steps: 1. Scope (value: Lab app/group only), 2. Action (value: Suppress notifications), and 3. Schedule (value: Short maintenance window). A 'Review + create' button is at the bottom right. A note at the bottom left says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: An alert is useful only if its evaluation and notification path work.

1. Open Monitor > Alerts > Alert processing rules > Create.
2. Scope only the lab app/group; choose suppression for a short maintenance window.
3. Observe that suppression affects actions rather than deleting the underlying metric.

7A.04 · ACTION 6 OF 6

AZURE PORTAL · Monitor → Alerts → Alert rules / Alert processing rules

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Monitor

Monitor > Alerts > Alert rules / Alert processing rules

Monitor

Alerts

Alert rules / Alert processing rules

1 Remove

lab-request-test and its temporary suppression

2 Keep

Evidence of test

Find the labels above in your own portal.

Disable or delete

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Disable/delete the temporary rule and suppression after testing.
2. For a useful operating alert use sustained errors/latency/availability thresholds tied to a service target.

You should see

You have evidence for condition evaluation and notification delivery.

If it does not work

Metric time grain, aggregation, threshold, window and evaluation frequency all affect firing. An action group is the destination, not the threshold. [R55]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

7A.05 Monitor connectivity and turn an incident into work

AZ-104 + AZ-400 · CORE

Why this task: Operational feedback should drive a concrete improvement.

7A.05 · ACTION 1 OF 6

AZURE PORTAL · Network Watcher → Connection monitor → Create

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Network Watcher

Network Watcher > Connection monitor > Create

Network Watcher

Connection monitor

Create

1 Source vm-ops

2 Destination Your app HTTPS hostname

3 Port 443

4 Result Latency and reachability

Find the labels above in your own portal.

Create

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Open Network Watcher > Connection monitor > Create.
2. Choose vm-ops as a supported source and your app HTTPS endpoint on port 443 as destination.

7A.05 · ACTION 2 OF 6**AZURE PORTAL · Create connection monitor → Test groups**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Create connection monitor

Create connection monitor > Test groups

Create connection monitor

Test groups

1 **Source** vm-ops

2 **Destination** App hostname

3 **Port** 443

4 **Extension** Install if requested

Find the labels above in your own portal. **Review + create**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Install/enable the required supported monitoring extension if the wizard requests it.
2. Configure a test group and a short test interval, select the workspace if required, then create.

7A.05 · ACTION 3 OF 6**AZURE PORTAL · Network Watcher → Connection monitor → Your monitor**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Network Watcher

Network Watcher > Connection monitor > Your monitor

Network Watcher

Connection monitor

Your monitor

1 **Read** Latency and reachability

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: Operational feedback should drive a concrete improvement.

1. Inspect latency and reachability.
2. A healthy TCP connection does not prove the application returns the correct content.

7A.05 · ACTION 4 OF 6**AZURE DEVOPS · Boards → Work items → New → Bug**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: Boards

Boards > Work items > New > Bug

Boards
Work items
New

1 Title OneLab intentional failure

2 Evidence Time, operation ID, impact and verified fix

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: Operational feedback should drive a concrete improvement.

1. Create a Boards bug from the intentional failure exercise.
2. Add timestamps, request/operation ID, observed impact, root cause and fix verification.

7A.05 · ACTION 5 OF 6**AZURE DEVOPS · Overview → Dashboards → Edit**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: Overview

Overview > Dashboards > Edit

Overview
Dashboards
Edit

1 Add Release failures, duration, recovery time

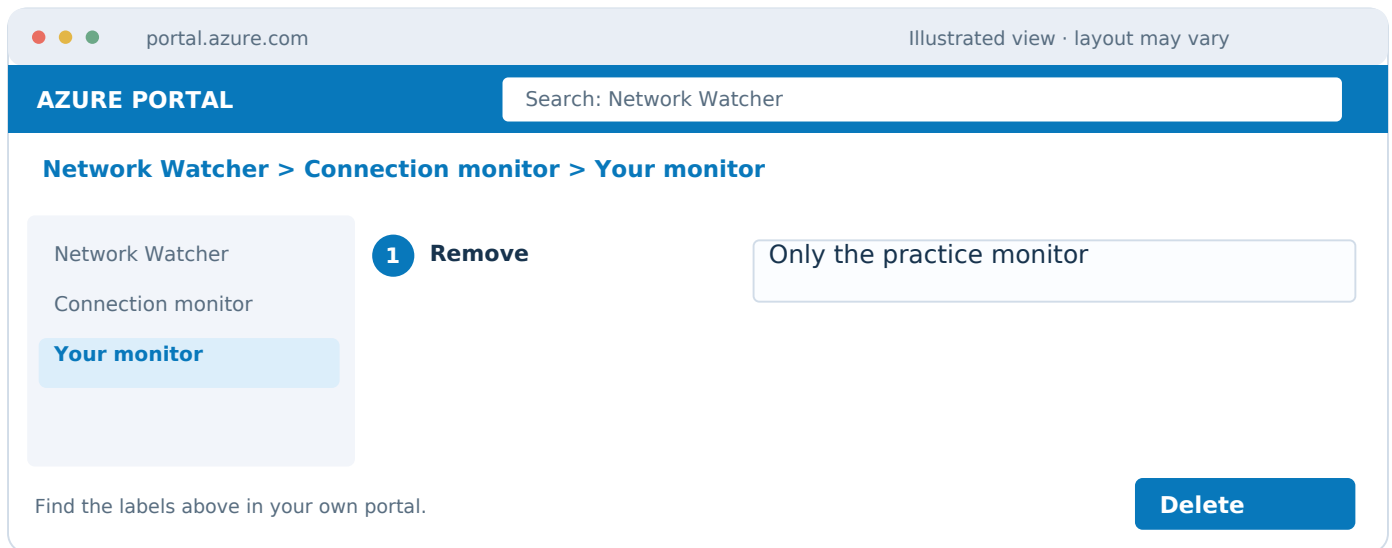
2 Notifications Subscribe to failed runs

Find the labels above in your own portal. **Save**

Illustrated navigation view · use your own saved names and IDs

Why this action: Operational feedback should drive a concrete improvement.

1. Update the dashboard with failed release count, deployment duration and the recovery time from Day 5, Block B.
2. Configure failed-run notifications for both Azure Pipelines and GitHub Actions.

7A.05 · ACTION 6 OF 6**AZURE PORTAL · Network Watcher → Connection monitor → Your monitor**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Delete the Connection monitor after the exercise if you do not need continuous tests.

You should see

An operational observation has a linked work item and measurable recovery outcome.

If it does not work

Use infrastructure CPU/memory/disk/network signals together with application error/latency data; a quiet CPU does not prove the application is healthy. [R56]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

7B · Restore a copy of the service

Planning estimate: 100 minutes. Finish with: A verified VM/file restore, disk restore and backup reporting evidence.

7B.01 Restore without overwriting the source

AZ-104 · EXTENSION - backup storage and restore costs

Why this task: A restored copy lets you prove recovery while preserving the working VM.

7B.01 · ACTION 1 OF 6

AZURE PORTAL · rsv-PREFIX → Backup items → vm-ops → Restore VM

portal.azure.comIllustrated view · layout may vary

AZURE PORTAL

Search: rsv-PREFIX

rsv-PREFIX > Backup items > vm-ops > Restore VM

rsv-PREFIX

Backup items

vm-ops

Restore VM

1 Recovery point

2 Restore type

3 Name

4 Target group

5 Network

Completed point

New VM if supported

vm-ops-restored

Separate lab restore group

Isolated restore network

Find the labels above in your own portal.

Restore

Illustrated navigation view · use your own saved names and IDs

Why this action: A restored copy lets you prove recovery while preserving the working VM.

1. Open the protected VM item > Restore VM.
2. Select the completed recovery point.

7B.01 · ACTION 2 OF 6**AZURE PORTAL · vm-ops → Backup → Restore VM → Restore configuration**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: vm-ops

vm-ops > Backup > Restore VM > Restore configuration

vm-ops
Backup
Restore VM

- 1 Route** Create new VM if supported
- 2 Group** rg-PREFIX-restore
- 3 Network** vnet-restore/restore

Find the labels above in your own portal. **Next**

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. Choose Create new virtual machine where supported, or Restore disks if the portal requires that route.
2. Use a separate lab restore group and an isolated subnet/network.
3. Before restore, use Resource groups > Create to make rg-PREFIX-restore in the primary region.
4. Use Virtual networks > Create to make vnet-restore there with 10.50.0.0/16 and subnet restore 10.50.1.0/24.
5. Do not peer this restore network.

7B.01 · ACTION 3 OF 6**AZURE PORTAL · Restore configuration → Review**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Restore configuration

Restore configuration > Review

Restore configuration
Review

- 1 New VM** vm-ops-restored
- 2 Original** Do not replace vm-ops

Find the labels above in your own portal. **Restore**

Illustrated navigation view · use your own saved names and IDs

Why this action: The network needs a valid path and a name that points to the intended destination.

1. For the new VM route, give it a distinct name such as vm-ops-restored.
2. Review storage, network and price, then restore.

7B.01 · ACTION 4 OF 6**VM TERMINAL IN AZURE PORTAL · Azure Portal → vm-ops-restored → Run command → RunShellScript**

The dark code panel below is the terminal/editor view for this action.

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Wait for restore completion, open the restored VM and use Run command to inspect `/var/www/html/index.html` and start/check nginx if needed.

USE IN: VM RUNSHELLSCRIPT · COPY ONLY THE CODE

```
sudo systemctl start nginx
cat /var/www/html/index.html
curl -f http://127.0.0.1
```

What it means: Run inside vm-ops-restored. Compare the saved text with the recovery-point time.

curl · Send an HTTP request. -f fails on HTTP error status; -s quiets the progress meter; -v shows connection details when used.

systemctl enable --now · Set the service to start at boot and start it now; stop/start act on the running service.

sudo · Run that Linux operation with administrator privileges inside the named VM.

7B.01 · ACTION 5 OF 6**AZURE PORTAL · rsv-PREFIX → Backup items → vm-ops → Restore VM**

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page header includes 'AZURE PORTAL' and a search bar containing 'rsv-PREFIX'. The breadcrumb navigation path is 'rsv-PREFIX > Backup items > vm-ops > Restore VM'. On the left, a sidebar lists the navigation items: 'rsv-PREFIX', 'Backup items', and 'vm-ops'. The main content area is divided into two sections. The first section, labeled '1 Locate', contains a text input field with the value 'Restore VM'. The second section, labeled '2 Action', contains a text input field with the value 'Confirm the restored text matches the backup-time state.' At the bottom right of the main content area, there is a blue button labeled 'View'. Below the main content area, a footer note states 'Find the labels above in your own portal.'

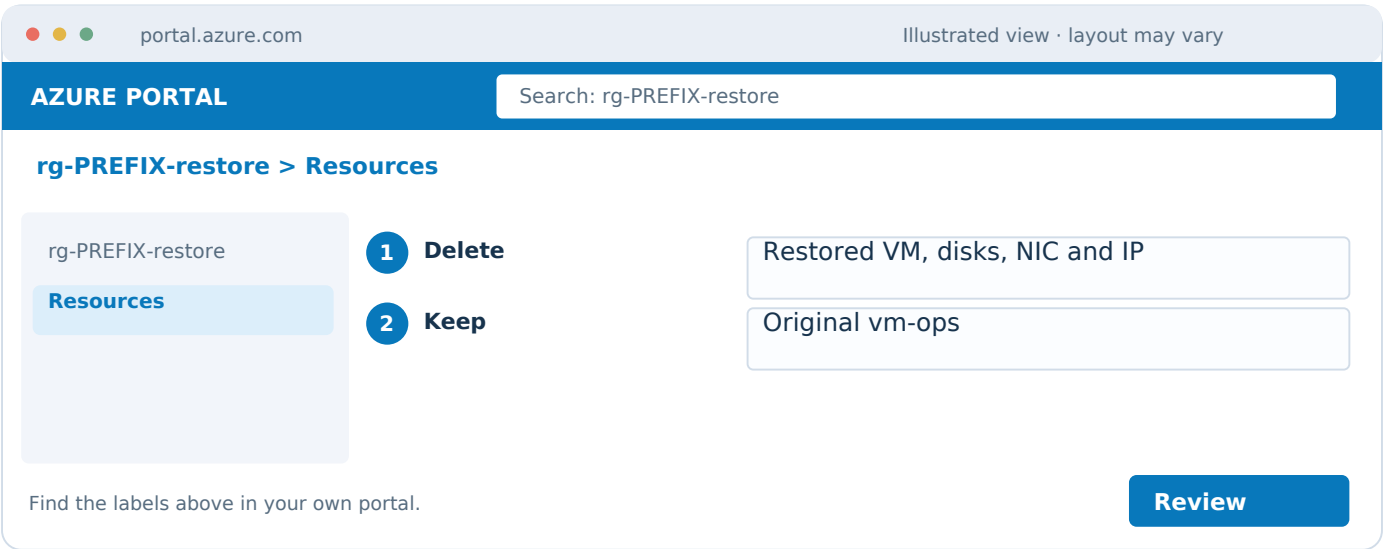
Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Confirm the restored text matches the backup-time state.
2. Record actual elapsed restore time and what data was recovered.

7B.01 · ACTION 6 OF 6

AZURE PORTAL · rg-PREFIX-restore → Resources



Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Delete the restored VM and its test disks/NIC/public IP when finished.
2. Keep the original VM for the Site Recovery extension.

You should see

The restored resource contains the expected file and can run the operations service.

If it does not work

A restore can incur compute, disk and network charges. A new resource might have different IPs or identity; recovery is not complete until its dependencies work. [R58]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED
Worked: ☐ Independently ☐ With the guide

7B.02 Use a Backup vault for the separate managed disk

AZ-104 · EXTENSION - backup storage and restore costs

Why this task: Backup vault and Recovery Services vault are different resource types supporting different workloads.

7B.02 · ACTION 1 OF 6

AZURE PORTAL · Backup vaults → Create

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL

Search: Backup vaults

Backup vaults > Create

Backup vaults

Create

1 Name

2 Group

3 Region

bv-PREFIX

rg-PREFIX-ops

Primary region

Review + create

Find the labels above in your own portal.

Illustrated navigation view · use your own saved names and IDs

Why this action: Backup vault and Recovery Services vault are different resource types supporting different workloads.

1. Search Backup vaults > Create.
2. Use rg-PREFIX-ops, name bv-PREFIX, primary region and the appropriate redundancy offered for the workload.

7B.02 · ACTION 2 OF 6**AZURE PORTAL · bv-PREFIX → Backup policies → Add**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: bv-PREFIX

bv-PREFIX > Backup policies > Add

bv-PREFIX
Backup policies
Add

1 Datasource Azure Disk

2 Name disk-lab

3 Retention Supported short policy

Find the labels above in your own portal. **Create**

Illustrated navigation view · use your own saved names and IDs

Why this action: Backup vault and Recovery Services vault are different resource types supporting different workloads.

1. Open Backup policies > Add.
2. Datasource type Azure Disk; name disk-lab; choose a supported snapshot schedule and short allowed retention.

7B.02 · ACTION 3 OF 6**AZURE PORTAL · bv-PREFIX → Configure backup**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: bv-PREFIX

bv-PREFIX > Configure backup

bv-PREFIX
Configure backup

1 Source disk-ops-data

2 Snapshot group rg-PREFIX-snapshots

Find the labels above in your own portal. **Validate**

Illustrated navigation view · use your own saved names and IDs

Why this action: Backup vault and Recovery Services vault are different resource types supporting different workloads.

1. Choose Configure backup and select disk-ops-data from Day 2, Block D, Day 6, Block A.
2. Select a dedicated snapshot resource group.
3. First create rg-PREFIX-snapshots in the primary region using Resource groups > Create.
4. Use that group when the wizard requests the snapshot group.

7B.02 · ACTION 4 OF 6**AZURE PORTAL · Source disk / snapshot group → Access control (IAM)**

The screenshot shows the Azure Portal interface for configuring access control. The breadcrumb navigation is 'Source disk / snapshot group > Access control (IAM)'. On the left, a sidebar lists 'Source disk / snapshot group' and 'Access control (IAM)'. The main area contains three numbered steps: 1. Vault identity (Backup vault managed identity), 2. Disk role (Disk Backup Reader), and 3. Snapshot group role (Disk Snapshot Contributor). A 'Review + assign' button is at the bottom right.

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Source disk / snapshot group

Source disk / snapshot group > Access control (IAM)

Source disk / snapshot group

Access control (IAM)

- 1 Vault identity** Backup vault managed identity
- 2 Disk role** Disk Backup Reader
- 3 Snapshot group role** Disk Snapshot Contributor

Find the labels above in your own portal. **Review + assign**

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. Grant the Backup vault managed identity Disk Backup Reader on the source disk and Disk Snapshot Contributor on the snapshot resource group, as requested by the wizard.
2. Retry validation after permissions propagate.

7B.02 · ACTION 5 OF 6**AZURE PORTAL · bv-PREFIX → Backup instances → Your disk**

The screenshot shows the Azure Portal interface for managing backup instances. The breadcrumb navigation is 'bv-PREFIX > Backup instances > Your disk'. On the left, a sidebar lists 'bv-PREFIX', 'Backup instances', and 'Your disk'. The main area contains three numbered steps: 1. First (Backup now, wait for recovery point), 2. Then (Restore to new disk), and 3. Target role (Disk Restore Operator as required). A 'Restore' button is at the bottom right.

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: bv-PREFIX

bv-PREFIX > Backup instances > Your disk

bv-PREFIX

Backup instances

Your disk

- 1 First** Backup now, wait for recovery point
- 2 Then** Restore to new disk
- 3 Target role** Disk Restore Operator as required

Find the labels above in your own portal. **Restore**

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Run Backup now, wait for a recovery point, then Restore to a new managed disk.
2. Grant Disk Restore Operator on the target group if required.

7B.02 · ACTION 6 OF 6

AZURE PORTAL · Disks → Your restored lab disk

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Disks

Disks > Your restored lab disk

Disks

Your restored lab disk

1 Inspect

2 After test

Expected new disk resource

Delete only that disk

Find the labels above in your own portal.

Delete

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Inspect the restored disk resource and delete it after the exercise.
2. This test protects a managed disk; it does not configure the full VM backup or its application consistency.

You should see

You have observed the policy, vault identity, snapshot storage and restore target for disk backup.

If it does not work

Not every backup vault workload stores a separate vaulted copy. Azure Disk Backup uses managed snapshots, so understand the protection model and snapshot charges. [R59]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

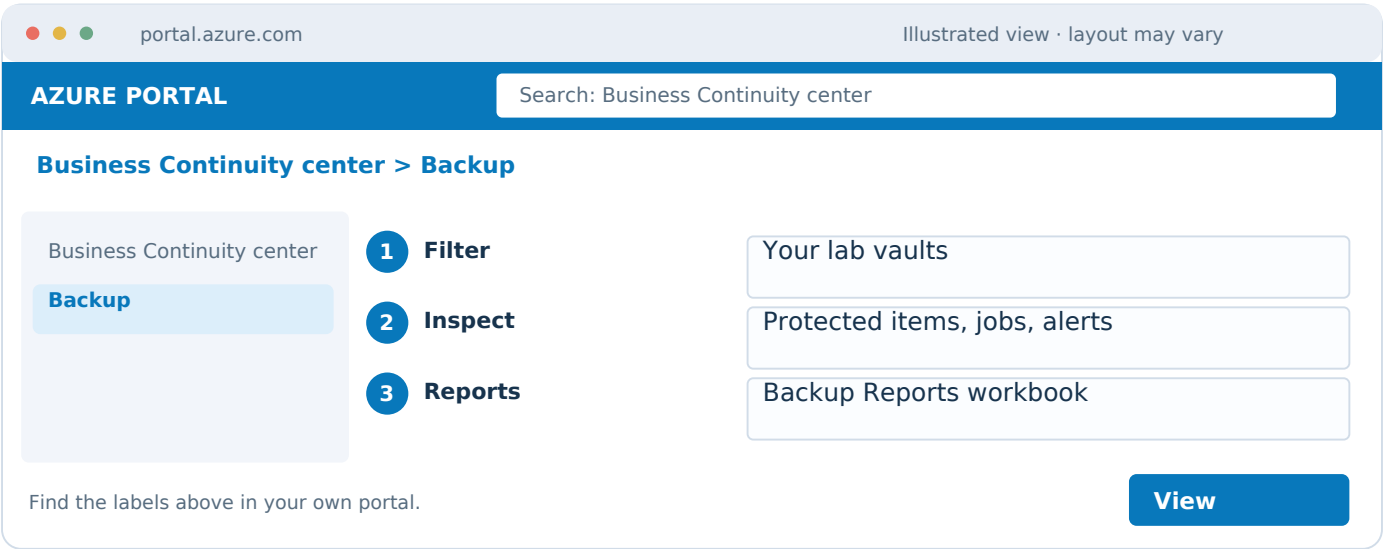
7B.03 Inspect backup reporting and alerts

AZ-104 · EXTENSION - backup storage and restore costs

Why this task: Failed or missing recovery points should be visible before an incident.

7B.03 · ACTION 1 OF 5

AZURE PORTAL · Business Continuity center → Backup



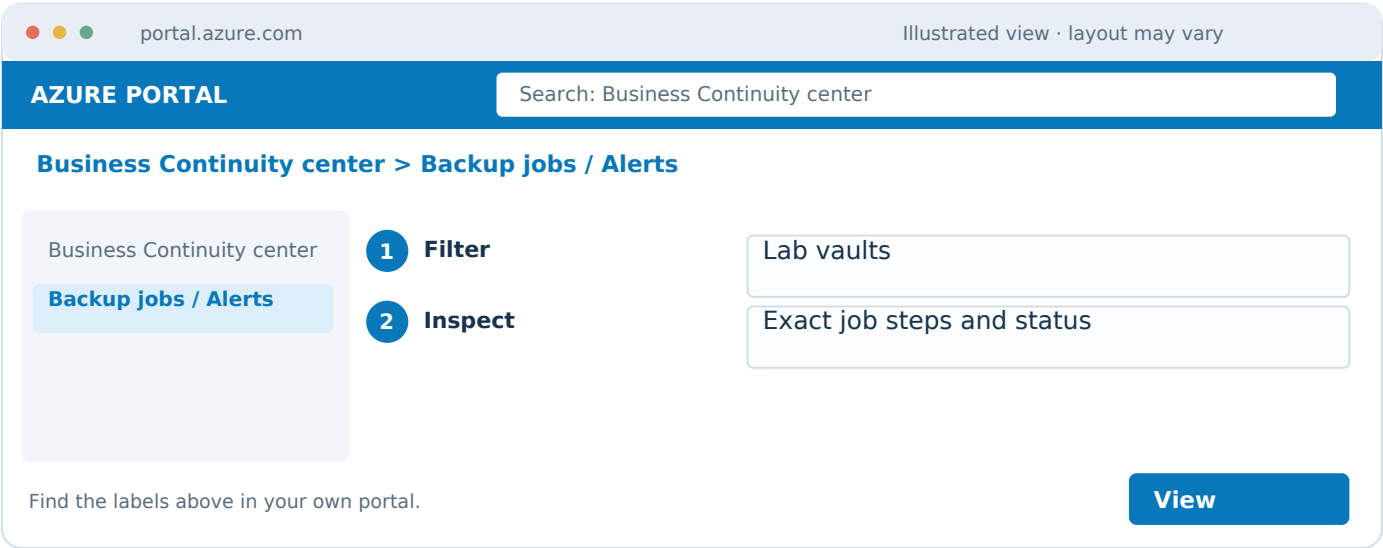
Illustrated navigation view · use your own saved names and IDs

Why this action: Failed or missing recovery points should be visible before an incident.

1. Open the current Backup center or Business Continuity center view and filter your lab vaults.

7B.03 · ACTION 2 OF 5

AZURE PORTAL · Business Continuity center → Backup jobs / Alerts



Illustrated navigation view · use your own saved names and IDs

Why this action: Failed or missing recovery points should be visible before an incident.

1. Inspect protected items, recent jobs and alerts.
2. Open a job's details to see the exact failed step if any.

7B.03 · ACTION 3 OF 5

AZURE PORTAL · Your vault → Monitoring → Diagnostic settings / Backup Reports

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL

Search: Your vault

Your vault > Monitoring > Diagnostic settings / Backup Reports

Your vault

Monitoring

Diagnostic settings / Backup Reports

1 Destination

2 Then

law-PREFIX

Read report after data arrives

Find the labels above in your own portal.

Save

Illustrated navigation view · use your own saved names and IDs

Why this action: Failed or missing recovery points should be visible before an incident.

1. On the vault configure supported diagnostic settings to the workspace and inspect the Backup Reports workbook when data is available.

7B.03 · ACTION 4 OF 5

AZURE PORTAL · Your vault → Monitoring → Alerts / Alert settings

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL

Search: Your vault

Your vault > Monitoring > Alerts / Alert settings

Your vault

Monitoring

Alerts / Alert settings

1 Action group

2 Inspect

ag-onelab

Condition and scope

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: Failed or missing recovery points should be visible before an incident.

1. Configure supported backup alert notification settings with your action group.
2. Read the alert condition and its scope.

7B.03 · ACTION 5 OF 5

AZURE PORTAL · Business Continuity center → Backup

The image shows a simulated Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com' and a note says 'Illustrated view · layout may vary'. The main header is 'AZURE PORTAL' with a search bar containing 'Business Continuity center'. Below the header, the breadcrumb 'Business Continuity center > Backup' is shown. On the left, a sidebar lists 'Business Continuity center' and 'Backup' (highlighted). In the center, there are two numbered steps: '1 Locate' and '2 Action'. To the right of step 2, there is a text box with the instruction: 'Record a practical check: last successful backup, oldest/newest recovery point, policy retention and last successful restore test.' Below this, there is a 'View' button. At the bottom left, a note says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Record a practical check: last successful backup, oldest/newest recovery point, policy retention and last successful restore test.

You should see

Your evidence describes recoverability rather than only saying backup is enabled.

If it does not work

Reports can have ingestion delay. An empty report immediately after setup does not prove there are no protected items.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

7C · Rehearse recovery and explain your work

Planning estimate: 140 minutes. Finish with: An isolated recovery test and practical understanding recorded before cleanup.

7C.01 Perform an isolated test failover

AZ-104 · PAID EXTENSION; DESIGN if unavailable

Why this task: A test failover proves the recovery process without intentionally taking down the primary service.

7C.01 · ACTION 1 OF 6

AZURE PORTAL · Site Recovery vault → Replicated items → vm-ops

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL

Search: Site Recovery vault

Site Recovery vault > Replicated items > vm-ops

Site Recovery vault

Replicated items

vm-ops

1 Operation

2 Recovery point

3 Network

4 After verification

Test failover

Suitable completed point

vnet-dr-test (isolated)

Cleanup test failover

Find the labels above in your own portal.

Test failover

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open the replicated item in the Site Recovery vault > Test failover.

7C.01 · ACTION 2 OF 6**AZURE PORTAL · vm-ops replicated item → Test failover → Settings**

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: vm-ops replicated item

vm-ops replicated item > Test failover > Settings

vm-ops replicated item

Test failover

Settings

1 Recovery point Suitable completed point

2 Network vnet-dr-test

Find the labels above in your own portal. **Start test**

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Choose a suitable recovery point and the isolated vnet-dr-test network, not a production-connected network.

7C.01 · ACTION 3 OF 6**VM TERMINAL IN AZURE PORTAL · Azure Portal → Test failover VM → Run command → RunShellScript**

The dark code panel below is the terminal/editor view for this action.

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Start the test and inspect the job.
2. When the test VM appears in the target region, use an authorized connection/Run command to inspect the OneLab health page.

USE IN: VM RUNSHELLSCRIPT · COPY ONLY THE CODE

```
sudo systemctl start nginx
cat /var/www/html/index.html
curl -f http://127.0.0.1
```

What it means: Run inside the isolated test-failover VM, not the original VM.

curl · Send an HTTP request. -f fails on HTTP error status; -s quiets the progress meter; -v shows connection details when used.

systemctl enable --now · Set the service to start at boot and start it now; stop/start act on the running service.

sudo · Run that Linux operation with administrator privileges inside the named VM.

7C.01 · ACTION 4 OF 6**AZURE PORTAL · Site Recovery vault → Replicated items → vm-ops**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: Site Recovery vault

Site Recovery vault > Replicated items > vm-ops

Site Recovery vault
Replicated items
vm-ops

1 Network vnet-dr-test (isolated)

Find the labels above in your own portal. **View**

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. Record startup time, recovered data timestamp, networking changes and any missing dependencies.

7C.01 · ACTION 5 OF 6**AZURE PORTAL · vm-ops replicated item → Cleanup test failover**

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL Search: vm-ops replicated item

vm-ops replicated item > Cleanup test failover

vm-ops replicated item
Cleanup test failover

1 Note Your test result

2 Confirm Testing completed

Find the labels above in your own portal. **OK**

Illustrated navigation view · use your own saved names and IDs

Why this action: The recovery service must close the rehearsal and remove its test resources cleanly.

1. Return to the replicated item > Cleanup test failover.

2. Add a note about the result, confirm testing is complete and run cleanup.

7C.01 · ACTION 6 OF 6

AZURE PORTAL · Site Recovery vault → Site Recovery jobs / Replicated items

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL

Search: Site Recovery vault

Site Recovery vault > Site Recovery jobs / Replicated items

Site Recovery vault

Site Recovery jobs / Replicated items

1 Cleanup

2 Test VM

3 Original protection

Completed

Removed

Healthy

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

- 1. Verify the test VM is removed and the original replication remains healthy.
- 2. Deleting only the test VM manually is not the complete Site Recovery cleanup workflow.

You should see

You have a documented test failover and completed its cleanup.

If it does not work

A test failover is not a production cutover and does not automatically prove DNS, identity and all dependencies work in the target region. [R61]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED
Worked: ☐ Independently ☐ With the guide

7C.02 Understand actual failover, commit and re protection

AZ-104 · PAID EXTENSION; DESIGN if unavailable

Why this task: A real cutover changes which site is active and how protection continues.

7C.02 · ACTION 1 OF 5

SEE THE IDEA · Real failover and failback design

A SMALL DECISION MAP

Real failover and failback design

- | | | |
|---|------------------|-------------------------------------|
| 1 | Validate | Recovered service and data |
| 2 | Commit | Finalize chosen recovery point |
| 3 | Reprotect | Reverse replication |
| 4 | Failback | Return after source region recovery |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: A real cutover changes which site is active and how protection continues.

1. DESIGN order: choose a recovery point, plan source shutdown where possible, initiate failover, validate the recovered service, update intended traffic routing, then commit the chosen point.

7C.02 · ACTION 2 OF 5

SEE THE IDEA · Real failover and failback design

A SMALL DECISION MAP

Real failover and failback design

- | | | |
|---|---------------|--------------------------------|
| 1 | Commit | Finalize chosen recovery point |
|---|---------------|--------------------------------|

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: A real cutover changes which site is active and how protection continues.

1. Committing finalizes the selected failover state.
2. Do not commit until validation and data-loss implications are understood.

7C.02 · ACTION 3 OF 5**SEE THE IDEA · Real failover and failback design****A SMALL DECISION MAP****Real failover and failback design**

- | | |
|--------------------|-------------------------------------|
| 1 Reprotect | Reverse replication |
| 2 Failback | Return after source region recovery |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Reprotect to reverse replication toward the original region.
2. Once healthy, plan failback and test it.
3. After failback, reprotect again in the normal direction.

7C.02 · ACTION 4 OF 5**SEE THE IDEA · Real failover and failback design****A SMALL DECISION MAP****Real failover and failback design**

- | | |
|--------------------|---------------------|
| 1 Reprotect | Reverse replication |
|--------------------|---------------------|

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. For an actual disposable-VM exercise, use the replicated item's Failover action instead of Test failover; review shutdown and recovery-point choices, then carry out the validation and reprotection sequence shown by the wizard.

7C.02 · ACTION 5 OF 5

SEE THE IDEA · Understand actual failover, commit and reprotection → Explain the decision

A SMALL DECISION MAP

Understand actual failover, commit and reprotection > Explain the decision

- | | | |
|----------|---------------|--|
| 1 | Locate | Explain the decision |
| 2 | Action | If you only completed test failover, mark actual failover/failback as DESIGN in the final checklist. |

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. If you only completed test failover, mark actual failover/failback as DESIGN in the final checklist.
2. Do not label it implemented.

You should see

You can explain test failover, real failover, commit, reprotect and failback in order.

If it does not work

This guide does not ask you to cut over an existing production VM. In a real incident, application dependencies and recovery plans determine the safe order. [R62]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

7C.03 Use one evidence sheet for both exams

Both exams · Practical assessment

Why this task: A list of completed clicks is weaker than proof you can diagnose and choose.

Do these tasks before final cleanup, or rebuild the relevant small part. First hide the instructions for that task. Then perform it and explain why. If you need to look up a step in this PDF, that is useful feedback: repeat the task after reviewing the explanation. These are practical checks, not a prediction of your exam score.

Challenge	Evidence you should produce
Identity	Create or identify a test group; grant Reader at the intended group scope; explain why it cannot download private blobs without a data role.
Governance	Identify a policy versus lock failure; locate budget alerts; explain why the budget did not stop compute.
Storage	Recover the previous welcome.txt content; issue a read-only short-lived SAS; explain network restrictions and its expiry.
Compute	Identify VM allocation state and disk purpose; change an App Service slot safely; explain when ACI or Container Apps fits.
Network	Locate the effective NSG rule and route; explain private endpoint DNS versus a service endpoint; name the five load-balancer checks.
IaC	Change a harmless Bicep tag, preview the difference and deploy it; explain parameters, resources and outputs.
Source control	Create a linked PR, resolve a conflict in the disposable repo, and explain a safe shared-branch revert.
Pipeline	Explain every stage and its inputs, find a failed test, fix it through a PR and prove production is protected.
Release	Promote the same artifact, show its version and rollback; explain why a database schema change needs extra planning.
Security	Identify a service principal, managed identity and federated trust; locate a dependency/secret/code/image scan result or an accurately labelled coverage gap.
Observability	Generate a failed request, find it with KQL, inspect a trace and demonstrate an alert reaches your own inbox.
Recovery	Show a restored file/VM and measured recovery time; explain backup versus replication and test versus actual failover.

7C.03 · ACTION 1 OF 6**YOUR COMPUTER · Your notes → Practical evidence**

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER Search: Your notes

Your notes > Practical evidence

Your notes

Practical evidence

1 Mark

2 Explain

3 Show

4 Repeat

DONE / WAITING / DESIGN / BLOCKED

Why this works

One verified result

An uncertain task

Find the labels above in your own portal.

Open

Illustrated navigation view · use your own saved names and IDs

Why this action: Your note lets you resume here and explain the result later without guessing.

1. For each challenge, record date, resource/run/PR identifier, expected result, actual result and one sentence explaining the reason.

7C.03 · ACTION 2 OF 6**YOUR COMPUTER · Your notes → Practical evidence**

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER Search: Your notes

Your notes > Practical evidence

Your notes

Practical evidence

1 Locate

2 Action

Practical evidence

Use status Independent, Needed guide, Design only, or Blocked.

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: A list of completed clicks is weaker than proof you can diagnose and choose.

1. Use status Independent, Needed guide, Design only, or Blocked.
2. Do not count Design only as a completed live operation.

7C.03 · ACTION 3 OF 6

YOUR COMPUTER · Your notes → Practical evidence

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER

Search: Your notes

Your notes > Practical evidence

Your notes

Practical evidence

1 Repeat

An uncertain task

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: A list of completed clicks is weaker than proof you can diagnose and choose.

1. Repeat each Needed guide task until you can do it without copying the solution.
2. Change one harmless input, such as the group name or flag value, so you demonstrate understanding.

7C.03 · ACTION 4 OF 6

YOUR COMPUTER · Your notes → Practical evidence

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER

Search: Your notes

Your notes > Practical evidence

Your notes

Practical evidence

1 Explain

Why this works

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: The correct identity needs permission at the place where it will do the work.

1. For each design-only feature, explain its purpose, prerequisites, permission boundary, cost consequence and one failure mode from this guide.

7C.03 · ACTION 5 OF 6**YOUR COMPUTER · Your notes → Practical evidence**

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER

Search: Your notes

Your notes > Practical evidence

Your notes

Practical evidence

1 **Locate**

2 **Action**

Practical evidence

Check the current exam objective links near your booking date because Microsoft can change scope.

Find the labels above in your own portal.

Run

Illustrated navigation view · use your own saved names and IDs

Why this action: A list of completed clicks is weaker than proof you can diagnose and choose.

1. Check the current exam objective links near your booking date because Microsoft can change scope.
2. You can use this PDF as the central practice guide without treating a single successful run as a passing guarantee.

7C.03 · ACTION 6 OF 6**YOUR COMPUTER · Your notes → Practical evidence**

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER

Search: Your notes

Your notes > Practical evidence

Your notes

Practical evidence

1 **Explain**

Why this works

Find the labels above in your own portal.

View

Illustrated navigation view · use your own saved names and IDs

Why this action: Azure jobs and permissions can take time. A submitted request is not a completed result.

1. Use DONE for verified hands-on work, WAITING for unfinished cloud jobs, DESIGN for explained-only work and BLOCKED for unavailable prerequisites.
2. Do not count a DESIGN or BLOCKED entry as a successful deployment.

You should see

Your evidence sheet makes strengths and remaining gaps visible.

If it does not work

If a scenario is unfamiliar, ask: what is the requirement, scope, identity, network path, cost/availability constraint and measurable success condition? This reasoning helps beyond memorizing commands.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Keep the idea connected to your result

Worked reasoning example: staging deploy succeeds but production cannot resolve its Key Vault reference after promotion. Start at the production slot identity, not the pipeline token. The identity is slot-specific; check its vault data role, URI and network path. A working staging identity does not prove production has permission. This is the kind of variation you should be able to solve independently.

7D · Save your work and remove the lab

Planning estimate: 60 minutes. Finish with: Saved evidence, removed automation and a checked resource/cost inventory.

7D.01 Save evidence and stop scheduled work

AZ-104 + AZ-400 · CORE - perform when finished

Why this task: Cleanup should remove infrastructure without losing the learning record.

7D.01 · ACTION 1 OF 5

YOUR COMPUTER · Private notes and project → Save a copy

Local file or desktop application

Illustrated view · layout may vary

YOUR COMPUTER

Search: Private notes and project

Private notes and project > Save a copy

Private notes and project

Save a copy

1 Save

2 Keep private

Safe source and evidence

Credentials and identifiers

Find the labels above in your own portal.

Download

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Download your notes, safe source, test results and a record of successful deployments.
2. Keep credentials and private identifiers out of any public portfolio.

7D.01 · ACTION 2 OF 5**AZURE DEVOPS · Pipelines / GitHub Actions → Each lab workflow → Settings**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: Pipelines / GitHub Actions

Pipelines / GitHub Actions > Each lab workflow > Settings

Pipelines / GitHub Actions

Each lab workflow

Settings

1 Stop

2 Cancel

3 Remove

Schedules and unwanted triggers

Queued lab runs

Test notifications / hooks

Find the labels above in your own portal.

Review

Illustrated navigation view · use your own saved names and IDs

Why this action: Cleanup should remove infrastructure without losing the learning record.

1. Disable scheduled workflows and unwanted CI triggers.
2. Cancel queued runs and remove temporary Teams/webhook subscriptions.

7D.01 · ACTION 3 OF 5**AZURE DEVOPS · User settings → Personal access tokens / Project settings → Service connections**

dev.azure.com / YOUR_ORG / OneLab

Illustrated view · layout may vary

AZURE DEVOPS Search: User settings

User settings > Personal access tokens / Project settings > Service connections

User settings

Personal access tokens / Project settings

Service connections

1 Revoke

2 Remove

Lab PATs

Unused lab connections and federated credentials

Find the labels above in your own portal.

Review

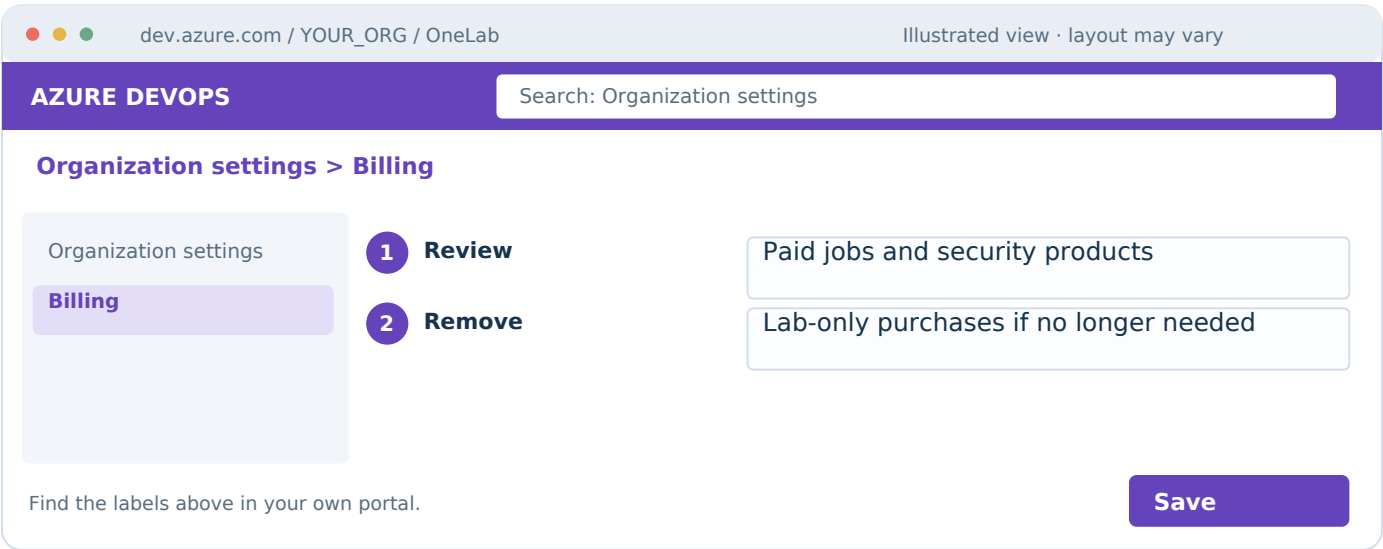
Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Revoke the short-lived Git/agent PATs and remove unused service connections or federated credentials.
2. Remove only identities you created for this lab.

7D.01 · ACTION 4 OF 5

AZURE DEVOPS · Organization settings → Billing



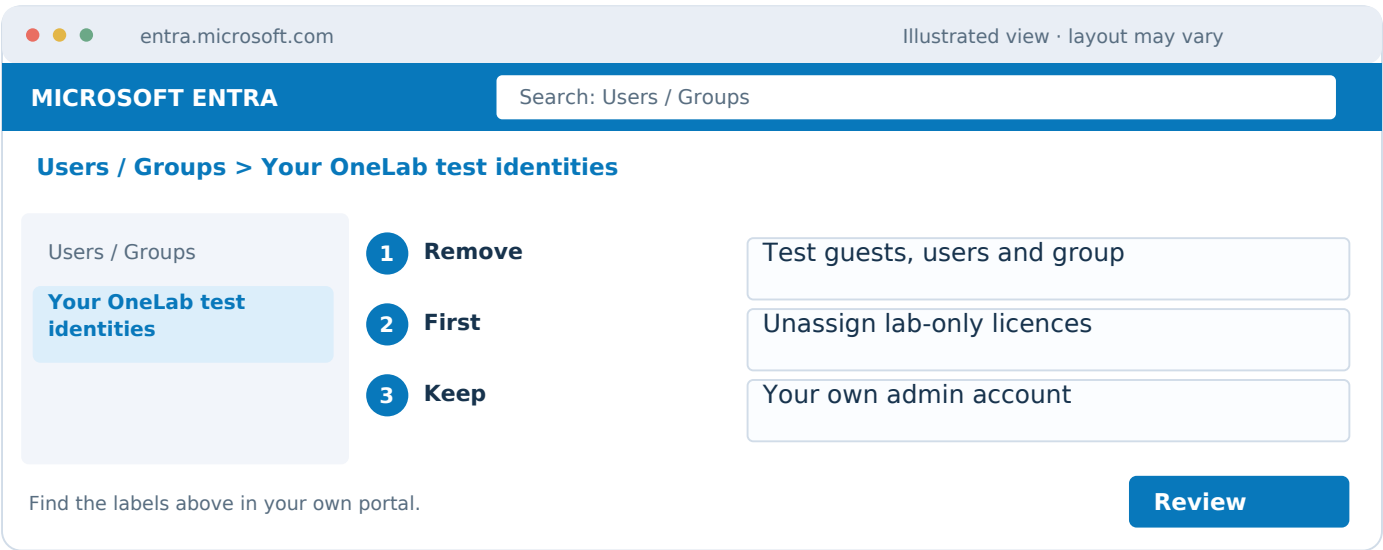
Illustrated navigation view · use your own saved names and IDs

Why this action: Cleanup should remove infrastructure without losing the learning record.

1. Check Azure DevOps Billing, Advanced Security products and paid parallel jobs; turn off lab-only purchases if they are no longer needed.

7D.01 · ACTION 5 OF 5

MICROSOFT ENTRA · Users / Groups → Your OneLab test identities



Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Remove test guest/users, groups and licence assignments only after confirming they are not used elsewhere.
2. Do not delete your own administrator identity.

You should see

Your evidence is saved and no automation will recreate resources during cleanup.

If it does not work

Deleting a resource group does not delete every tenant identity, GitHub secret, Azure DevOps connection or paid organization setting.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

7D.02 Remove recovery protection before deleting vaults

AZ-104 + AZ-400 · CORE - perform when finished

Why this task: Vaults and retained recovery data can block ordinary group deletion.

7D.02 · ACTION 1 OF 5

AZURE PORTAL · Site Recovery vault → Replicated items → vm-ops

The screenshot shows the Azure Portal interface. At the top, the browser address bar displays 'portal.azure.com'. The page title is 'Illustrated view · layout may vary'. The main header is 'AZURE PORTAL' with a search bar containing 'Search: Site Recovery vault'. Below the header, the breadcrumb navigation path is 'Site Recovery vault > Replicated items > vm-ops'. On the left, a sidebar shows the navigation menu with 'Site Recovery vault', 'Replicated items', and 'vm-ops' (highlighted). The main content area displays a list of actions to perform, numbered 1 to 3: '1 First', '2 Then', and '3 After'. The actions are: 'Cleanup test failover', 'Disable replication', and 'Inspect leftover target resources'. At the bottom right, there is a blue button labeled 'Disable replication'. Below the actions, a note says 'Find the labels above in your own portal.'

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. In Site Recovery, complete any pending test cleanup, then disable replication for the lab item.
2. Inspect target replica disks, cache account, networks and test VMs afterward.

7D.02 · ACTION 2 OF 5

AZURE PORTAL · Each vault → Backup items / Backup instances → Your item

portal.azure.com

Illustrated view · layout may vary

AZURE PORTAL

Search: Each vault

Each vault > Backup items / Backup instances > Your item

Each vault

Backup items / Backup instances

Your item

1 Action

2 Data

Stop backup

Choose retain or delete after reviewing recovery need

Find the labels above in your own portal.

Review

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. In each backup vault open protected items.
2. Use Stop backup with the appropriate retain/delete-data choice after confirming you no longer need the lab recovery points.

7D.02 · ACTION 3 OF 5

SEE THE IDEA · Retained recovery data may outlive the lab

A SMALL DECISION MAP

Retained recovery data may outlive the lab

1 Respect

2 Record

Soft delete and immutability

Remaining items, charges and removal date

Explain this using your own lab result. This is not a completed cloud operation.

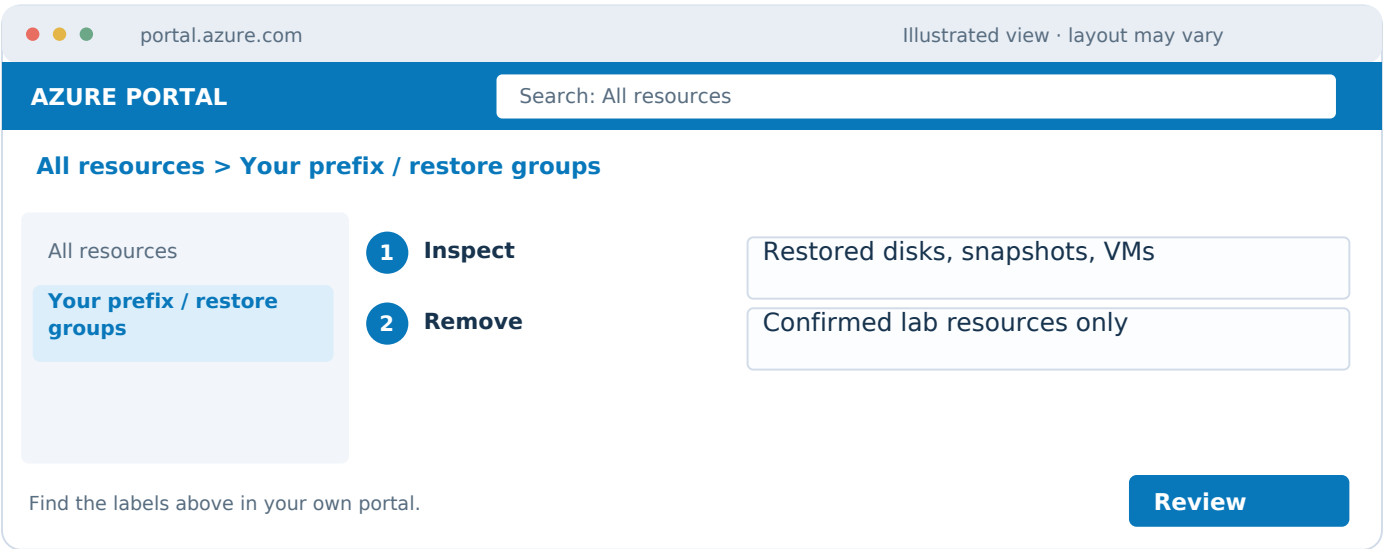
Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Respect soft-delete and immutability retention.
2. Some retained items prevent immediate vault deletion; do not weaken a company retention policy just to remove a lab.

7D.02 · ACTION 4 OF 5

AZURE PORTAL · All resources → Your prefix / restore groups



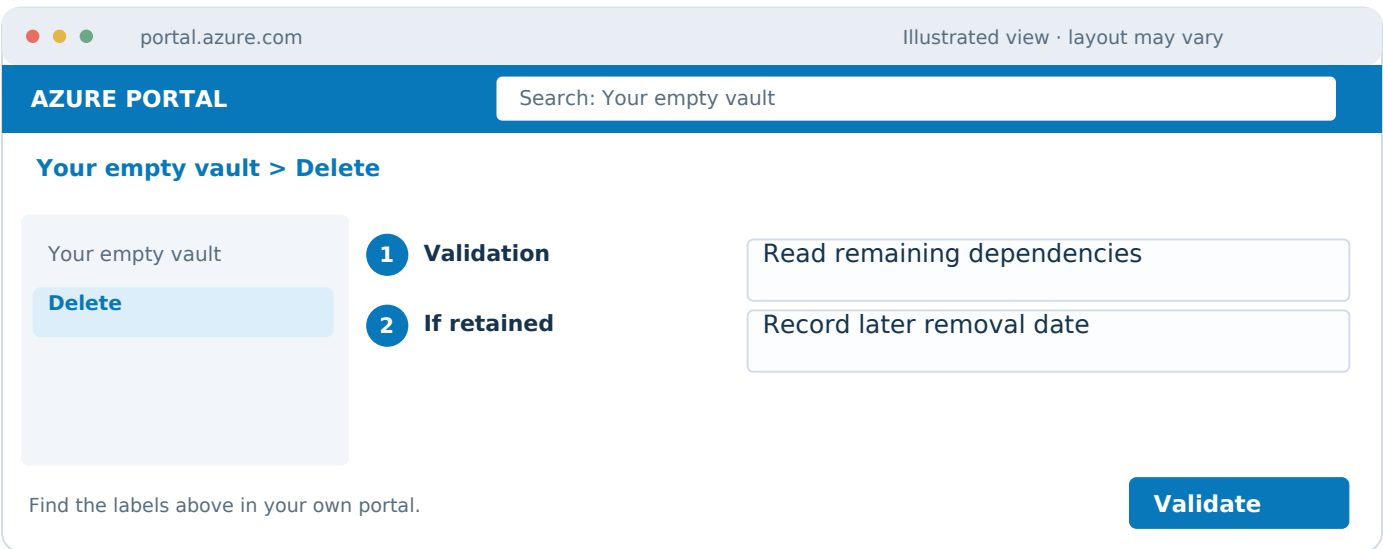
Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Delete restored disks, snapshots and restored VMs that were created outside the main groups.

7D.02 · ACTION 5 OF 5

AZURE PORTAL · Your empty vault → Delete



Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Use the vault's Delete validation to identify any remaining dependencies.
2. Record retained resources and the date they can be removed if immediate deletion is not allowed.

You should see

You understand and account for retained recovery data and resources.

If it does not work

Deleting an original VM does not automatically remove all protected backups or replicated resources.
[R63]

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

7D.03 Review groups and delete only your lab

AZ-104 + AZ-400 · CORE - perform when finished

Why this task: A resource-group delete is broad and irreversible for many resources.

7D.03 · ACTION 1 OF 6

AZURE PORTAL · Resource groups → Exact OneLab group → Delete resource group

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: Resource groups

Resource groups > Exact OneLab group > Delete resource group

Resource groups
Exact OneLab group
Delete resource group

- 1 Group**
- 2 Resources**
- 3 Locks**
- 4 Shell group**

Review its exact name
Inspect every item
Remove only your lab lock
Keep until downloads are finished

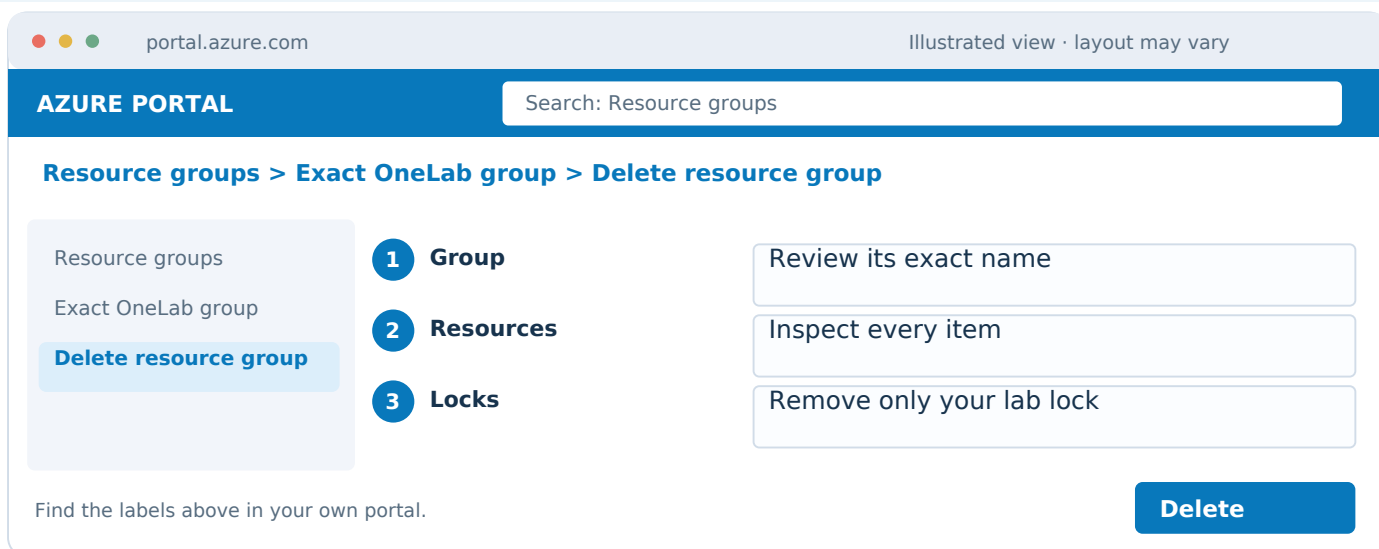
Delete resource group

Find the labels above in your own portal.

Illustrated navigation view · use your own saved names and IDs

Why this action: This check proves what happened. A successful Save button alone does not prove the whole service works.

1. Open each exact OneLab group and review every listed resource.
2. Confirm none belongs to other work.

7D.03 · ACTION 2 OF 6**AZURE PORTAL · Resource groups → Exact OneLab group → Delete resource group**

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Remove only the lab locks you created.
2. Delete the temporary groups and resources, including App Service plan, Bastion, private endpoints, registry, monitoring workspace and recovery targets when permitted.
3. Also review rg-PREFIX-restore, rg-PREFIX-snapshots and any move-test group separately.

7D.03 · ACTION 3 OF 6**CLOUD SHELL · Cloud Shell → Bash → ~/clouddrive/one-lab**

The dark code panel below is the terminal/editor view for this action.

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. To use the terminal, first run the list command below and compare the result to your naming record.
2. Then run delete for a confirmed group and answer its confirmation prompt.

USE IN: BASH · COPY ONLY THE CODE

```
az resource list -g "$RG" --query '[].{Name:name,Type:type}' -o table
# Run only after reviewing this exact group's contents:
az group delete --name "$RG"
# Repeat separately for confirmed OPS/DR/restore groups as needed.
```

What it means: No --yes is used: the Azure CLI asks you to confirm the exact group. The shell group is intentionally separate so you can finish saving your files.

--query · Select or transform returned fields with a JMESPath expression; this does not itself change the resource.

--name · The name of this resource or object.

-g · Short form of --resource-group.

-o · Azure CLI output format: table, tsv or none.

\$NAME / \${NAME} · Read a variable from this Bash session. Source lab.env first when these are lab names.

source FILE · Load settings or activate an environment in the current Bash shell.

7D.03 · ACTION 4 OF 6

SEE THE IDEA · Review groups and delete only your lab → Explain the decision

A SMALL DECISION MAP

Review groups and delete only your lab > Explain the decision

- 1 **Locate** Explain the decision
- 2 **Action** Do not use a wildcard script to delete all groups.

Explain this using your own lab result. This is not a completed cloud operation.

Illustrated navigation view · use your own saved names and IDs

Why this action: Remove the temporary item only after checking its name and saving what you need.

1. Do not use a wildcard script to delete all groups.
2. Keep rg-onelab-shell until your last notes/source download is complete.

7D.03 · ACTION 5 OF 6

AZURE PORTAL · All resources → Filter

portal.azure.com Illustrated view · layout may vary

AZURE PORTAL Search: All resources

All resources > Filter

All resources
Filter

- 1 **Filter**
- 2 **Check**

Find the labels above in your own portal. Apply

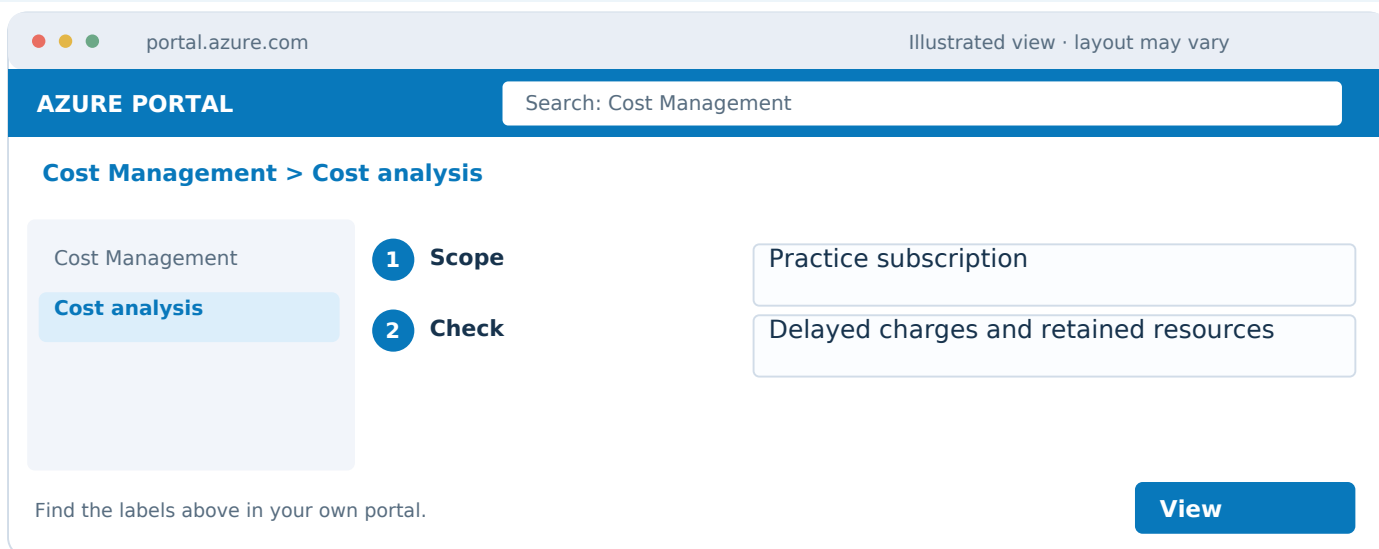
Illustrated navigation view · use your own saved names and IDs

Why this action: A resource-group delete is broad and irreversible for many resources.

1. After deletion finishes, search All resources and filter by your prefix.
2. Check both regions and any automatically created workspace or managed group.

7D.03 · ACTION 6 OF 6

AZURE PORTAL · Cost Management → Cost analysis



Illustrated navigation view · use your own saved names and IDs

Why this action: A resource-group delete is broad and irreversible for many resources.

1. Return to Cost analysis later to check delayed charges and any remaining resources.
2. A cleared browser window is not proof of stopped billing.

You should see

No unwanted lab compute remains and any retained items are explicitly recorded.

If it does not work

Soft-deleted Key Vaults retain names; reuse can fail even after deleting the resource group. Use recovery or a new name according to the retention policy, not blind purge.

Record your result: ☐ DONE ☐ WAITING ☐ DESIGN ☐ BLOCKED

Worked: ☐ Independently ☐ With the guide

Finish Day 7

Save your place and check cost

Download your evidence and source. Confirm the lab compute is removed. Record any recovery data retained by policy, its expected charge and its later removal date.

Say it in your own words: Why is a completed backup job alone insufficient proof that you can recover?

Check your explanation: You must restore a usable copy and test its files, service and dependencies. Record the time and data recovered.

Coverage map: where each skill family is taught

This map groups the published objective families in plain language. The detailed skills appear in the linked tasks. C = core implementation, E = connected extension with cost/permission prerequisites, D = design explanation or conditional implementation. Read the task labels: a topic being included is not the same as having performed it in your account. [R01, R02]

AZ-104 area	Day / Block	Practice
People, membership, guests, licences, reset	1C	C/E
Permission levels and inheritance	1C, 3C	C
Organization, policies, locks, tags, spending	1A, 1D	C/E
Storage authorization and network boundaries	2A, 2B, 3A	C/E
Account resilience, encryption, copying	2A, 2B	C/E
Object/share lifecycle and recovery	2A, 2B	C/E
Declarative deployment and conversion	5A, 6A	C
VM lifecycle, disks, placement and fleets	2D, 6A	C/E/D
Registries and managed container hosts	6B	E
Web hosting, scale, names, TLS and slots	3B, 3C, 5B	C/E
Addressing, peering, routing and diagnosis	2C, 2D, 3A, 6A	C/E
Traffic filtering and private service paths	2C, 2D, 3A, 6A	C/E
Naming and traffic distribution	2C, 3A, 3B, 3C	C/E
Metrics, logs, notifications and connectivity	3D, 7A	C/E
Vaults, recovery policies, restore and regional continuity	2D, 7B, 7C	E/D
AZ-400 area	Day / Block	Practice
Work flow, traceability and feedback	1A, 4A, 4C, 5C	C
Measures, queries and communication	1A, 3D, 4A, 5C, 7A	C/E
Branching, review and repository maintenance	4A, 4B, 5C	C/D
Package distribution and release identifiers	4D	C/E
Tests, coverage and quality controls	3B, 4C, 6C	C
Pipeline authorship, execution and reuse	4B, 4C, 5C, 5D, 6D	C/E
Release patterns, recovery and database sequencing	3B, 3C, 5B	C/D

AZ-400 area	Day / Block	Practice
Infrastructure and machine configuration	5A, 6A	C/E/D
Pipeline reliability, retention and migration	5D, 6D	C/D
Automation identities and project access	1A, 1C, 3C, 4A, 4B, 5C	C
Confidential configuration and files	3C, 4B, 6C	C/E
Source, dependency, credential and image checks	6C	C/E
Telemetry collection and feedback	3D, 7A	C/E/D
Performance interpretation and tracing	3D, 7A	C

Explicit practical gaps if you follow only CORE: licence-backed identity features, identity-based SMB, custom domain ownership, dedicated Bastion, full two-node load balancing, some VM/scale-set moves, container hosts, paid security integrations, machine policy enforcement, Deployment Environments, backup/restore and regional recovery. Core alone is not equivalent to having practised all AZ-104 operations. The extension steps are part of this same guide and project; complete the ones your account permits, and keep the rest labelled.

Additional explained alternatives include git-fat/Scalar, classic pipeline migration, real database change design, ring/canary/A-B releases, Automation State Configuration and AKS Container Insights. They are not presented as completed live deployments. This is deliberate coverage honesty, not a requirement to find random outside projects.

Troubleshooting: find the failing boundary

Symptom	Check in order
Subscription missing	Profile directory > subscription filter > account identity > subscription access. Compare portal with az account show.
AuthorizationFailed	Correct subscription/resource scope > actual principal > required management role > deny assignment/policy > propagation delay.
Storage 403	Correct auth method > Blob/Files data role > expiry/permissions of SAS > firewall/public access > private DNS and routing.
Quota/SKU error	Region > VM family/total vCPU quota > subscription offer restrictions > current capacity > allowed alternative.
Policy denied creation	Read the policy assignment and failing property in the deployment error. Correct the property or use an authorized scope; do not delete organization policy.
App 502/503 or startup timeout	Deployment/build log > ZIP root > requirements installation > runtime version > gunicorn app:app > listening on 0.0.0.0:8000.
Secret remains a reference string	Production versus staging identity > correct Key Vault Secrets User role > versionless URI > vault network access > reference refresh.
Git push denied	Exact org/project/repo URL > Git permission > token scope/expiry > branch policy > remote history. Use a PR when main is protected.
Pipeline YAML error	Indentation and spaces > correct nesting (stages/jobs/steps) > template path/case > variable name > task inputs.
Pipeline waits forever	Agent parallel-job entitlement > available pool agent > pending resource authorization > approval/check timeout.
PR does not build	Azure Repos branch Build validation policy, not YAML pr. For GitHub inspect workflow event and target branch.
Federated login denied	Tenant/client ID > exact issuer/subject/audience > GitHub environment versus branch entity > service connection state > Azure role assignment.
Image pull fails	Registry hostname/tag/digest > runtime identity attachment > registry role/mode > network path > architecture compatibility.
VM cannot reach Internet	Explicit outbound method > NSG outbound rules > effective route > DNS > service endpoint/firewall constraints.
Private endpoint seems public	Client is actually in linked network > zone name > VNet link > expected DNS record > endpoint approval/subresource.
Load balancer backend unhealthy	Application listening > correct probe/port/path > NSG probe allowance > backend NIC membership > route and client test location.
KQL table not found	Selected workspace > instrumentation/diagnostic setting/DCR destination > generated traffic > ingestion delay > actual table schema.

Symptom	Check in order
Alert does not send email	Condition fired? > time window/aggregation > action group attached/tested > processing-rule suppression > inbox/junk.
Vault cannot be deleted	Protected items > replica/test cleanup > soft-deleted items > immutability/retention > locks > remaining dependencies.

Plain-English glossary

Word	Meaning in this lab
Tenant	An identity directory containing users, groups and application identities.
Subscription	A billing and access boundary that contains Azure resources.
Resource group	A container for resources that you manage together.
Region / zone	A geographic Azure area / a separate data-center location within a supported region.
Principal	The user, group or application identity receiving permission.
Scope	The place where a role or rule applies, such as a subscription or one resource.
RBAC	Role-based access control: assign permitted actions to an identity at a scope.
Control plane / data plane	Manage the service itself / use the data inside the service.
SKU	A product tier or size, with associated capability and cost.
Endpoint	A network address where a service can be reached.
DNS	The system that translates a name to a network address.
NSG / route	A traffic allow/deny rule set / a next-hop decision for a packet.
Repository / branch	Versioned project history / a line of development within it.
Commit / tag	A recorded snapshot / a named label pointing to a commit.
Pull request	A request to review and merge a proposed change.
CI / CD	Automatically validate changes / automate delivery or deployment of validated changes.
Agent / runner	The machine/process that executes pipeline jobs.
Stage / job / step	A major pipeline phase / unit assigned to an agent / individual task or command.
Artifact / package	A produced file set / a reusable component distributed with a version.
Environment / slot	A named release target/check boundary / a separate App Service deployment endpoint.
Managed identity	An Azure-managed identity that can get tokens without an application password.
OIDC / federation	Trusting a signed identity token from another system to obtain short-lived access.
PAT	A personal access token; a scoped credential with expiry that must stay private.

Word	Meaning in this lab
Secret / certificate / key	Confidential value / identity or TLS proof with key material / cryptographic or access material.
Idempotent	Repeating the operation keeps the intended state instead of duplicating or breaking it.
Drift	Actual settings have moved away from the declared configuration.
Metric / log / trace	A number over time / an event record / the path and timing of a request.
RPO / RTO	Acceptable recent data loss / acceptable time until service recovery.
Failover / restore	Move service to a recovery location / retrieve an earlier saved copy.
Rollback / roll forward	Return to a known-good release / deploy a corrective new release.

Your final learning goal: explain the reason, perform the action, verify the result and recover from a small failure. Keep all four together as you practise.

Source files: everything needed to run the project

The recommended route is OneLab-GitHub-Project.zip, supplied separately, downloadable from the HTML and attached to this PDF. It includes these runtime and workflow files plus all seven illustrated daily notes. Extract it at the repository root as shown on Day 3.

For a manual recovery copy, create the parent folder, run nano PATH, paste one complete file, press Ctrl+O then Enter to save, and Ctrl+X to exit. YAML indentation uses spaces. Do not type page headers into files. The source is training software: /fail is intentionally broken, and the migration uses a local SQLite database.

.dockerignore

USE IN: FILE: .DOCKERIGNORE · COPY ONLY THE CODE

```
.git
.venv
lab.env
*.zip
*.db
__pycache__
.pytest_cache

notes.txt
private-evidence
docs
*.sarif
training-certificate.txt
```

What it means: Save as .dockerignore at the project root or the shown subfolder. The ZIP already includes this file.

.github/dependabot.yml

USE IN: FILE: .GITHUB/DEPENDABOT.YML · COPY ONLY THE CODE

```
version: 2
updates:
- package-ecosystem: pip
  directory: /
  schedule:
    interval: weekly
- package-ecosystem: github-actions
  directory: /
  schedule:
    interval: weekly
- package-ecosystem: docker
  directory: /
  schedule:
    interval: weekly
```

What it means: Save as .github/dependabot.yml at the project root or the shown subfolder. The ZIP already includes this file.

.github/workflows/ci.yml

USE IN: FILE: .GITHUB/WORKFLOWS/CI.YML · COPY ONLY THE CODE

```
name: OneLab CI
on:
  push:
    branches: [main]
  pull_request:
    branches: [main]
  workflow_dispatch:
permissions:
  contents: read
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v7
      - uses: actions/setup-python@v7
        with:
          python-version: '3.12'
      - run: pip install -r requirements-dev.txt
      - run: pytest --junitxml=results.xml --cov=app --cov-report=xml
      - uses: actions/upload-artifact@v7
        if: always()
        with:
          name: test-results-${{ github.run_id }}
          path: |
            results.xml
```

Code continues below; keep the lines in this order.

```
coverage.xml
```

What it means: Save as .github/workflows/ci.yml at the project root or the shown subfolder. The ZIP already includes this file.

.github/workflows/codeql.yml

USE IN: FILE: .GITHUB/WORKFLOWS/CODEQL.YML · COPY ONLY THE CODE

```
name: CodeQL in a container
on:
  push:
    branches: [main]
  pull_request:
    branches: [main]
  workflow_dispatch:
permissions:
  contents: read
jobs:
  analyze:
    runs-on: ubuntu-latest
    container:
      image: python:3.12-bookworm
    permissions:
      contents: read
      security-events: write
      actions: read
    steps:
      - uses: actions/checkout@v7
      - uses: github/codeql-action/init@v4
        with:
          languages: python
          build-mode: none
      - uses: github/codeql-action/analyze@v4
```

What it means: Save as `.github/workflows/codeql.yml` at the project root or the shown subfolder. The ZIP already includes this file.

.github/workflows/deploy.yml**USE IN: FILE: .GITHUB/WORKFLOWS/DEPLOY.YML · COPY ONLY THE CODE**

```
name: OneLab manual Azure deployment
on: workflow_dispatch
permissions:
  contents: read
  id-token: write
jobs:
  deploy:
    runs-on: ubuntu-latest
    environment: github-staging
    steps:
      - uses: actions/checkout@v7
      - uses: actions/setup-python@v7
        with:
          python-version: '3.12'
      - run: |
          pip install -r requirements-dev.txt
          pytest
          python -m zipfile -c app.zip app.py requirements.txt
      - uses: azure/login@v3
        with:
          client-id: ${vars.AZURE_CLIENT_ID}
          tenant-id: ${vars.AZURE_TENANT_ID}
          subscription-id: ${vars.AZURE_SUBSCRIPTION_ID}
      - name: Deploy only to the staging slot
        env:
```

Code continues below; keep the lines in this order.

```

RG: ${vars.AZURE_RESOURCE_GROUP}
APP: ${vars.AZURE_APP_NAME}
run: |
  az webapp deploy -g "$RG" -n "$APP" --slot staging \
    --src-path app.zip --type zip
  HOST=$(az webapp show -g "$RG" -n "$APP" --slot staging \
    --query defaultHostName -o tsv)
  bash scripts/smoke.sh "https://$HOST"

```

What it means: Save as `.github/workflows/deploy.yml` at the project root or the shown subfolder. The ZIP already includes this file.

.github/workflows/image-scan.yml

USE IN: FILE: `.GITHUB/WORKFLOWS/IMAGE-SCAN.YML` · COPY ONLY THE CODE

```

name: OneLab image scan
on:
  push:
    branches: [main]
  pull_request:
    branches: [main]
  workflow_dispatch:
permissions:
  contents: read
jobs:
  scan:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v7
      - run: docker build -t onelab:scan .
      - uses: anchore/scan-action@v7
        id: imagecheck
        with:
          image: onelab:scan
          fail-build: true
          severity-cutoff: high
          output-format: sarif
      - uses: actions/upload-artifact@v7
        if: always()
        with:

```

Code continues below; keep the lines in this order.

```

name: image-scan-${github.run_id}
path: ${steps.imagecheck.outputs.sarif}

```

What it means: Save as `.github/workflows/image-scan.yml` at the project root or the shown subfolder. The ZIP already includes this file.

.gitignore

USE IN: FILE: .GITIGNORE · COPY ONLY THE CODE

```
lab.env
.venv/
__pycache__/
.pytest_cache/
.coverage
coverage.xml
results.xml
*.zip
*.db
*.pem
*.key
*.pfx
reports/
site-packages/

notes.txt
private-evidence/
*.sarif
training-certificate.txt
```

What it means: Save as .gitignore at the project root or the shown subfolder. The ZIP already includes this file.

Dockerfile

USE IN: FILE: DOCKERFILE · COPY ONLY THE CODE

```
FROM python:3.12-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app.py .
RUN useradd --uid 10001 --create-home appuser
USER 10001
EXPOSE 8000
CMD ["gunicorn", "--bind", "0.0.0.0:8000", "app:app"]
```

What it means: Save as Dockerfile at the project root or the shown subfolder. The ZIP already includes this file.

app.py

USE IN: FILE: APP.PY · COPY ONLY THE CODE

```
"""OneLab training application. Never put passwords in this file."""
import os
from flask import Flask, jsonify

if os.getenv("APPLICATIONINSIGHTS_CONNECTION_STRING"):
    from azure.monitor.opentelemetry import configure_azure_monitor
    configure_azure_monitor()

app = Flask(__name__)

@app.get("/")
def home():
    return jsonify(service="OneLab", version="1.0.0",
                  new_banner=os.getenv("NEW_BANNER", "false").lower() == "true")

@app.get("/health")
def health():
    return jsonify(status="healthy"), 200

@app.get("/secret-status")
def secret_status():
    value = os.getenv("DEMO_MESSAGE", "")
    resolved = bool(value) and not value.startswith("@Microsoft.KeyVault")
    return jsonify(secret_resolved=resolved), 200 if resolved else 503
```

Code continues below; keep the lines in this order.

```
@app.get("/dependency")
def dependency():
    import requests
    r = requests.get("https://example.com", timeout=5)
    return jsonify(upstream_status=r.status_code)

@app.get("/fail")
def fail():
    # A controlled error for the monitoring exercise; no real customer data.
    return jsonify(error="Intentional lab error"), 500

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=8000)
```

What it means: Save as app.py at the project root or the shown subfolder. The ZIP already includes this file.

azure-pipelines.yml

USE IN: FILE: AZURE-PIPELINES.YML · COPY ONLY THE CODE

```
trigger:
  branches:
    include: [main]
pr: none
pool:
  vmImage: ubuntu-24.04
variables:
- group: onelab-settings
stages:
- stage: Build
  jobs:
  - job: TestAndPackage
    steps:
    - checkout: self
      fetchDepth: 0
    - template: pipelines/templates/test.yml
    - bash: |
        set -euo pipefail
        python -m zipfile -c app.zip app.py requirements.txt
        git log -10 --pretty=format:'%h %s' > release-notes.txt
      displayName: Package only runtime files
    - publish: app.zip
      artifact: web-$(Build.BuildId)
    - publish: release-notes.txt
      artifact: release-notes
```

Code continues below; keep the lines in this order.

```
- stage: Staging
  dependsOn: Build
  condition: and(succeeded(), eq(variables['Build.SourceBranch'], 'refs/heads/main'))
  jobs:
  - deployment: StageWeb
    environment: onelab-staging
    strategy:
      runOnce:
        deploy:
          steps:
          - checkout: self
          - download: current
            artifact: web-$(Build.BuildId)
          - task: AzureCLI@2
            inputs:
              azureSubscription: sc-onelab
              scriptType: bash
              scriptLocation: inlineScript
              inlineScript: |
                set -euo pipefail
                test "$(az account show --query id -o tsv)" = "$EXPECTED_SUB"
                az webapp deploy -g "$RG" -n "$APP" --slot staging \
                  --src-path "$PACKAGE" --type zip
                HOST=$(az webapp show -g "$RG" -n "$APP" \
                  --slot staging --query defaultHostName -o tsv)
```

Code continues below; keep the lines in this order.

```

        bash scripts/smoke.sh "https://$HOST"
    env:
        EXPECTED_SUB: $(subscriptionId)
        RG: $(resourceGroup)
        APP: $(appName)
        PACKAGE: $(Pipeline.Workspace)/web-$(Build.BuildId)/app.zip
- stage: Production
  dependsOn: Staging
  jobs:
  - deployment: PromoteWeb
    environment: onelab-production
    strategy:
      runOnce:
        deploy:
          steps:
            - checkout: self
            - task: AzureCLI@2
              inputs:
                azureSubscription: sc-onelab
                scriptType: bash
                scriptLocation: inlineScript
                inlineScript: |
                  set -euo pipefail
                  az webapp deployment slot swap -g "$RG" -n "$APP" \
                    --slot staging --target-slot production

```

Code continues below; keep the lines in this order.

```

        HOST=$(az webapp show -g "$RG" -n "$APP" \
          --query defaultHostName -o tsv)
        bash scripts/smoke.sh "https://$HOST"
    env:
        RG: $(resourceGroup)
        APP: $(appName)

```

What it means: Save as azure-pipelines.yml at the project root or the shown subfolder. The ZIP already includes this file.

ci-only.yml

USE IN: FILE: CI-ONLY.YML · COPY ONLY THE CODE

```

trigger: none
pr:
  branches:
    include: [main]
pool:
  vmImage: ubuntu-24.04
steps:
- checkout: self
- template: pipelines/templates/test.yml

```

What it means: Save as ci-only.yml at the project root or the shown subfolder. The ZIP already includes this file.

infra/storage.bicep

USE IN: FILE: INFRA/STORAGE.BICEP · COPY ONLY THE CODE

```
param storageName string
param location string = resourceGroup().location
@allowed(['Standard_LRS', 'Standard_ZRS'])
param storageSku string = 'Standard_LRS'

resource archive 'Microsoft.Storage/storageAccounts@2023-05-01' = {
  name: storageName
  location: location
  kind: 'StorageV2'
  sku: { name: storageSku }
  tags: { Project: 'OneLab', Purpose: 'IaC-exercise' }
  properties: {
    supportsHttpsTrafficOnly: true
    minimumTlsVersion: 'TLS1_2'
    allowBlobPublicAccess: false
  }
}

output storageId string = archive.id
```

What it means: Save as infra/storage.bicep at the project root or the shown subfolder. The ZIP already includes this file.

pipelines/iac.yml

USE IN: FILE: PIPELINES/IAC.YML · COPY ONLY THE CODE

```
trigger: none
pr: none
parameters:
- name: deploy
  type: boolean
  default: false
pool:
  vmImage: ubuntu-24.04
variables:
- group: onelab-settings
steps:
- checkout: self
- task: AzureCLI@2
  inputs:
    azureSubscription: sc-onelab
    scriptType: bash
    scriptLocation: inlineScript
    inlineScript: |
      set -euo pipefail
      az bicep build --file infra/storage.bicep
      az deployment group validate -g "$RG" \
        --template-file infra/storage.bicep \
        --parameters storageName="$STORAGE_NAME"
      az deployment group what-if -g "$RG" \
        --template-file infra/storage.bicep \
```

Code continues below; keep the lines in this order.

```

    --parameters storageName="$STORAGE_NAME"
  if [ "${DO_DEPLOY,,}" = true ]; then
    az deployment group create -g "$RG" -n pipeline-storage \
      --template-file infra/storage.bicep \
      --parameters storageName="$STORAGE_NAME"
  fi
env:
  RG: $(resourceGroup)
  STORAGE_NAME: $(iacStorageName)
  DO_DEPLOY: ${ parameters.deploy }

```

What it means: Save as pipelines/iac.yml at the project root or the shown subfolder. The ZIP already includes this file.

pipelines/packages.yml

USE IN: FILE: PIPELINES/PACKAGES.YML · COPY ONLY THE CODE

```

trigger: none
pr: none
pool:
  vmImage: ubuntu-24.04
steps:
- checkout: self
- bash: |
    mkdir -p "$(Build.ArtifactStagingDirectory)/package"
    cp app.py requirements.txt "$(Build.ArtifactStagingDirectory)/package/"
  displayName: Collect only the release files
- task: UniversalPackages@0
  inputs:
    command: publish
    publishDirectory: '$(Build.ArtifactStagingDirectory)/package'
    feedsToUsePublish: internal
    vstsFeedPublish: OneLab/onelab
    vstsFeedPackagePublish: onelab-web
    versionOption: custom
    versionPublish: '1.0.$(Build.BuildId)'
    packagePublishDescription: OneLab teaching application

```

What it means: Save as pipelines/packages.yml at the project root or the shown subfolder. The ZIP already includes this file.

pipelines/templates/test.yml

USE IN: FILE: PIPELINES/TEMPLATES/TEST.YML · COPY ONLY THE CODE

```
steps:
- task: UsePythonVersion@0
  inputs:
    versionSpec: '3.12'
- bash: |
  set -euo pipefail
  python -m pip install -r requirements-dev.txt
  python -m pytest --junitxml=results.xml --cov=app --cov-report=xml
  python scripts/migrate.py
  python scripts/migrate.py
  displayName: Test the application and repeat the migration
- task: PublishTestResults@2
  condition: succeededOrFailed()
  inputs:
    testResultsFormat: JUnit
    testResultsFiles: results.xml
    failTaskOnFailedTests: true
- task: PublishCodeCoverageResults@2
  condition: succeededOrFailed()
  inputs:
    summaryFileLocation: '$(System.DefaultWorkingDirectory)/coverage.xml'
```

What it means: Save as pipelines/templates/test.yml at the project root or the shown subfolder. The ZIP already includes this file.

requirements-dev.txt

USE IN: FILE: REQUIREMENTS-DEV.TXT · COPY ONLY THE CODE

```
-r requirements.txt
pytest==8.4.2
pytest-cov==7.0.0
```

What it means: Save as requirements-dev.txt at the project root or the shown subfolder. The ZIP already includes this file.

requirements.txt

USE IN: FILE: REQUIREMENTS.TXT · COPY ONLY THE CODE

```
Flask==3.1.2
unicorn==23.0.0
azure-monitor-opentelemetry==1.8.1
requests==2.32.5
```

What it means: Save as requirements.txt at the project root or the shown subfolder. The ZIP already includes this file.

scripts/migrate.py

USE IN: FILE: SCRIPTS/MIGRATE.PY · COPY ONLY THE CODE

```

"""An idempotent, local SQLite migration demonstration; not a shared cloud DB."""
import sqlite3
import sys

path = sys.argv[1] if len(sys.argv) > 1 else 'lab.db'
with sqlite3.connect(path) as db:
    db.execute('CREATE TABLE IF NOT EXISTS tickets (id INTEGER PRIMARY KEY)')
    columns = {r[1] for r in db.execute('PRAGMA table_info(tickets)')}
    if 'status' not in columns:
        db.execute("ALTER TABLE tickets ADD COLUMN status TEXT DEFAULT 'open'")
    print([r[1] for r in db.execute('PRAGMA table_info(tickets)')])

```

What it means: Save as scripts/migrate.py at the project root or the shown subfolder. The ZIP already includes this file.

scripts/smoke.sh**USE IN: FILE: SCRIPTS/SMOKE.SH · COPY ONLY THE CODE**

```

#!/usr/bin/env bash
set -euo pipefail
: "${1:?Pass the complete HTTPS app URL}"
curl --fail --silent --show-error --retry 12 --retry-delay 10 \
  --retry-all-errors "$1/health"

```

What it means: Save as scripts/smoke.sh at the project root or the shown subfolder. The ZIP already includes this file.

test_app.py**USE IN: FILE: TEST_APP.PY · COPY ONLY THE CODE**

```

from app import app

def test_home():
    assert app.test_client().get("/").json["service"] == "OneLab"

def test_health():
    assert app.test_client().get("/health").status_code == 200

def test_flag(monkeypatch):
    monkeypatch.setenv("NEW_BANNER", "true")
    assert app.test_client().get("/").json["new_banner"] is True

def test_secret_missing(monkeypatch):
    monkeypatch.delenv("DEMO_MESSAGE", raising=False)
    assert app.test_client().get("/secret-status").status_code == 503

def test_secret_reference(monkeypatch):
    monkeypatch.setenv("DEMO_MESSAGE", "@Microsoft.KeyVault(SecretUri=example)")
    assert app.test_client().get("/secret-status").status_code == 503

def test_secret_private(monkeypatch):
    monkeypatch.setenv("DEMO_MESSAGE", "dummy-lab-value")
    r = app.test_client().get("/secret-status")
    assert r.status_code == 200
    assert "dummy-lab-value" not in r.get_data(as_text=True)

```

Code continues below; keep the lines in this order.


```
def test_failure():  
    assert app.test_client().get("/fail").status_code == 500  
  
def test_missing_page():  
    assert app.test_client().get("/missing").status_code == 404
```

What it means: Save as test_app.py at the project root or the shown subfolder. The ZIP already includes this file.

Publishing the prepared GitHub repository

The repository ZIP is ready to upload; it has not been published to your account. Day 5, Block C illustrates the normal lab route: Azure Repos is origin and GitHub is a deliberate second remote.

For an early standalone upload, the ZIP includes docs/publish-to-github.md with private-repository creation, token scopes, Git commands and verification. Use a fresh extraction for this early upload. For the Azure Repos-first weekly lab, keep that early-upload folder separate so its origin remote cannot silently point at the wrong service.

USE IN: BASH · FRESH EXTRACTION ONLY · COPY ONLY THE CODE

```
git init -b main
git config user.name "Your Name"
git config user.email "your-email@example.com"
git config credential.helper ""
git status --short
git add .
git diff --cached --stat
git commit -m "Add OneLab visual practice project"
git remote add origin 'https://github.com/REPLACE_OWNER/onelab.git'
git push -u origin main
```

What it means: Create an empty private GitHub repository named onelab first, with no extra README. Replace author and URL values. Use a short-lived fine-grained token for this repository with Contents and Workflows Read and write. Enter it at the invisible password prompt, never in the URL. Follow the illustrated account steps in Day 5.

Validation and source notes

Local verification covers the supplied application, scripts and generated artifacts. Eight baseline tests, two migration runs, Bash syntax and YAML structure are checked; the original Bicep template was compiled, and packaging and guide navigation are checked. No Azure resources or online repository were created to author this edition. Tenant permissions, deployment jobs, billing, notifications and restores require your own recorded results.

All 108 original exercises remain in the same seven-day plan. This edition adds one action-specific visual for each of the 583 action groups, places runnable code beside the action that needs it, adds missing verification instructions, and corrects naming, Git remote and command sequencing errors.

The vector screens are original illustrated views. They make navigation labels and lab fields easy to see; they do not assert exact pixel placement in every portal rollout. The 24 Microsoft reference images were extracted unchanged from the supplied illustrated HTML. Their sample values are not the OneLab values.

Microsoft exam scope checked 22 September 2026: AZ-104 effective 17 April 2026; AZ-400 effective 27 July 2026. The following primary references verify product behavior. You do not need to visit them to get missing project files or the normal lab steps.

R01 · AZ-104 official study guide

<https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/az-104>

R02 · AZ-400 official study guide

<https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/az-400>

R03 · Create and manage Azure budgets

<https://learn.microsoft.com/en-us/azure/cost-management-billing/costs/tutorial-acm-create-budgets>

R04 · Cloud Shell Bash quickstart

<https://learn.microsoft.com/en-us/azure/cloud-shell/quickstart>

R05 · Licence assignment through Microsoft 365 admin center

<https://learn.microsoft.com/en-us/entra/identity/users/licensing-admin-center>

R06 · Enable self-service password reset

<https://learn.microsoft.com/en-us/entra/identity/authentication/tutorial-enable-sspr>

R07 · Azure Policy overview

<https://learn.microsoft.com/en-us/azure/governance/policy/overview>

R08 · Blob versioning and recovery

<https://learn.microsoft.com/en-us/azure/storage/blobs/versioning-overview>

R09 · Shared access signatures

<https://learn.microsoft.com/en-us/azure/storage/common/storage-sas-overview>

R10 · AzCopy getting started

<https://learn.microsoft.com/en-us/azure/storage/common/storage-use-azcopy-v10>

R11 · Azure Files and Entra Kerberos prerequisites

<https://learn.microsoft.com/en-us/azure/storage/files/storage-files-identity-auth-hybrid-identities-enable>

R12 · Configure Blob object replication

<https://learn.microsoft.com/en-us/azure/storage/blobs/object-replication-configure>

R13 · Virtual network concepts

<https://learn.microsoft.com/en-us/azure/virtual-network/virtual-networks-overview>

R14 · Private DNS zones

<https://learn.microsoft.com/en-us/azure/dns/private-dns-overview>

R15 · Explicit outbound networking changes

<https://learn.microsoft.com/en-us/azure/virtual-network/ip-services/default-outbound-access>

R16 · Azure VM availability options

<https://learn.microsoft.com/en-us/azure/virtual-machines/availability>

R17 · Storage virtual-network access rules

<https://learn.microsoft.com/en-us/azure/storage/common/storage-network-security>

R18 · Private endpoints for Azure Storage

<https://learn.microsoft.com/en-us/azure/storage/common/storage-private-endpoints>

R19 · Virtual network routing

<https://learn.microsoft.com/en-us/azure/virtual-network/virtual-networks-udr-overview>

R20 · Azure Load Balancer overview

<https://learn.microsoft.com/en-us/azure/load-balancer/load-balancer-overview>

R21 · Python App Service configuration

<https://learn.microsoft.com/en-us/azure/app-service/configure-language-python>

R22 · ZIP deployment to App Service

<https://learn.microsoft.com/en-us/azure/app-service/deploy-zip>

R23 · App Service slots and swapping

<https://learn.microsoft.com/en-us/azure/app-service/deploy-staging-slots>

R24 · App Service networking

<https://learn.microsoft.com/en-us/azure/app-service/networking-features>

R25 · Key Vault Azure RBAC

<https://learn.microsoft.com/en-us/azure/key-vault/general/rbac-guide>

R26 · App Service Key Vault references

<https://learn.microsoft.com/en-us/azure/app-service/app-service-key-vault-references>

R27 · Bicep deployment and what-if

<https://learn.microsoft.com/en-us/azure/azure-resource-manager/bicep/deploy-what-if>

R28 · ACR Tasks quick build

<https://learn.microsoft.com/en-us/azure/container-registry/container-registry-tutorial-quick-task>

R29 · ACI image pull through managed identity

<https://learn.microsoft.com/en-us/azure/container-instances/using-azure-container-registry-mi>

R30 · Container Apps managed identity image pull

<https://learn.microsoft.com/en-us/azure/container-apps/managed-identity-image-pull>

R31 · Azure Pipelines parallel-job entitlement

<https://learn.microsoft.com/en-us/azure/devops/pipelines/licensing/concurrent-jobs?view=azure-devops>

R32 · Lead time and cycle time widgets

<https://learn.microsoft.com/en-us/azure/devops/report/dashboards/cycle-time-and-lead-time?view=azure-devops>

R33 · Azure DevOps PATs

<https://learn.microsoft.com/en-us/azure/devops/organizations/accounts/use-personal-access-tokens-to-authenticate?view=azure-devops>

R34 · Azure Repos pipeline triggers

<https://learn.microsoft.com/en-us/azure/devops/pipelines/repos/azure-repos-git?view=azure-devops>

R35 · Azure Resource Manager service connections

<https://learn.microsoft.com/en-us/azure/devops/pipelines/library/connect-to-azure?view=azure-devops>

R36 · Azure Pipelines approvals and checks

<https://learn.microsoft.com/en-us/azure/devops/pipelines/process/approvals?view=azure-devops>

R37 · Azure Artifacts Universal Packages

<https://learn.microsoft.com/en-us/azure/devops/pipelines/artifacts/universal-packages?view=azure-devops>

R38 · Azure App Configuration feature CLI

<https://learn.microsoft.com/en-us/cli/azure/appconfig/feature?view=azure-cli-latest>

R39 · GitHub workflow syntax

<https://docs.github.com/en/actions/reference/workflows-and-actions/workflow-syntax>

R40 · GitHub OIDC to Azure

<https://docs.github.com/en/actions/how-to-secure-your-work/security-harden-deployments/oidc-in-azure>

R41 · Connect Azure Boards to GitHub

<https://learn.microsoft.com/en-us/azure/devops/boards/github/connect-to-github?view=azure-devops>

R42 · Azure DevOps Teams integration and events

<https://learn.microsoft.com/en-us/azure/devops/service-hooks/services/teams?view=azure-devops>

R43 · Azure Pipelines from GitHub repositories

<https://learn.microsoft.com/en-us/azure/devops/pipelines/repos/github?view=azure-devops>

R44 · Dependabot configuration options

<https://docs.github.com/en/code-security/reference/supply-chain-security/dependabot-options-reference>

R45 · CodeQL in a container

<https://docs.github.com/en/code-security/tutorials/customize-code-scanning/run-in-a-container>

R46 · Configure Azure DevOps security products

<https://learn.microsoft.com/en-us/azure/devops/repos/security/configure-github-advanced-security-features?view=azure-devops>

R47 · Defender for Cloud DevOps security

<https://learn.microsoft.com/en-us/azure/defender-for-cloud/defender-for-devops-introduction>

R48 · Azure Pipelines secure files

<https://learn.microsoft.com/en-us/azure/devops/pipelines/library/secure-files?view=azure-devops>

R49 · Automation State Configuration retirement and Machine Configuration

<https://learn.microsoft.com/en-us/azure/governance/machine-configuration/whats-new/migrating-from-azure-automation>

R50 · Deployment Environments setup and retirement notice

<https://learn.microsoft.com/en-us/azure/deployment-environments/quickstart-create-and-configure-devcenter>

R51 · Self-hosted Linux Azure Pipelines agent

<https://learn.microsoft.com/en-us/azure/devops/pipelines/agents/linux-agent?view=azure-devops>

R52 · Pipeline retention

<https://learn.microsoft.com/en-us/azure/devops/pipelines/policies/retention?view=azure-devops>

R53 · Azure Monitor OpenTelemetry configuration

<https://learn.microsoft.com/en-us/azure/azure-monitor/app/opentelemetry-configuration>

R54 · Azure VM monitoring and DCRs

<https://learn.microsoft.com/en-us/azure/azure-monitor/vm/vm-enable-monitoring>

R55 · Alert processing rules

<https://learn.microsoft.com/en-us/azure/azure-monitor/alerts/alerts-processing-rules>

R56 · Network Watcher Connection monitor

<https://learn.microsoft.com/en-us/azure/network-watcher/connection-monitor-overview>

R57 · VM backup in Azure Portal

<https://learn.microsoft.com/en-us/azure/backup/quick-backup-vm-portal>

R58 · Restore a VM through Azure Portal

<https://learn.microsoft.com/en-us/azure/backup/backup-azure-arm-restore-vms>

R59 · Back up managed disks

<https://learn.microsoft.com/en-us/azure/backup/backup-managed-disks>

R60 · Enable Azure-to-Azure recovery

<https://learn.microsoft.com/en-us/azure/site-recovery/azure-to-azure-tutorial-enable-replication>

R61 · Test failover and cleanup

<https://learn.microsoft.com/en-us/azure/site-recovery/azure-to-azure-tutorial-dr-drill>

R62 · Actual failover and reprotection

<https://learn.microsoft.com/en-us/azure/site-recovery/azure-to-azure-tutorial-failover-failback>

R63 · Delete Recovery Services vaults and dependencies

<https://learn.microsoft.com/en-us/azure/backup/backup-azure-delete-vault>

R64 · Official Anchore container-scanning action

<https://github.com/anchore/scan-action>

R65 · Official GitHub Python setup action

<https://github.com/actions/setup-python>

R66 · Official GitHub artifact upload action

<https://github.com/actions/upload-artifact>

R67 · Official Azure login action

<https://github.com/Azure/login>

R68 · Harmless Azure Policy audit assignment

<https://learn.microsoft.com/en-us/azure/governance/policy/concepts/recommended-policies>